FUNDAMENTALS OF PHYSICS

DARLINGTON NAOSA

The author

Darlington Naosa	*Head of Department - Natural Sciences*
Address	*Namushakende Secondary School* *P.O Box 910197* *Mongu* *Western Province* *Zambia*
Email	*naosadarlington@yahoo.com*
Contact numbers	*+260955997821* *+260966997821* *+260977997821*

CONTENT **PAGE**

PREFACE

I love to write. I get real pleasure from taking a 'perceived complicated subject' turning it around until I see it clearly, and then explaining it in simple words. I write to explain physics to students today the way I wish it had been explained to me years ago.

I have made every effort to make this book as effective, clear, and readable as possible; to show the beauty and logic of physics; and to make physics enjoyable to learn. The book adopts an investigative approach to the learning and teaching of physics. It encourages a learner centered approach and it is gender sensitive. 'Personally, I am always ready to learn, although I do not always like being taught; but I believe that 'to teach is to learn twice'.

This book provides a modern, comprehensive and systematic treatment of the core physics required at school certificate or ordinary level. It is structured and written in a manner which facilitates easy access to the theory and knowledge of physics needed at grades 10 – 12 levels. It is also designed to stimulate and sustain students' interest in the subject. It further forms the basis for college level physics for those who would like to peruse physics as a teaching subject at teacher education level. The special feature of this chemistry book is that besides giving the fundamental principles of physics, stress has been laid on the applied part and the use of physics in daily life. Sample questions are given to guide students through their study. At every stage the student is encouraged to think and understand, rather than simply memorizing the facts. Considering that most syllabi are constructed on the assumption that practical activities form an integral part of the course, a number of experiments have been included throughout the book with instructions kept very brief for easy understanding. The students will find this book very easy to follow and useful to the demands of the current physics examinations.

To my physics students, if you can't swim, at least try to float. But if you can't swim and you can't float, then there is very little I can do to help you. Remember, you cannot fail but you can only discover ways which cannot work. Believe with all of your heart that you will do what you were made to do. There is a difference between school and life. In school you are taught a lesson and then you are given a test. In life, you are given a test that teaches you a lesson. In addition, life and time are the world's best teachers. Life teaches you the use of time and time teaches you the value of life. Therefore, eat your food as your medicines. Otherwise you have to eat medicines as your food.

To my fellow teachers, there is no system in the world or any school in the country that is better than its teachers. Teachers are the lifeblood of the success of schools. Teachers are not common people and common people are not teachers. Please don't choose to become a teacher until you are worth it. Teaching is a commitment to make a difference in a life of a learner. Being teacher does not only involve standing in front of students. It involves a selfless act of changing *less* to *more* and *nothing* to *something*. If you want to walk fast, walk alone. But if you want to walk far, walk together. If I have seen further than others, it is because I stood on the heads of the giants.

To the parents, teach the child to preserve the power of knowledge from generation to generation. Don't educate your children to be rich. Educate them to be happy. So when they grow up they will know the value of things not the price. Not everything that counts can be counted and not everything that is counted truly counts. Remember that our eyes are placed in front because it is more important to look ahead than to look back.

APPRECIATION OF SAFETY IN THE PHYSICS LABORATORY

Safety is the state of non-exposure to hazards or to danger. It can also be described as the state of being safe.

Safety rules in the physics laboratory

1. Enter a laboratory only when a teacher says so.
2. Always wear closed shoes. No one wearing open footwear such as slippers and sandals enters the laboratory. This is in order to reduce the chance of occurrence of foot injuries.
3. Wear protective clothes.
4. Do not run or play in the laboratory.
5. Do not perform any experiment without permission from the teacher, and always follow the instructions carefully. Avoid handling any unfamiliar equipment in the laboratory.
6. Do not drink, eat or taste anything in the laboratory except when allowed to by the teacher. The food might be contaminated with chemicals which are harmful to human beings. When you suspect poisoning, note the suspected poisoning agent and call your teacher immediately.
7. Always add acid to water and not water to acid. Never add water to concentrated acid as doing so may result into an accident since the little water coming into contact with the acid may boil immediately splashing the acid into your face.
8. Accidents and breakages must be immediately reported to the teacher.
9. Never point the mouth of a test tube containing a substance being heated towards another person or yourself.
10. Do not hold very hot objects with your hand. Hold them with a test tube holder, tongs or a piece of cloth or place them on a heat proof mat.
11. When smelling a substance, do not hold it very near the nose. Hold it about 20cm from the nose and with the hand wave the vapour towards the nose and sniff carefully.
12. Use specified or small amounts of substances in reactions to avoid waste and reactions which cannot be controlled.
13. Make sure you know the substances being used unless you are advised to use it as unknown
14. Any chemical accidentally taken into the mouth or spilt onto any part of the body should be washed off immediately with water and reported to the teacher. Seek medical attention.
15. Do not use broken glass-ware. Glassware should frequently be checked. Broken pieces of glassware should be put a vessel such as a bucket and kept securely for later disposal.
16. Do not bring flammable substances near a flame. If fires breaks out accidentally, quickly turn off the gas, electricity or water if necessary. Electrical installations in the laboratory should be checked for faults on a daily basis. This is in order to avoid the incidence of such accidents as fire resulting from a short circuit.
17. Wear eye protection when you are told to and keep it on until you are told to take it off when the practical is finished. Where a foreign matter enters the eye, flush with plenty of water. Use an eye wash bottle or fountain.
18. When you are told to use a Bunsen burner, make sure hair, cardigans, scarves, ties etc. are tied back or tacked in to keep them well away from the frame.
19. When you are working with liquids, always stand up and never sit. That way you can move out of the way easily if something spills.
20. Always put any waste solids in the correct liter bin and not in the sink.
21. Bottles should be never held by the neck.
22. Be careful that the name or label on the bottle is exactly the same as that of the chemical you require. Avoiding use of unlabeled chemicals. Any of such should be treated as potentially dangerous.
23. Before leaving the laboratory, clean the apparatus, work surface and your hands well. Nothing must be taken from the laboratory.
24. Make sure that no piece of apparatus is placed on the edge of a work bench. Apparatus that are not in use should be stored in the correct designated places. Those that are in use should be placed far from the bench edges
25. Gas taps should be kept closed at all times other than when gas burners are in use. It is also important to ensure that there are no leaking points in the gas pipes.

26. Avoiding overcrowding work benches with such things as bags and pieces of apparatus which are not in use.
27. When one suffers from burns, apply cold water. Call your teacher immediately.
28. When one has cuts and bruises, stop any bleeding by applying direct pressure. Cover cuts with a clean dressing. Call your teacher immediately. Due to possibility of infection, disposable gloves should be worn whenever there is a chance of contact with body fluids such as blood.
29. When one faints, leave the person laying down. Loosen any tight clothing and keep clowds away. Call your teacher immediately.
30. Any spills on skin, flush with large amounts of water or use safety shower. Call your teacher immediately.

Reasons why laboratory accidents may occur

- Lack of awareness
- Lack of control
- Lack of knowledge
- Lack of right attitude

UNIT 1: GENERAL PHYSICS

Physics is the branch of science which deals with the properties and interaction of matter and energy.

Properties of matter

The properties of matter are called physical quantities.

Physical quantities are measurable features or properties of objects.

International system of units

Short form: SI units

This is a system of units which is universally agreed to be used in measurements of quantities worldwide.

Types of physical quantities

There are two types of physical quantities:

- Base quantities
- Derived quantities

1. **Base quantities**

   These are quantities with only one SI unit. There are seven basic quantities in use in physics.

Base unit	Symbol for base unit	For measuring
Metre	m	Length
Kilogram	Kg	Mass
Second	s	Time
Ampere	A	Electric current
Kelvin	K	Temperature
Mole	mol	Amount of substance
Candela	cd	Luminous intensity

2. **Derived quantities**

   These are quantities which are expressed by combining two or more base units.

Derived quantity	SI unit	Symbol for derived units
Speed	Metre per second	m/s
Acceleration	Metre per second squared	m/s^2
Density	Kilogram per cubic metre	Kg/m^3
Force	Newton	Kgm/s^2
Energy	Joule	Kgm^2/s^2
Electricity	Coulomb	As

Prefixes

Sometimes a physical quantity is too big or too small to be conveniently expressed in SI units. Then some symbols are used as the prefixes instead of Zeros or many places. Prefixes are multiples or decimals of ten. The following table shows some prefixes.

Prefixes	Symbol	Exponent	Meaning
Pico	P	10^{-12}	1/1000000000000 (= 0.000000000001)
Nano	N	10^{-9}	1/1000000000 (= 0.000000001)
Micro	µ	10^{-6}	1/1000000 (= 0.000001)
Milli	M	10^{-3}	1/1000 (= 0.001)
Centi	C	10^{-2}	1/100 (= 0.01)
Deci	D	10^{-1}	1/10 (= 0.1)
Giga	G	10^{9}	1,000,000,000
kilo	K	10^{3}	1,000
Mega	M	10^{6}	1,000,000

Scientific notation

Scientific notation is also called standard form.

Scientific notation is a method of expressing a number in the form: a x 10^n, where $1 \leq a < 10$ and n is an integer. This is where numbers are expressed in the power of ten

1. Locate the Decimal Point
2. Move the decimal point to the **right** of the non-zero digit in the largest place
   The new number is now between 1 and 10
3. Multiply the new number by 10^{n} where ***n*** is the number of places you moved the decimal point

- Determine the sign on the exponent, *n*

- If the decimal point was moved left, *n* is positive (+**)**
- If the decimal point was moved right, *n* is negative (–**)**
- If the decimal point was not moved, *n* is zero (0**)**

Examples

1. Express the following in standard form
    (a) 3000000
    (b) 4200
    (c) 600
    (d) 0.0016
    (e) 0.235
    (f) 0.2001
    (g) 0.2000

Solution

(a) 3×10^{6}
(b) 4.2×10^{3}
(c) 6×10^{2}
(d) 1.6×10^{-3}
(e) 2.35×10^{-1}
(f) 2.001×10^{-1}
(g) 2×10^{-1}

Exercise

1. Write down the standard form of;
    (a) 6423
    (b) 5200
    (c) 60003
    (d) 0.03
    (e) 0.3002
    (f) 0.004010

Significant figures in measurements

Scientists report measurements in significant figures. The significant in a measurement include all the digits that can be known precisely plus a last digit that must be estimated.

The rules for determining which digits in a measurement are significant are as follows:

1. *Every non-zero digit in a recorded measurement is significant*.
   The measurements 24.7, 0.743, and 714 all have three significant figures.
2. *Zeros appearing between non-zero digits are significant*.
   The measurements 7003, 40.79, 1.503 all have four significant figures.
3. *Zeros appearing in front of all non-zero digits are not significant*. *They are acting as place holders*.
   The measurements 0.0071, 0.42 and 0.000099 all have two significant figures.
4. *Zeros at the end of a number and to the right of a decimal point are significant*.
   The measurements 43.00, 1.010m and 9.00 all have four significant figures.
5. *Zeros at the end of a measurement and to the left of the decimal point can be confusing. They are not significant if they just serve as place markers to show the magnitude of the number*.
   The zeros in the measurements 300, 7000 and 27210 are probably not significant, but some of them may be. We cannot tell any difference. *If these zeros were measured then they are significant.* To avoid ambiguity, the measurements should then be written in standard exponential form 3.00×10^{2}, 7.000×10^{3} and 2.7210×10^{4}. In these examples, the number of significant figures is three, four and five respectively.

When calculations are done with scientific measurements, we sometimes end up with an answer with more digits than can be justified as significant. Such numbers must be rounded off to make them consistent with the data they represent. *Any answer cannot be more precise than the least precise measurement.*

Rules for rounding off

1. *If the digit immediately following the last digit is less than 5, then all the digits after the last significant place are dropped.*
    - *If the digit is 5, or greater, the value of the digit in the last significant place is increased by 1.*
      In a series of calculations, carry the extra digits to the final result and *then* round off

- Don't forget to add place-holding zeros if necessary to keep value the same!!

Examples

1. Round off the following numbers to four significant figures
   (a) 56.212
   (b) 56.216

Solution

   (a) 56.21
   (b) 56.22

2. Round off the following numbers according to the specifications:
   (a) 683 to the nearest ten
   (b) 683 to nearest hundred
   (c) 786 to the nearest ten
   (d) 9.3 to the nearest whole number
   (e) 5.7 to the nearest whole number
   (f) 9.9 to the nearest whole number

Solution

   (a) 680
   (b) 700
   (c) 790
   (d) 9
   (e) 6
   (f) 10

3. Round off the following according to the decimal places specified
   (a) 6.83 correct to one decimal place
   (b) 1.057 correct two decimal places
   (c) 0.0863648 correct to two decimal places
   (d) 0.95 correct to one decimal place

Solution

   (a) 6.8
   (b) 1.06
   (c) 0.09
   (d) 1.0

Exercise

1. Round off the following according to the decimal places specified.
   (a) 4.38 correct to one decimal place
   (b) 2.065 correct to two decimal places
   (c) 0.004689 correct to three decimal places.

Fundamental quantities

There are three fundamental quantities upon which all measurements are based. These are;

- Length
- Time
- Mass

Length

Symbol: L

SI unit: metre [m]

Definition: Length is distance between two or more points.

Instruments used to measure length

- Rule
- Vernier calipers
- Micrometer screw gauge

The rule

Accuracy: 1mm

Quantity measured: Length

Common types of rules

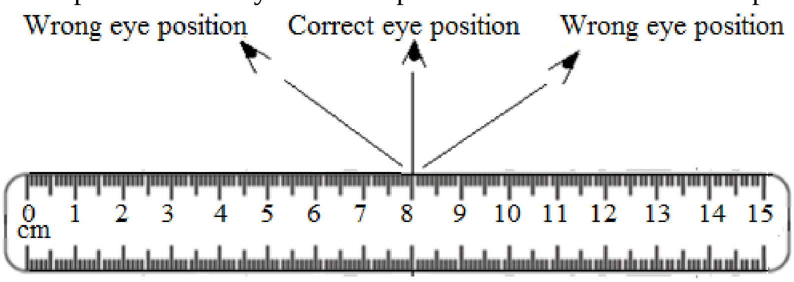
- metre rule (100cm rule)

- 30cm rule
- 15cm rule

Meter rule
The metre rule is used to measure length of more than 1mm
It is usually graduated in centimeters
It has sub- divisions in millimeters

Correct use of a rule

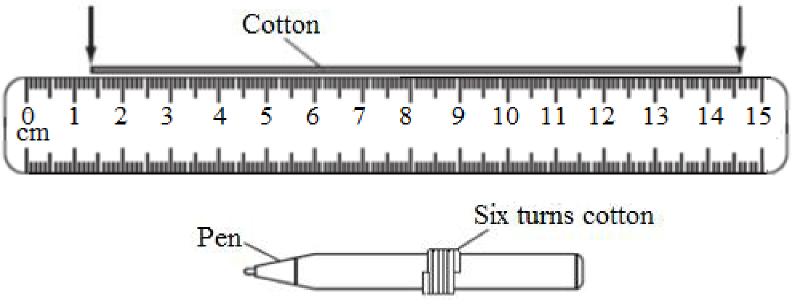
- The eye should be placed vertically above the point to be measured to avoid parallax error



- If a rule has no zero edge, it means you cannot use this point. Therefore, to take a reading, start slightly inwards say at 1cm and remember to subtract from the final reading

Example

1. A piece of cotton is measured between two points on a ruler



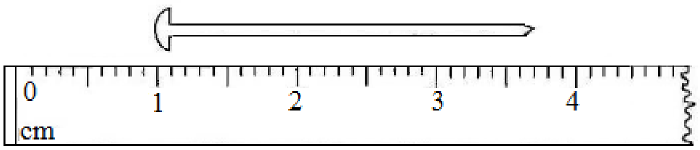
When the length of cotton is wound closely around a pen, it goes round six times.
What is the length of the cotton?

Solution

Length of cotton = 14.6cm – 1.4cm
=13.2cm

Exercise

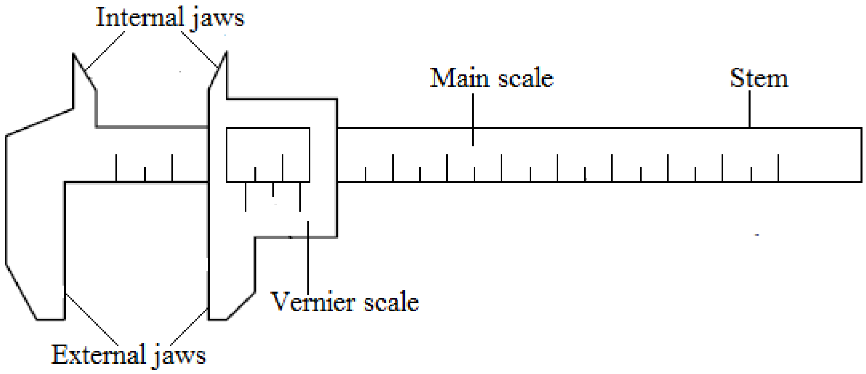
1. The digram below shows part of a ruler used to find the length of a nail.



What is the length of the nail?

Vernier Calipers
Accuracy: 0.01cm
Quantity measured: Length
Use: It is used to measure the length of solids where an ordinary rule cannot be used.
The Vernier calipers can also be used to measure the diameter of balls and cylinders.

Structure of the vernier calipers



Main scale

The main scale is on the stem and fixed.
It is marked in centimeters, cm.

Vernier scale

The vernier scale is movable and slides on the main scale.
It is marked in millimeters, mm.
It has an accuracy of up to $\frac{1}{10}^{th}$ of a millimeter
The vernier scale has ten divisions that correspond to nine divisions of the main scale.

Internal jaws

They measure internal diameter of objects

External jaws

They measure external diameter of objects.

Precautions when using vernier calipers

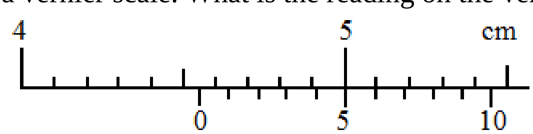
- Zero the instrument before taking a reading
- Clean the instrument so that it is free from dust particles.

How to read vernier calipers

1. Find the value on the main scale that appears just before the zero of the vernier scale in centimeters, cm.
2. Find the value of the line on the vernier scale that coincides with a line on the main scale and multiply it by 0.01cm in order to convert it into cm.
3. Add main scale reading and vernier scale reading.

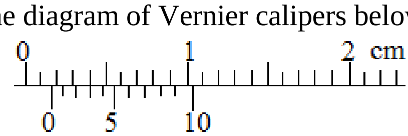
Example

1. The diagram shows part of a vernier scale. What is the reading on the vernier scale?



Main scale reading = 4.5cm
Vernier scale reading = 5 x 0,01cm = 0.05
Vernier calipers reading = 4.5cm + 0.05cm
= 4.55cm

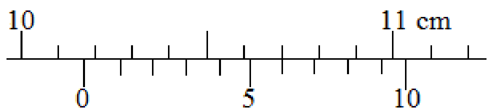
2. State the reading shown in the diagram of Vernier calipers below.

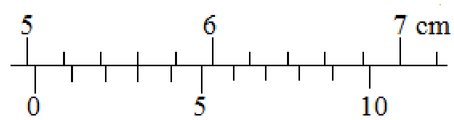


Main scale reading = 0.10cm
Vernier scale reading = 3 x 0.01cm = 0.03cm
Vernier calipers reading = 0.10cm + 0.03cm
= 0.13cm

Exercise

1. Find the readings registered by the vernier calipers below.

(a) 

(b) 

Experiment

To take measurements, using vernier calipers:

(a) Length and width of a small wooden block.
(b) The internal and external diameters of a small test tube.

Object	**Measurement**	**Reading on main scale**	**Reading on vernier scale**	**Final reading = main scale reading + vernier scale reading**
Wooden cube	Length			
Cuboid	Length			
	Breadth			
	Height			
Test tube	Internal diameter			
	External diameter			

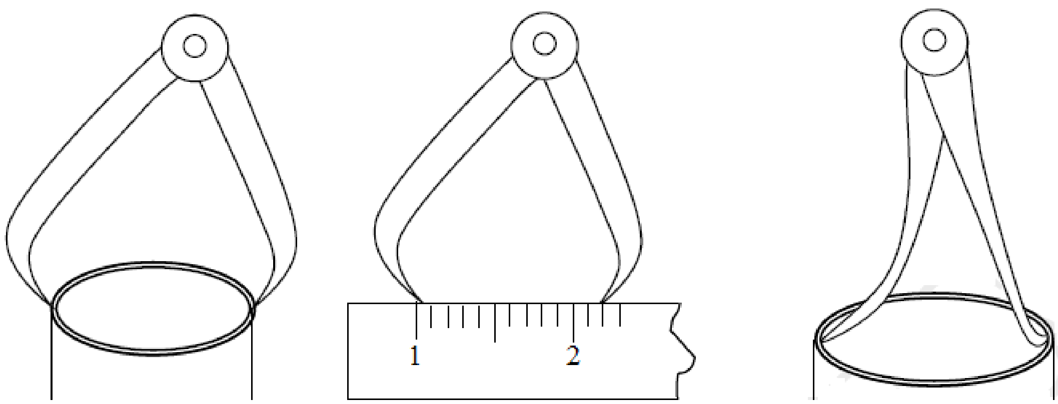
Engineer's Caliper

The engineer's caliper is used to measure the internal and external diameters of hollow objects such as pipes. The object to be measured is clapped in the in the jaws of the calipers.

The instrument has no scale attached to it. The calipers are locked in position once they have been used to mark the dimension and the measurements are read off by laying the caliper points onto a scale ruler.

Outside calipers for external diameter

Inside calipers for internal diameter



Micrometer screw gauge

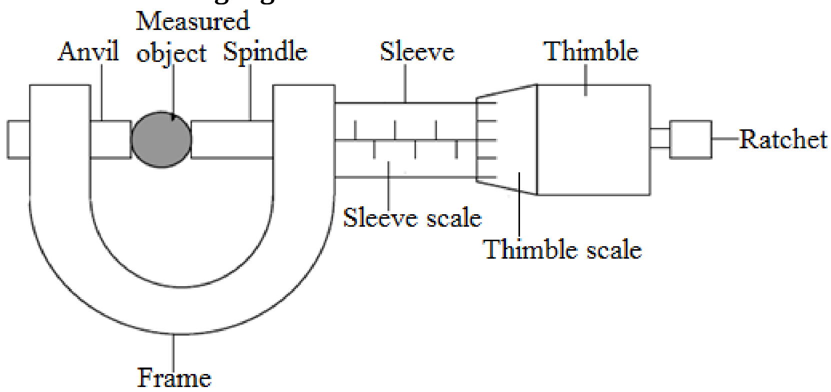
Accuracy: 0.01mm

Quantity measured: Length

Use: It is used to measure smallest size of length such as;

- the thickness of a hair,
- the diameter of a wire,
- the thickness of a piece of paper,
- the thickness of a coin,
- the thickness of a razor blade.

Structure of the micrometer screw gauge



Important parts of the micrometer screw gauge

Sleeve

This is the part that bears the sleeve scale.
The sleeve scale is graduated in millimeters, mm.
The sleeve scale measures correct to 0.5mm

Thimble

This is the part that bears the thimble scale.
The thimble scale measures correct to a 100th of a millimeter or 0.01mm.
A thimble scale has 50 divisions and each division represents 0.01mm.

Anvil and Spindle

These two parts hold the object that is being measured by the instrument.

Ratchet

This is a part used to move the spindle towards or away from the anvil in order to hold the object.

Measurement using the micrometer screw gauge

The two parts or scales are considered, namely;

- thimble scale
- sleeve scale

Precautions when using a micrometer screw gauge

- Zero the instrument before making any measurement
- Clean the anvil and spindle before making any measurement
- Turn the ratchet gently.
- Wipe the object to be measured

How to read the micrometer screw gauge

1. Find the value on the sleeve scale which appears just before the edges of the thimble. The value above the horizontal line gives the whole numbers.
   The value below the horizontal line but in front of the whole number obtained is a mark of 0.5mm and is added to the whole number.
2. Find the value on the thimble scale which is in line with the horizontal line of the sleeve scale and multiply it by 0.01mm.

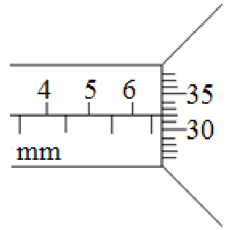
Note

- If there isn't any mark in line, but the horizontal line or point is in between the mark, the highest mark is taken and then multiplied by 0.01mm.

3. Add the sleeve scale reading and thimble scale reading.

Example

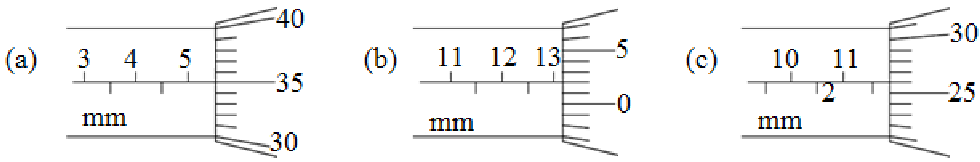
1. State the measurement shown in the diagram of the micrometer screw below.



Sleeve scale reading	= 6.5mm
Thimble scale reading	= 32 x 0.01mm
	= 0.32mm
Micrometer reading	= 6.5mm + 0.32mm
	= 6.82mm

Exercise

1. State the measurements shown in the diagrams of the micrometers below



Experiment

To take measurements using micrometer screw gauge:

(a) thickness of a coin

(b) diameters of a coin and a metal wire.

Object	Measurement	Sleeve scale reading [mm]	Reading on thimble = Number of divisions x 0.01 mm	Final reading = sleeve scale reading + thimble scale reading [mm]
1 kwacha Coin	Thickness			
5 Ngwee Coin	Diameter			
Metal wire	Diameter			

Time

Symbol: t

SI unit: Second [s]

Definition: It is the measure of how long matter occupies a given space.

A time measurement enables us to determine the interval between the beginning and the end of an event.

Other units for time

- Minutes
- Hours
- Days
- Months
- Years
- Centuries

Conversion of the units of time

- 60 seconds = 1 minute
- 60 minutes = 1 hour
- 24 hours = 1 day-night
- 7 days= 1 week
- 4 weeks = 1 month
- 12 months = 1 year = 365 days

Summary on measure of time

1year = 365days = 8760hours = 525600minutes = 31536000s
1 day = 24hours = 1440minutes = 86400s
1hour = 60minutes = 3600s
1minute = 60s

Exercise

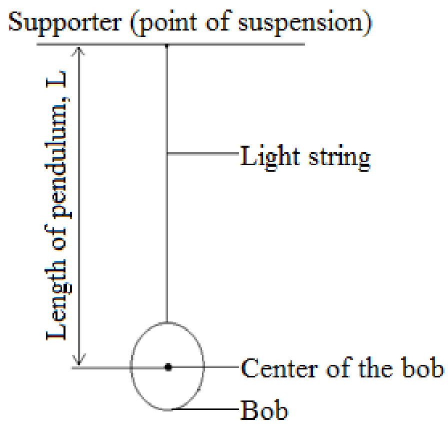
1. Convert the following to the stated units.
   (a) 120 seconds to minutes
   (b) 30 minutes to hours

Instruments for measuring time

- Simple pendulum
- Stop watch
- Ticker tape timer
- Oscilloscope (C.R.O)

The simple pendulum

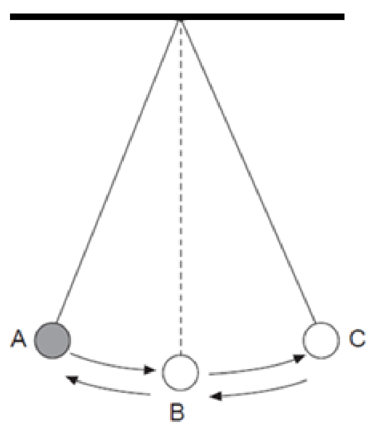
A simple pendulum is a small heavy bob suspended by a light inextensible string.
This consists of a string tied to a horizontal support. A bob is suspended at the lower end of the string.



Terms used to describe a simple pendulum

Oscillation

Alternative term: *Vibration / cycle / swing*
Definition: An oscillation is a complete to and fro movement of the bob.



Note

A swing from;

- A to B = $\frac{1}{4}$ or 0.25 oscillations.
- A to C = $\frac{1}{2}$ or 0.5 oscillations
- A to C and back to B = $\frac{3}{4}$ or 0.75 oscillations

- A to C and back to A = 1 complete oscillation

Amplitude

Symbol: A

SI unit: metre [m]

Definition: Amplitude is the maximum displacement of the bob from the rest position.

Length of the pendulum

Symbol: L

SI unit: Metre [m]

Definition: Length of the pendulum is the distance from the supporter (point of suspension) to the centre of the bob.

Period of a pendulum

Symbol: T

SI unit: Second [s]

Definition: Period of the pendulum is the time taken by the bob to make a complete oscillation.

Factors affecting the period of the pendulum

- Length of the pendulum, L
- Acceleration due to gravity, g

(a) Length of the pendulum, L

Period of the pendulum becomes;

- longer if the length of the pendulum is increased
- shorter if the length of the pendulum is reduced

(b) Acceleration due to gravity, g

When;

- gravity is high, period reduces
- gravity is low period increases

Factors that do not affect the period of a simple pendulum

- The amplitude of swing of a simple pendulum
- The mass of the bob. Period **does not** depend on the **mass or material** of the bob. Mass of the bob has no effect on the period of the pendulum.

Precautions when using a simple pendulum

- The location of the pendulum should be in a place where there is less or no wind blowing.
- Provide small amplitude to reduce effects of air resistance.
- Use length of pendulum to reduce its frequency and easy counting of oscillations.

Relationship between period and time

- $t = n \times T$
- $T = \frac{t}{n}$
- $n = \frac{t}{T}$

Note

- t = time interval in seconds, s
- n = number of oscillations (swings/cycles/times)
- T = period of the pendulum in seconds, s

Frequency

Symbol: f

SI unit: Hertz [Hz]

Definition: Frequency is the number of oscillations in one second.

Relationship between frequency and period

$T = \frac{1}{f}$

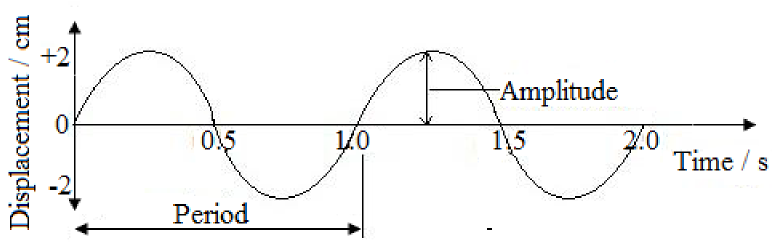
$f = \frac{1}{T}$

$f = \frac{n}{t}$

- T = period of the pendulum [s]
- t = time [s]
- f = frequency [Hz]

- n = number of oscillations

Displacement- time graph for a simple pendulum



Note

- *Amplitude = 2cm*
- *Period of the pendulum = 1.0s*
- *When length of the pendulum increases, period also increases but frequency reduces.*
- *When length of the pendulum reduces, period also reduces but frequency increases.*

Determining (measuring) period of the pendulum

- Set the pendulum oscillating
- Note the time, t, and the number of oscillations, n.
- Calculate the period, T, using the formula; $T = \frac{t}{n}$

Measuring time interval using a simple pendulum

- Set the pendulum oscillating.
- Note the number of oscillations, n.
- Calculate time by using the formula; t = n x T

Note

- *A number of runs are done and the average is taken to minimize error.*

Experiment

Aim: To determine the relationship between the length (L) and period (T) of the pendulum

Materials

- Bob
- Clamp and stand
- String
- Stop watch

Method

- Measure and record the length of the string from the point of support to the centre of the bob.
- Pull the bob to one side with angular amplitude of less than 10^o.
- Release the bob so that it starts swinging.
- When the bob reaches the maximum displacement, start the stop watch and start counting
- Record the time taken for 20 complete oscillations
- Repeat the experiment with different lengths (L). Record values in the table.

Results

	Length of string (cm)	**Time taken for 20 complete oscillations (s)**	**Period (s)**
1.	30cm		
2.	20cm		
3.	10cm		

Conclusion

Period of the pendulum depends on the length of the pendulum and acceleration due to gravity.

Examples

1. In an experiment to measure the period of the pendulum, the time taken for 50 complete oscillations was found to be one minute. What is the period of the pendulum?

Data	Solution
T =? t = 60 seconds n = 50	$T = \frac{t}{n}$ $T = \frac{60s}{50}$ T = 1.2s

2. What is the period of a pendulum that makes 50 oscillations in 9s?

Data	Solution
T =? t = 9s n = 50	$T = \frac{t}{n}$ $T = \frac{9s}{50}$ T = 0.18s

3. A pendulum has period 0.6s. Calculate the time it takes to make 75 oscillations?

Data	Solution
t =? n = 75 T = 0.6s	t = n x T t = 75 x 0.6s t = 45s

4. How many oscillations are made by a pendulum whose period is 1.2s in 30s?

Data	Solution
n =? T = 1.2s t = 30s	$n = \frac{t}{T}$ $n = \frac{30s}{1.2s}$ n = 25 cycles

5. A pendulum makes 96 oscillations in 4.8s.What is its frequency?

Data	Solution
f =? n = 96 t = 4.8s	$f = \frac{n}{t}$ $f = \frac{96}{4.8s}$ f = 20Hz

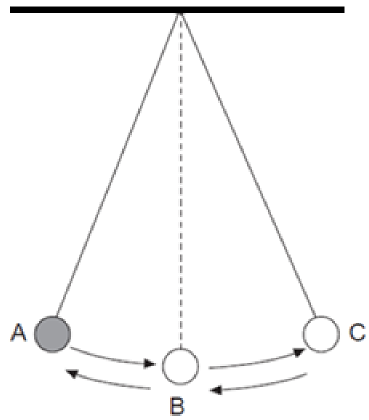
6. A pendulum's frequency is 15Hz.How many oscillations does it make in 3.6s?

Data	Solution
n =? f = 15Hz t = 3.6s	n = f x t n = 15Hz x 3.6s n = 54 cycles

7. What is the time taken for a pendulum of frequency 25Hz to make 40 oscillations?

Data	Solution
t =? n = 40 f = 25Hz	$t = \frac{n}{f}$ $t = \frac{40}{25Hz}$ t = 1.6s

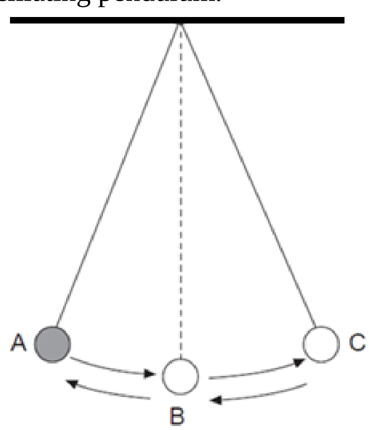
8. The figure below shows a simple pendulum that oscillates between position A and C.



(a) If it takes 2 seconds for the bob to move from A to C and back to B, find the number of oscillations.
(b) Calculate the period of the pendulum.
(c) Calculate the frequency of the pendulum

	Data	**Solution**
a	A to C back to B	n = 0.75 oscillations
b	T =? t = 2s n = 0.75	$T = \frac{t}{n}$ $T = \frac{2s}{0.75}$ T = 2.67s
c	f =? T= 2.67s	$f = \frac{1}{T}$ $f = \frac{1}{2.67s}$ f = 0.37Hz

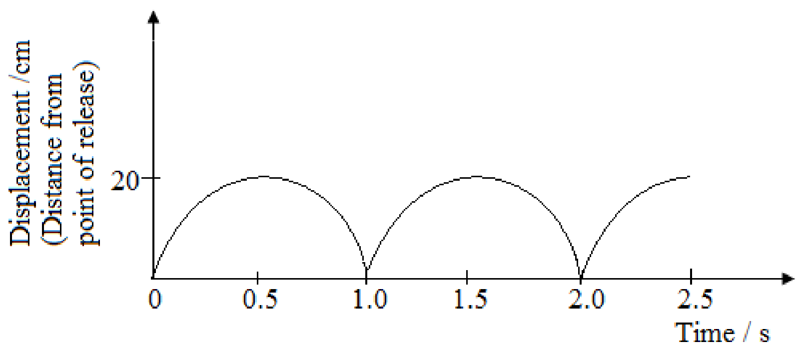
9. The diagram below shows an oscillating pendulum.



If the period of the pendulum is 0.4s, find the time taken for the pendulum to swing from;
(a) A to C
(b) A to B
(c) A to C and back to B

	Data	Solution
a	t =? n = 0.5 oscillations T = 0.4s	t = n x T t = 0.5 x 0.4s t = 0.2s
b	t =? n = 0.25 oscillations T = 0.4s	t = n x T t = 0.25 x 0.4s t = 0.1s
c	t =? n = 0.75 oscillations T = 0.4s	t = n x T t = 0.75 x 0.4s t = 0.3s

10. The bob of a simple pendulum is pulled to one side and released. The motion during its swing is shown in the graph.

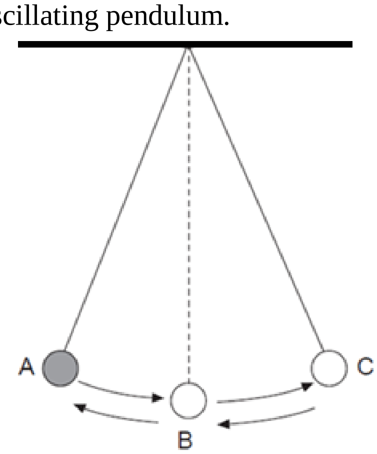


(a) What is the amplitude of the pendulum?
(b) What is the value of the period of the pendulum?
(c) Calculate the frequency of the pendulum
(d) What would you do in order to change the periodic time of the same pendulum to 1.5s?

	Data	Solution
(a)		A = 20cm
(b)		T = 2.0s
(c)	f =? T = 2.0s	$f = \frac{1}{T}$ $f = \frac{1}{2.0s}$ f = 0.5Hz
(d)		By reducing the length of the pendulum.

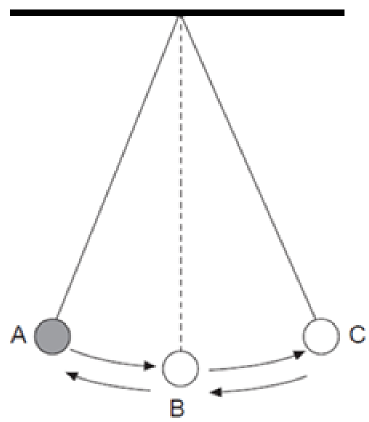
Exercise

1. Find the period of the pendulum if it oscillates 15 times for 45 seconds.
2. The diagram below shows an oscillating pendulum.



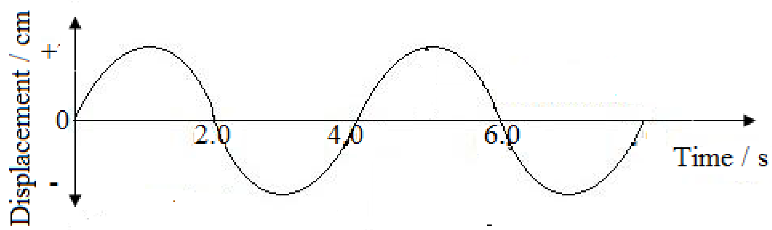
(a) If it takes 3 seconds for the bob to move from A to C, find the period of the pendulum.
(b) Find the time taken for 12 complete oscillations.

3. The bob of the pendulum shown below takes 0.25s to swing from A to C.



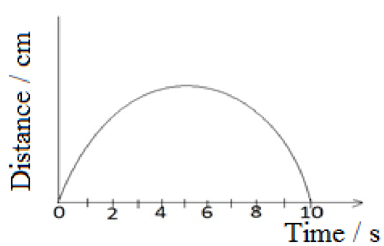
(a) If A and C are extreme points, determine;
   (I) the period of the pendulum
   (II) the frequency of the pendulum
(b) State whether the frequency of oscillations will increase, decrease or remain the same if;
   (I) the length of the string is increased
   (II) the mass of the bob is increased
   (III) the distance between A and C is increased.
(c) Briefly describe how the period of the pendulum would be measured.

4. Study the displacement- time graph for a simple pendulum.



   (a) What is value of the period of the pendulum?
   (b) State the maximum displacement of the pendulum.
   (c) Naosa carried out an experiment to determine the time Kakula took to finish drinking one litre of castle using a simple pendulum. The period of the pendulum was 1.5 seconds and its length was 0.8m.
   (I) Calculate the time taken for Kakula to finish drinking one litre of castle if the number of oscillations were 50.
   (II) State what will happen to the frequency of pendulum if the length was;
   (a) reduced to 0.5m
   (b) Increased by 0.5m.

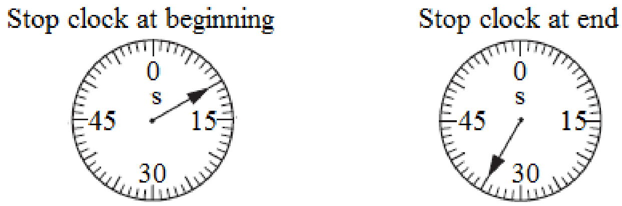
5. The graph below is for a pendulum bob which was pulled to one side and then released to swing. Assume that there is no friction of any sort as the bob swings.



(a) What do you understand by period of the pendulum?
(b) After how long does the pendulum bob reach the maximum distance of travel?
(c) If the pendulum bob swings at the rate of 5m/s, how far from the starting position is it at 8 seconds later from the time it started swinging?
(d) Explain why this pendulum would be suitable for keeping or measuring time.

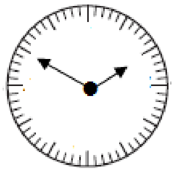
Stop clocks
Exercise

1. The diagrams show the times on a stop clock at the beginning and at the end of an experiment.



How long did the experiment take?
A 10 s
B 25 s
C 35 s
D 45 s

2. One side of the main bedroom has a modern clock while the opposite side had a large dressing mirror. A child enters this room and sees the image of the clock in the mirror as shown below.



What is the correct time shown by the actual clock?
A 10:10 hours
B 11:10 hours
C 13:50 hours
D 14:50 hours

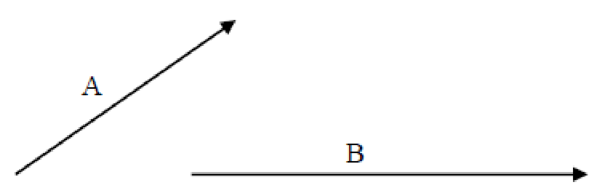
UNIT 2: MECHANICS
Mechanics is the study of the motion of objects.
Kinematics
Kinematics is the science of describing the motion of objects using words, diagrams, numbers, graphs and equations. Kinematics is a branch of mechanics.
Scalar quantity
It is a quantity which has magnitude (size) with no direction
Scalar quantities can easily be added and subtracted.
Examples of scalar quantities

- Distance
- Speed
- Mass
- Volume
- Temperature

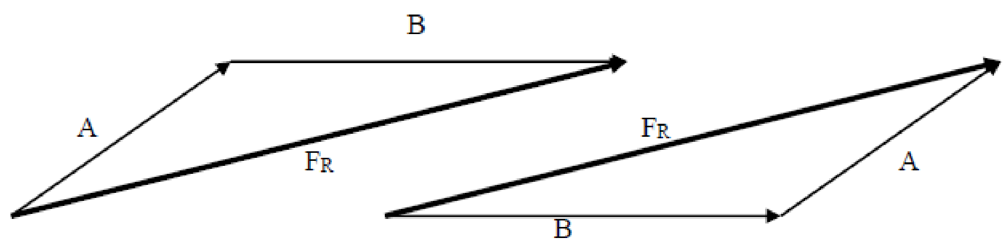
Vector quantity
It is a quantity which has both magnitude (size) and direction.
Vector quantities are mainly represented graphically or an arrow with a point (→).
Vectors can be represented by (in a particular direction), straight line with an arrow on a diagram. The length of the arrow is proportional to the magnitude (size) of the vector (that is, we chose a scale) while the direction of the arrow is the direction of the vector.
Examples of Vector quantities

- Displacement
- Velocity
- Acceleration
- Force
- Weight
- Momentum

Composition or Addition of Vectors
Several vectors can be added to get one **resultant vector.** The vectors which are added are called **components** of the resultant vector.
When adding vectors, the magnitude and direction of the components has to be taken into consideration. There are two methods used when adding vectors. Suppose we are given vectors A and B shown below find their resultant.

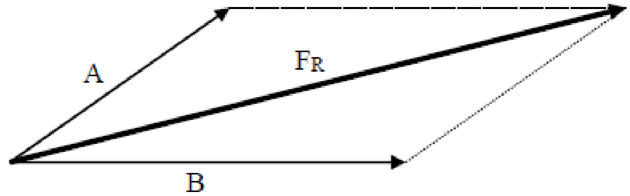


(a) Head to tail method
Join the head of the first vector to the tail of the second vector. The resultant is from the tail of the first vector to the head of the second vector



(b) Parallelogram

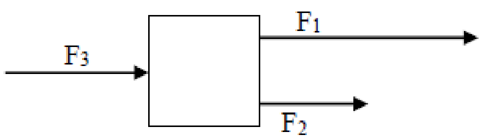
Join the tails of the two vectors and complete the parallelogram



The resultant is the diagonal from the tails of the component vectors

Adding vectors in same direction

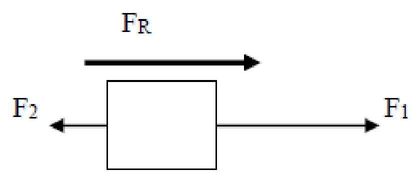
If the vectors are acting in the same direction find algebraic sum of the vectors



$F_R = F_1 + F_2 + F_3$

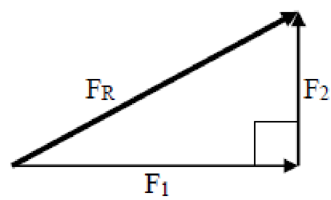
Vectors in the opposite direction

If the vectors are in the opposite direction find the difference and the resultant is in the direction of the larger vector



$F_R = F_1 – F_2$

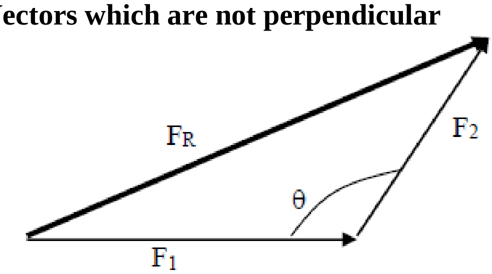
Vectors acting perpendicular to each other



Use the Pythagoras theorem

$F_R^2 = F_1^2 + F_2^2$

Vectors which are not perpendicular

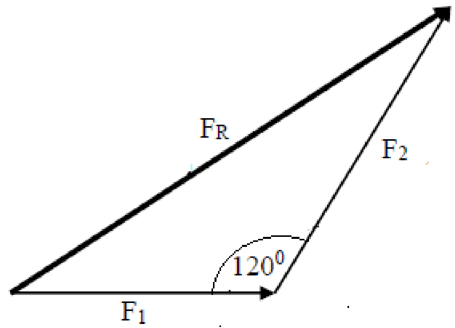


Use the cosine rule

$F_R^2 = F_1^2 + F_2^2 - 2F_1F_2\cos\theta$

Examples

1. Find the resultant of 100N and 150N force acting at an angle of 120º to each other.

Data	**Solution**
$F_R = ?$ $F_1 = 100N$ $F_2 = 150N$	 $F_R^2 = F_1^2 + F_2^2 - 2F_1F_2\cos\theta$ $F_R^2 = 100^2 + 150^2 - 2(100)(150)120^o$ $F_R^2 = 1000 + 22500 - 30000(-0.5)$ $F_R^2 = 32500 + 15000$ $F_R^2 = 47500$ $F_R = \sqrt{47500}$ $F_R = 217.94N$ $= 218N$

Motion
Motion is the change of position of an object in a given direction.
Types of motion
1. **Linear motion**

This is the movement of an object along a straight line or path e.g. a car travelling along a straight road.

2. **Circular motion (Rotational motion)**

This is the movement in a circle about the centre or an axis e.g. a spinning wheel or rotating fan.

3. **Oscillatory motion**

This is the movement where an object moves to and fro about a fixed position e.g. the swinging of the bob of the pendulum.

4. **Random motion**

This is the movement of an object in a disorderly manner e.g. in the case of gaseous particles.

Linear motion
Four parameters are required to describe motion in a straight line. These are;
- Distance or displacement
- Speed or velocity
- Acceleration
- Time

Distance
Symbol: s
SI unit: Metre [m]
Definition: Distance is the length between two or more points. It can also be defined as the actual path travelled by an object from its initial position to the final position.
Other units for distance
Millimeters, mm
Centimeters, cm
Kilometers, Km
Relationship of units
- 10mm = 1cm
- 100cm = 1m
- 1000m = 1Km
- 1Km = 100000cm = 1000000 mm

Example
1. Convert 30cm into m

Solution

100cm $\rightarrow$ 1m

30cm $\rightarrow x$

$x = \frac{30\text{cm x 1m}}{100\text{cm}}$

x = 0.3m

Exercise

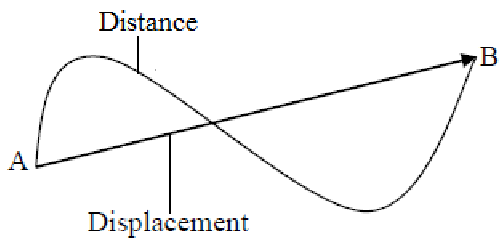
1. Convert the following to the stated units
   (a) 8.0Km to m
   (b) 0.8cm to m
   (c) 500m to Km
   (d) 13m to mm

Displacement

Symbol: s

SI unit: metre [m]

Definition: Displacement is the distance travelled in a specified direction.



Similarity between distance and displacement

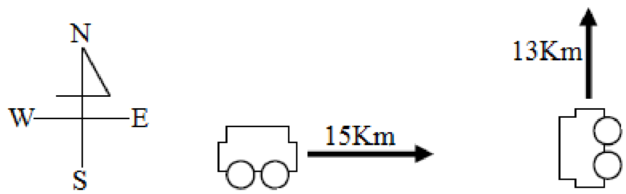
- The SI unit for both distance and displacement is the metre, m.

Differences between distance and displacement

- Distance is a scalar quantity while displacement is a vector quantity.
- Distance is the length between two points while displacement is the distance travelled in a specified direction.

Examples

1. A car moves 15Km to the East and 13Km to the North.



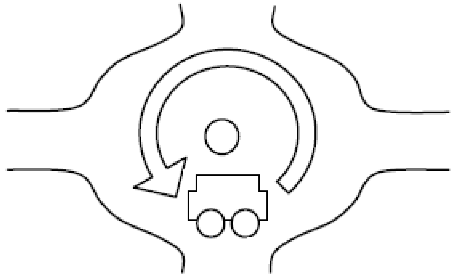
Find the

(a) Distance covered by the
(b) Displacement of the car.

Solution

(a) Distance = 15Km + 13Km
= 28Km
(b) Displacement = 15Km East and 13Km North.

2. The circumference of a round bout is 50m and the car turns it once.



Find the

(a) Distance covered by the car
(b) Displacement of the car.

Solution

(a) Distance = 50m
(b) Displacement = 0m because the car came back to the starting point.

3. A boy walks forward 25m and backward 15m.

Find the

(a) Distance covered by the boy
(b) Displacement of the boy.

Solution

(a) Distance = 25m + 15m
= 40m
(b) Displacement = 25m – 15m
= 10m forward

Speed

Symbol: V
SI unit: metre per second [m/s] or [ms^{-1}]
Definition: Speed is the rate of change of distance with time.

Formula: Average speed = $\frac{\text{total distance covered}}{\text{total time taken}}$

$v = \frac{s}{t}$

Example

1. A car travels from Lusaka to Mongu 600Km away in 8hours. Find the average speed of the car in Km/h.

Data	**Solution**
v =? s = 600Km t = 8h	$v = \frac{s}{t}$ $v = \frac{600\text{Km}}{8\text{h}}$ v = 75Km/h

2. A cheetah runs at a speed of 20m/s in 50 seconds. Calculate distance covered by the cheetah.

Data	**Solution**
s =? v= 20m/s t = 50s	s = v x t s = 20m/s x 50s s = 100m

3. A bus takes 2400s to complete its 24000m route. Calculate its average speed in m/s.

Data	**Solution**
v =? s = 24Km = 24000m t = 40min = 2400s	$v = \frac{s}{t}$ $v = \frac{24000\text{m}}{2400\text{s}}$ v = 10m/s

Note

- 1m/s = 3.6Km/h

Examples

1. Express
   (a) 72Km/h in m/s
   (b) 10m/s in Km/h

Solution

(a) $= \frac{(72 \text{ x } 1000)\text{m}}{(60 \text{ x } 60)\text{s}}$

$= 20\text{m/s}$

OR

$3.6\text{Km/h} \rightarrow 1\text{m/s}$

$72\text{Km/h} \rightarrow \text{x}$

$\text{x} = \frac{72\text{Km/h X } 1\text{m/s}}{3.6\text{Km/h}}$

$\text{x} = 20\text{m/s}$

(b) $1\text{m/s} \rightarrow 3.6\text{Km/h}$

$10\text{m/s} \rightarrow x$

$x = \frac{10\text{m/s x } 3.6\text{Km/h}}{1\text{m/s}}$

$x = 36\text{Km/h}$

Velocity

Symbol: V

SI unit: metre per second [m/s] or [ms^{-1}]

Definition: Velocity is the rate of change of displacement with time.

Formula: Average velocity $= \frac{\text{displacement}}{\text{time taken}}$

$$v = \frac{s}{t}$$

Similarity between speed and velocity

- Both speed and velocity are measured in meters per second, m/s.

Differences between speed and velocity

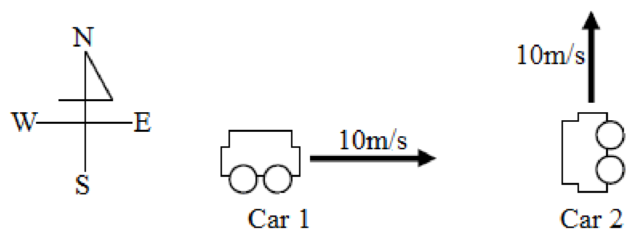
- Speed is a scalar quantity while velocity is vector quantity.
- Speed is the rate of change of distance with time while velocity is the rate of change of displacement with time.

Note

- Speed is called velocity when it has direction and velocity is called speed when it has no direction.

Example

1. Car 1 moves 10m/s east and car 2 moves 10m/s north. Find the speed and velocity of the two cars.



Solution

Both car 1 and 2 have the same speed of 10m/s

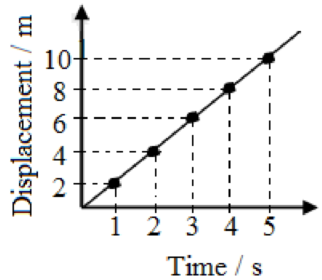
Car 1 has a velocity of 10m/s east while car 2 has a velocity of 10m/s north.

Constant or uniform velocity

Constant velocity is when a body travels equal displacements in equal times.

e.g.

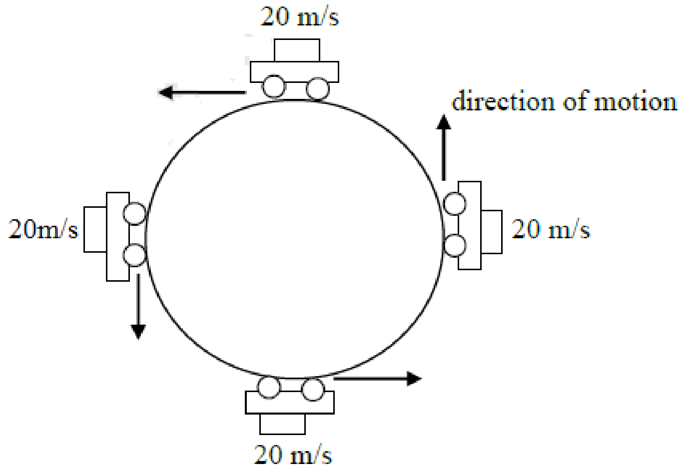
Displacement (m)	0	2	4	6	8	10
Time (s)	0	1	2	3	4	5
Velocity (m/s)	0	2	2	2	2	2



Constant velocity = 2 m/s

Velocity round a curved path

The velocity is not uniform for a body which moves in a curved path at constant speed. This is because its direction of motion would be continuously changing, hence velocity would be changing too although the speed remains constant. Therefore such a body would be **accelerating**.



Acceleration

Symbol: a

SI unit: metre per second squared [m/s^2] or [ms^{-2}]

Definition: Acceleration is the rate of change of velocity with time.

Formula: Average acceleration = $\frac{\text{Final velocity} - \text{initial velocity}}{\text{time taken}}$

$$a = \frac{V-U}{t}$$

Types of acceleration

1. **Positive acceleration**
   This is when velocity is increasing with time.
   It is always given a positive sign.
2. **Negative acceleration**
   This is when velocity is decreasing with time. It is also called retardation or deceleration
   It is always given a negative sign
3. **Uniform acceleration**
   This is when the rate of velocity is constant
   Under uniform acceleration, velocity is changing continuously but at the same rate.
   Uniform acceleration is also called constant acceleration
4. **Non uniform acceleration**
   This is the acceleration in which the rate of change of velocity is not constant.
   The rate of change of velocity keeps on changing.

Note

- *Negative acceleration is called retardation or deceleration*
- *When speed or velocity is constant, acceleration, a = $0m/s^2$*
- *From rest, initial velocity, u = 0m/s*
- *Moving at/travelling at/moving with, initial velocity, u = given velocity in m/s*

- *To rest, final velocity, $v = 0m/s$*

Examples

1. A car starting from rest increases its velocity uniformly to 15m/s in 3 seconds. What is the acceleration?

Data	**Solution**
a =? v = 15m/s u = 0m/s t = 3s	$a = \frac{v-u}{t}$ $a = \frac{15m/s-0m/s}{3s}$ $a = \frac{15m/s}{3s}$ $a = 5m/s^2$

2. A car slows down from 36m/s to rest in 12s. Calculate the retardation.

Data	**Solution**
a =? v = 0m/s u = 36m/s t = 12s	$a = \frac{0m/s-36m/s}{12s}$ $a = \frac{-36m/s}{12s}$ $a = -3m/s^2$

Equations of uniformly accelerated linear motion

- $v = u + at$
- $s = ut + \frac{1}{2}at^2$
- $v^2 = u^2 + 2as$
- $s = \frac{(v+u)t}{2}$

Examples

1. A car travelling at 10m/s accelerates at $2m/s^2$ for 3 seconds. What is its final velocity?

Data	**Solution**
v =? u = 10m/s t = 3s $a = 2m/s^2$	$v = u + at$ $v = 10m/s + (2m/s^2 \text{ x } 3s)$ $v = 10m/s + 6m/s$ $v = 16m/s$

2. A car starts from rest accelerates at $3m/s^2$. How far does it travel in4 seconds?

Data	**Solution**
s =? u = 0m/s t = 4s $a = 3m/s^2$	$s = ut + \frac{1}{2}at^2$ $s = 0m/s \text{ x } 4s + \frac{1}{2} \text{x } 3m/s^2 \text{ x } (4s)^2$ $s = 24m$

3. A car accelerates from rest to a velocity of 8m/s over a distance of 200m. How long does it take to accelerate from rest to 8m/s?

Data	**Solution**
t =? s = 200m v = 8m/s u = 0m/s	$s = \frac{(v+u)t}{2}$ $t = \frac{2s}{v+u}$ $t = \frac{2 \text{ x } 200m}{8m/s+0m/s}$ $t = 50s$

4. A car accelerates uniformly from rest until it reaches a velocity of 10m/s in 5s. Calculate the distance covered 5s?

Data	Solution
s =? v =10m/s u = 0m/s t = 5s	s = $\frac{(v+u)t}{2}$ s = $\frac{(10m/s+0m/s)5s}{2}$ s = 25m

Exercise

1. A car travelling at 20m/s accelerates at the rate of 2m/s^2 for 30 seconds.

Calculate;

(a) the final velocity of the car

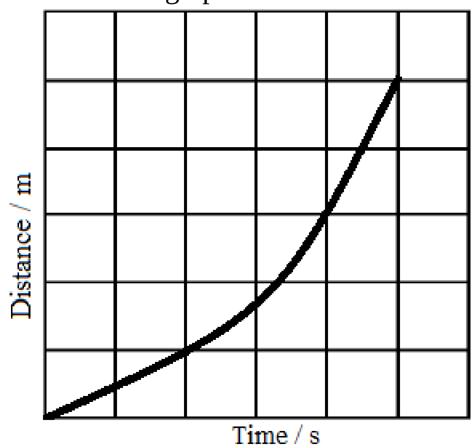
(b) the distance travelled by the car.

Time graphs

Distance-time graphs

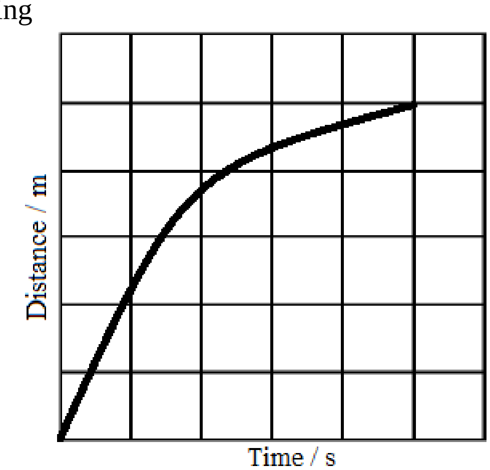
A distance time - graph is a graph where distance is plotted against time.

The diagrams below represent the distance time graphs for the motion of an object.



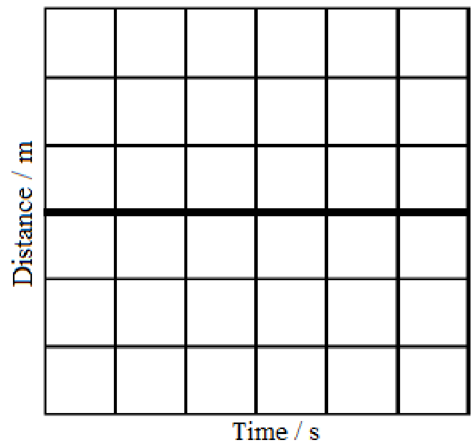
Description

- The object was accelerating



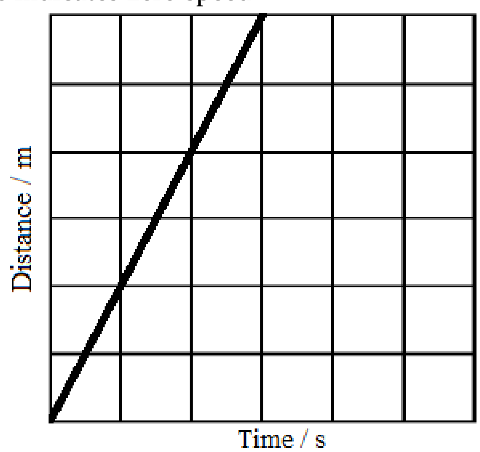
Description

- The object was decelerating or retarding



Description

- The object stopped moving. (was at rest)
- The horizontal straight line indicates zero speed



Description

- The object was moving with constant velocity. It travelled a distance of 10m in 6s.
- The slope on the distance-time graph represents velocity.

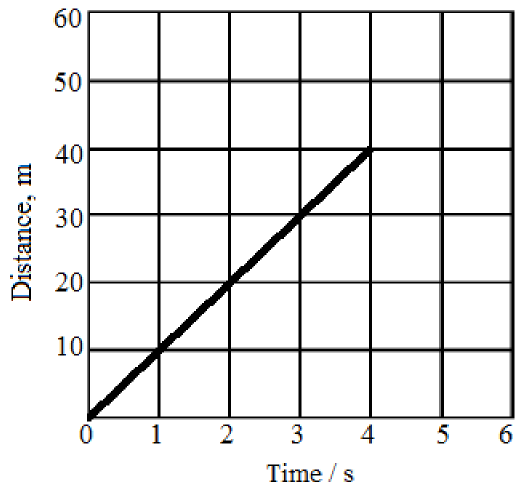
  Velocity = $\frac{\text{Distance}}{\text{Time}}$

Example

1. An object travelled a distance of 40m in 4 seconds.
   (a) Sketch the distance- time graph to interpret the information above.
   (b) Calculate the velocity of an object.

Solution

(a)



(b) Velocity $= \frac{\text{Distance}}{\text{Time}}$

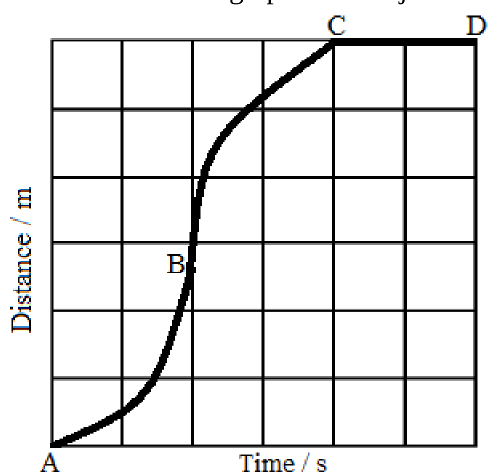
$v = \frac{s}{t}$

$v = \frac{40\text{m}}{4\text{s}}$

v = 10m/s

Exercise

1. The diagram below shows the distance-time graph of an object.



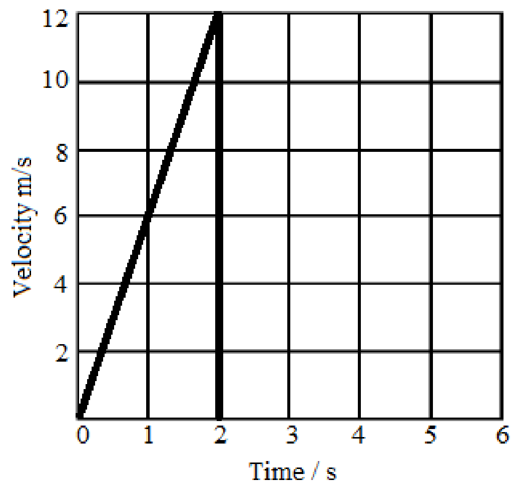
Describe the motion of an object from:

(a) A to B
(b) B to C
(c) C to D

Velocity (speed) – time graphs

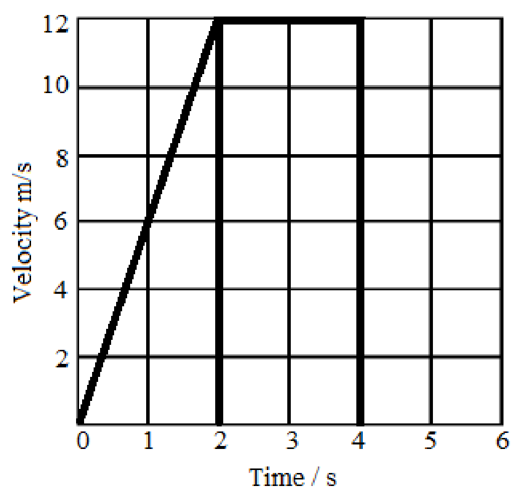
Velocity-time graph is a graph where velocity is plotted against time.

The diagrams below show the velocity-time graphs for the motion of an object.



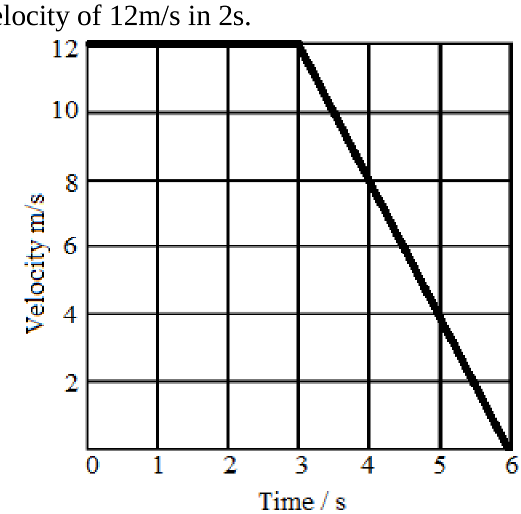
Description

- The object was moving from rest with constant acceleration to a velocity of 12m/s in 2s.
- The slope indicates constant acceleration.
- Increasing (uniform) velocity
- Constant or uniform acceleration



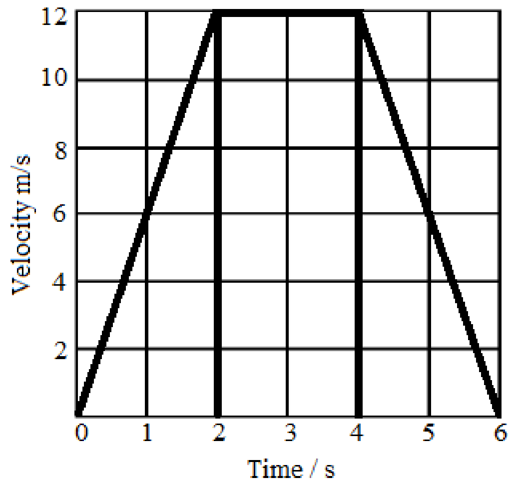
Description

- The object was moving from rest with constant acceleration to a velocity of 12m/s in 2s. It then moved with constant velocity of 12m/s in 2s.



Description

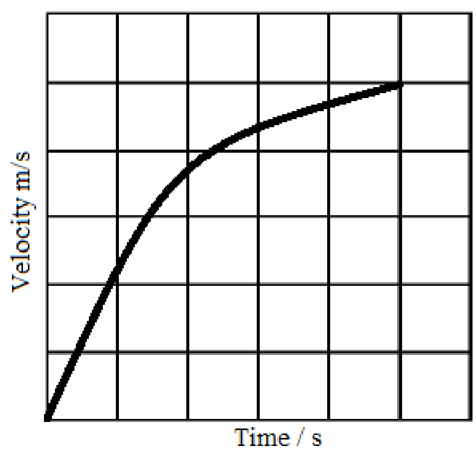
- The object was moving at a constant velocity of 12m/s in 3s and then decelerated uniformly to rest in 3s.



Description

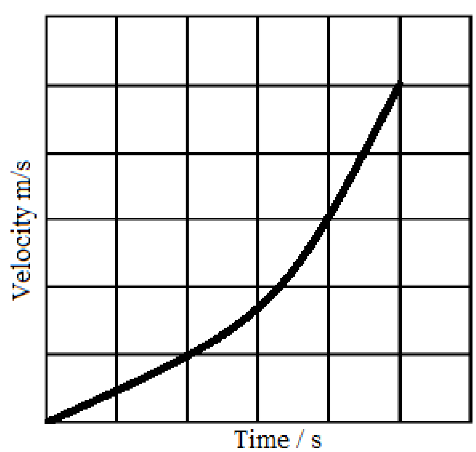
- The object was moving from rest with constant (uniform) acceleration to a velocity of 12m/s in 2s and then it moved with constant velocity of 12m/s in 2s and finally decelerates uniformly to rest in 2s.

Summary of velocity-time graphs



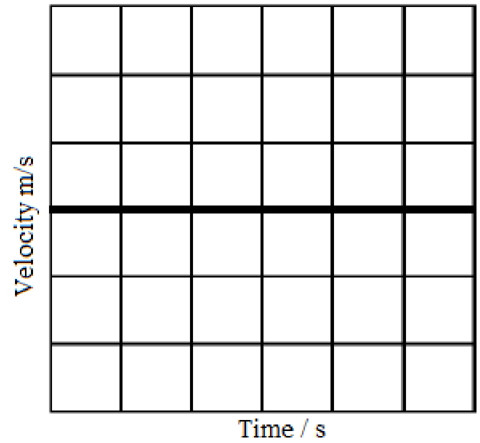
Description

- Non-uniform acceleration



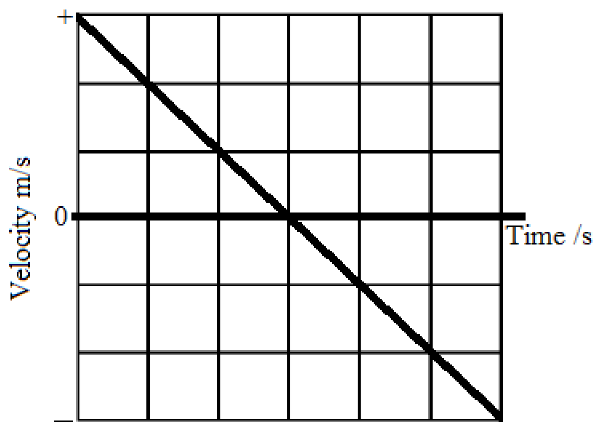
Description

- Non-uniform deceleration or retardation



Description

- Constant velocity
- Horizontal line represents zero acceleration



Description

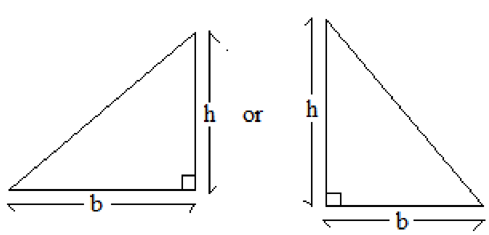
- Negative velocity shows that the object was dropping or falling.
- Constant (uniform) deceleration

Note

- *The slope (gradient) on the velocity-time graph represents acceleration.*
  $a = \frac{v-u}{t}$
- *The area under the velocity-time graph represents the distance covered.*

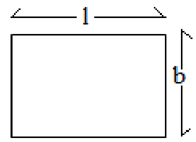
For a;

(a) triangle,



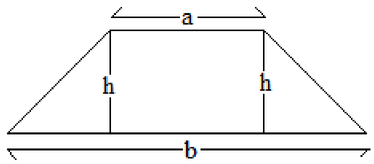
Distance, s = $\frac{1}{2}$bh

(b) rectangle,



Distance, s = l x b

(c) trapezium,

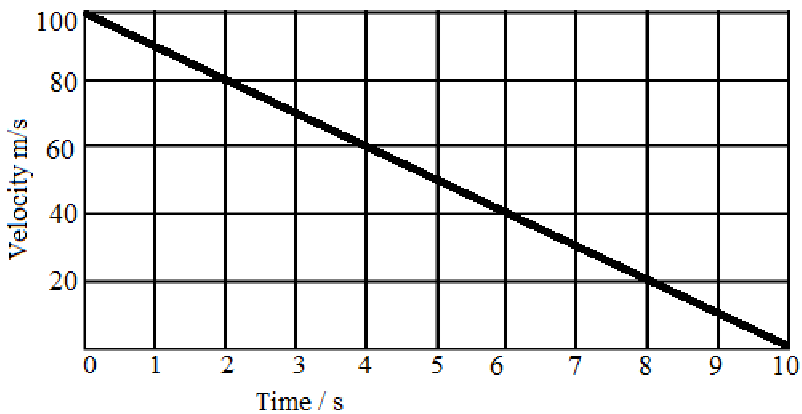


Distance, s = $\frac{1}{2}(a + b)h$

- Velocity of a body must be changing when the body is accelerating.

Example

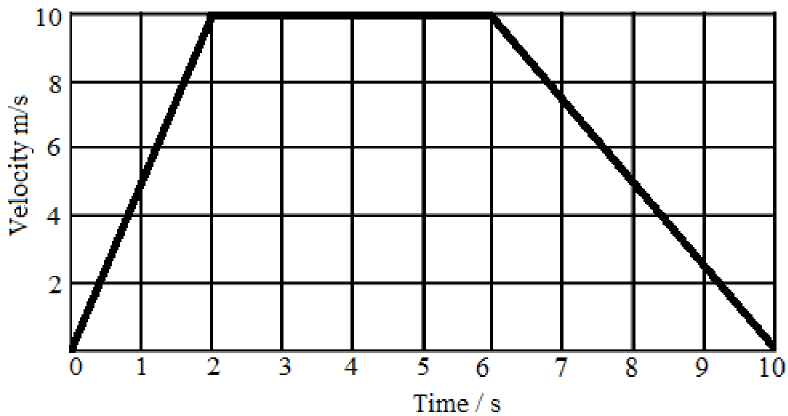
1. The diagram below shows a speed versus time graph for an arrow which was shot vertically upwards.



(a) At what speed was the arrow shot?
(b) How long did it take the arrow to reach its highest point?
(c) Determine how high the arrow rose.

	Data	**Solution**
(a)		u = 100m/s
(b)		t = 10s
(c)	s =? b = 10s h = 100m/s	s = $\frac{1}{2}$bh s = $\frac{1}{2}$ x 10s x 100m/s s = 500m

2. The figure below shows a velocity-time graph for a car travelling along a straight road in 10s.



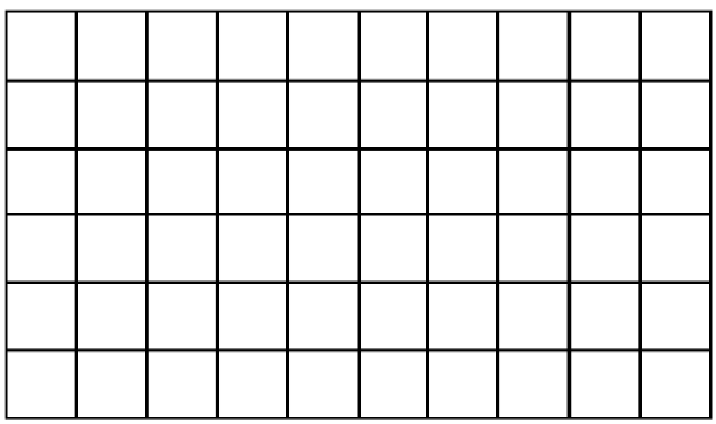
(a) Describe the motion of the car in 10s.
(b) Find the acceleration of the car in the first 2s.
(c) Find the acceleration of the car between 2 and 6 seconds of the journey.
(d) Calculate the acceleration of the car in the last 4s of its motion.
(e) Find the distance travelled by the car in the first 2s.
(f) Calculate the distance travelled by the car during the constant velocity (between 2 and 6 seconds)
(g) Find the total distance travelled by the car.

	Data	**Solution**
(a)		The car was moving from rest with constant acceleration to a velocity of 10m/s in 2s and then it moved with constant velocity of 10m/s in 4s and finally decelerates uniformly to rest in 4s.
(b)	a =? v = 10m/s u = 0m/s t = 2s	$a = \frac{v-u}{t}$ $a = \frac{10m/s - 0m/s}{2s}$ $a = 5m/s^2$
(c)	a =? v = 10m/s u = 10m/s t = 4s	$a = 0m/s^2$ because velocity is constant. or $a = \frac{v-u}{t}$ $a = \frac{10m/s - 10m/s}{4s}$ $a = \frac{0m/s}{4s}$ $a = 0m/s^2$
(d)	a =? v = 0m/s u = 10m/s t = 4s	$a = \frac{v-u}{t}$ $a = \frac{om/s - 10m/s}{4s}$ $a = -2.5m/s^2$
(e)	s =? b = 2s h = 10m/s	$s = \frac{1}{2}bh$ $s = \frac{1}{2}\ x\ 2s \text{ x } 10m/s$ s = 10m
(f)	s =? l = 4s b = 10m/s	s = l x b s = 4s x 10m/s s = 40m
(g)	s =? a = 4s b = 10s h = 10m/s	$s = \frac{1}{2}(a + b)h$ $s = \frac{1}{2}(4s + 10s)10m/s$ $s = \frac{1}{2} \text{ x } 14s \text{ x } 10m/s$ s = 70m

Exercise

1. A car moving from rest acquires a velocity of 20m/s with uniform acceleration in 4s. It then moves with this velocity for 6s and again accelerates uniformly to 30m/s in 5s. It travels for 3s at this velocity and then comes to rest with uniform deceleration in 12s.
   (a) Draw a velocity-time graph
   (b) Calculate the total distance covered.
   (c) Calculate the average speed.
2. A car starting from rest accelerates uniformly to 20m/s in 5s. And it accelerates more to 40m/s in 2s and then decelerates until it stops 8s later.
   (a) Draw the speed-time graph
   (b) Calculate the retardation
   (c) Calculate the total distance travelled
   (d) Calculate the average speed.
3. A car accelerated uniformly from 10m/s to 20m/s. It travelled a distance of 50m during this time.
   (a) What the acceleration of the car?
   (b) How long does it take to travel this distance?
4. A car stating from rest accelerates uniformly at $5m/s^2$ in 3s.
   (a) Calculate the final velocity
   (b) Calculate the distance covered.
5. A man drives a car at 5Km/h. He brakes and stops in 3s. Calculate the retardation.
6. A man rides a bicycle. He accelerates from rest to a velocity of 8m/s in 5s. What is the acceleration?
7. An object moving at a velocity of 10m/s comes to rest in 4s.
   (a) Sketch the velocity-time graph for the motion of this object.
   (b) Using your graph, calculate the acceleration of the object.
8. The table below shows the readings obtained by a group of pupils performing an experiment to determine variation of velocity with time for a car starting from rest.

Velocity, m/s	0	10	20	20	20
Time, s	0	2	4	6	8

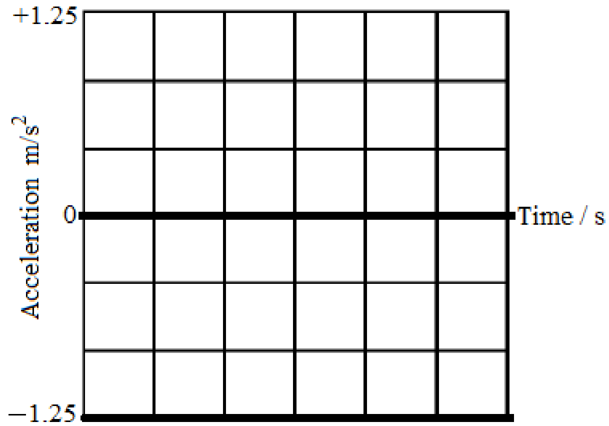


   (a) On the axes above, draw the velocity-time graph.
   (b) Calculate the acceleration of the car for the first 4 seconds.

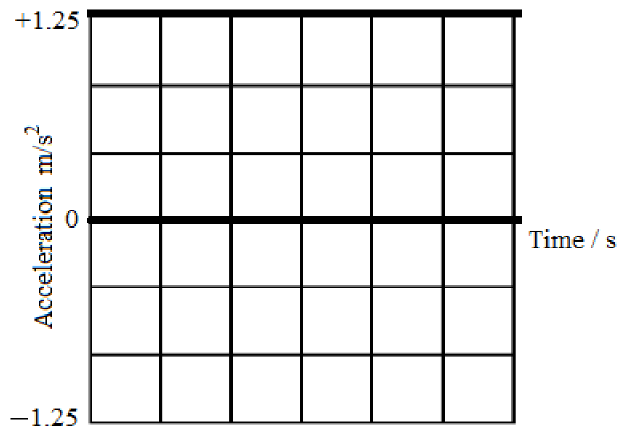
Acceleration-time graphs

Acceleration-time graph is a graph where acceleration is plotted against time.

The diagrams below represent the acceleration- time graphs.

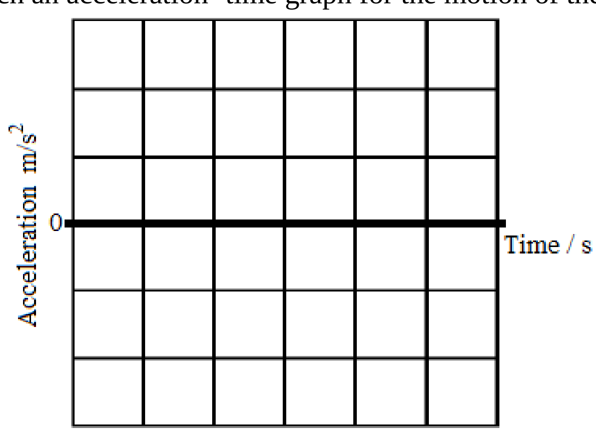


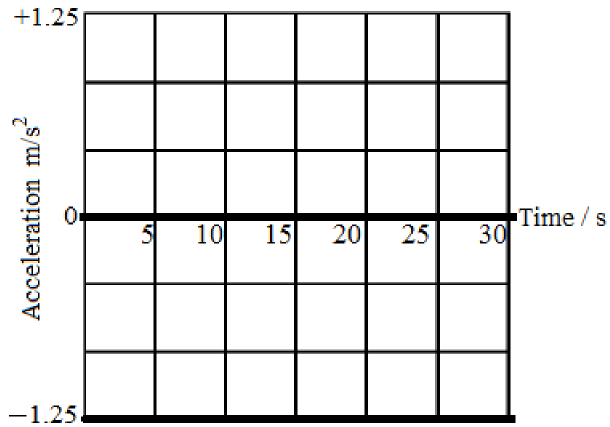
Description
- Constant deceleration
- Decreasing velocity



Description
- Constant acceleration
- Increasing velocity

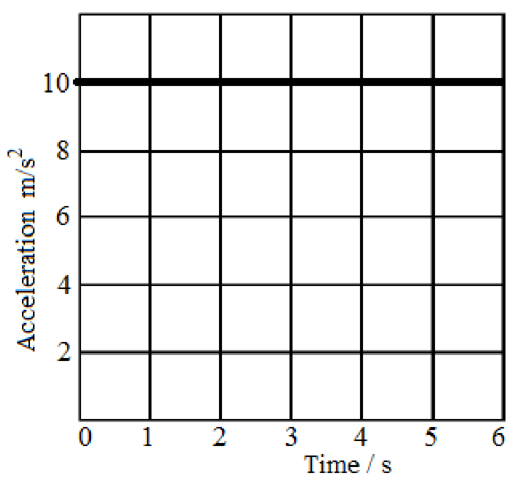
Example
1. As it went past an observer standing by the road side, a bus decelerated at 1.25m/s^2. Thirty seconds later, the bus stopped.
   (a) How far from the observer has the bus moved when it stopped?
   (b) What was the speed of the bus as it went past the observer?
   (c) On the axis below, sketch an acceleration- time graph for the motion of the bus.



	Data	**Solution**
(a)	s =? u = 0m/s t = 30s a = −1.25m/s^2	$s = ut + \frac{1}{2}at^2$ $s = 0m/s \times 30s + \frac{1}{2} \times 1.25m/s^2 \times (30s)^2$ $s = 0m/s \times 30s + \frac{1}{2} \times 1.25m/s^2 \times 30s \times 30s$ s = 562.5m
(b)	u =? v = 0m/s t = 30s a = −1.25m/s^2	v = u + at u = v – at u = 0m/s – (−1.25m/s^2) x 30s u = 37.5m/s
(c)		

Exercise

1. Starting from rest at t = 0s, a car moves in a straight line with an acceleration given by the graph below.



What is the speed of the car at t = 3s?

Acceleration due to gravity: free fall

Symbol: g

SI unit: metre per second squared [m/s^2]

Definition: It is the acceleration of free falling objects.

All objects accelerate uniformly downwards on the earth if air resistance is ignored. This is called acceleration due to gravity.

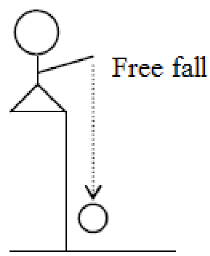
Objects fall because of the gravitational attraction between the objects and the earth.

If an object is dropped from the top of the building, it accelerates uniformly downwards.

- If an object is released without applying force, it starts from rest. This is called free fall.

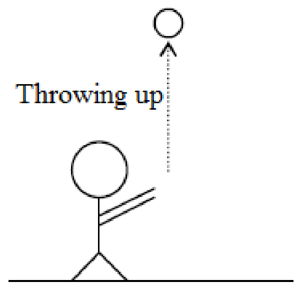
Free fall (dropping)

- u = 0m/s
- g = 10m/s^2

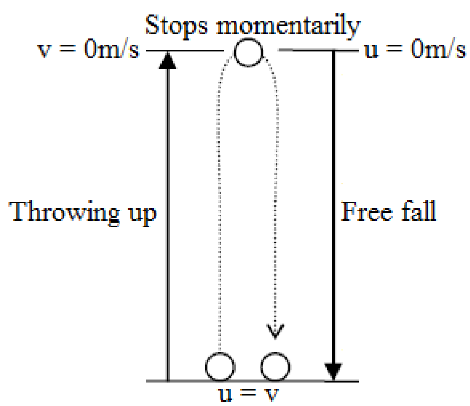


- If an object is thrown vertically upwards, it decelerates to the top. Then the object stops momentarily on the top and then it starts to fall freely.

Throwing up
- v = 0m/s
- $g = -10m/s^2$



Summary



- The time taken for an object thrown vertically upwards to rise is equal to the time it will take to drop, $t_1 = t_2$
  t_1 = time taken from the ground to the top.
  t_2 = time taken from the top to the ground.
  Total time, $t = t_1 + t_2$
- The equations of motion can be applied to free falling objects using "g" instead of "a" and "h" instead of "s".
  - $v = u + gt$
  - $h = ut + \frac{1}{2}gt^2$
  - $v^2 = u^2 + 2gh$
  - $h = \frac{(v+u)t}{2}$

Note
- g = acceleration due to gravity [$10\ m/s^2$]
- h = height or distance [m]

Examples
1. A stone is thrown upwards with an initial velocity of 20m/s. Air resistance is ignored.
   (a) How far does it reach to the top?

(b) How long does it take to the top?
(c) What is its velocity just before reaching the ground?
(d) How long does it take to the ground?

	Data	Solution
(a)	h=? u = 20m/s v = 0m/s g = −10m/s^2	$h = \frac{v^2-u^2}{2g}$ $h = \frac{0^2-20^2}{2\,x(-10)}$ $h = \frac{-400}{-20}$ h = 20m
(b)	t_1 =? u = 20m/s v = 0m/s g = −10m/s^2	$t_1 = \frac{v-u}{g}$ $t_1 = \frac{0-20}{-10}$ t_1 = 2s
(c)	u = v	v = 20m/s (Final velocity is equal to initial velocity, the velocity with which it was thrown)
(d)	t =? t_1 = 2s $t_1 = t_2$ = 2s	$t = t_1 + t_2$ t = 2s + 2s t = 4s

2. A body falls freely from rest. Air resistance is ignored.
(a) What is its velocity after 1s?
(b) How far does it reach in 1s?

	Data	Solution
(a)	v =? u = 0m/s g = 10m/s^2 t = 1s	v = u + gt v = 0m/s + (10m/s^2 x 1s) v = 0m/s + 10m/s v = 10m/s
(b)	h =? u = 0m/s t = 1s g = 10m/s^2	$h = ut + \frac{1}{2}gt^2$ $h = 0m/s \times 1s + \frac{1}{2} \times 10m/s^2 \times 1s \times 1s$ h = 5m

3. A ball is thrown vertically upwards with a velocity of 10m/s. Calculate,
(a) the maximum height that the ball reaches.
(b) the total time the ball is in the air
(c) the velocity with which the ball hits the ground.

	Data	Solution
(a)	h =? u = 10m/s v = 0m/s g = -10m/s^2	$h = \frac{v^2-u^2}{2g}$ $h = \frac{0^2-10^2}{2\,x\,(-10)}$ $h = \frac{-100}{-20}$ h = 5m
(b)	t_1 =? $t_1 = t_2$ t =? h = 5m u = 10m/s v = 0m/s	$t_1 = \frac{2h}{v+u}$ $t_1 = \frac{2\,x\,5m}{0m/s+10m/s}$ $t_1 = \frac{10m}{10m/s}$ t_1 =1s $t_1 = t_2$ = 1s $t = t_1 + t_2$ t = 1s + 1s = 2s
(c)	u = v	v = 10m/s (Final velocity is equal to initial velocity, the velocity with which it was thrown.)

Exercise

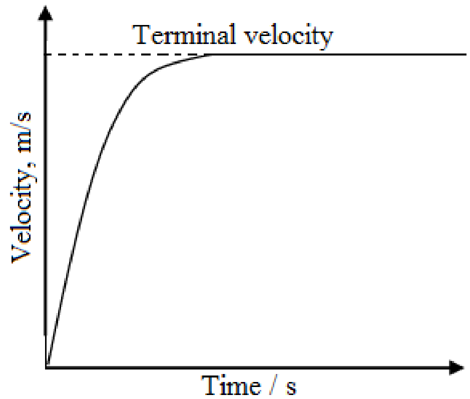
1. A stone is released from the top of a building and takes 3s to reach the ground. The air resistance is ignored.
   (a) What was the final velocity of the stone?
   (b) How tall is the building?
2. A ball is thrown vertically upwards with an initial velocity of 40m/s.

(a) Find the maximum height the ball reaches.
(b) How long does the ball remain in the air? (assuming air resistance is ignored)

Terminal velocity

Terminal velocity is a constant maximum velocity reached by a falling body when the air resistance acting upwards on it equals the downward pull (weight) on the object.

Graph of velocity against time



When an object falls in air, the air resistance (fluid friction) opposing its motion increases as its velocity increases, thus reducing its acceleration. Eventually, air resistance acting upward equals the weight of object acting downwards. At this point the resultant force on the object is zero (since two opposing forces balance) and the acceleration of the body is zero.

Every falling object experiences some air resistances which increase with speed. When a falling object acquires a high speed such that air resistance becomes equal to the weight of the object, the object stops accelerating and falls with constant velocity. This constant velocity is called terminal velocity.

Summary

Bodies which are falling freely in a fluid experiences three forces. These are

(a) ***The upthrust (U)***
This acts vertically upwards and it is equal to the weight of the displaced fluid. Because the volume of the object is constant it follows that the volume and the weight of the displaced fluid will be constant. Therefore the upthrust is also constant.

(b) ***Weight (W)***
Acts vertically downwards and if the gravitational field strength is constant it follows that weight is also constant.

(c) ***Fluid resistance (R)***
Acts vertically upwards it is also called viscosity. This is directly proportional to the velocity of the body. For a body accelerating downwards the weight (W) is greater than the total upward force (R + U). But because the velocity is increasing the value of R also increases until the sum R + U will be equal to W (W = R + U). When this condition is reached the resultant force acting on the body will be zero.

When the body is falling freely in a fluid

(a) ***Acceleration*** decreases until it reaches zero
(b) ***Resultant force*** decreases until it reaches zero [$F_R = W - (U + R)$]
(c) ***Velocity*** increases until it reaches the constant terminal velocity

Factors (conditions) that affect terminal velocity

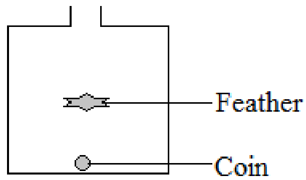
- Size of the object
- Shape of the object
- Weight of the object

- Mass of the object

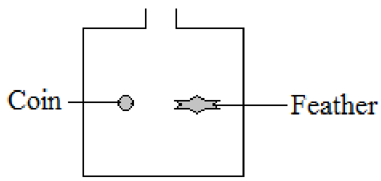
An object of low density but large surface area reaches terminal velocity e.g. a feather.

A man who jumps out of a helicopter has a high terminal velocity, but when he opens the parachute to his advantage, terminal velocity reduces due to increased air resistance.

- If a coin and a feather are enclosed in a long tube which contains air and the tube is inverted, the coin falls much faster than a feather. A feather falls more slowly because it has a low density and large surface area.
  Terminal velocity is reached where there is air.



- If air is pumped out of the tube with a vacuum pump and the tube is inverted, both the feather and the coin fall at the same time and the same acceleration called acceleration due to gravity.
  Terminal velocity is not reached in a vacuum.



Example

1. Fig. 1 shows a free-fall parachutist falling vertically downwards. Fig. 2 shows how the speed of the parachutist varies with time.

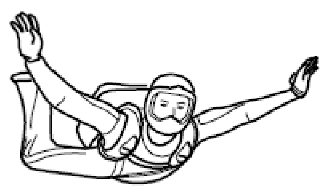


Fig 1

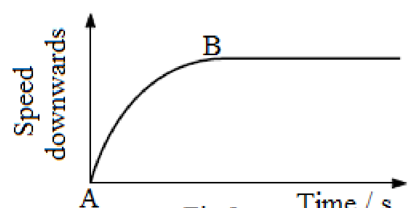


Fig 2

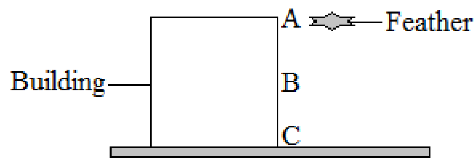
(a) State the name of the downward force acting on the parachutist.
(b) State the name of one upward force acting on the parachutist.
(c) State the initial value of the acceleration of the parachutist. Give the unit of your answer.
(d) Explain why the acceleration decreases from A to B.
(e) Explain why the parachutist falls at a constant speed after B.

Solution

(a) Weight
(b) Air resistance
(c) 10 m/s^2
(d) The increase in air resistance as speed increases decreases acceleration
(e) Constant speed is reached when air resistance balances weight, or when the resultant force is zero

Exercise

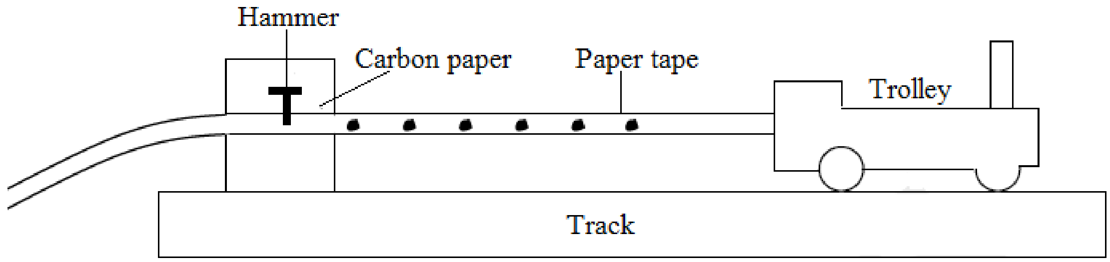
1. Give an example where a person uses terminal velocity to his advantage.
2. Explain a reason why a piece of paper falls more slowly than a stone, although both of them are on earth and are supposed to have the same acceleration of 10m/s^2.
3. The figure below shows a feather, dropped from the top of a building which reaches terminal velocity at point B.



The velocity of the feather at B is 30m/s. If time taken for the feather to move from B to C is 3s, what is its velocity at C?

Recording motion using a ticker tape timer

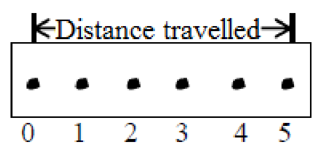
A ticker tape timer is a device that can be used to record motion of an object.
A ticker tape makes dots on a paper tape.
When a paper tape is pulled through the timer, a dot is marked on the tape every 0.02s.
In one second, 50 dots are made on the paper tape. This implies that a dot is made in $\frac{1}{50s}$ or 0.02s, no matter how far apart these dots may be.



Speed of the trolley= $\frac{\text{Total distance}}{\text{Total time}}$

From the tape, you can record both the distance moved and the time taken. Thus the distance between any two successive dots is the distance the object (trolley) has moved in 0.02 seconds.

Analysis of the tape



- Time interval between any two successive dots = 0.02s
- Time taken = number of dot spaces x 0.02s
- Speed = $\frac{\text{Distance}}{\text{Time}}$

  $v = \frac{s}{t}$

Note

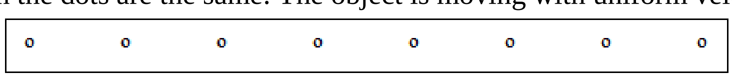
- *Usually there is a mess on the start of tape so measure the time and distance from, say, the tenth dot*
- *The appearance of the dots on the tape gives important immediate information about the movement of tape*

Implications (interpretations) of ticker tape dot pattern

Constant speed

- Equally or evenly spaced dots show that equal distances are traveled in equal times, i.e. the tape is moving with uniform or constant speed.

  Spaces between the dots are the same. The object is moving with uniform velocity.



Accelerating

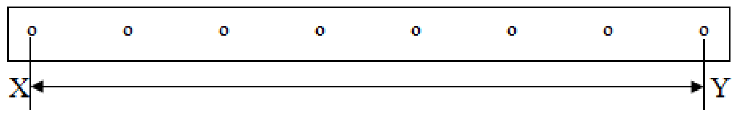
- When the distance between the ticks increases, the tape is accelerating.
  The space between two consecutive dots increases with time. This is for an accelerating body.

Decelerating

- When the distance between the dots decreases, the tape is decelerating.
  The space between consecutive dots is decreasing.

Example

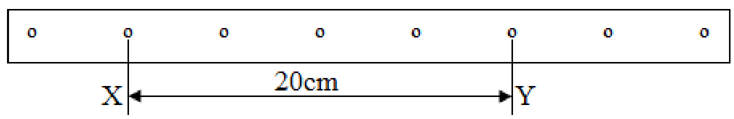
1. Determine the time interval between x and y.



Solution

t = number of dot spaces x 0.02s
t = 7 x 0.02s
t = 0.14s

2. From the ticker tape shown below, work out the speed.



Solution

t = number of dot spaces x 0.02s
t = 4 x 0.02s
t = 0.8s

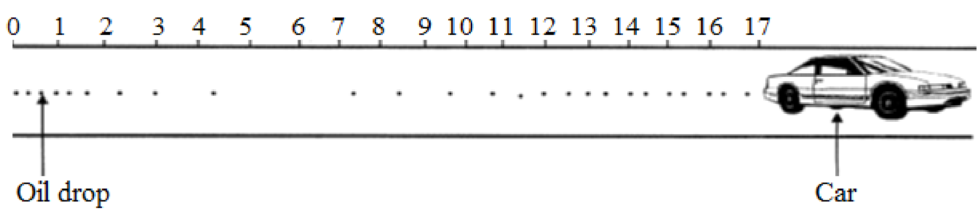
Speed = $\frac{\text{Distance}}{\text{Time}}$

$v = \frac{s}{t}$

$v = \frac{0.2\text{m}}{0.8\text{s}}$

v = 2.5m/s

3. A car with an oil leak travelling along the road loses oil exactly one drop per second. The oil drops on the road as shown below.



(a) Explain what happens on the portion of the road where,
(I) the spaces between the dots are increasing
(II) the dot are equally spaced
(III) the spaces between the dots are decreasing
(b) If the time taken to move from the 4m to 8m marks on the road was 0.5 second, calculate the fastest speed at which the car was travelling.

Solution

(a) (I) the car was accelerating
(II) the car was moving with a constant velocity
(III) the car was decelerating

(b) Speed = $\frac{\text{Change in distance}}{\text{Time}}$

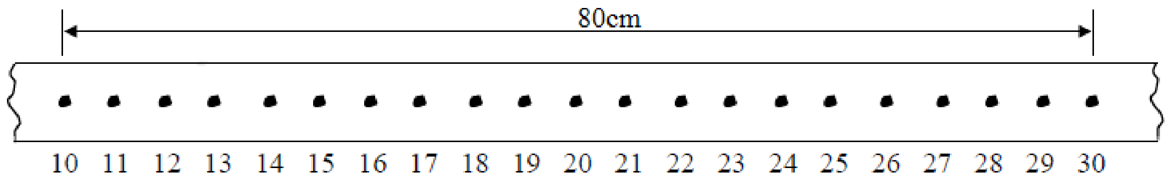
Speed = $\frac{8\text{m}-4\text{m}}{0.5\text{s}}$

Speed = $\frac{4\text{m}}{0.5\text{s}}$

Speed = 8m/s

Exercise

1. Calculate the speed of the object which pulls the paper strip through the ticker timer that the distance between the tenth dots and the thirtieth dot is 80 cm.



Mass
Symbol: m
SI unit: Kilogram [Kg]
Definition: Mass is the quantity of matter contained in a substance.
The mass of an object is also the measure of its inertia.
Measuring instruments

- Electronic balance
- Beam balance

Mass of an object is constant (same) everywhere the object is taken e.g. if the stone on earth is 75Kg, its mass on the moon will also be 75Kg.

Conversion of units

1 Kg = 1000g
1 tonne = 1000Kg
1 tonne = 1000000g

Examples

1. Convert 200 kg to g

Solution

1Kg → 1000g
200Kg → x

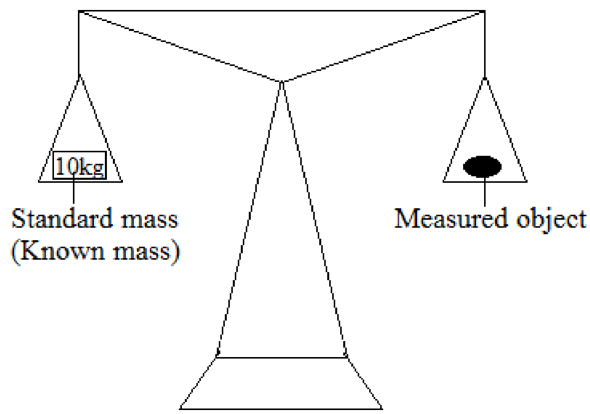
$$x = \frac{200\text{Kg x } 1000\text{g}}{1\text{Kg}}$$

x = 200000g

Measurement of mass

Comparing masses using a beam balance

When measuring the mass of a substance, we compare the mass of the measured object with standard masses (known masses)



Procedure

- Place the standard mass (e.g. 10kg) on one pan.(Standard mass of a substance of mass 10kg is needed)
- Place the measured object on the other pan until the object and standard mass balances.
- When the two balances, it means they have the same mass or weight.

Precautions

- Clean the pans and beams
- Adjust the zeroing screw so that the pointer coincides with the zero mark.
- Read the mass of the known mass object when the beam is balanced.

Determining the mass of a liquid

Experiment

Aim: To find the mass of the liquid, m

Materials

- Electronic balance
- Beaker
- Liquid

Method

- Place a dry empty beaker on the electronic balance and record its mass, m_1.
- Pour the liquid into the beaker. Measure and record the mass of the liquid and beaker, m_2.
- Find the mass of the liquid using the formula; $m = m_2 – m_1$.

Conclusion

Mass of liquid = mass of beaker and liquid – mass of empty beaker

Precaution

- The beaker should be cleaned and dried before the experiment.

Example

1. In an experiment to determine the mass of a certain volume of paraffin, the mass of the beaker was found to be 20g. When the paraffin was poured into the beaker, the mass increased to 42.5g. What was the mass of paraffin?

Solution

Mass of paraffin = mass of beaker and liquid – mass of empty beaker

$m = m_2 - m_1$

$m = 42.5g - 20g$

$m = 22.5g$

Exercise

1. Briefly describe how the mass of a liquid can be determined. Show how the final result can be calculated.

Determining the mass of air

Experiment

Aim: To find the mass of air, m

Materials

- Bottle with air
- Electronic balance
- Vacuum pump

Method

- Place the bottle filled with air on the electronic balance and record the mass, m_1
- Remove the air from the bottle using the vacuum pump. Measure and record the mass of the empty bottle, m2
- Find the mass of the air using the formula; $m = m_1 – m_2$

Conclusion

Mass of air = mass of bottle with air – mass of empty bottle

Example

1. The mass of the bottle filled with air is 60.65g. When the air is removed from the bottle, the mass of the empty bottle is 60g. Calculate the mass of air.

Solution

Mass of air = mass of bottle filled with air – mass of empty bottle

$m = m_1 - m_2$

$m = 60.65g - 60g$

$m = 0.65g$

Exercise

1. A bottle filled with air with mass 32g has a mass of 63.2g. Find the mass of the empty bottle.

Weight

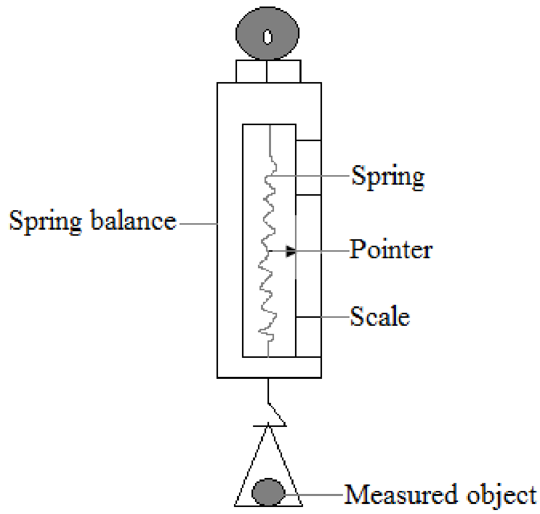
Symbol: W

SI unit: Newton [N]

Definition: Weight is the force of gravity acting on an object.

Instruments for measuring weight

- Spring balance
- Bathroom scale



The weight of an object varies from place to place i.e. from the earth to the moon.
Weight is less on the moon and more on the earth.
There is no weight in the outer space. (Weight is equal to zero newtons)

Factors that affect the weight of an object

1. **Mass of an object**
   The greater the mass of an object, the greater its weight.
2. **Acceleration due to gravity, g**
   When **g** is high, weight is also high and when g is low, weight is also low.
3. **Distance of an object from the centre of the earth**
   As the object is moved further away from the centre of the earth, weight reduces because g keeps reducing. On the earth's surface, **g** varies depending on how far the object is from the centre of the earth. Nearer to the centre of the earth, **g** is high and further away from the centre of the earth, **g** is low. A place at the south or North Pole where g is zero, weight is also zero.
   As the object is moved closer to the centre of the earth, g increases and weight also increases.

Example

1. Explain why;
   (a) The weight of the miner increases as he goes along a deep mine?
   (b) The weight of the space craft reduces as it moves upwards?
   (c) The rocket's weight is zero?

Solution

(a) Because g increases as the miner goes closer to the centre of the earth and this results in an increase in the weight.
(b) Because g reduces as an object moves away from the centre of the earth and weight also reduces.
(c) Because in the space g is equal to zero and also results in weight to be zero.

Differences between mass and weight

Mass	**Weight**
Mass is the quantity of matter contained in a substance	Weight is the force of gravity acting on an object
Mass is a scalar quantity	Weight is a vector quantity.
Mass is measured using a beam balance	Weight is measured using a spring balance
Mass does not change	Weight varies slightly from place to place
Mass is a basic physical quantity	Weight is a derived physical quantity
SI unit for mass is the kilogram	The SI unit for weight is the Newton

Note

- *A mass of 1Kg weighs approximately 10N*

Relationship between mass and weight

Weight = mass x acceleration due to gravity

W = mg
Where;
W = weight [N]

m = mass [Kg]
g = acceleration due to gravity [N/Kg]

Note

- *The value of g on earth is 10N/Kg*
- *The value of g on the moon is 1.6N/Kg*

Example

1. The mass of a man is 70kg. What is his weight on the moon?

Data	**Solution**
W =? m = 70kg g = 10N/kg	W = mg W = 70kg x 10N/kg W = 700N.

2. The weight of an on object is 300N.
   (a) What is its mass on earth?
   (b) What is its
      (I) mass on the moon
      (II) Weight on the moon?
   (c) What is its
      (I) mass in the outer space
      (II) Weight in the outer space?

	Data	**Solution**
(a)	m =? W = 300N g = 10N/kg	$m = \frac{W}{g}$ $m = \frac{300N}{10N/kg}$ m = 30Kg
(b)(I)	Mass does not change	m = 30kg
(II)	W =? m = 30kg g = 1.6N/kg	W = mg W = 30kg x 1.6N/kg W = 48N
(c) (I)	Mass does not change	m = 30kg
(II)	g = 0N/kg m = 30kg	W = mg W = 30kg x 0N/kg W = 0N

Exercise

1. A stone of mass 20kg is placed on earth where gravitational strength is 10N/kg.
   (a) Find the weight of the stone on earth.
   (b) What is the weight of the same stone on the moon?
2. A block of mass 5000g is found at a place on earth where g is 10N/kg.
   (a) Find its weight at this place.
   (b) What is its mass when it is taken down into the mine?
3. An astronaut with a mass of 75kg on earth travels to the moon whose gravitational strength is 1.6N/kg.
   (a) What is meant by mass?
   (b) What is the mass of an astronaut on the moon?
   (c) What is his weight on the moon?
4. State the differences between mass and weight

Centre of gravity

Alternative term: *Centre of mass*

Centre of gravity of an object is the point through which its whole weight appears to act.

Centre of gravity can also be defined as a point within an object where its total mass seems to originate from.

How to determine the centre of gravity of an irregular object

Aim: To find out (locate) the centre of gravity of an irregularly shaped object (plane lamina)

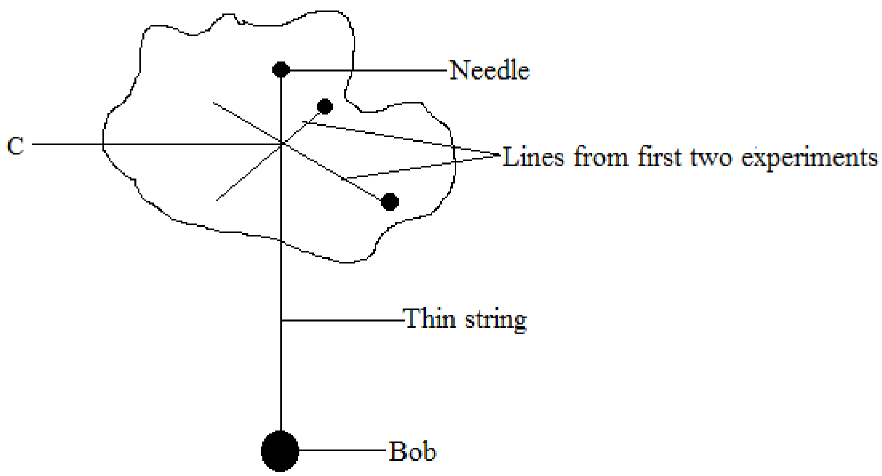
Materials

- String

- Plane lamina/ paper
- Pen/pencil/ruler with a knife edge
- Pin
- Bob

Method

- Make a small hole near the edge of a flat plane lamina
- Hang the plane lamina by a needle and make sure that it can swing freely.
- Hang the plumb line from the same needle and again make sure that it is also free to turn
- Mark the position of the plumb line on the plane lamina (to do this accurately, make a point near the bottom edge of the plane lamina over which the string passes)
- Draw a straight line from the needle to this point to represent the position of the plumb line [the centre of gravity lies somewhere along this line]
- Make two other holes near the edge of the plane lamina so that all the three holes are as far as possible.
- Repeat the experiment and draw two other lines.



Observation

The irregular shaped plane lamina balances at point C.

Conclusion

Since the centre of gravity lies on each of the lines, their intersection locates the centre of gravity.

Stability

Stability of an object is defined as the ability of an object to regain its original position after it has been displaced slightly.

Stability can also be defined as a condition in which an object is not moving and cannot fall.

A stationary object can either be stable or unstable

Something stable is an object which cannot easily fall when slightly pushed or tilted.

Something unstable is an object which can easily fall when slightly pushed or tilted.

Conditions (factors) for stability

- Low centre of gravity
- Wide base

1. **Low centre of gravity**
   The position of centre of gravity affects the stability of an object.
   The centre of gravity should be as low as possible.
2. **Wide base**
   The base area should be as large as possible
   The wider the base, the more stable an object will be.
   The mass of an object should be concentrated at the base.

Equilibrium

Equilibrium is a condition of an object in which the sum of all forces acting on it is zero e.g. resultant force is zero.

Objects which are in equilibrium are;

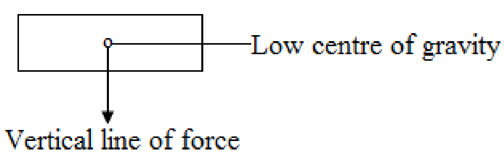
- those that are stationary i.e.at rest
- those that are moving with constant velocity

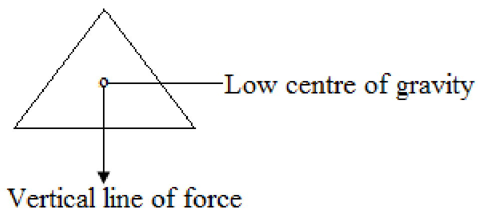
A stationary object can either be in a stable equilibrium, unstable equilibrium or neutral equilibrium.

1. **Stable equilibrium**

   An object is said be in stable equilibrium if when slightly pushed or tilted goes back to its original position.

Examples





The objects above are more stable because they have;

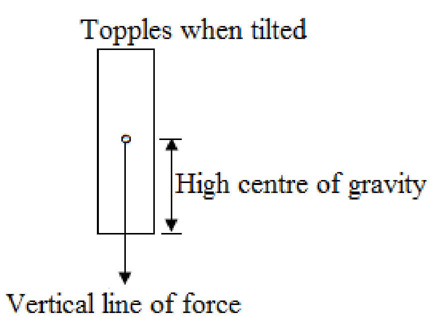
- Low centre of gravity
- Wider base

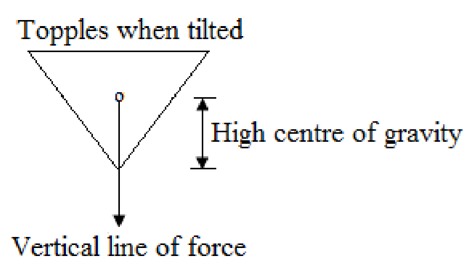
Objects in stable equilibrium do not easily fall when slightly pushed because the vertical line of force from the centre of gravity does not easily fall on the other side of base.

2. **Unstable equilibrium**

   An object is said to be in unstable equilibrium if when slightly pushed or tilted falls off i.e.it does not go back to its original position.

Examples





The objects above are unstable because they have;

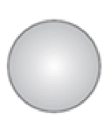
- High centre of gravity
- Smaller or narrow base

Objects in unstable equilibrium fall off easily because when slightly pushed or tilted, the vertical line of force easily falls off on the other side of the base.

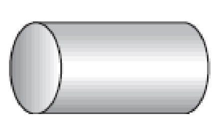
3. **Neutral equilibrium**

   An object is said to be in neutral equilibrium if it stays in its new position after it has been pushed slightly.

Example



Ball



Cylinder

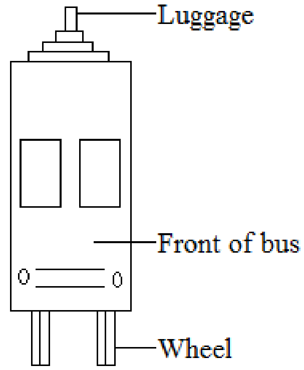
When a ball and a cylinder are rolled, they come to rest in a new stable equilibrium.

Note

- *It is not advisable to put a heavy luggage on the roof of a minibus because it can topple over at the corner when it is moving fast.*

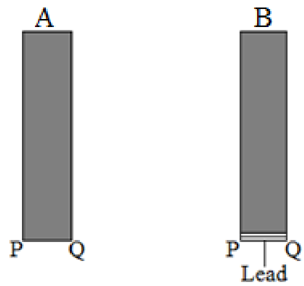
Exercise

1. The figure below shows a bus



State three modifications that should be made in the design of the bus to make it more stable.

2. The diagram below shows two identical rectangular wooden blocks A and B. Block B has a layer of lead attached to its base. The blocks were tilted about edges PQ as shown in the diagram below.



(a) Explain why A topples over at a smaller angle of tilt than B
(b) State two conditions which can help to prevent a truck toppling over when tilted.
(c) What two factors will make an object stable?

Volume

Symbol: V
SI unit: cubic metre [m^3]
Definition: Volume is the amount of space occupied by an object.

Other units for volume

- Cubic centimeters, cm^3
- Milliliters, ml
- Litres, L

Relationship of units

- $1ml = 1cm^3$
- $1L = 1000ml = 1000cm^3$
- $1m^3 = 1000L = 1000000cm^3$

Note

- *In the laboratory, we usually use cubic centimeters because the cubic metre is a very large unit.*

Instruments for measuring volume of liquids

- Measuring cylinder
- Pipette
- Burette
- Flasks

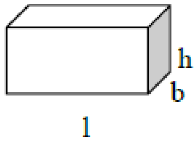
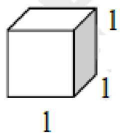
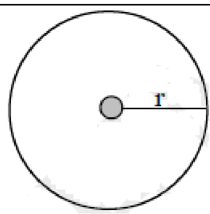
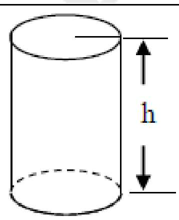
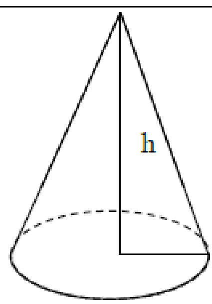
Volume of regular solids

An irregular solid is an object whose sides can be measured easily.

Procedure

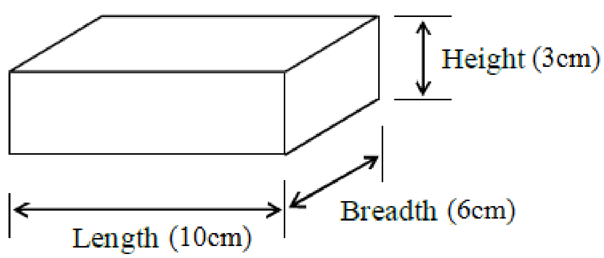
- Measure the length of an object using a ruler or vernier calipers or micrometer screw gauge

- Use the appropriate formula to find the volume.

Shape of object	Name of shape	Formula for volume (m^3)
	Cuboid (rectangle)	$V = l \times b \times h$ $V = A \times h$
	Cube (square)	$V = l^3$
	Sphere (circle)	$V = \frac{4}{3}\pi r^3$
	Cylinder (wire or pipe)	$V = \pi r^2 h$ $V = A \times h$
	Cone	$V = \frac{1}{3}\pi^2 h$

Examples

1. Find the volume of the block which has the following measurements; length = 10cm, breadth = 6cm, height = 3cm.



Data	Solution
V =? l = 10cm b = 6cm h = 3cm	$V = l \times b \times h$ $V = 10cm \times 6cm \times 3cm$ $V = 180cm^3$

2. Find the volume of a cube of sides 4cm.

Data	Solution
V =? l = 4cm	$V = l^3$ $V = l \times l \times l$ $V = 4cm \times 4cm \times 4cm$ $V = 64cm^3$

3. Calculate the volume of the sphere of radius 6cm.

Data	**Solution**
V =? $\pi = \frac{22}{7}$ r = 6cm	$V = \frac{4}{3}\pi r^3$ $V = \frac{4}{3} \times \frac{22}{7} \times 6cm \times 6cm \times 6cm$ $V = 905cm^3$

Exercise

1. Calculate the volume of the pipe of cross section area $30cm^2$ and 50cm long.
2. Find the volume of a wire of diameter 0.2cm and height 7cm.

Volume of liquids

Liquids take the shape of the container in which they are placed.

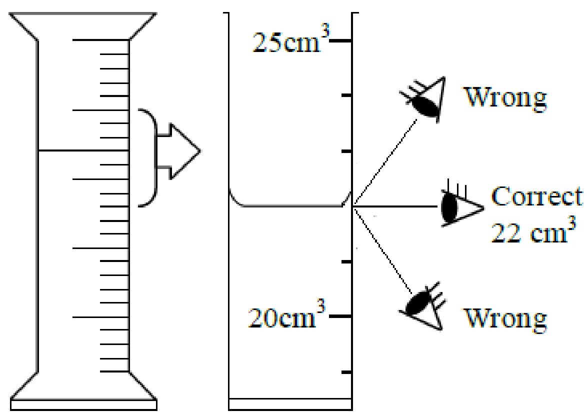
If a container is filled to its capacity, its volume can be determined by pouring the contents into the measuring cylinder.

How to read volumes of liquids

- When a liquid is poured into a measuring cylinder, it forms a curved surface on the upper part of the liquid.
- The curve could be concave or convex depending on the properties of the liquid.
- The curved surface is called meniscus and is caused by the attraction between the liquid particles and the container.
- When the meniscus is convex (i.e. curving upwards) it is read from the top and when it is concave (i.e. curving downwards) it is read from the bottom.

How to use the measuring cylinder

- Pour the measured liquid into the measuring cylinder.
- Read the scale at the flat surface of liquid.



Precautions

- Place the measuring cylinder on the horizontal flat surface
- Place the eye level with the flat surface of liquid.

  (The surface of liquid is curved where it meets the glass. This surface is called the meniscus.)

Volume of irregular solids

An irregular solid is an object whose sides cannot be measured easily.

An irregular solid has no specific dimensions e.g. a stone

The volume of small solids is measured by the **displacement method** using;

- A measuring cylinder
- An over flow can

(a) Using a measuring cylinder

Experiment

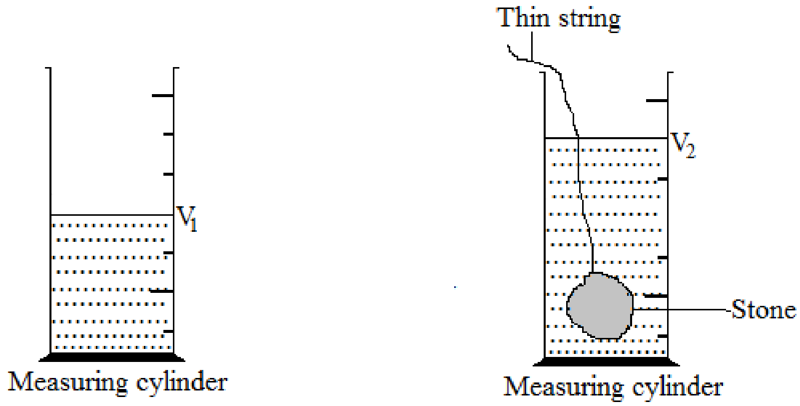
Aim: To find the volume of the stone, V

Materials

- Measuring cylinder
- Water
- Stone
- Thin string

Method

- Pour water into a measuring cylinder and record the initial water level, V_1
- Tie a piece of thin string around a small stone and slowly lower the stone into the measuring cylinder until it is fully submerged. Record the final water level, V_2



- Find the volume of the stone using the formula, $V = V_2 – V_1$

Conclusion

Volume of water displaced by the stone is equal to the volume of the stone.

(b) Using an over flow can

Experiment

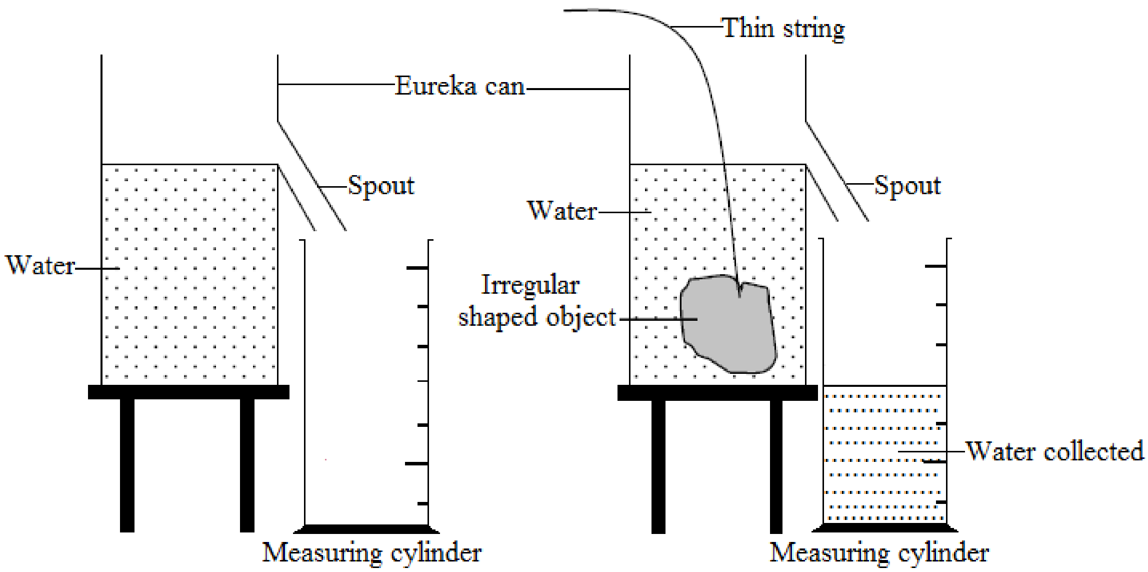
Aim: To find the volume of the stone, V

Materials

- Eureka can (Over flow can)
- Water
- Measuring cylinder
- Tripod stand
- Small stone
- Thin string

Method

- Place an over flow can on a tripod stand
- Pour water into an over flow can until it begins to flow from the spout.
- Leave the can until the water stops over flowing (dripping)
- Place an empty measuring cylinder under the spout
- Tie a piece of thin string around a small stone and slowly lower the stone into the can until it is fully submerged.
- Water from the can is collected in a measuring cylinder. Water collected in the cylinder is the volume of the stone.



Conclusion

The water collected in the measuring cylinder is called displaced water and its volume is equal to the volume of the stone lowered in the can.

Precautions

- Use a thin string to reduce the amount of water displaced by it.
- Use a solid that does not react or dissolve in the liquid.
- Lower the irregular solid gently to avoid the splashing of the liquid.
- Place the measuring cylinder on the flat or horizontal surface
- Tap the measuring cylinder to remove any amount of air bubbles.
- Place the eye level with the flat surface of the liquid [in case of water, read from the bottom of the meniscus]

Example

1. $100cm^3$ of water is poured into a measuring cylinder. A block of copper wire is gently lowered into the measuring cylinder and the water level rises to $183cm^3$ mark.

(a) What is the volume of the copper block?

(b) If the height of the block is 10cm, what is the cross sectional area?

	Data	**Solution**
(a)	$V = ?$ $V_2 = 183cm^3$ $V_1 = 100cm^3$	$V = V_2 - V_1$ $V = 183cm^3 - 100cm^3$ $V = 83cm^3$
(b)	$A = ?$ $V = 83cm^3$ $h = 10cm$	$A = \frac{V}{h}$ $A = \frac{83cm^3}{10cm}$ $A = 8.3cm^2$

Volume of a small irregular floating solid

Experiment

Aim: To find the volume of an irregular floating solid, V

Materials

- Cork (floating object)
- Stone
- Water
- Thin string
- Measuring cylinder

Method

- Pour water into the measuring cylinder.
- Tie a thin string around a small stone and gently lower the stone into the measuring cylinder until it is fully submerged. Record this initial water level, V_1.

- Then tie a floating object together with the stone and then lower them into the same measuring cylinder. Water level rises and record the this final water level, V_2
- Find the volume of the floating object using the formula, $V = V_2 - V_1$

Conclusion

Volume of floating object = final volume – initial volume

Note

- *The stone is used to make the floating object to sink or submerge*
- *Anything that sinks can be used in place of a stone.*

Density

Symbol: D

SI unit: Kilogram per cubic metre [Kg/m^3]

Definition: Density is defined as mass per unit volume of a substance

Formula: Density = $\frac{\text{Mass}}{\text{Volume}}$

$D = \frac{m}{v}$

D = Density [kg/m^3] or [g/cm^3]

m = mass [kg] or [g]

V = volume [m^3] or [cm^3]

Relationship of units

- 1kg/m^3 = 0.001g/cm^3
- 1g/cm^3 = 1000kg/m^3

Example

1. Convert
   (a) 3kg/m^3 into g/cm^3
   (b) 5g/cm^3 into kg/m^3

Solution

(a) 0.001g/cm^3 $\rightarrow$ 1kg/m^3
$x \rightarrow$ 3kg/m^3

$$x = \frac{0.001\text{g/cm}^3 \text{ x } 3\text{kg/m}^3}{1\text{kg/m}^3}$$

x = 0.003g/cm^3

(b) 1g/cm^3 $\rightarrow$ 1000kg/m^3
5g/cm^3 $\rightarrow$ x

$$x = \frac{5\text{g/cm}^3 \text{ x } 1000\text{kg/m}^3}{1\text{g/cm}^3}$$

x = 5000kg/m^3

Exercise

1. Convert
   (a) 3g/cm^3 into Kg/m^3
   (b) 5gK/m^3 into g/cm^3

Simple determination of density

- Find the mass of an object using electronic balance
- Find the volume of an object
- Find the density of an object using the formula; Density = $\frac{\text{Mass}}{\text{Volume}}$

Density of irregular solids

Experiment

Aim: To find the density of an irregular object, D

Method

- Measure and record the mass of the object, m
- Pour water in the measuring cylinder and record the initial volume of water, V_1

- Slowly, lower the object into a measuring cylinder using a thin string and record the final volume of water, V_2.
- Find the density of the object by using the formula; $D = \frac{m}{V_2 - V_1}$

Examples

1. A body of mass 500g was suspended in 100cm^3 of water by a piece of cotton. The level rises to 150cm^3 . What is its density?

Data	**Solution**
D =? m = 500g V_1= 100cm^3 V_2 = 150cm^3	$D = \frac{m}{V_2 - V_1}$ $D = \frac{500g}{150cm^3 - 100cm^3}$ $D = \frac{500g}{50cm^3}$ D = 10g/cm^3

Exercise

1. A material has density of 9.0g/cm^3 and volume 50cm^3. What is its mass?
2. A metal has mass of 225g and volume of 30cm^3. What is its density?

Density of liquids

Experiment

Aim: To find the density of a liquid, D

Materials

- Measuring cylinder
- Electronic balance
- Liquid

Method

- Measure and record the mass of an empty cylinder, m_1.
- Pour the liquid into the measuring cylinder. Measure and record the mass of the cylinder and water, m_2.
- Record the volume of the liquid in the measuring cylinder, V
- Find the density of the liquid by using the formula; $D = \frac{m_2 - m_1}{V}$

Example

1. A container of mass 200g and contains160cm^3 of liquid. The total mass of the container and liquid is 520g. What is the density of the liquid?

Data	**Solution**
D =? m_1 = 200g m_2 = 520g v = 160cm^3	$D = \frac{m_2 - m_1}{V}$ $D = \frac{520g - 200g}{160cm^3}$ $D = \frac{320g}{160cm^3}$ D = 2.0g/cm^3

Exercise

1. A stone of mass 20g and density 0.5g/cm^3 was immersed into water in a measuring cylinder whose initial volume was 30cm^3. Find the final volume of the water in the measuring cylinder.
2. A tin containing 5000cm^3 of paint has a mass of 7.0kg.
   (a) If the mass of the empty tin including the lid is 0.5kg, calculate the density of the paint.
   (b) If the tin is made of a metal which has a density of 7800kgm^{-3}, calculate the volume of metal used to make the tin and the lid.

Relative density

Alternative term: *Specific gravity*

Symbol: RD

Definition: Relative density is the ratio of the mass of a substance to the mass of water.
It is also the ratio of the density of a substance to the density of water

Formula: Relative density = $\frac{\text{Mass of liquid}}{\text{Mass of water}}$

Relative density = $\frac{\text{Density of a substance}}{\text{Density of water}}$

Units: Relative density has no units.

Note

- Density of water = $1g/cm^3$ or $1000kg/m^3$

Example

1. Find the relative density of a liquid of mass 300g if it has the same volume as 100g of water.

Data	Solution
RD=? Mass of liquid = 300g Mass of water = 100g	RD = $\frac{\text{Mass of liquid}}{\text{Mass of water}}$ = $\frac{300g}{100g}$ = 3

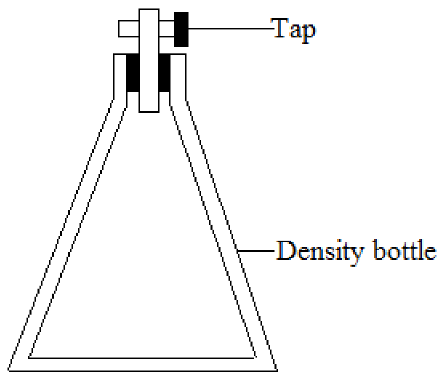
Exercise

1. The density of mercury is $13600kg/m^3$. The density of water is $1000kg/m^3$.

Calculate the relative density of mercury.

Density bottle

A density bottle is used to determine the relative density of a liquid



Experiment

Aim: To find the relative density of a liquid using the density bottle.

Method

- Measure and record the mass of the density bottle
- Measure and record the mass of the density bottle containing the water
- Measure and record the mass of the density bottle containing the liquid under investigation
- Find the relative density by using the formula

  Relative density of liquid = $\frac{\text{Mass of liquid}}{\text{Mass of water}}$

NB

- *The density of a liquid is then found by multiplying relative density of the liquid by the density of water.*
- *Density of liquid = relative density* x *density of water*

Precautions when using a density bottle

- The density bottle must be thoroughly dried.
- The water outside the density bottle must be dried completely with a dry cloth
- The density bottle must be held by the neck to avoid expansion of the liquid [if the bottle itself is held in the hands, the heat will cause expansion to the liquid]
- Remove the water from the top of the stopper with a blotting paper.

Example

1. An empty relative density bottle weighs 25g. The mass of the relative density bottle and with a liquid is 65g. The mass of the relative density bottle with water is 75g.

(a) Calculate the mass of the liquid
(b) Calculate the mass of water
(c) Calculate the relative density of the liquid
(d) Calculate the density of the liquid

Solution

(a) Mass of liquid = 65g – 25g
= 40g

(b) Mass of water = 75g – 25g
= 50g

(c) Relative density of the liquid = $\frac{\text{Mass of liquid}}{\text{Mass of water}}$

Relative density of the liquid = $\frac{40\text{g}}{50\text{g}}$
= 0.8

(d) Density of liquid = relative density x density of water
= 0.8 x 1g/cm^3
= 0.8g/cm^3

Exercise

1. In an experiment, the results below were obtained
   Mass of empty bottle = 50.2g
   Mass of bottle filled with ethanol = 130.2g
   Mass of bottled filled water = 150.2g
   (a) Calculate the mass of the liquid
   (b) Calculate the mass of water
   (c) Calculate the relative density of the liquid
   (d) Calculate the density of the liquid
2. An empty relative density bottle has a mass of 25g. When filled with a liquid of relative density 0.92, its mass becomes 85g.
   Calculate
   (a) The mass of the bottle when filled with water.
   (b) The capacity of the bottle
3. An empty relative density bottle has a mass of 35g. When filled with water, its mass becomes 85.
   Calculate the
   (I) Mass of water
   (II) The volume of the bottle (take density of water to be 1g/cm^3)

Density of air

Experiment

Aim: To find the density of air

Materials

- Electronic balance
- Bottle / container with a top and tube

Note

- *A tube of the container can be connected to a suction pump which draw air in or suck air out of the container.*

Method

- Measure and record the mass of the container filled with air
- Remove all the air from the container using a suction pump and then close the tap. Measure and record the mass of the container without air. (Empty container)
- **NB**: The volume, V, of the container should be known.
- Next, open the container and fill it with water. Close the container tightly and make sure all the air has been replaced by water
- Measure and record the mass of the container filled with water
- Find the density of air by using the formula:

  Density of air = $\frac{\text{Mass of container with air} - \text{Mass of empty container}}{\text{Volume of the air}}$

Note

- *Volume of container = Volume of air = Volume of water*
- *The volume of air depends very much on the temperature and pressure of the surrounding. It is therefore important to take note of the temperature and atmospheric pressure during the experiment.*

Example

1. Mr. Naosa D.K, a physics teacher at Namushakende Secondary School did an experiment to find the density of air and he obtained the following results:
   Mass of container = 265.12g
   Mass of container and air = 265.42g
   Mass of container and water = 515.12g
   Take density of water to be 1g/cm^3

Calculate

(a) The mass of air
(b) The mass of water
(c) The volume of the container
(d) The density of air

Solution

(a) Mass of air = mass of container with air – mass of empty container
= 265.42g – 265.12g
= 0.3g

(b) Mass of water = mass of container with water – mass of empty container
= 515.42g – 265.12g
= 250g

(c) Volume of container = volume of water

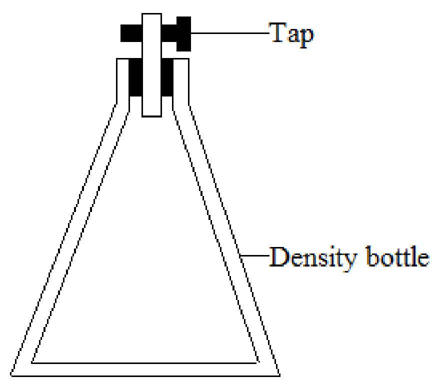
$$\text{Volume} = \frac{\text{Mass}}{\text{Density}}$$
$$= \frac{250\text{g}}{1\text{g/cm}^3}$$
$$= 250\text{cm}^3$$

(d) Volume of air = volume of container

$$\text{Density of air} = \frac{\text{Mass}}{\text{Volume}}$$
$$= \frac{0.3\text{g}}{250\text{cm}^3}$$
$$= 0.0012\text{g/cm}^3$$

Exercise

1. An experiment was carried out by Darlington Naosa junior to determine the density air contained in a thick walled bottle as shown below.



The following results were obtained:
Mass of empty bottle = 309g
Mass of bottle filled with air = 310g
Mass of bottle filled with water = 1050g
Take density of water to be 1g/cm^3

(a) What was the mass of water?
(b) What was the initial volume of the bottle?
(c) What was the mass of air?
(d) Calculate the density of air.

Density of a mixture

Method

- Add the mass of the components to find the total mass
- Add the volume of the components to find the total volume
- Find the density of the mixture by using the formula: Density = $\frac{\text{Total mass of mixture}}{\text{Total volume of mixture}}$

Example

1. 30g of alcohol of volume $38cm^3$ is mixed in a jug with water of volume $20cm^3$ with mass 20g. Find the density of the mixture.

Solution

Total mass of mixture = 30g + 20g

= 50g

Total volume of mixture = $38cm^3 + 20cm^3$

= $58cm^3$

Density = $\frac{\text{Total mass of mixture}}{\text{Total volume of mixture}}$

= $\frac{50\text{g}}{58\text{cm}^3}$

= $0.86g/cm^3$

Exercise

1. 32g of kerosene of density $0.80g/cm^3$is mixed with 8g of water.
   (a) Find the total mass of the mixture
   (b) Find the volume of kerosene
   (c) Find the volume of water
   (d) Calculate the total volume of the mixture
   (e) Calculate the density of the mixture
2. $300cm^3$ of water is mixed with $300cm^3$ of pure alcohol. Calculate the density of the mixture if the relative density of alcohol is 0.79.

Note

- *When impurities of pollutants are added to a substance, its density increases e.g. the density of water is $1g/cm^3$, but when salt is added to it, density increases depending on the amount of impurities.*
- *An egg sinks in pure water because an egg is denser than pure water and an egg floats in salt water because salt water is denser than an egg.*

Force
Symbol: F
SI unit: Newton [N]
Definition: Force is the push or pull exerted on an object
Measuring instrument**:** Spring balance.

Examples of forces

- Weight
- Friction
- Tension
- Up thrust
- Magnetic force
- Electric force
- Constant force

Effects of force on an object

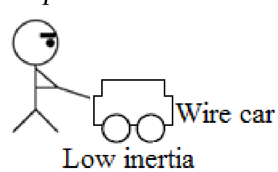
1. Force can change the size and shape of an object
2. Force can change the motion of an object. Force can change the motion of an object in the following ways:
- It makes an object to start moving
- It makes an object accelerates either uniformly or non-uniformly i.e. makes an object accelerates uniformly if the force is constant and makes an object accelerates non uniformly if the force varies.
- It makes an object decelerates
- It makes an object change direction
3. Force can make an object to turn about the point (pivot). It can also make an object to rotate.

Newton's laws of motion
There are three basic laws of motion given by **Sir Isaac Newton**

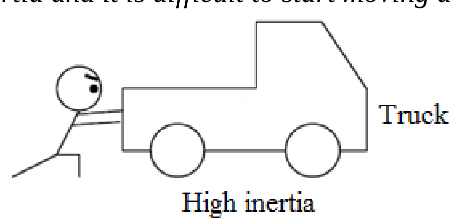
Newton's first law of motion
The law states that: *Any given body continues in its state of rest or uniform motion in a straight line unless it is compelled to change that by an external force exerted on it.*
Newton's first law of motion is also called the law of **inertia.**
Inertia is the property of a body that resists a change to its motion.
Inertia is not a force but a property of an object.
Inertia depends on the mass of an object.
If something has a high resistance [high mass] to the change of motion, its inertia is said to be high.

Note

- *A wire car is easier to start and easier to stop*



- *A heavy truck has high inertia and it is difficult to start moving and difficult to stop.*



Every day effects of inertia

- When a fast moving bus stops suddenly, the passengers tend to be thrown forward to maintain their forward motion. Similarly, when a bus suddenly starts moving, the passengers are thrown backwards; they tend to remain behind due to inertia.
- When a block is placed on a smooth card on the table and the card is suddenly pulled away horizontally, the block remains behind.

Example

1. Use inertia to explain or discuss each of the following situations:
   (a) You fall forward if, while walking, you trip over a log on the ground.
   (b) The driver of the moving car is thrown forward through the wind screen if a car hits the brick wall head on.
   (c) A heavy, rolling pram is more difficult to stop than a light, rolling pram.
   (d) The head of a passenger in a stationary car is jerked backwards when the car he or she is in is struck from behind by a truck

Solution

(a) Due to inertia, you tend to keep on moving in a straight line at a constant velocity, since your mass resists change in its motion.
(b) Due to inertia, you tend to keep on moving in a straight line at a constant velocity, since your mass resists change in its motion.
(c) Because it has a large mass, it has a large inertia since inertia depends on the mass.
(d) Due to inertia, you tend to remain in your state of rest, as a result you feel as though you move backwards.

Newton's second law of motion

The law states that: *An unbalanced force acting on a body produces an acceleration in the direction of the force.*

This acceleration is directly proportional to the force but inversely proportional to the mass of the body.

- $a \propto F$ and $a \propto \frac{1}{m}$

Force = mass x acceleration

F = ma

Note

$a = \frac{F}{m}$

$m = \frac{F}{a}$

- F = force [N]
- m = mass [kg]
- a = acceleration [m/s^2] or [N/kg]ra

Example

1. A horizontal force of 5N was applied to a brick of mass 2kg resting on a frictionless table. What was the acceleration of the brick?

Data	**Solution**
a =? F = 5N m = 2kg	$a = \frac{F}{m}$ $a = \frac{5N}{2kg}$ a = 2.5N/kg

Exercise

1. A man pushes an 8kg luggage on the smooth floor. It starts from rest and reaches the final velocity of 15m/s in 5seconds.
   (a) Calculate the acceleration
   (b) What was the force acting on the luggage?

Newton's third law of motion

The law states that: *To every action, there is an equal and opposite reaction.*

Resultant force

Symbol: R_f

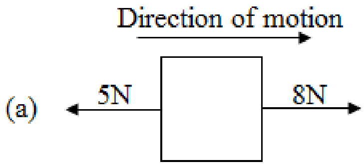
Definition: Resultant force is the sum of all forces acting on a body.

Formula: Resultant force = sum of forward forces – sum of backward forces

Resultant force = horizontal force – friction

Example

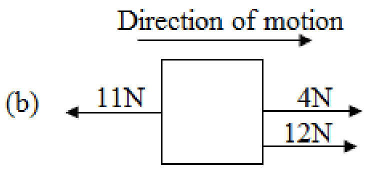
1. Find the resultant force of each of the following:



Solution

$R_f = 8N - 5N$

$= 3N$

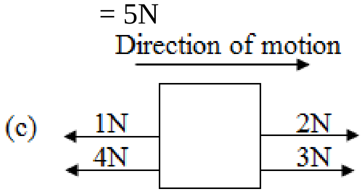


Solution

$R_f = (4N + 12N) - 11N$

$= 16N - 11N$

$= 5N$



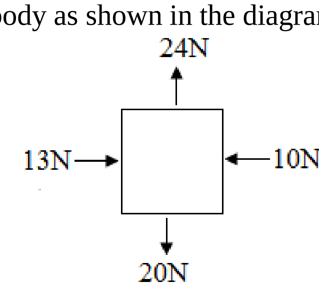
Solution

$R_f = (2N + 3N) - (1N + 4N)$

$= 5N - 5N$

$= 0N$

2. A number of forces are acting on a body as shown in the diagram below

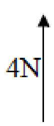


What is the magnitude of the resultant force acting on the body?

Solution

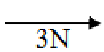
$R_f = 24N - 20N$

$= 4N$

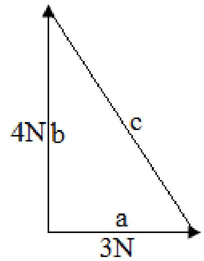


$R_f = 13N - 10N$

$= 3N$

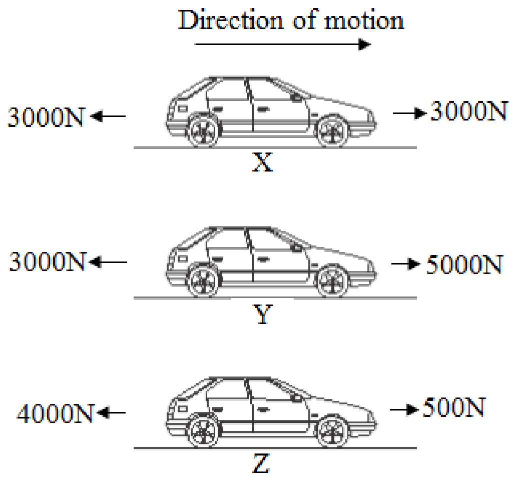


Since force is a vector, we find $c^2 = a^2 + b^2$



$c^2 = 3^2 + 4^2$
$c^2 = 9 + 16$
$c^2 = 25$
$c = \sqrt{25}$
$c = 5N$
The magnitude of the resultant force acting on the body is 5N

3. The figure below shows the total forces acting forwards and backwards on a car at different times X, Y and Z during a journey.

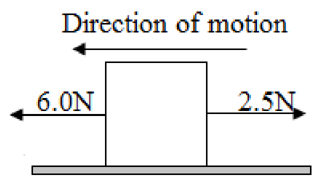


In each case, the car is moving forwards. The mass of the car is 1000kg.
(a) State the name of one of the forces that is acting in the opposite direction to the motion of the car.
(b) State whether the speed of the car is changing at time X.
Explain your answer.
(c) State whether the speed of the car at time Y is increasing, decreasing or is constant.
Explain your answer.
(d) Calculate the acceleration of the car at time Y.

	Data	**Solution**
(a)		Friction
(b)	Forward force = Backward force	The speed is not changing Reason: Because the resultant force is zero (i.e. $R_{f} = 3000N – 3000N = 0N$)
(c)	Forward force > Backward force	Increasing Reason: Because the forward force is greater than the backward force.
(d)	a =? F = 5000N – 3000N = 2000N m = 1000kg	$a = \frac{F}{m}$ $a = \frac{2000N}{1000kg}$ $a = 2m/s^2$

Exercise

1. The figure below shows an object of mass 0.7kg resting on a horizontal surface.



If the object is pulled to the left by a force of 6.0 N and to the right by a force of 2.5N and assuming that no other forces act on the object,

Calculate

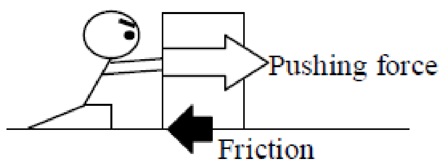
(a) The resultant force
(b) The acceleration produced by the resultant force in (a)
(c) Explain why in practice the actual acceleration for the object may be lower than your answer in (b) above.

Friction

Friction is the force which opposes the motion of two touching surfaces.

Friction acts in the opposite direction to the motion of an object.

For example, if you push a luggage, friction is caused between the luggage and the floor. Its direction is opposite to pushing force and it resists moving. If you push it on a smooth floor like ice, friction reduces and it is easy to push.



Application of friction

- It enables us to walk without slipping
- It enables us to hold or grip something
- It helps a vehicle to run and stop.

Problems (consequences) of friction

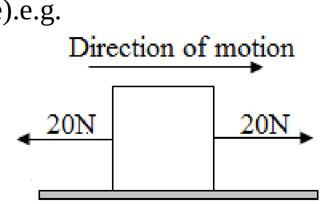
- It produces unnecessary heat and reduces the efficiency of machines
- It causes the wearing and tearing of surfaces in contact

How to reduce friction

- Lubrication of surfaces in contact using grease or oil
- Putting ball bearings between movable surfaces in contact

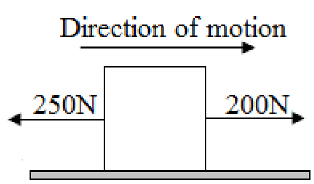
Friction and its effects on motion

1. It causes an object to move with constant velocity. In this case, horizontal force (forward force) is equal to friction (backward force).e.g.



$R_f = 20N - 20N$
$= 0N$

2. It causes an object to come to rest or decelerates. In this case, friction is greater than forward force. e.g.



$R_f = 200N - 250N$
$= -50N$

3. It causes the change in the direction of motion

Exercise

1. A man pushes a packing car having a total mass of 400kg across a floor at a constant speed of 0.5m/s by exerting a horizontal force of 100N.
    (a) How big was the force of friction acting on the car?
    (b) What was the resultant force on the car?

Force and motion in a circular path

There are a number of objects which move round in circular motion.

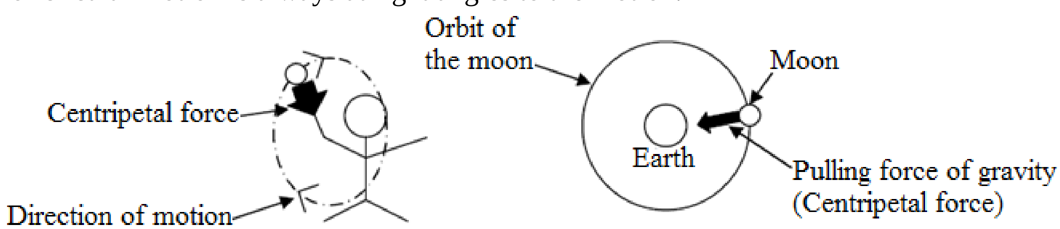
Examples

- The moon goes round the earth
- The earth goes round the sun in an orbit
- In the laboratory, a mass tied to a string can be made to swing round.

Centripetal force

Centripetal force is a force where the direction of the force is always directed towards the Centre of the circle.

The force of circular motion is always at right angles to the motion.



The acceleration caused by the centripetal force is called centripetal acceleration

Effects of force on the shape and size of an object

Force changes the shape and size of an object

The change of shape and size of an object is called deformation

Force can change the size and shape in the following ways;

- It compresses the object, hence reduces it in size
- It stretches the object, hence makes it longer
- It twists the object, hence changes its shape

Elastic material

It is a substance which regains its original shape and size when the force applied has been removed

Examples of elastic materials

- Spring
- Rubber

Elasticity

Definition: Elasticity is the ability of an elastic material to regain its original shape and size after the applied force has been removed

Elastic limit of the spring

Definition: Elastic limit of the spring is the maximum force that can be applied to a spring without stretching it permanently

Original length

Alternative term: Neutral length

Definition: Original length is the length of the spring before being stretched

Formula: Original length = new length – extension

New length

Definition: New length is the length the spring reaches when it is stretched

Formula: New length = original length + extension

Extension

Definition: Extension is the difference between the new length and original length of the spring

Formula: Extension = new length – original length

Experiment

Title: Hooke's law

Aim: To find the relationship between loads and extensions on a spring

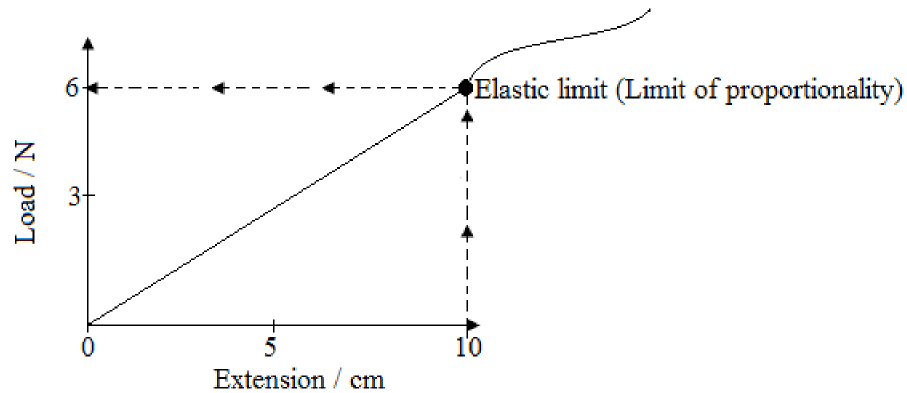
Materials
- Spring
- Loads (standard masses)
- Clamp and stand

Method
- Support the spring vertically by means of a clamp and stand. Place a pan on the lower end of the spring.
- Measure the original length of the spring
- Hang a load (standard mass) on the lower end of the spring
- Calculate the new length of the spring
- Calculate the extension of the spring
- Repeat the experiment by adding loads
- Calculate the spring constant by using the formula;

Constant = $\frac{\text{Load}}{\text{Extension}}$

K = $\frac{F}{E}$

Graph of load against extension



Elastic limit = 6N

Constant = $\frac{\text{Load}}{\text{Extension}}$

K = $\frac{F}{E}$

Constant = $\frac{6\text{N}}{10\text{cm}}$

Constant = 0.6N/cm

Conclusion

The extension of the loaded spring is directly proportional to the force applied, provided the elastic limit is not exceeded. This is called Hooke's law.

Example

1. A load of 1N extends a spring by 5mm. What load extends it by 10mm?

Solution

1N → 5mm

x→ 10mm

$x = \frac{1\text{N x }10\text{mm}}{5\text{mm}}$

x = 2N

2. Calculate the extension of a spring that would be produced by a 20N load if a 15N load extends the spring by 3cm?

Solution

15N → 3cm

20N → x

$x = \frac{20\text{N x }3\text{cm}}{15\text{N}}$

x = 4cm

3. What is the force constant of a spring which is stretched
   (a) 2mm by a force of 4N
   (b) 4cm by a mass of 200g

	Data	Solution
(a)	K =? F = 4N e = 2mm = 0.002m	$K = \frac{F}{e}$ $K = \frac{4N}{0.002m}$ K = 2000N/m
(b)	K =? e = 4cm = 0.04cm m = 200N = 0.2Kg a = 10N/kg	F = ma F = 0.2Kg x 10N/kg F = 2N $K = \frac{F}{e}$ $K = \frac{2N}{0.04m}$ K = 50N/m K = 2000N/m

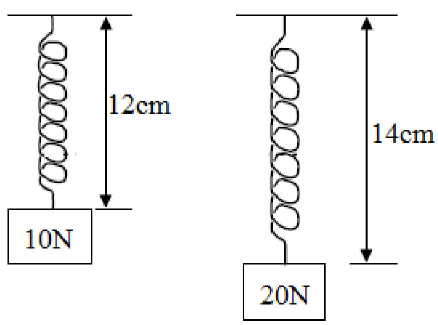
Exercise

1. A load of 4N extends a spring by 10mm. What load would extend it by 15mm?
2. A steel spring obeys Hooke's law. A force of 8N extends a spring by 10mm. Calculate the extension of the spring that would be produced by a force of 10N
3. Use the data below to answer this question;
   Original length = 20cm
   New length = 25cm
   Load = 50N
   (a) Find the extension of the spring
   (b) Calculate the elastic limit
   (c) Find the extension caused by the 100N load that the elastic is not exceeded
   (d) Find the new length when the spring is stretched by the 100N force.
4. In an experiment to verify Hooke's law, standard masses were placed on the pan which was attached to a suspended spring at the lower end. The corresponding lengths of the stretched spring were recorded as shown below.

Load/N	0.0	1.0	1.5	2.0	2.5	3.0	3.5	3.8	3.4	3.1
Length of the spring/mm	500	505	510	515	520	525	528	530	530	530
Extension/mm										

   (a) Complete the table by filling in the values of extension
   (b) Plot the graph of load against extension
   (c) Show clearly on the graph the elastic limit
   (d) Use your graph to determine the spring constant

5. In the figure below, the length of the spring with 10N force hang on it is 12cm and with 20N is 14cm.



What would be the length of the spring with 12N hang on it if the spring obeys Hooke's law?

Moments

Symbol: Γ

SI: Newton metre [Nm]
Definition: Moment is the turning effect of the force about the pivot

Moment of a force

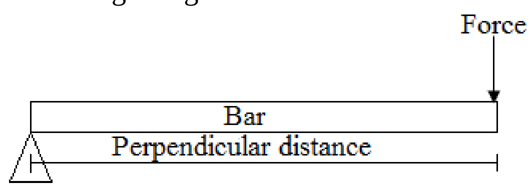
Moment of a force about a pivot is the product of the force and perpendicular distance from the point to the line of action of the force.

Moment = force x perpendicular distance

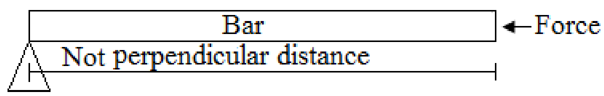
$\Gamma = F \times d$

Γ = moment [Nm]
F = force [N]
d = perpendicular distance[m]

- Perpendicular distance must be distance from the pivot to the force
- Perpendicular distance must be at right angle to the force



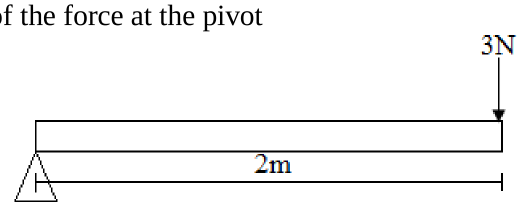
In this case, there is a moment because the force is perpendicular to the bar.
The force can produce the turning effect.



In this case, there is no moment because force is in the same direction of distance
The force doesn't produce the turning effect.

Example

1. Calculate the moment of the force at the pivot



Data	**Solution**
Γ =?	Γ= F x d
F =3N	Γ= 3N x 2m
d = 2m	Γ = 6Nm

Principle of moments

The law states that: For a body in equilibrium, the sum of clockwise moment is equal to the sum of anticlockwise moment about the same point.

Total anticlockwise moment = Total clockwise moment

Experiment

Title: Moments

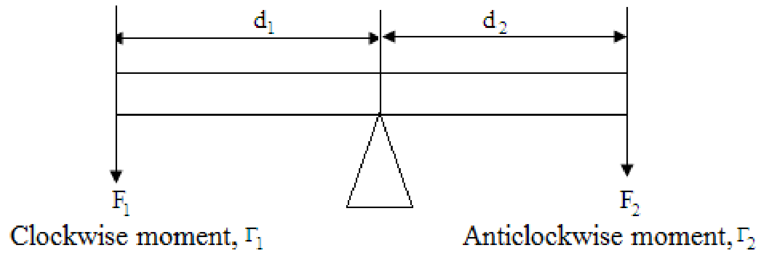
Aim: To verify the principle of moment

Materials

- Long ruler (30cm or more)
- 3 string
- Loads

Method

- Hang a ruler by a string at the centre of mass and make it balanced
- Hang some loads at a certain point from the pivot
- Find the position where other loads are hanging to balance the ruler and measure the length from the pivot to the position.
- Calculate the clockwise moment and the anticlockwise moment
- Repeat the experiment with different pairs of loads and distances



$\Gamma_1 = \Gamma_2$

$F_1 \times d_1 = F_2 \times d_2$

Application of the principle of moments

- Opening or closing a door
- Pair of scissors in use
- See saw in use
- Wheel barrow in use

Conclusion

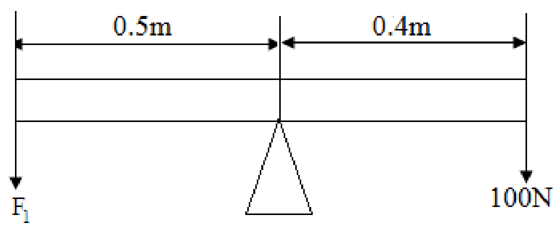
If a body is balanced, then the total clockwise moment is equal to the total anticlockwise moment.

Note

- *A mechanic prefer to use a longer spanner to loosen a nut than a shorter one because a longer spanner would give a greater turning effect.*

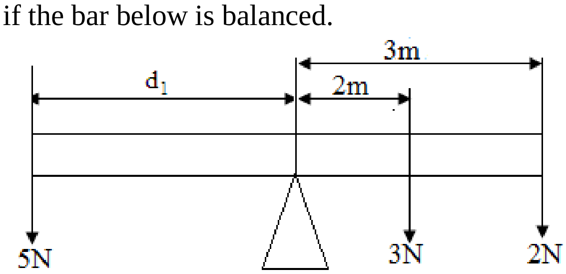
Example

1. Find the force, F_1, if the bar below is balanced



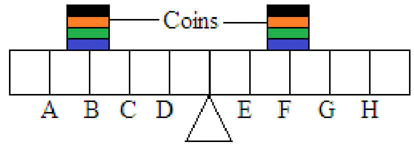
Data	Solution
$\Gamma_1 = \Gamma_2$	$F_1 \times d_1 = F_2 \times d_2$
$F_1 = ?$	$F_1 \times 0.5m = 100N \times 0.4m$
$d_1 = 0.5m$	$F_1 \times 0.5m = 40Nm$
$F_2 = 100N$	$F_1 = \frac{40Nm}{0.5m}$
$d_2 = 0.4m$	$F_1 = 80N$

2. Find the distance, d_1, if the bar below is balanced.



Data	Solution
$\Gamma_1 = \Gamma_2 + \Gamma_3$	$F_1 \times d_1 = F_2 \times F_2 + F_3 \times d_3$
$d_1 = ?$	$5N \times d_1 = 3N \times 2m + 2N \times 3m$
$F_1 = 5N$	$5N \times d_1 = 6Nm + 6Nm$
$d_2 = 2m$	$5N \times d_1 = 12Nm$
$F_2 = 3N$	$d_1 = \frac{12Nm}{5N}$
$d_3 = 3m$	$d_1 = 2.4m$
$F_3 = 2N$	

3. Below is the diagram of a uniform beam suspended on a pivot. Four coins of equal masses are put on points B and F as shown below.



(a) What happens to the beam if left to move freely?
Give a reason for your answer

(b) Which position on the beam would you put one coin to balance the beam? Mark the position with letter P

Solution

(a) The beam will tilt anticlockwise
Reason: The anticlockwise moments are more than the clockwise moments

(b) P should be at H

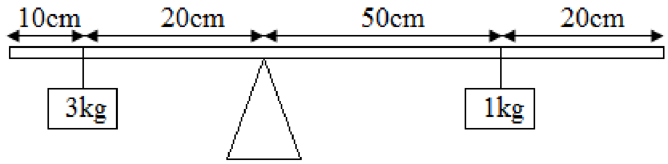
Determining mass using the principle of moments

Hint

- Weight of a body acts through its centre of gravity
- Centre of gravity of a body with a uniform cross section is at mid points
- For a uniform body, its mass / weight is found in the mid-point, e.g a uniform meter rule has its mass or weight calculated from the mid-point.

Example

1. The figure below is a uniform plank of length 100cm kept in equilibrium by a 3kg and 1kg mass placed 10cm and 20cm from each other respectively.



(a) Calculate the
(I) Moment of the 3kg mass
(II) Weight of the plank

(b) State four applications of the above set up

Solution

(a) (I) d = 20cm = 0.2m
To change mass into weight, multiply that mass by 10N/kg
W = mg
W = 3kg x 10N/kg
W = 30N
Γ = F x d
Γ = 30N x 0.2m
Γ = 6Nm

(II) To change mass into weight, multiply that mass by 10N/kg
W_1 = mg
W_1 = 3kg x 10N/kg
W_1 = 30N
W_3 = 1kg x 10N/kg
W_3 = 10N
Weight of the plank
Applying the principle of moments;
W_1 = 30N
d_1 = 20cm
W_2 =?
d_2 = 20cm

$W_3 = 10N$
$d_3 = 50cm$
$W_1 \times d_1 = W_2 \times d_2 + W_3 \times d_3$
$30N \times 20cm = W_2 \times 20cm + 10N \times 50cm$
$600Ncm = W_2 \times 20cm + 500Ncm$
$W_2 \times 20cm = 600Ncm - 500Ncm$
$W_2 \times 20cm = 100Ncm$
$W_2 = \frac{100Ncm}{20cm}$
$W_2 = 5N$

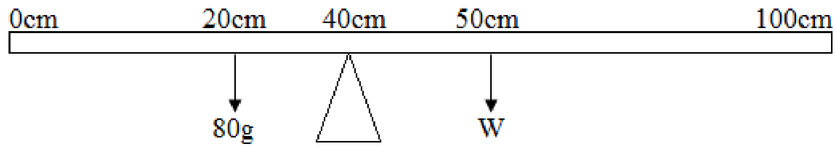
Note

To find the distance of W_2 from the pivot, an assumption was made that 10cm + 20cm which is 30cm was subtracted from 50cm where W_2 is hanging to find 20cm to the pivot.

(b)
- Opening or closing a door
- Pair of scissors in use
- See saw in use
- Wheel barrow in use

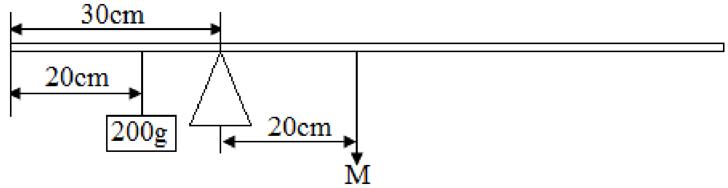
2. A meter rule pivoted at the 40cm mark is balanced by an 80g placed at the 20cm mark. Find the mass of the rule.

Solution



$20cm \times 0.8N = 10cm \times W$
$W = \frac{20cm \times 0.8N}{10cm}$
$W = 1.6N$
$m = \frac{W}{g}$
$m = \frac{1.6N}{10N/kg}$
Mass of the rule = 0.16kg = 160g

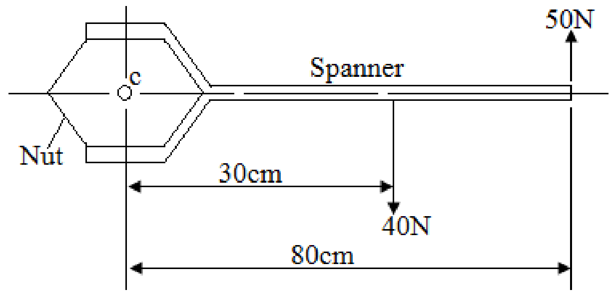
3. The figure below shows a uniform meter rule pivoted at the 30cm mark. It is balanced when a mass of 200g is hung from the 20cm mark, what is the mass, M of the ruler?



Solution

The mass M can be found using the principle of moments
$M_1 \times d_1 = M_2 \times d_2$
$M_1 = 200g$
$d_1 = 30cm - 20cm = 10cm$
$d_2 = 20cm$
$200g \times 10cm = M_2 \times 20cm$
$M_2 = \frac{200g \times 10cm}{20cm}$
$M_2 = 100g$

4. The diagram below shows two forces of 40N and 50N applied to a spanner fitted to a nut.



What is the turning effect of the spanner due to the two forces about c, the center of the nut?

Solution

Turning effect = Anticlockwise moment – clockwise moment

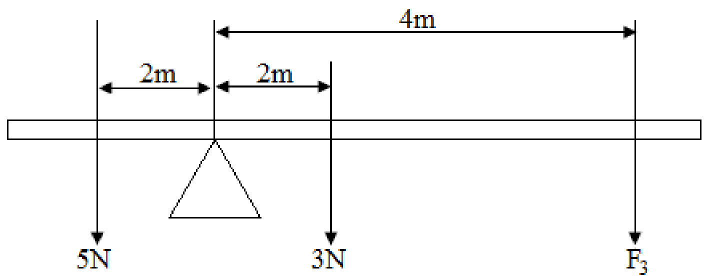
= 50N x 8cm – 40N x 30cm

= 4000Ncm – 1200Ncm

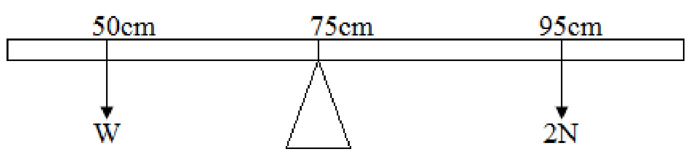
= 2800Ncm

Exercise

1. Calculate the force, F_3, if the bar below is balanced.



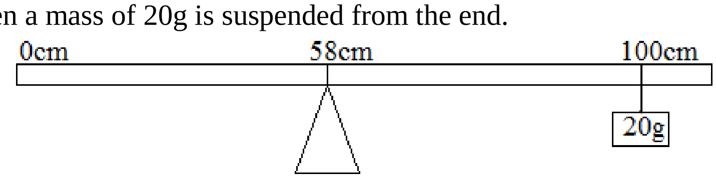
2. The diagram below shows a uniform rule, weight, W, pivoted at the 75cm mark and balanced by a force of 2N acting at the 95cm mark.



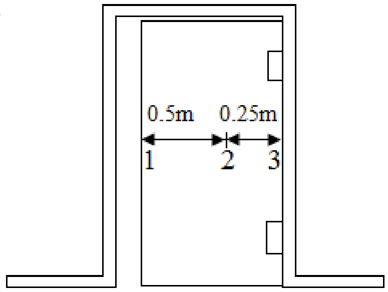
(a) Calculate the moment of the 2N force about the pivot

(b) Use the principle of moments to calculate the value of W.

3. A metre rule hangs by a string at the 80cm mark and a mass of 140g hangs at 95cm mark. The weight of the ruler appears on the centre of mass.
   (a) Where is the pivot
   (b) What is the weight of the 140g mass?
   (c) Calculate the weight of the ruler, W
   (d) Calculate the mass of the ruler
4. The diagram below shows the uniform metre rule balanced horizontally on a knife-edge placed at the 58cm mark when a mass of 20g is suspended from the end.

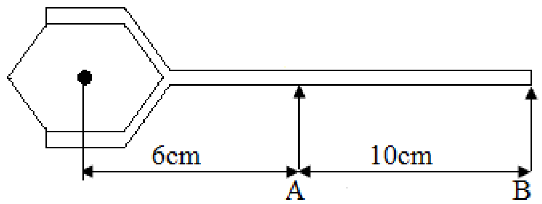


   (a) Find the mass of the rule.
   (b) What is the weight of the rule (taking $g = 10m/s^2$)
   (c) A candle stand has a wide heavy base. Explain why the base has both heavy mass and wide area.
5. The figure below shows a door well secured on the door frame.



(a) What is meant by moment of force?
(b) Calculate the moment of force if a force of 10N is applied at point 1 to open or close the door.
(c) Explain why it is easier to open or close the door if the handle is fixed at point 1 than at point 2 or 3.

6. The diagram below shows a spanner being used to unscrew a nut from a bolt.



(a) Explain why it is better to push the spanner at B rather than at A
(The arrows A and B shows the direction of the applied forces)
(b) A force of 120N is applied at B. Calculate the moment of this force about the center of the bolt.
(c) The nut and bolt are so rusty that cannot be turned. What effect may be produced on the nut and bolt by the application of the force?

Simple machines
A machine is mechanical device which uses an effort to overcome a load
Terms associated with simple machines
Effort
Symbol: E
SI unit: Newton [N]
Definition: Effort is the applied force
Load
Symbol: L
SI unit: Newton [N]
Definition: Load is the force which the effort overcomes.
Load can also be defined as the force an object pulls or pushes on a machine.
Mechanical advantage
Symbol: M.A
Definition: Mechanical advantage is the ratio of the load to the effort.
Formula: M.A = $\frac{\text{Load}}{\text{Effort}}$
Units: M.A has no units since it is a ratio whose units cancel each other.
A machine with greater mechanical advantage is more efficient than the one with a lower mechanical advantage.
Velocity ratio
Alternative term*: Ideal mechanical advantage or speed ratio*
Symbol: V.R
Definition: Velocity ratio is the ratio of the distance moved by the effort to the distance moved by the load
Formula: V.R = $\frac{\text{Distance moved by effort}}{\text{Distance moved by load}}$
Units: V.R has no units since it is a ratio whose units cancel each other.
Velocity ratio depends on the geometry of the machine and not on friction.
Efficiency of a machine
Symbol**:** η
Definitions
Efficiency is the ratio of the useful energy output to the energy input multiplied by 100%.
Efficiency is the ratio of the power output to the power input multiplied by 100%.
Formulae: $\eta = \frac{\text{Energy out put}}{\text{Energy in put}} \times 100\%$

$$\eta = \frac{\text{Power out put}}{\text{Power in put}} \times 100\%$$

$$\eta = \frac{\text{M.A}}{\text{V.R}} \times 100\%$$

Note
- Efficiency of a machine can never be more than 100% because the energy output (work done by a machine) is never more than energy in put (work done on the machine)
- $\eta \leq 100\%$
- M.A≤ V.R
- Generally, in an **ideal situation**, the efficiency of any machine is equal to100% and this just theoretical. This means that M.A = V.R or energy output = energy input.
- Efficiency of a machine cannot be 100% because;
  - *Some energy is used to overcome friction*
  - *Some energy is used to move parts of the machine*

1. Prove that M.A ≤ V.R

Solution

$$\eta = \frac{\text{M.A}}{\text{V.R}} \times 100\%$$
$$\eta \leq 100\%$$

$\frac{M.A}{V.R}$ x 100% ≤ 100%

$\frac{1}{100\%}$ x $\frac{M.A}{V.R}$x 100% ≤ 100% x $\frac{1}{100\%}$

$\frac{M.A}{V.R} \leq 1$

M.A ≤ V.R, hence proved

2. Find the efficiency of an electric motor that is capable of pulling a 50kg mass through a height of 15m after consuming 30,000J of electric energy.

Solution

Energy output = mgh

= 50kg x 10N/kg x 15m

= 7,500J

Energy in put = 30,000J

$\eta = \frac{\text{Enrgy out put}}{\text{Energy in put}}$ x 100%

$= \frac{7{,}500J}{30{,}000J}$ x 100%

= 25%

3. A machine with a velocity ratio of 5 requires 1000J of work to raise a load of 500N through a vertical distance of 1.5m.

Find

(a) The efficiency of the machine

(b) The mechanical advantage of the machine

Solution

(a) Work output = F x d

= 500N x 1.5m

= 750J

Work input = 1000J

$\eta = \frac{\text{Work out put}}{\text{Work in put}}$ x 100%

$= \frac{750J}{1{,}000J}$ x 100%

= 75%

(b) $\eta = \frac{M.A}{V.R}$ x 100%

75% = $\frac{M.A}{5}$ x 100%

M.A = $\frac{75\% \text{ x } 4}{100\%}$

= 3.75

Types of simple machines

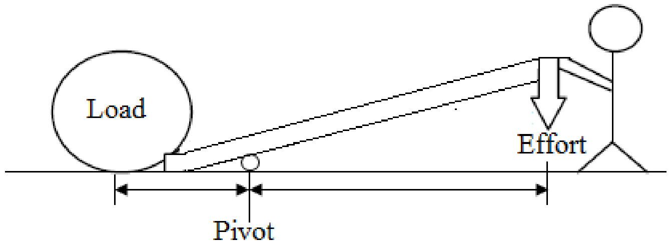
- Levers
- Pulleys
- Inclined planes
- Gears
- Wheel and axle
- Hydraulic machines
- Screws

A. Levers

A lever is a simple machine consisting of a beam or rigid rod pivoted at a fixed hinge, or fulcrum. It is a rigid body capable of rotating on a point on itself.

The simplest form of a lever is a crow bar, but the term lever may be applied to any rigid body which is pivoted about an axis called the *fulcrum*. Levers are based on the principle of moments.

Levers are used to lift heavy weights with least amount of effort.



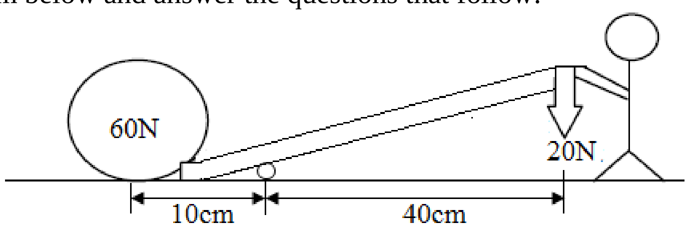
V.R = $\frac{\text{Distance from pivot to effort}}{\text{Distance from pivot to load}}$

Some examples of levers

- Wheel barrow
- Claw hammer
- Table knife
- Scissors
- Bore hole
- Cooking stick
- Hoe
- Slasher

Example

1. Study the diagram below and answer the questions that follow.



Calculate

(a) The mechanical advantage
(b) The velocity ratio
(c) The efficiency

Solution

(a) M.A = $\frac{\text{Load}}{\text{Effort}}$

M.A = $\frac{60\text{N}}{20\text{N}}$

M.A = 3

(b) V.R = $\frac{\text{Distance from pivot to effort}}{\text{Distance from pivot to load}}$

V.R = $\frac{40\text{cm}}{10\text{cm}}$

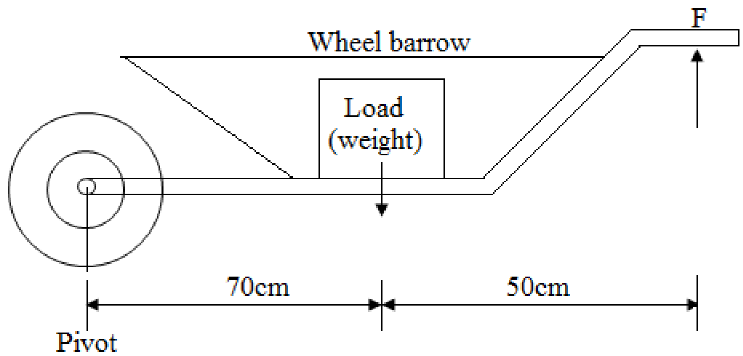
V.R = 4

(c) Efficiency = $\frac{\text{M.A}}{\text{V.R}}$ x 100%

Efficiency = $\frac{3}{4}$ x 100%

Efficiency = 75%

2. A load is to be moved using a wheelbarrow. The total mass of the load and wheelbarrow is 60kg. The gravitational field strength is 10N/kg



What is the size of force, F, needed just to lift the loaded wheelbarrow?

Data	Solution
F =? m = 60kg g = 10N/kg d_1 = 70cm d_2 = 120cm (70cm + 50cm)	W = mg W = 60kg x 10N/kg W = 600N 600N x 70cm = F x 120cm $F = \frac{600N \times 70cm}{120cm}$ F = 350N

B. Pulleys

A pulley is a wheel with a grooved rim mounted on a block

The effort is applied to a rope which passes over the pulley to change the direction of forces and for producing larger force from a small force.

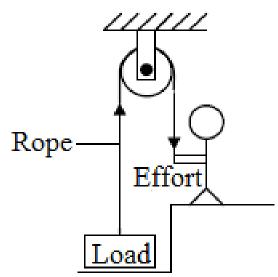
Types of pulleys

- Single fixed pulleys
- Single moving pulleys
- Block and tackle

(I) Single fixed pulley

A single fixed pulley is often used to raise small loads to higher positions.

The pulley has a fixed support, only the rope moves. The tension in the rope is the same throughout. This type of pulley changes the direction of the force but does not change its size.



Load = Effort

M.A = 1

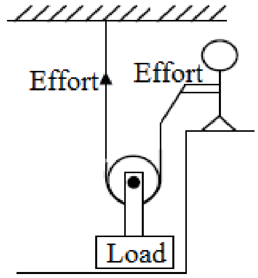
Load distance = effort distance

V.R = 1

(II) Single moving pulley

The single moving pulley is made up of a pulley that is not fixed to any support. It moves freely.

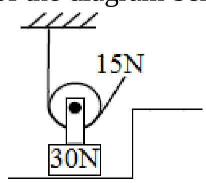
The tension in the string is not normally equal to the effort. Therefore, the upward force on the pulley is twice the effort. This means that the effort is half the load.



Load is twice effort
M.A = 2
V.R = 2

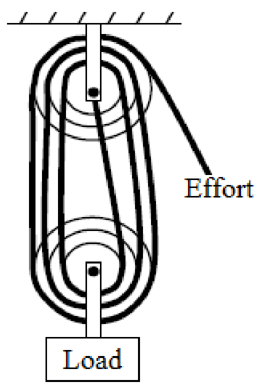
Exercise

1. Calculate the mechanical advantage of the diagram below.



(III) Block and tackle

The block and tackle consists of two sets of pulleys, each with one or more pulleys. The load is carried by one set of the pulleys. A single rope passes round all the pulleys. If an effort is applied to the free end of the rope, this effort is shared between the sections of the rope supporting the lower set of pulleys, which incidentally is equal to the total number of pulleys in the two sets.



The velocity ratio of the pulley system is equal to the number of ropes supporting the lower block.
To find the velocity ratio of the pulley system:

- Count the number of lines connected to moving pulley or
- Count the number of pulley wheels.

V.R = 6

Note

- *If greater loads need to be lifted, more pulleys are used thus increasing M.A. However, increasing the loads increases friction and since more pulleys have to be lifted, efficiency is reduced.*

Example

1. The block and tackle pulley system has 5 pulley wheels and it is used to lift a load of 200kg.
   (a) State the velocity ratio of the pulley system
   (b) If the machine has an efficiency of 80%, calculate the effort applied.
   (Take g = 10N/kg)

Solution

(a) V.R = 5

(b) $\eta = \frac{M.A}{V.R} \times 100\%$

$80\% = \frac{M.A}{5} \times 100\%$

$$M.A = \frac{80\% \times 5}{100\%}$$
$$= 4$$

W = mg
= 200kg 10N/kg
= 2000N

Therefore load = 2000N

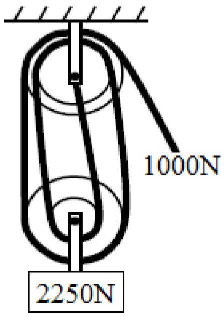
$$M.A = \frac{Load}{Effort}$$

$$4 = \frac{2000N}{Effort}$$

$$Effort = \frac{2000N}{4}$$
$$= 500N$$

Exercise

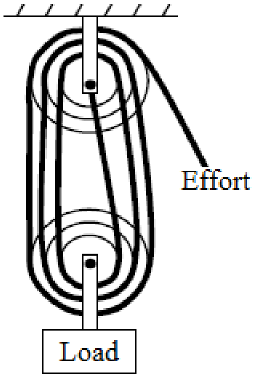
1. The diagram below shows the pulley system.



Find

(a) The mechanical advantage
(b) The velocity ratio
(c) The efficiency

2. The figure below is an ideal system of four pulleys

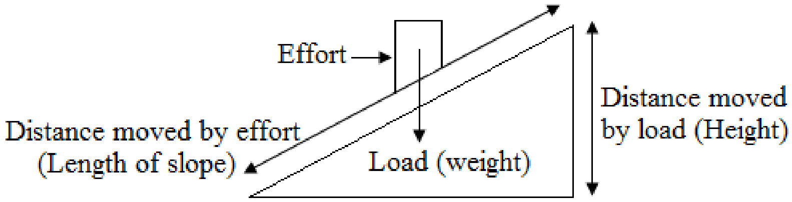


What is the mechanical advantage of this system?

C. Inclined plane

An included plane is a simple machine that is used to raise a heavy load from the ground to a higher platform by way of putting the load than lifting it.

An inclined plane is used to raise heavy loads along a sloping surface.



$$V.R = \frac{\text{Distance moved by effort}}{\text{Distance moved by load}}$$
$$V.R = \frac{\text{Length of slope}}{\text{Height}}$$
$$V.R = \frac{1}{\sin \theta}$$

The load is pulled up an inclined plane in order to raise it through a longer distance to the length of the inclined plan (L). It must be understood that because the weight of the load acts vertically downwards, the distance through the load is overcome is (h) and not (L).

The magnitude of the effort needed to pull or push the load over an inclined plane depends on the

- the length of the plane
- the angle (θ) of inclined plane i.e. of the plane
- the friction force between the plane and the load

The effort needed is smaller when the angle of the inclination θ is very small i.e. the length of the plane is big and when the friction force between the plane and the load is negligible.

Therefore the (MA) of the inclined plane is increased by decreasing the angle of inclination through the use of a longer plane and decreasing the friction between the load (body) and plane

Example

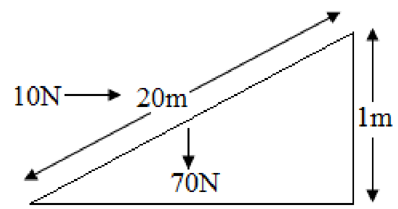
1. An inclined plane making an angle of 30^o to the horizontal is used to as a simple machine. Calculate the V.R of the machine.

Solution

$$V.R = \frac{1}{\sin \theta}$$
$$V.R = \frac{1}{\sin 30^0}$$
$$V.R = \frac{1}{0.5}$$
$$V.R = 2$$

Exercise

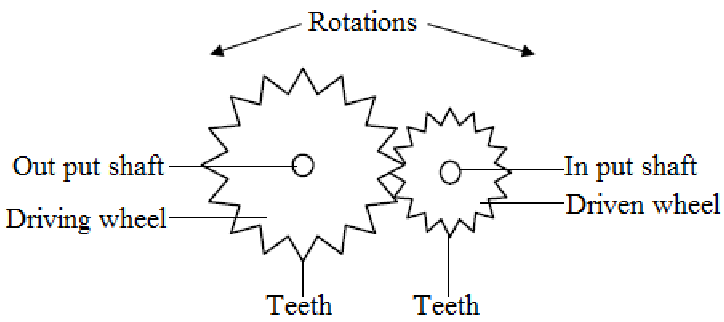
1. The diagram below shows an inclined plane.



Find
(a) The mechanical advantage
(b) The velocity ratio
(c) The efficiency

D. Gears

A gear is a wheel which can rotate around its center and has equally spaced teeth around it. Gears transmit motion from one wheel to another. The driving wheel provides the effort while the driven wheel is the wheel on which the load acts.



$$V.R = \frac{\text{Number of teeth in driving wheel}}{\text{Number of teeth in drven wheel}}$$

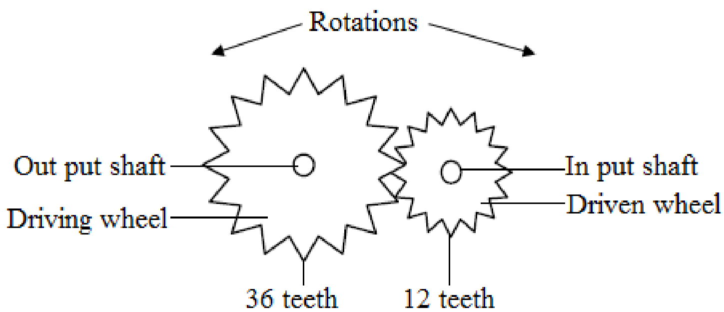
$$V.R = \frac{\text{Effort distance}}{\text{Load distance}}$$

Note

- $Number\ of\ rotations\ in\ driving\ wheel = \dfrac{Number\ of\ rotations\ in\ driven\ wheel}{V.R}$

Exercise

1. The figure below shows the diagram of rotating gear wheels. The driving wheel has 36 teeth and the driven wheel has 12 teeth.

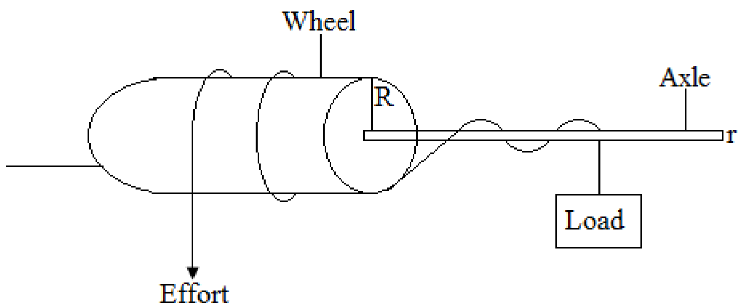


(a) Find the velocity ratio

(b) If the driven wheel makes 15 rotations, how many rotations would the driving wheel make

E. Wheel and axle

The wheel is fixed to the axle. The axle has a smaller diameter and the rope attached to it is wound in the opposite direction to that of the wheel. The load to be lifted is attached to the end of the axle.



The effort therefore moves a distance equal to the circumference of the wheel while the load moves a distance equal to the circumference of the axle.

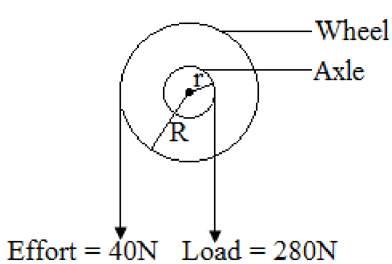
$$V.R = \frac{\text{Effort distance}}{\text{Load distance}}$$

$$V.R = \frac{2\pi R}{2\pi r}$$

$$V.R = \frac{R}{r}$$

Exercise

1. The diagram below shows a wheel and axle (not drawn to scale) used to raise a load of 280N by a force of 40N.



If the radii of the wheel (R) and axle (r) are 70cm and 5cm respectively.

(a) Calculate the mechanical advantage

(b) What is the velocity ratio of the wheel and axle?

(c) Find the efficiency of the wheel and axle

F. Screws

The pitch (P) of a screw is the distance between two successful threads (the turns of the groove). For one complete revolution, a screw moves through a distance equal to its pitch.

$$V.R = \frac{\text{Effort distance}}{\text{Load distance}}$$

$$V.R = \frac{\text{Circumference}}{\text{Pitch}}$$

$$V.R = \frac{2\pi R}{\text{Pitch}} = \frac{2\pi R}{P}$$

Where **R** is the radius of rotation of the effort.

Example

1. The handle of a screw jack is 35cm long and the pitch of the screw is 0.5cm.
   (a) What is the velocity ratio of the system?
   (b) What force must be applied at the end of the handle when lifting a load of 200N if the efficiency of the jack is 40%?

Solution

(a) $V.R = \frac{\text{Effort distance}}{\text{Load distance}}$

$= \frac{2\pi R}{\text{Pitch}}$

$= 2 \times \frac{22}{7} \times \frac{35cm}{0.5\text{cm}}$

$= 440$

(b) Efficiency $= \frac{M.A}{V.R} \times 100\%$

$40\% = \frac{M.A}{440} \times 100\%$

$M.A = \frac{40\% \times 440}{100\%}$

$M.A = 176$

$M.A = \frac{\text{Load}}{\text{Effort}}$

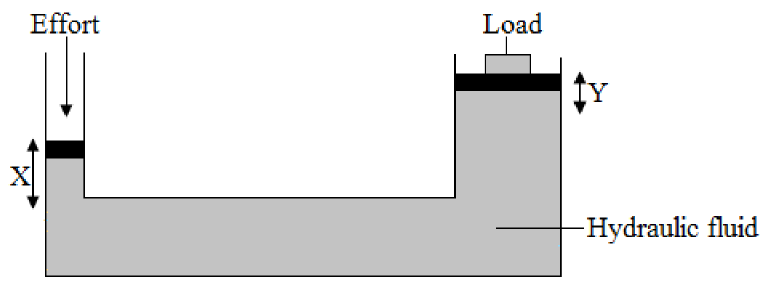
$176 = \frac{200N}{E}$

$E = \frac{200N}{176}$

$E = 1.12N$

G. Hydraulic machines

This machine is used to obtain large force from a small force with the help of a fluid. The hydraulic press applies Pascal's principle which states that in a confined fluid, an externally applied pressure is transmitted equally in all directions.



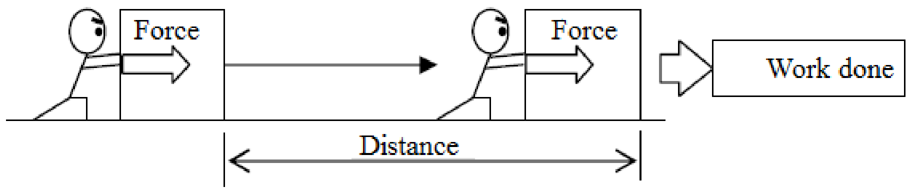
$$V.R = \frac{\text{Effort distance}}{\text{Load distance}}$$

Work
Symbol: W
SI unit: Joule [J]
Definition: Work is the product of the force and the distance moved in the direction of the force.
Formula: Work = force x distance
W = F x d
W = mgh
Note

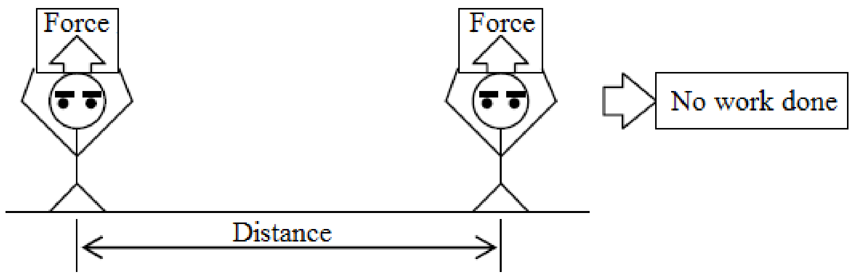
W = work [J]
F = force [N]
d = distance [m]
h = height [m]
m = mass [kg]
g = acceleration due to gravity [$10m/s^2$] or [10N/kg]

Joule is the work done when the point of application of a force of 1 Newton moves through 1 metre in the direction of the force.
1 Newton-metre is equal to 1 joule of work.
Work is said to have been done when we push an object through a certain point or when we lift an object from the ground.

- If a man pushes an object on the floor, he does the work on the object because the distance is in the same direction of the force.



- But if a woman carries a container on her head, she does no work on the container because the distance is in the different direction of the force. (In this case perpendicular to the force)



Relationship of units
- 1KJ = 1000J
- 1MJ = 1000KJ = 1000000J = 10^6J

Note
- KJ = kilo joule
- MJ = mega joule

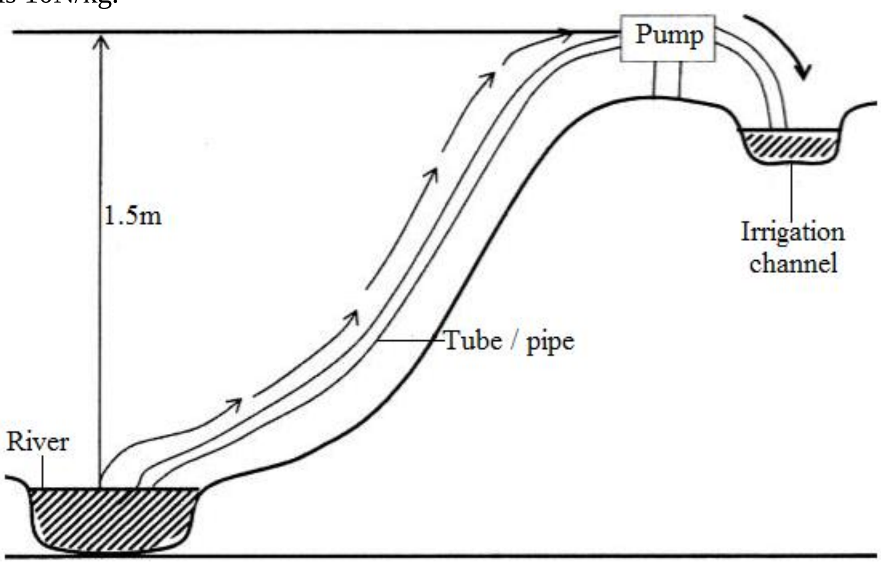
Example
1. A force of 5N acts on a 2kg brick, moving it 8m horizontally from rest. Find the work done by the force.

Data	**Solution**
W =? F = 5N d = 8m	W = F x d W = 5N x 8m W = 40Nm W = 40J

2. A hawk picks a 2kg chicken and lifts it up to a branch of a tree 15m from the ground. How much work has it done on the chicken? g = 10N/kg

Data	Solution
W =? m = 2kg g = 10N/kg h = 15m	W = mgh W = 2kg x 10N/kg x 15m W = 300Nm W = 300J

3. The figure below shows water being pumped from a river into an irrigation channel. The water is lifted to a height of 1.5m.The pump is able to lift 50kg of water each second. The gravitational field strength is 10N/kg.



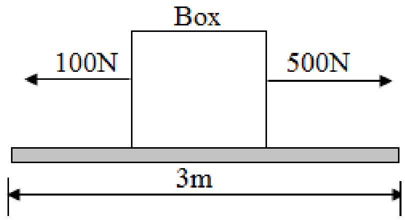
(a) Calculate the work done when 50kg of water is lifted to a height of 1.5m.
(b) The pump used 1200J of energy to lift 50kg of water to a height of 1.5m. Explain the difference between this value and the value calculated in (a).
(c) Calculate the efficiency of the pump.

	Data	Solution
(a)	W =? m = 50Kg g = 10N/kg h = 1.5m	W = F x d W = mgh W = 50Kg x 10N/kg x 1.5m W = 750J (Work in put)
(b)	The calculated value is smaller than the used energy	Some of the energy is used to overcome friction force as the water is being pumped up
(c)	Efficiency =? Work input = 750J Work output = 1200J	Efficiency = $\frac{\text{Work in put}}{\text{Work out put}}$ x 100% Efficiency = $\frac{750\text{J}}{1200\text{J}}$ x 100% = 62.5%

Exercise

1. A car of mass 1000kg is accelerated at $2m/s^2$ from rest in 20 seconds.

Calculate
   (a) The force acting on the car.
   (b) The final velocity
   (c) The distance travelled by the car
   (d) The work done by the car.
2. A crane lifts a weight of 200N through 50m. Find the work done by the crane.
3. A crane lifts a car of mass 500kg through 5m. Find the work done by the crane.
4. A person exerts a horizontal force of 500N on a box, which also experiences a friction force of 100N.



How much work is done against friction when the box moves a horizontal distance of 3m?

Energy
Symbol: E
SI unit: Joule [J]
Definition: Energy is the ability to do work.

Potential energy
Symbol: P. E or E_P
SI unit: Joule [J]
Definition: Potential energy is the energy which the body possess by virtual of its position.
Formula: P.E = mgh
P.E = potential energy [J]
m = mass [kg]
g = acceleration due to gravity [$10m/s^2$] or [10N/kg]
h = height [m]

Example

1. A 2kg object is raised to a height of 5m. What is its potential energy?

Data	Solution
P.E =? m = 2kg g = 10N/kg h = 5m	P.E = mgh P.E = 2kg x 10N/kg x 5m P.E = 100Nm P.E = 100J

2. A rock of mass 10kg is on top of the hill. Calculate the height of the hill if the potential energy of the rock is 5000J. (Take g to be 10N/kg)

Data	Solution
h =? P.E =? m = 2kg g = 10N/kg	P.E = mgh $h = \frac{P.E}{mg}$ $h = \frac{500J}{10Kg \times 10Kg/N}$ h = 5m

Exercise

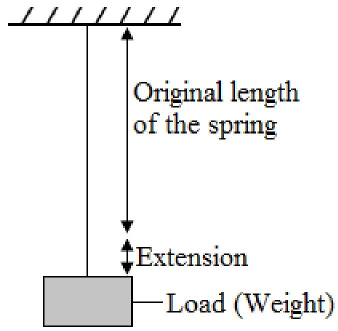
1. A book which has a mass of 1.2kg is put on the desk of height 0.8m. Calculate the potential energy. (Take g to be 10N/kg)

Types of potential energy

- Gravitational potential energy
- Elastic potential energy

Gravitational potential energy
It is the work done in raising an object against the earth's gravitational attraction.

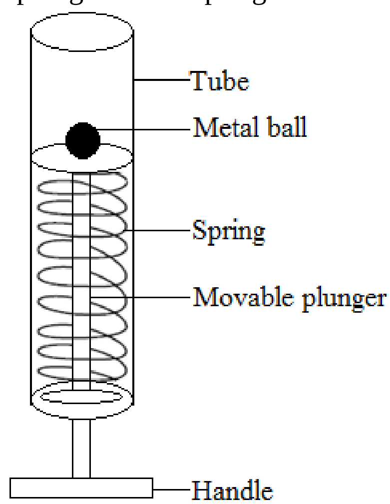
Elastic potential energy
Elastic potential energy is the energy possessed by a stretched spring, wire or rubber band.



All elastic materials gain elastic potential energy whenever their shape is changed. The change in shape of the elastic material causes a gain in elastic potential energy.

Example

1. The figure below is a diagram of a spring catapult designed by a pupil for a science project. The catapult consists of a movable plunger with a spring attached to it.



A metal ball of mass 0.2kg was placed on a metal pale and the handle of the catapult pulled to fully compress the spring. On release of the handle, the ball was projected 1.5m vertically.

(a) Name the type of energy stored in a compressed spring
(b) Calculate the maximum potential energy acquired by the ball from the catapult
(c) Give one reason why the potential energy you have calculated in (a) is less than the original stored energy of the spring

Solution

(a) Elastic potential energy
(b) P.E = mgh
= 0.2kg x 10N/kg x 1.5m
= 3J
(c) Because of friction between the spring and the tube.

Kinetic energy

Symbol: K.E or E_k
SI unit: Joule [J]
Definition: Kinetic energy is the energy the body has due to its motion.
Formula: K.E = $\frac{1}{2}mv^2$

K.E = kinetic energy [J]
m = mass [kg]
v = velocity [m/s]

Example

1. A 2kg stone is thrown vertically with a velocity of 5m/s. What is the kinetic energy?

Data	Solution
K.E =? m = 2kg v =5m/s	K.E = $\frac{1}{2}$mv^2 K.E = $\frac{1}{2}$ x 2kg x (5m/s)^2 K.E = $\frac{1}{2}$ x 2kg x 5m/s x 5m/s K.E = 25J

Exercise

1. A car of mass 500kg moves with a velocity of 20m/s. Find the kinetic energy.
2. A 60kg pupil runs 600m in one minute uniformly.
   (a) Calculate his velocity
   (b) Calculate his kinetic energy.

The law of conservation of energy

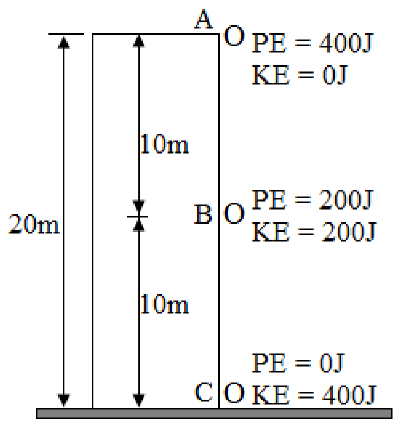
The law states that: Energy cannot be created or destroyed but can only be changed from one form to another.

Energy transformations

Each energy can be changed but the total energy is constant
When there is only P.E and K.E, then;
P.E + K.E = Constant

Applications

(I) ***Conservation of mechanical energy of a falling body***



A. Before a ball is released, its potential energy is 400J and the kinetic energy is 0J because it is not moving
B. At the mid-point of its journey, the potential energy drops to 200J but the kinetic energy increases to 200J. At height 10m, P.E becomes equal to K.E.
   The total energy is still 400J.
C. Just before hitting the ground, the potential energy becomes 0J but the kinetic energy increases to 400J. All P.E becomes K.E
   There is no change in the total energy throughout its falling.

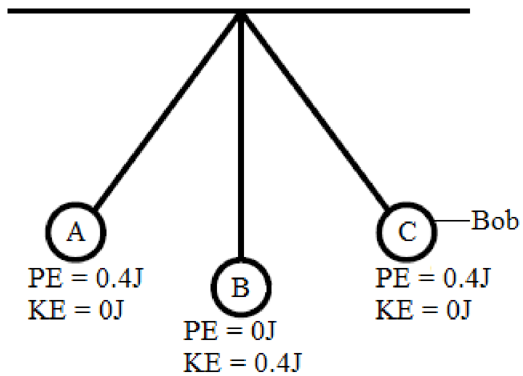
Example

1. A 2kg stone is dropped from the top of a 20m building.
   (a) What potential energy does it possess?
   (b) At what height does its potential energy becomes equal to its kinetic energy?
   (c) What is its kinetic energy just before it hits the ground?
   (d) With what velocity does it reach the ground?

	Data	Solution
(a)	P.E =? m = 2kg g = 10N/kg h = 20m	P.E = mgh P.E = 2kg x 10N/kg x 20m P.E = 400Nm P.E = 400J
(b)		At height 10m
(c)	All P.E becomes K.E	K.E = 400J

(d)	v =? K.E = 400J m = 2kg	$K.E = \frac{1}{2}mv^2$ $400 = \frac{1}{2} \times 2 \times v^2$ $v^2 = 400$ $v = \sqrt{400}$ $v = 20m/s$

(II) ***Conservation of mechanical energy in an oscillating pendulum***



A. The pendulum bob is pulled to position A. Before it is released, its potential energy is 0.4J and kinetic energy is 0J because it is at rest.
B. As the bob moves from A to B, it loses potential energy and gains kinetic energy of 0.4J because of reducing the height and increasing the velocity. It has maximum velocity at B
C. Moving from B to C, the bob slows down losing kinetic energy but gaining potential energy. If air resistance is ignored, the height of A is the same as the height of C because the potential energies must be the same.

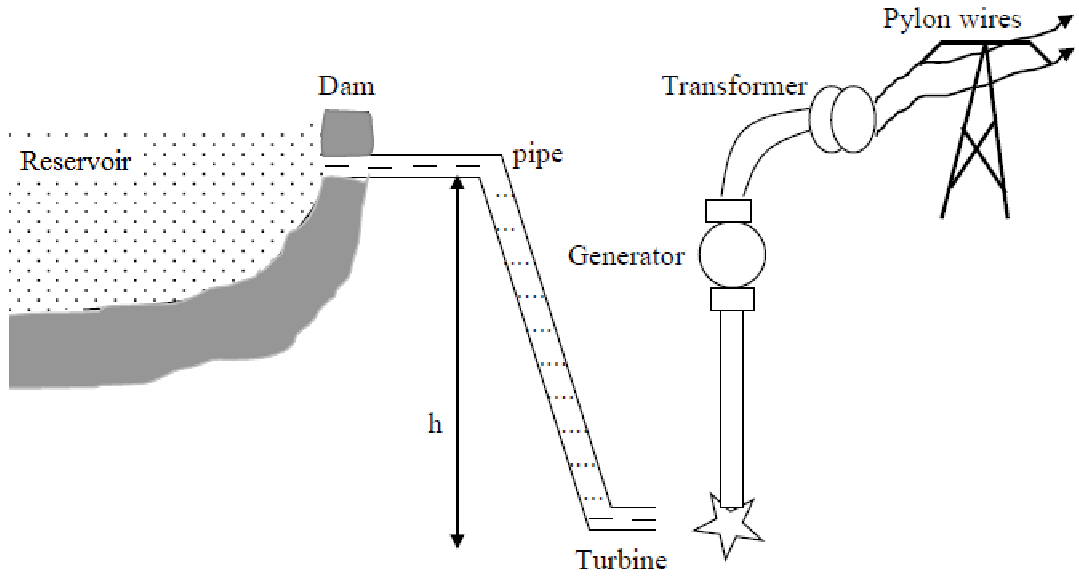
Example

1. A pendulum bob of mass 0.1Kg is raised to a height of 0.4m above its lowest point. It is then released.
   (a) What is its potential energy at this height?
   (b) What is its kinetic energy at its lowest height?
   (c) What is its maximum velocity?

	Data	**Solution**
(a)	P.E =? m = 0.1kg g = 10N/kg h = 0.4m	P.E = mgh P.E = 0.1Kg x 10N/kg x 0.4m P.E = 0.4Nm P.E = 0.4J
(b)	All P.E becomes K.E	K.E = 0.4J
(c)	v =? m = 0.1kg K.E = 0.4J	$K.E = \frac{1}{2}mv^2$ $0.4 = \frac{1}{2} \times 0.1 \times v^2$ $v^2 \times 0.1 = 0.4 \times 2$ $v^2 \times 0.1 = 0.8$ $v^2 = \frac{0.8}{0.1}$ $v^2 = 8$ $v = \sqrt{8} = 2.83m/s$

(III) ***Hydro – electric power generation***

In hydro – electric power generation, water is trapped in the reservoir by building a dam across a river. From the dam the water goes into pipes which fall through a great height ‘h’ in order to increase the potential energy.



When water falls through the height ‘h’, the potential energy is converted to kinetic energy which will rotate the turbine connected to the generator. The generator converts the mechanical energy (PE + KE) into electrical energy which is transmitted to the consumers.

Exercise

1. A 25kg bag of Mealie meal is lifted from the ground to the top of a wall 1.8m high in 0.6 seconds.
   (a) What type of energy has the Mealie meal bag gained?
   (b) If the bag is released from the wall, with what velocity does it strike the ground?
   (c) Calculate the power which developed
   (d) On striking the ground, into what form is the energy of the bag converted?
2. A tin of mass 64g fell from a height of 11.25m.
   (a) Work out the speed of the tin at the moment it struck the ground.
   (b) Calculate the kinetic energy of the tin when it was just hitting the ground.

Power

Symbol: P
SI unit: watt [W]
Definition: Power is the rate of doing work.
Formula: Power = $\frac{\text{Work done}}{\text{Time taken}}$

$P = \frac{W}{t}$

$P = \frac{mgh}{t}$

P = power [W]
W = wok [J]
t = time [s]
m = mass [kg]
g = acceleration due to gravity [10m/s^2] or [10N/kg]
h = height [m]

Example

1. A machine can lift 200kg to a height of 100m in 20 seconds. Find the useful power of the machine.

Data	**Solution**
P =? m = 200kg g = 10N/kg h = 100m t = 20s	$P = \frac{mgh}{t}$ $P = \frac{200\text{kg x } 10\text{N/kg x } 100\text{m}}{20\text{s}}$ P = 10,000W

2. A boy whose mass is 40kg finds that he can ran up a flight of 45 steps each 16cm high in 5 seconds. Calculate the power.

Data	Solution
P =? m = 40kg g = 10N/kg h = 45 x 16cm = 720cm = 7.2m t = 5s	$P = \frac{mgh}{t}$ $P = \frac{40kg \ X\ 10N/kg\ X\ 7.2m}{5s}$ $P = \frac{2880J}{5s}$ P = 576W

Exercise

1. A pupil of mass 50kg runs up a flight of 20 stairs each 5cm high in a time of 20 seconds.
   [Take g = 10N/kg]
   Calculate
   (a) The pupil's gain in potential energy
   (b) The useful power developed by the pupil in climbing the stairs.
2. A force of 1000N is needed to push a mass of 30kg through a distance of 40m to raise an inclined plane to a height of 5m.
   Calculate
   (a) The weight of the object
   (b) The mechanical advantage
   (c) The velocity ratio
   (d) The efficiency of the inclined plane.
   (e) The energy at the height of 5m
   (f) The work done by the force of 1000N.

Pressure
Symbol: P
Units:
- Newton per metre squared [N/m^2]
- Pascal [Pa]
- Millibars
- Atmospheres [atm]
- Torres [torr]

Note
- $1\text{atm} = 760\text{mmHg} = 760 \text{ torr} = 101325 \text{ Pa}$
- $1 \text{ Pascal} = 1\text{N/m}^2$
- $1000\text{Pa} = 1\text{Kpa}$

Definition: Pressure is force per unit area
Formula: Pressure = $\frac{\text{Force}}{\text{Area}}$

$$P = \frac{F}{A}$$

Gas pressure is as a result of the collisions of the gas molecules with the walls of the container vessel
Pressure of a gas depends on:
- Frequency of collisions
- Speed of molecules

Frequency	Pressure
High	High
Low	Low

Speed	Pressure
High	High
Low	Low

Examples
1. If a 70N is applied over an area of $0.8m^2$, how much pressure is exerted?

Data	Solution
P =? F = 70N $A = 87.5\text{m}^2$	$P = \frac{F}{A}$ $P = \frac{70\text{N}}{0.8\text{m}^2}$ $P = 87.5 \text{ N/m}^2$

2. When a force acted over an area of $16cm^2$, a pressure of 20 Kpa was established. Determine the magnitude of the force.

Data	Solution
F =? P = 20Kpa = 20,000Pa $A = 16\text{cm}^2 = 0.0016\text{m}^2$	$P = \frac{F}{A}$ F = PA $F = 20000\text{Pa x } 16 \text{ x } 10^{-3}\text{m}$ F = 32N

3. Explain why
   (a) A sharp knife cuts well
   (b) Low heeled shoes are more comfortable than high heeled shoes

Solution
   (a) A sharp knife cuts well because it exerts less pressure
   (b) Low heeled shoes exerts less pressure on the feet compared to high heeled shoes

4. A block measuring 0.1m x 0.2m x 0.8m has a mass of 20Kg.

What is the maximum and minimum pressure it can exert on the ground?

Data	Solution
P =? W = weight m = 20Kg g = 10N/Kg L = 0.1m B = 0.2m	$P = \frac{F}{A}$ $P = \frac{W}{A}$ W = mg $P = \frac{mg}{\text{L x B}}$

	$P = \frac{20Kg \times 10N/Kg}{0.1m \times 0.2m}$ $P = \frac{200N}{0.02m^2}$ $P = 10000N/m^2$

Exercise

1. A person weighing 1200N is supported on an inflated air pillow. The total area of soles of his shoes is $0.1m^2$. Calculate the minimum pressure of the air inside the pillow.

Pressure due to a liquid column

Formula: $P = \rho hg$

Note

P = pressure
ρ = density
h = height or depth
g = acceleration due to gravity
Pressure due to a liquid column increases with depth
Pressure due to a liquid column increases with density of the liquid

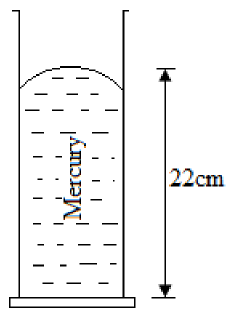
Example

1. Density of water is $1000Kg/m^3$. Determine the pressure due to a liquid at the bed of a 6m deep river.

Solution

$P = \rho hg$
$P = 1000Kg/m^3 \times 6m \times 10N/Kg$
$P = 60000N/m^2$

2. Refer to the diagram below



Density of mercury is $13.6g/cm^3$. Calculate the pressure exerted by the mercury at the base area of the measuring cylinder.

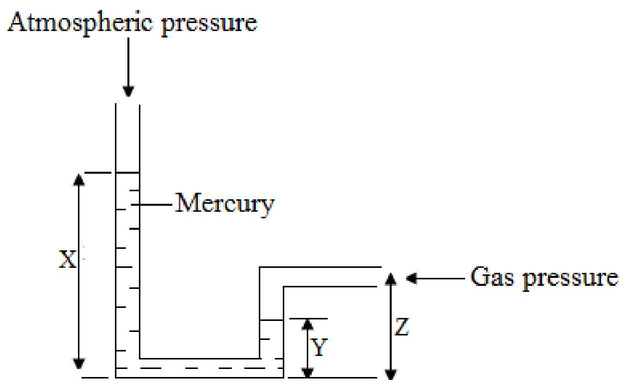
Solution

$P = \rho hg$
$P = 13.6g/cm^3 \times 22 \times {}^{-2}m \times 10N/Kg$
$P = 13600Kg/m^3 \times 0.22m \times 10N/Kg$
$P = 29920$ Pa

Measurement of pressure using a manometer

A manometer can be used to measure pressure. It applies the following relationship

$P_1 + P_2 = P_3$
P_1 = atmospheric pressure
P_2 = pressure due to a liquid column
P_3 = pressure due to a gas supply



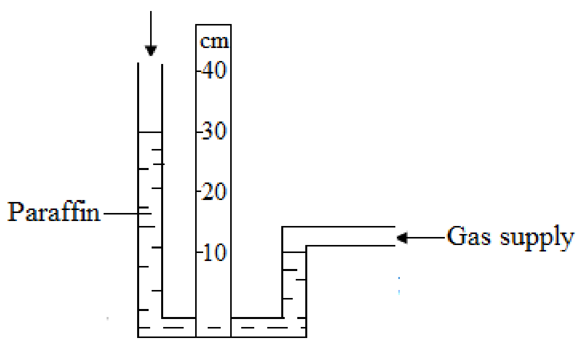
Note

- h = X – Y

Atmospheric pressure = 76cmHg = 760mmHg = 1×10^5 Pa

Example

1. Refer to the diagram below



The density of paraffin is 790Kg/m^3 and atmospheric pressure is 1×10^5 Pa, determine

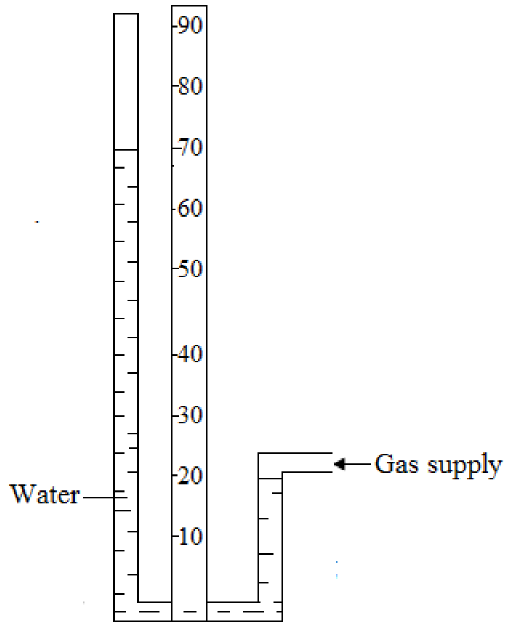
(a) Pressure due the liquid head of the paraffin column
(b) The pressure of the gas supply

Solution

(a) P = ρhg
P = 790Kg/m^3 x (0.3m – 0.1m)x 10N/Kg
P = 790Kg/m^3 x 0.2m x 10N/Kg
P = 1580 Pa

(b) $P_3 = P_1 + P_2$
P_3 = 1×10^5 Pa + 1580 Pa
P_3 = 101580 Pa

2. Refer to the diagram below



The density of water is $1000Kg/m^3$ and atmospheric pressure is 1×10^5 N/Kg

(a) Calculate pressure due to the head of the water column
(b) Calculate pressure of the gas supply
(c) If ethanol (density $8 \times 10^2 Kg/m^3$) was used in place of water, what would have been the difference in the liquid columns in the two arms of the manometer? Assume the pressure of the gas supply is unchanged.

Solution

(a) $P = \rho hg$

$P = 1000Kg/m^3 \times (0.7m - 0.2m) \times 10N/Kg$

$P = 1000Kg/m^3 \times 0.5 \times 10N/Kg$

$P = 5000$ pa

(b) $P_3 = P_1 + P_2$

$P_3 = 1 \times 10^5 \text{ Pa} + 5 \times 10^3 \text{ Pa}$

$P_3 = 105000$ Pa

(c) $P = \rho hg$

$h = \frac{P}{\rho g}$

$h = \frac{5000 \text{ Pa}}{800Kg/m^3 \times 10N/Kg}$

$h = 0.625m$

Transmission of pressure in a fluid

Pascal's law

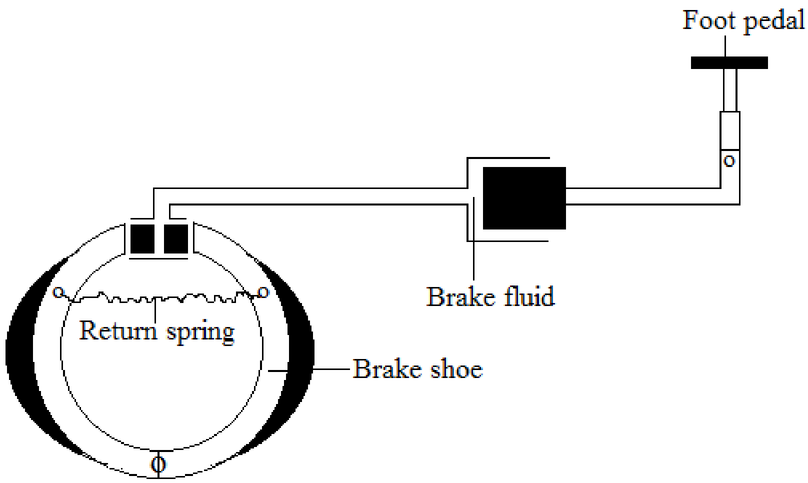
The law state that: Pressure of a fluid is transmitted undiminished to every part of a fluid and every part of the container

Application

1. **Hydraulic press**

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

2. Hydraulic brakes



Example

1. Refer to the diagram below

Calculate the force, F

Solution

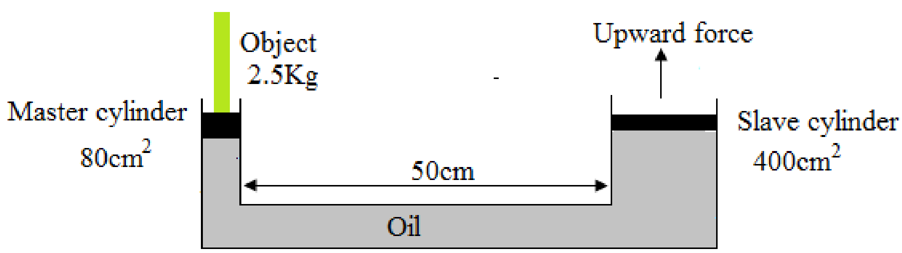
$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$F_2 = \frac{F_1 A_2}{A}$$

$$F_2 = \frac{200N \text{ x } 20m^2}{4m^2}$$

$$F_2 = 1000N$$

2. Figure below shows two cylinders. In each cylinder there is a piston and the space below each piston if full of oil. The object of 2.5kg is placed on the master cylinder as shown below.



The area of the master cylinder is $80cm^2$ and the area on the Slave cylinder is $400cm^2$. [$1m^2$ =$10,000cm^2$]

(I) What is the weight of the object?
(II) Determine the downward force acting on the master cylinder?
(III) Calculate the pressure created in the oil.
(IV) Calculate the upward force.

Solution

(I) $W = mg$

$= 2.5Kg \times 10N/Kg$

$= 25N$

(II) Down ward force = 25N

(III) $P = \frac{F}{A}$

$P = \frac{25N}{0.008m^2}$

$= 3125Pa$

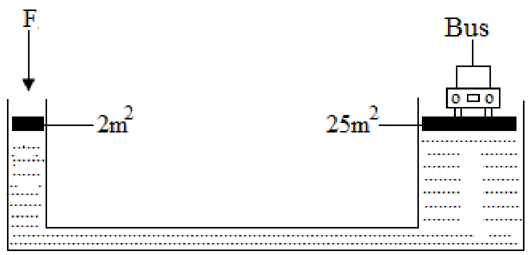
(IV) $F = PA$

$=3125Pa \times 0.04m^2$

$=125N$

Exercise

1. State Pascal's law.
2. Refer to the figure below



The mass of the bus is 4 tonnes

Calculate

(a) The weight of the bus

(b) The minimum force, F, required to lift the bus

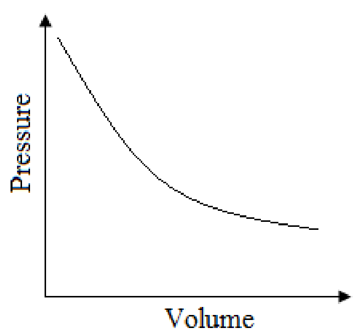
Gas laws

Boyle's law

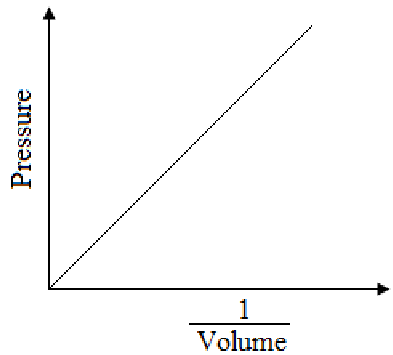
The law state that *the pressure of a fixed mass of gas at constant temperature is inversely proportional to its volume.*

Formula: $P_1V_1 = P_2V_2$

Graph of pressure against volume



Graph of pressure against $\frac{1}{\text{Volume}}$



Example

1. A sample of gas has volume 1000cm^3 and pressure 8 x 10^{8}Pa. Assuming temperature remains constant,

Calculate

(a) The pressure of the gas when its volume is doubled
(b) The volume of the gas when its pressure is 1 x 10^5Pa

	Data	Solution
A	P_2= ? $P_1 = 8\ X\ 10^3\ Pa$ V_1 = 1000cm^3 V_2 = 2000cm^3	$P_1V_1 = P_2V_2$ $P_2 = \frac{P_1V_1}{V_2}$ $P_2 = \frac{8 \text{ x } 10^3 \text{ x } 1000\text{cm}^3}{2000\text{cm}^3}$ $P_2 = 4 \text{ x } 10^3$Pa
B	V_2= ? $P_1 = 8 \text{ x } 10^3\ Pa$ V_1 = 1000cm^3 $P_2 = 1 \text{ x } 10^5 Pa$	$P_1V_1 = P_2V_2$ $V_2 = \frac{P_1V_1}{P_2}$ $V_2 = \frac{8 \text{ x } 10^3\text{Pa x } 1000\text{cm}^3}{1 \text{ x } 10^5 Pa}$ $V_2 = 80\text{cm}^3$

Exercise

1. A gas occupies a volume of 10cm^3 at 100Kpa. If the pressure is increased to 200Kpa, what is the new volume if the temperature remains the same at 27^{o}C?
2. In an experiment to verify Boyle's law, a gas syringe containing some air was connected to a bourdon pressure gauge. The pressure, P, was measured for different volumes, V. the results are tabulated below

V (cm^3)	10	12	14	16	18
P (Kpa)	198	167	142	125	112

(a) Plot a graph of volume against pressure
(b) State the relationship between pressure and volume of a gas at a constant temperature.
(c) Using your graph, find the pressure when the volume is 15cm^3.
(d) Find the volume of air at a pressure of 102Kpa.

3. The table below shows the relationship between pressure and volume.

V (cm^3)	120	110	100	90	80
P (Kpa)	11	16	19	23	28

(a) Calculate the value of $\frac{1}{V}$ for each pair of the readings
(b) Plot the graph of :
(I) P against V
(II) P against $\frac{1}{V}$
(c) Estimate the volume when pressure is 109Kpa

(d) Estimate the pressure when volume is $20cm^3$.

Absolute temperature

Definition: Absolute temperature is the temperature expressed in kelvins

Conversions

$T_K = (T_C^o + 273)K$

$T_C^o = (T_K - 273)K$

T_K = Temperature in kelvins

T_C^o = Temperature in degrees Celsius

Examples

1. Convert
   (a) 20^oC to K
   (b) 300K to ^oC
   (c) 0^oC to K

Exercise

1. Convert
   (a) 30^oC to K
   (b) -50K to ^oC
   (c) 450K to ^oC

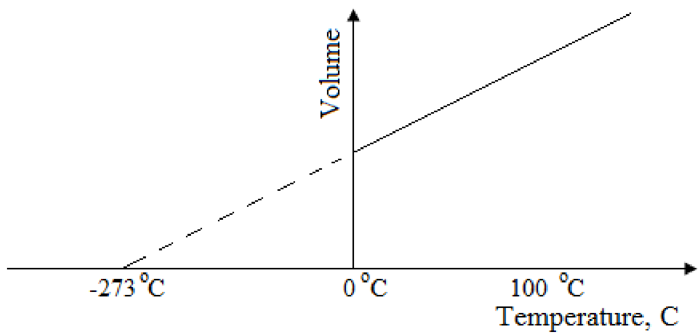
Gas laws

Charles law

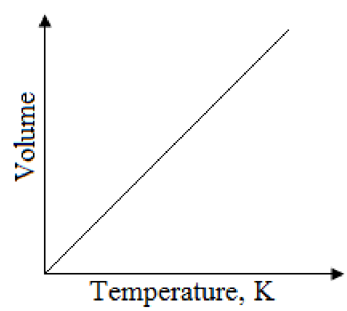
The law states that: The volume of a fixed mass of a gas at constant pressure is directly proportional to the absolute temperature

Formula: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

Graph of volume against temperature,^{o} C



Graph of volume against temperature, K



Note

- The temperature value of -273^oC or 0K is called absolute zero.

Examples

1. At 20^oC, a sample of gas occupies a volume of $250cm^3$. Assuming pressure remains unchanged, determine the volume which the gas sample would occupy at 30^oC.

Data	Solution
V_2= ? V_1 = 250cm^3 T_1 = 20°C = 293K T_2 = 30°C = 303K	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $V_2 = \frac{V_1 T_2}{T_1}$ $V_2 = \frac{250cm^3 \text{ x } 303K}{293K}$ $V_2 = 258.5cm^3$

2. A gas at 27°C extends from a volume of 5cm^3 to 7.5cm^3 at constant pressure. Find its final temperature.

Data	Solution
T_2= ? V_1 = 5cm^3 T_1 = 27°C = 300K V_2 = 7.5cm^3	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $T_2 = \frac{V_2 T_1}{V_1}$ $T_2 = \frac{7.5cm^3 \text{ x } 300K}{5cm^3}$ T_2= 450K

Exercise

1. At what temperature will a mass of gas occupying 200cm^3 at 0°C have a volume of 300cm^3 if pressure remains constant?

Combinations of Boyle's law and Charles law

Alternative term: *General gas law*

In general, all the three quantities; volume, pressure and temperature, vary simultaneously. In such cases, we cannot use either Charles' law or Boyle's law alone. We need an equation that combines all of them.

Formula: $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

Examples

1. 15cm^3 of a gas at pressure 70N/m^2 and temperature of 27°C. Find the volume at a temperature of 127°C and pressure of 35N/m^2.

Data	Solution
V_2=? V_1 = 15cm^3 P_1 = 70N/m^2 P_2 = 35N/m^2 T_1 = 27°C = 300K T_2 = 127°C = 400K	$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$ $V_2 = \frac{P_1 V_1 T_2}{T_1 P_2}$ $V_2 = \frac{70N/m^2 \text{ x } 15m^2 \text{ x } 400K}{300K \text{ x } 35N/m^2}$ V_2 = 40m^3

Exercise

1. A mass of gas occupies a volume of 200cm^3 at a temperature of 27°C and a pressure of 1atm. Calculate the new volume when the pressure is 2atm and the temperature is 37°C.
2. A gas at 7°C and 100Kpa occupies 20L. The gas is heated to 27°C at a pressure of 120Kpa. Find the new volume.

Summary

Constant	Law applied
Temperature	Boyle's law
Pressure	Charles law
None	General gas law

UNIT 3: THERMAL PHYSICS

Matter exists in the three states of Solid, Liquid and Gas. The physical difference between the three states of matter depends on the arrangement and behaviour of the molecules in each particular state. This difference can be explained in terms of the Kinetic Theory, model which states that;

- Matter is made up of very small particles called molecules.
- These molecules are not stationary but are constantly moving.
- The degree of movement of the molecules depends on their temperature

State of matter

Alternative term: *Physical form of matter*

State of matter is the form in which matter exists. Matter exists in three forms. These are solids, liquids and gases

Examples of solids

- Stone
- Glass block
- Wooden block
- Copper block

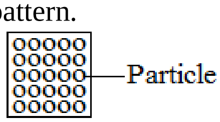
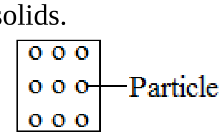
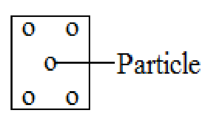
Examples of liquids

- Water
- Cooking oil
- Paraffin
- Petrol

Examples of gases

- Oxygen
- Hydrogen
- Carbon dioxide
- Carbon monoxide

Characteristic properties of the three states of matter

Property	**Solids**	**Liquids**	**Gases**
Shape	Solids have a fixed shape. Particles in solids are arranged in a regular manner. Therefore, they cannot move, but vibrate in a fixed position; hence their shapes are always fixed.	Liquids have no fixed shape. They take the shape of the container in which they are placed.	Gases have no fixed shape. Particles of gases are relatively far apart and have virtually no forces of attraction between each other. As a result, they do not have a fixed shape.
Volume	Liquids have a fixed volume.	Liquids have a fixed volume.	Gases have no fixed volume. Particles spread to fill the space available.
Arrangement of particles	In solids, particles are closely packed and arranged in a regular pattern. 	In liquids, particles are slightly further apart than in solids. 	In gases, particles are much further apart from each other. 

Forces of attraction between particles	In solids, the particles are held together by strong electrostatic forces of attraction called cohesive forces.	In liquids, the paticles are held together by weak electrostatic forces of attraction.	In gases, the forces which hold the particles together are negligible.
Movement of particles	In solids, particles move by vibrating in their fixed positions.	In liquids, particles move by vibrating rapidly over short distances. Particles move from one position to the other. Particles slide past each other randomly.	In gases, particles move at random. Particles move to fill any space available.
Compressibility	Solids can not be compressed because their particles are close together.	Liquids can not be compressed because their particles are close together.	Gases can be compressed because the gas particles are far apart from each other and can be forced to move closer by exerting pressure.
Fluidity	Solids do not flow.	Liquids generally flow easily	Gases flow easily.

Solids and liquids are difficult to compress because their particles are close together. However, gases can easily be compressed because the gas particles are far apart from each other and can be forced to move closer by exerting pressure. Strictly, a vapour is a gas which can be compressed into a liquid without cooling.

Exercise

1. Explain why solids and liquids are difficult to compress but gases are compressible.
2. Matter is classified as solid, liquid or gas. State two physical properties of each of the following:
   (a) Solid
   (b) Liquid
   (c) Gas

Changes of state

Alternative term: *Physical changes*

Changes of state are physical changes that occur when the particles of a substance absorb or lose energy.

A physical change is a change in a substance that does not involve a change in the identity of the substance

A substance can change from one state to another when it is either heated or cooled. When a state of matter gains or loses heat, it undergoes a change.

[A] Heating

Heating involves the addition or supply of heat to a substance. A gain in heat is called an **endothermic change.** An endothermic reaction is a chemical reaction which absorbs or takes in heat energy from the surroundings

As a substance is heated, it absorbs energy and changes from a solid to a liquid and finally to a gas. The kinetic energy possessed by its particles increases and they move vigorously.

Effects of heating substances

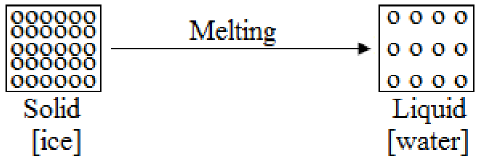
1. Melting

Alternative term: *Liquefaction*

Melting is change of state from solid to liquid.

When a solid is being heated, the temperature suddenly stops rising. Instead of getting hotter, the solid melts.

For example, ice changes to water when heated.



Melting takes place when the particles of a solid absorb energy to overcome the forces holding them in fixed positions and move. They rearrange themselves to form a liquid.

The temperature at which a substance changes from solid to liquid is called melting point. At melting point, the particles of solid **lose their means position** and their arrangement. The solid collapses and turns to liquid.

Note

- *Solid ice loses its shape when it melts. Particles in solid ice vibrate in a fixed position. However, as the ice melts, it gains heat which breaks the strong forces that hold the particles together. This results in particles being able to slide over each other and move in different directions, resulting in loss of shape.*

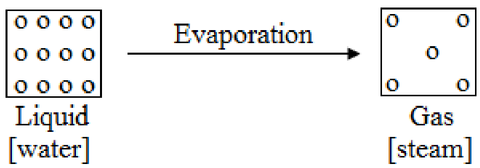
Uses of melting

- It is used in welding.
- It is used in cutting and shaping of metals in the industry.

2. Evaporation

Alternative term: *Vapourisation*

Evaporation is the change of state of a liquid to a gas at the surface. Evaporation is the escape of more energetic molecules from the liquid phase to vapour phase.



The constant temperature at which a substance changes from liquid to gas is called boiling point.

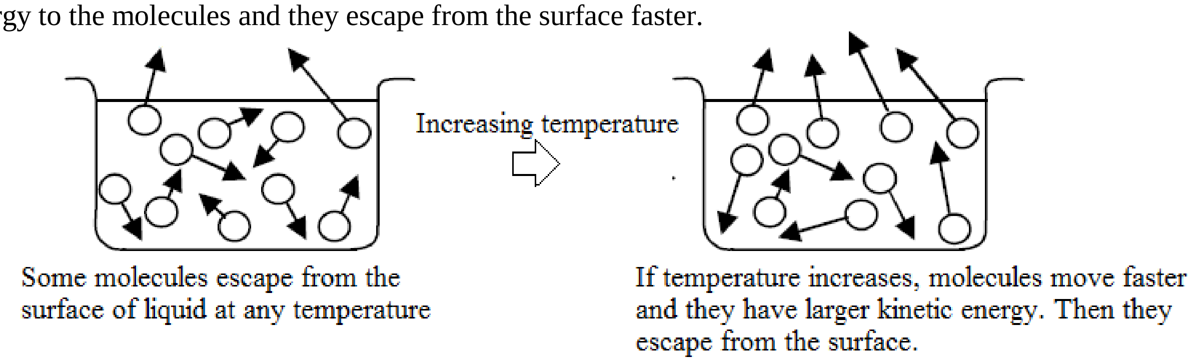
A liquid open to air slowly evaporates. At the same time, the liquid gets colder. The higher the temperature, the faster the liquid evaporates.

Heat increases the kinetic energy of particles in a liquid. As a result, they move further from each other, making the attractive forces between them weaker, they eventually become loosely packed and the liquid attain a gaseous state.

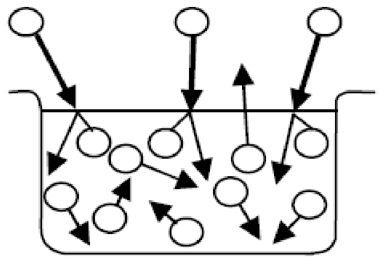
Evaporation occurs at any temperature but more rapidly at high temperature because heat gives more kinetic energy to the molecules and they escape from the surface faster.

Increased pressure on the surface of the liquid reduces

It occurs at any temperature but occurs more rapidly at higher temperature because heat gives more kinetic energy to the molecules and they escape from the surface faster.



Increased gas pressure on the surface of the liquid reduces the rate of evaporation because more collisions occur between the evaporating liquid molecules and the gas molecules, and some of the evaporated liquid molecules bounce back into the liquid.



The molecules that have the largest kinetic energy escape from the liquid. Then, the average kinetic energy of molecules in the liquid is reduced, and also the temperature of liquid reduces. This is called the *cooling effect of evaporation*.

Uses of evaporation

- It is used in obtaining crystals from solutions.
- It is used in drying clothes.

Note

- *Wet clothes dry up faster on warm days than on cold days. This is because water particles in wet clothes gain more kinetic energy from high temperatures on a warm day and evaporate faster from the clothes, resulting in quick drying of the clothes.*
  The higher the temperature, the faster the movement of particles on average

Factors that affect rate of evaporation

- Surface area
- Wind current
- Humidity
- Temperature

Note

- *Boiling occurs when the particles in a liquid absorb enough energy to overcome the forces holding them together and begin to move apart to form a gas.*
- *Evaporation and boiling are both physical processes that change a liquid into a gas. The liquid absorbs heat energy during these physical changes in state.*

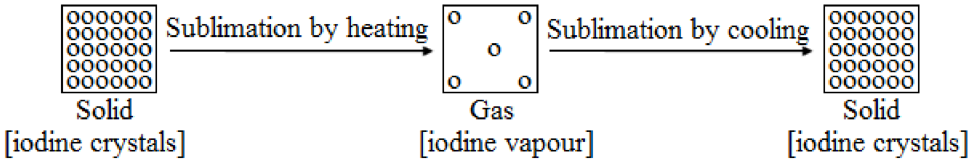
Differences between evaporation and boiling

Evaporation	**Boiling**
Occurs at any temperature below boiling	Occurs at boiling point
Occurs only at the surface of the liquid	Occurs throughout the liquid
No bubbles are observed	Bubbles are observed
Occurs slowly	Occurs rapidly
Heat is absorbed by substance from the surroundings	Heat is supplied to substance by an energy source

3. **Sublimation**

   Sublimation is the direct change of state from solid to gas by heating or from gas to solid by cooling without passing through the liquid state.

   For example, iodine crystals change to iodine vapour when heated and iodine vapour changes to iodine crystals when cooled.



Examples of substances that can sublime

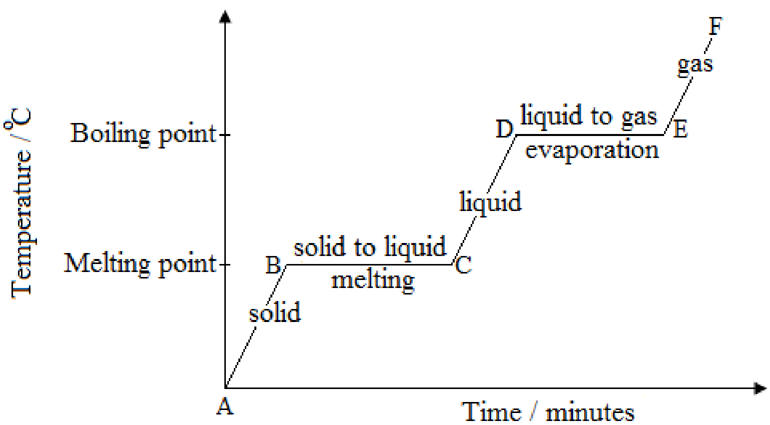
- Iodine
- Ammonium chloride
- Ammonium sulphate
- Carbon dioxide (upon cooling to form ice)
- Naphthalene
- Benzoic acid

Summary of the physical effect of heating the three states of matter

- *Solids* expand slightly. With more heating, solids melt to become a liquid.
- *Liquids* expand slightly. With more heating, liquids boil to become gases.
- *Gases* expand a great deal.

The heating curve

The heating curve is a graph showing changes in temperature with time for a substance being heated.



Slope sections of the heating curve

As a substance is heated, it absorbs heat energy and its temperature rises, then it changes from solid to liquid and finally to gas.

Flat sections of the heating curve

The flat section shows the melting point and boiling point. Here the temperature remains constant over a period of time as energy being absorbed is used to change the state of a substance (change of state points).

Note

- *A pure substance has a fixed temperature. It has an exact boiling point and melting point.*
- *Impurities raise the boiling point and lower the melting point.*

Slope sections of the heating curve

As a substance is heated, it absorbs heat energy and its temperature rises, then it changes from solid to liquid and finally to gas.

Flat sections of the heating curve

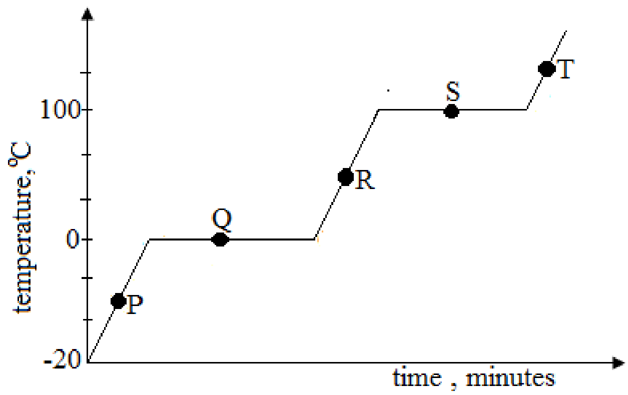
The flat section shows the melting point and boiling point. Here the temperature remains constant over a period of time as energy being absorbed is used to change the state of a substance (change of state points).

Note

- *A pure substance has a fixed temperature. It has an exact boiling point and melting point.*
- *Impurities raise the boiling point and lower the melting point.*

Example

1. The graph below shows changes in temperature a solid substance X underwent when it was heated.



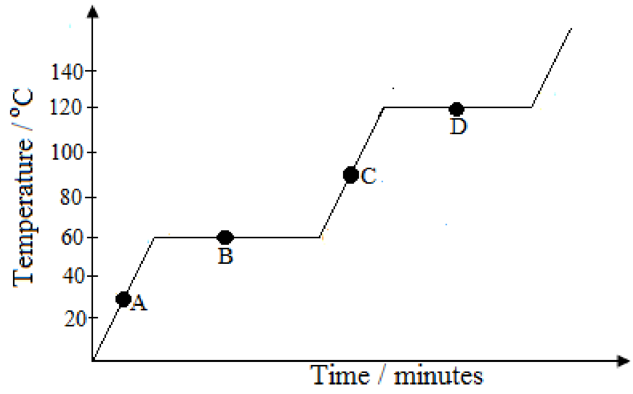
(a) Which letter on the graph represents melting point?
(b) Which letter on the graph represents boiling point?
(c) Give a reason for your answers in (a) and (b) above.
(d) Identify the stage in which the substance X would be in liquid state only
(e) At what stage would substance X exist in two different states of solid and liquid at the same time?
(f) What is the term given to change of state of substance X from R to T?
(g) What is the term given to change of state of substance X from P to T?

Solution

(a) Q
(b) S
(c) Temperature remains constant / Change of state point(s)
(d) Stage R
(e) Stage Q
(f) Evaporation
(g) Sublimation

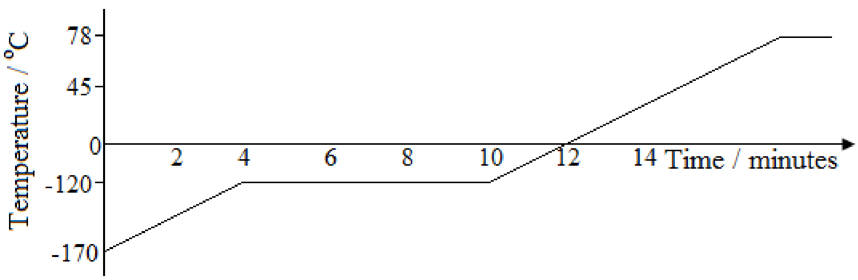
Exercise

1. The graph below shows a heating curve for a pure substance. The temperature rises with time as the substance is heated.



(a) What physical state(s) is the substance in at A and C?
(b) Describe the movement of the particles of the substance at A and C?
(c) What is the melting point and boiling point of the substance?
(d) Describe what happens to the temperature while the substance is changing its physical state.

2. The graph below shows how the temperature of a sample of ethanol varied with time.



(a) What is the melting point of ethanol?
(b) What is the boiling point of ethanol?
(c) For how long did the sample of ethanol melt?
(d) How are you able to tell from the graph that the sample of ethanol is pure?

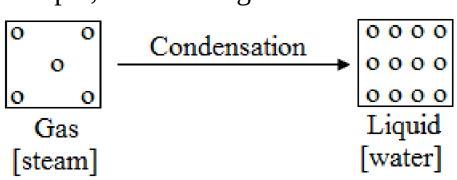
[B] Cooling

Cooling involves the removal of heat from a substance. A loss in heat is called an **exothermic change.**
An exothermic reaction is a chemical reaction which gives out heat energy to the surroundings.

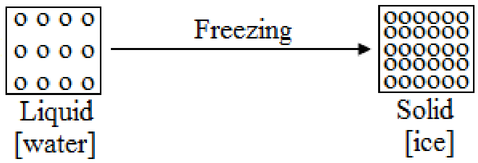
Effects of cooling substances

1. Condensation

Alternative term: *Liquefaction*
Condensation is the change of state from gas to liquid.
For example, steam changes to water when cooled.



2. Freezing
Alternative term: *Solidification*
Freezing is the change of state from liquid to solid.
For example, water changes to ice when cooled.



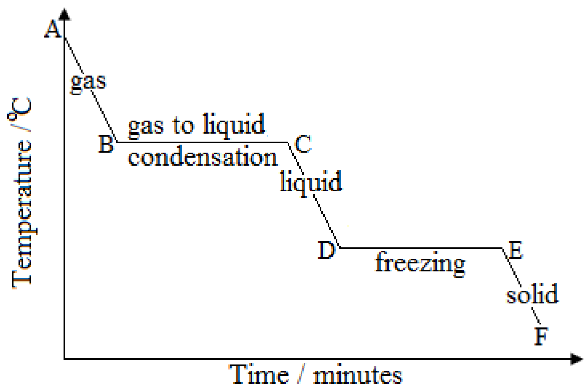
The constant temperature at which a liquid changes into a solid is called freezing point.
Freezing point is also called the melting point for a pure substance.

Summary of the physical effect of cooling the three states of matter

- *Gases* contract a great deal. With more cooling, gases condense to become a liquid.
- *Liquids* contract slightly. With more cooling, liquids freeze to become a solid.
- *Solids* contract slightly.

The cooling curve

The cooling curve is a graph showing changes in temperature with time for a substance being cooled.



Slope sections of the cooling curve

As a substance is cooled, it loses heat energy and its temperature falls, then it changes from gas to liquid and finally to solid.

Flat sections of the cooling curve

Here the temperature remains constant over a period of time as energy being lost is used to change the state of a substance.

Example

1. During a marathon race, the runner shown in the diagram is very hot

(a) At the end of the race, evaporation and convection cool the runner.
 (I) Explain how evaporation helps the runner to lose energy. Use ideas about molecules in your answer.
 (II) Explain why hot air rises around the runner at the end of the race.
(b) At the end of the race, the runner is given a shiny foil blanket, as shown in Figure below wearing the blanket stops the runner from cooling too quickly.

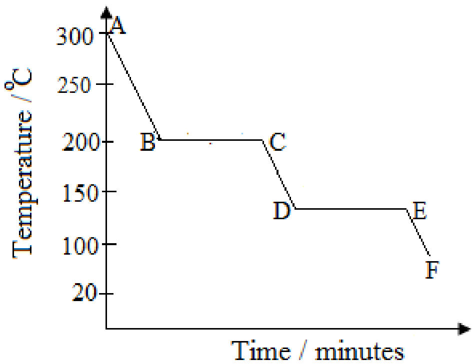
Explain how the shiny foil blanket helps to reduce energy losses.
Use ideas about conduction, convection and radiation in your answer.

Solution

(a) (I) This is because as particles of sweat evaporate, they takes away energy from the body, hence, leaving the body at a lower temperature.
(II) Hot air rises because it expands and become less dense
(b) The trapped air between the foil prevents heat loss by conduction since air is a poor conductor of heat. The shiny foil prevents heat loss by radiation by reflecting heat.

Exercise

1. The graph below shows a cooling curve of a substance as its temperature falls from 300°C to 20°C.



(a) In what state of matter is the substance between A and B?
(b) In what state of matter is the substance between B and C?
(c) What name is given to the point labelled B?
(d) What was the boiling point of the substance?
(e) What was the melting point of the substance?
(f) Explain the reason why the thermometer reading remained constant between points B and C or D and E.
(g) Explain what happens during cooling in relation to the heat content of the substance.

[A] Brownian motion

Brownian motion is the term used to for the continuous random motion of microscopic particles, particularly of gases and liquids. This movement is caused by collisions with the molecules of the surrounding gas or liquid. This random movement supports the kinetic molecular theory of matter.
This phenomenon was first observed by Robert Brown in 1827 who, while studying pollen grains under water, he observed that the pollen grains were moving about in a random way. This same phenomenon can be observed by studying smoke particles in air.

Experiment

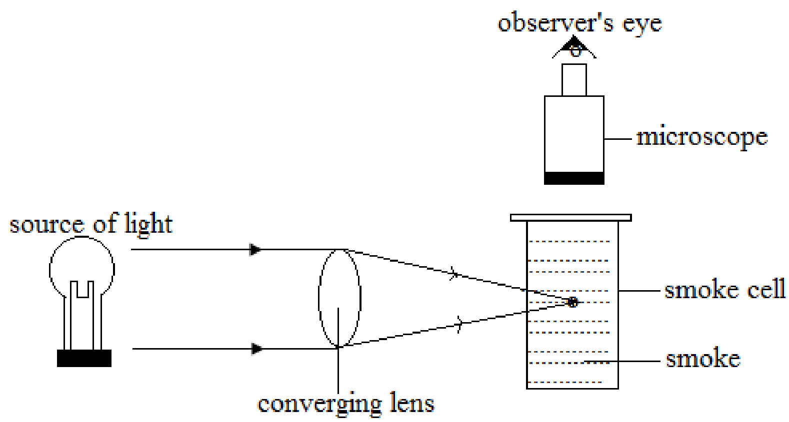
Aim: To study the random motion of smoke particles / to observe Brownian motion in a smoke cell.

Apparatus

- Glass cell
- Source of light
- Microscope
- Converging lens
- Source of smoke

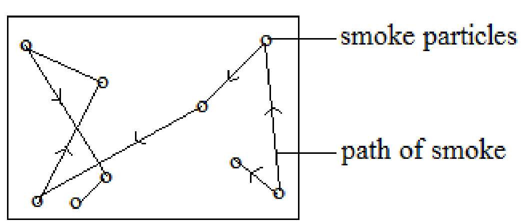
Method

- A small glass cell (smoke cell) in which smoke has been trapped is viewed through a microscope. A microscope is used because the smoke particles are too tiny to be seen using the naked eye. A converging lens is used to focus light from the lamp into the smoke cell. The experimental arrangement is shown below.



Observation

When light strikes the smoke particles, they appear as bright points of light under the microscope moving randomly in a zig – zag path. The smoke particles appear as spots of light because they reflect some of the light from the source of light towards the microscope.



Explanation

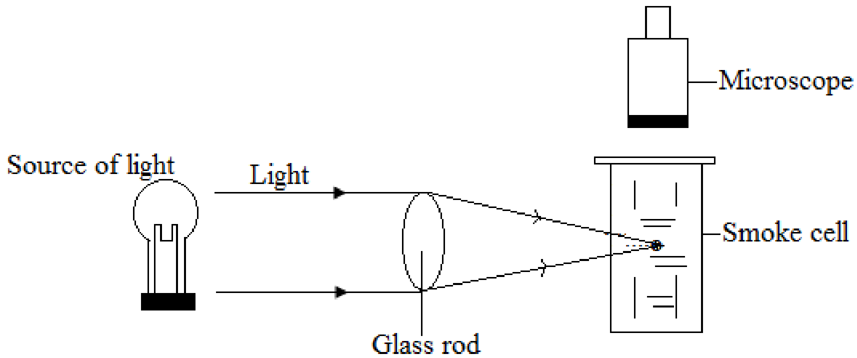
The zig – zag movement is due to the collision of the smoke particles with invisible air molecules that move about randomly in the smoke cell. This is called Brownian motion.

Conclusion

The air molecules are in a continuous random motion colliding with the smoke particles and the walls of the smoke cell.

Exercise

1. The figure below shows one of the forms of an apparatus used to observe Brownian motion of smoke particles in air. Ruth Naosa looking through the microscope sees tiny bright specks which she describes as 'dancing about'.



(a) What are bright specks?

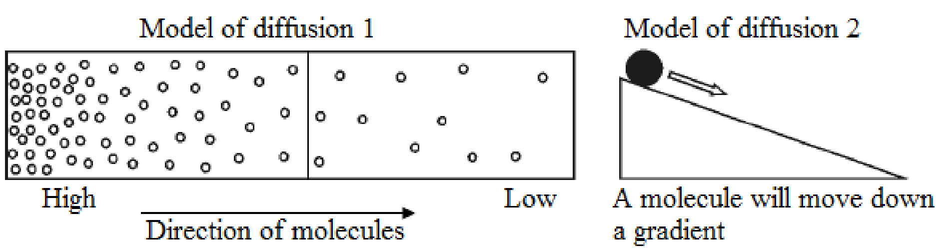
(b) Why are the specks 'dancing about'?

(c) State the conclusion that can be drawn from the Brownian motion?

[B] Diffusion

Definition: Diffusion is the *movement of particles from the region of high concentration to the region of low concentration.*

Diffusion can also be defined as the *spreading movement of the particles due to the motion of their own molecules or ions.* It is the spreading out and mixing process seen mainly in gases and liquids. The particles of one substance mingle with, and move through the particles of another. Diffusion goes on until the mixture is uniform.



Note

- *If you open a bottle of perfume, the aroma soon reaches the nose of anyone nearby. Obviously, the molecules of the fragrance have diffused from the bottle. The particles move from a region of high concentration to region of low concentration. This diffusion is simple evidence that molecules move.*
- *Diffusion is very strong evidence for the mobility of particles*.

Rate of diffusion

Rate of diffusion is the amount of gas or liquid diffusing in a unit of time.

Factors that affect the rate of diffusion

1. Temperature

Rate of diffusion is faster is if the temperature is high and slower if the temperature is low.

2. Concentration

Rate of diffusion is faster if there is a large difference in the concentration of particles between two regions.

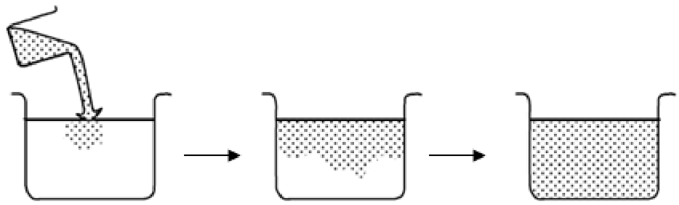
3. Size of particles / mass of particles

Rate of diffusion is faster if size of particles is small and slower if the size of particles is large. A particle that has a smaller mass will diffuse at the faster rate and a particle that has a larger mass will diffuse at a slow rate. Small, light particles diffuse faster than large, heavier particles. Diffusion in gases is faster than in liquids. Diffusion occurs slowly in liquids but faster in gases because in gases, molecules move examples randomly.

Examples

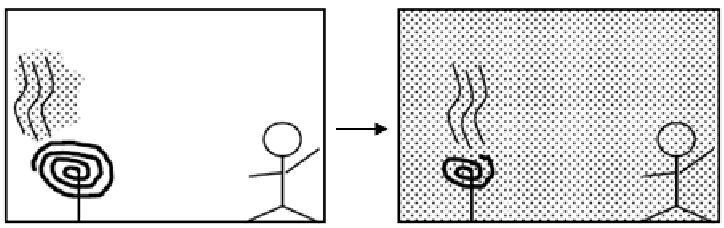
1. **Diffusion of ink in water**

   The colour of water is changed by the diffusion of ink particles. (Without mixing)



2. **Diffusion of mosquito coil**

   The smell of mosquito coil reaches to your place by the diffusion of its smoke particles.



Temperature

Definition: Temperature is the measure of how hot or cold an object is as compared to a particular scale.

Units:

Degrees Celsius [°C]

Kelvins [K]

Temperature is measured instrument called thermometer

Principles of thermometry

Temperature is measured using some physical properties which vary with temperature

Thermometer

The thermometer is an instrument used to measure temperature. There are different types of thermometers.

They make use of a physical property which changes continuously with temperature to show the temperature.

The most widely used temperature scale is the Celsius scale (°C).

The thermometer which utilizes the expansion of a liquid to measure temperature is called ***Liquid-in-glass thermometer***.

Thermometers and those physical properties that change with temperature

Thermometer	Physical property
Liquid-in-glass Thermometer	Volume of a fixed mass of liquid (length of liquid column)
Thermocouple	Electromotive force (e.m.f.)
Resistance thermometer	Resistance of piece of metal
Constant volume gas thermometer	Pressure of a fixed mass of gas at constant volume

Thermometric liquid

- Mercury
- Alcohol

The temperature is read at the top of the mercury or alcohol thread.

Properties of Mercury and Alcohol

Mercury	Alcohol
It has a high freezing point (−39ºC).	It has a low freezing point (−112ºC)
It has a high boiling point (357ºC)	It has a low boiling point (78ºC)
It is silvery coloured and doesn't allow light through. (It is easy to see.)	It is colourless but is made visible by adding colouring.
It expands uniformly. But its expansion is not very large.	It doesn't expand uniformly. However, it expands about six times more than mercury.
It does not wet glass. (It doesn't stick to glass.)	It stick to the wall of the capillary tube when the thread is falling.
It doesn't vaporize at room temperature onto the upper parts of the tube.	It easily vaporizes.
It is a good conductor of heat and therefore responds to change in temperature.	It is not a good conductor of heat.
It is poisonous.	It is safe.
It is expensive.	It is cheap.

Reasons why mercury is suitable for use in thermometers

1. It is easy to see through the glass
2. It does not wet the glass
3. It expands uniformly
4. It is a liquid over a wide range of temperature
5. It conducts heat rapidly and therefore more sensitive to temperature variations

Advantages of mercury over alcohol

1. It is coloured (so it is easily seen through the glass)
2. It is a good thermal conductor (so it expands evenly)
3. It is highly cohesive (so it does not wet the glass)
4. It has a high boiling point and a low freezing point (so it is used over a wide temperature range)

Advantages of alcohol over mercury

1. It is cheaper
2. It is safe
3. It expands more than mercury
4. It has a low freezing point

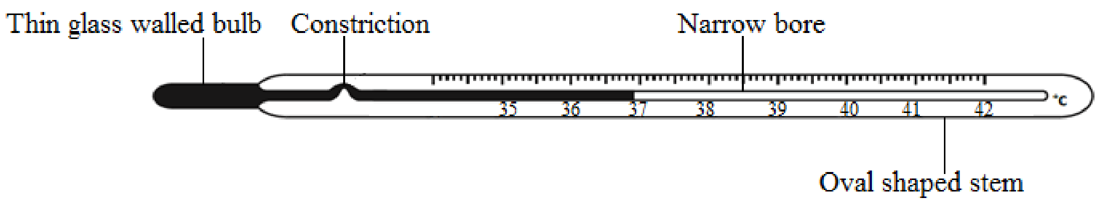
Types of thermometers

- Clinical thermometer
- Laboratory thermometer
- Thermocouple thermometer

[A] Clinical thermometer

The clinical thermometer is used to measure the temperature of the human body.

Structure of the clinical thermometer



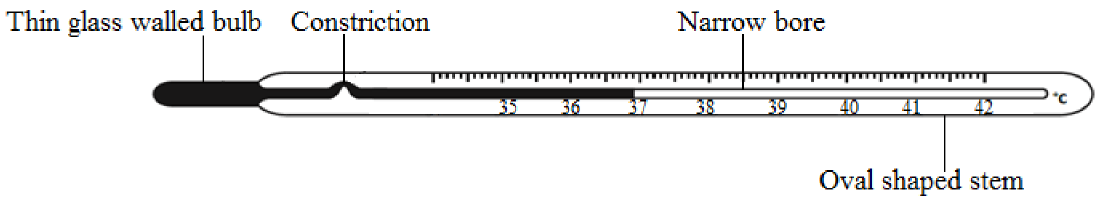
Digital clinical thermometer

Features of the clinical thermometer

1. **Constriction**
   The constriction is a sharp bend in bore at the bottom of the scale.
   The constriction helps to prevent the mercury thread from flowing back before the reading is taken. It can record the maximum temperature.
2. **Short range**
   It has a short range of temperature calibration because it measures body temperature which fluctuates within a narrow range. The range is from $35^{o}\mathrm{C}$ to $42^{o}\mathrm{C}$
   The short range also gives greater accuracy.
3. **Narrow bore**
   The narrow bore makes the thermometer have a greater precision and sensitivity
4. **Thin glass walled bulb**
   The thin walled bulb makes the thermometer sensitive to temperature changes and for quick responsiveness.
5. **Oval shaped glass stem**
   Convenience for replacement in the armpit.

Exercise

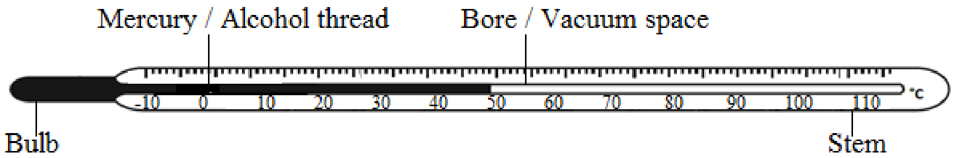
1. The figure below shows a diagram of a clinical thermometer some features labeled.



Explain why it has each of the following features.

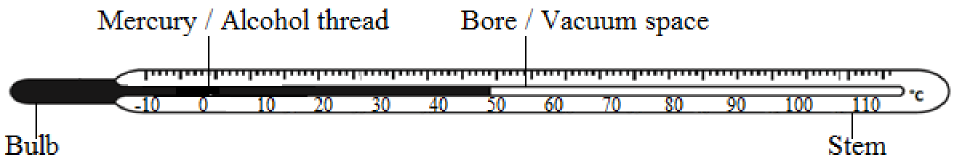
(a) A thin glass walled bulb.
(b) A constriction.
(c) A short range of temperature calibration.
(d) A narrow bore.
(e) An oval shaped stem.

[B] Laboratory thermometer
Structure of the laboratory thermometer



Example

1. The student uses the mercury-in-glass thermometer shown in the diagram below.
   He does not detect any temperature rise in the water in the beaker.



(a) Describe how you would check the 0°C and 100°C points on the thermometer.
(b) Explain why the thermometer is not sensitive enough to detect the temperature rise.
(c) State and explain one change that will make a mercury-in-glass thermometer more Sensitive.

Solution

(a) Pure melting ice for 0°C.
Pure boiling water/steam above boiling water (at 1 atmosphere) for 100° C
(b) Each division on thermometer is too small described in some way e.g. does not expand far up tube (not bore too thin, not enough mercury).
(c) Change: Use more mercury or use smaller bore.
Reason: More expansion or further distance up tube (for same expansion).

Graduating a thermometer – temperature scale

When a mercury – in – glass thermometer is produced, the temperature scale must be marked on the stem. Then, two known temperatures are needed for marking the scale. These temperatures are called fixed points.

Fixed point

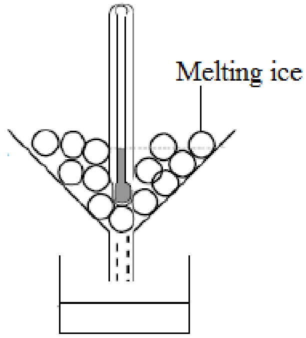
Definition: Fixed point is a reference temperature chosen because it is readily reproducible
Fixed points are important for calibration of thermometers.

Lower fixed point

Alternative term: *Ice point*
Value: 0ºC
Definition: Ice point is the temperature of melting pure ice.
Ice and water are present together at the lower fixed point.

Determining lower fixed point

- Place the thermometer bulb in the melting ice.



- Measure the length of the mercury thread when it has stabilized. Mark it. It is the lower fixed point.

Upper fixed point

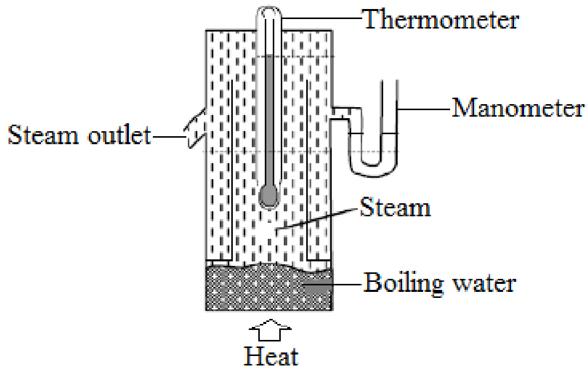
Alternative term: *Steam point*
Value: 100ºC

Definition: Steam point is the temperature of steam from water boiling under standard atmospheric pressure. Water and steam are present together at the upper fixed point.

Determining steam point

This is done by using a double walled vessel called a ***Hypsometer***. Water is steadily boiled in the lower part of the hypsometer thus keeping the bulb surrounded by pure water vapour at atmospheric pressure.

- Place the thermometer bulb in the steam from boiling water.



- Measure the length of the mercury thread when it has stabilized. Mark it. It is the upper fixed point.

Measurement of temperature using uncalibrated thermometer

Formula: $\Theta = \left(\frac{L_\Theta - L_0}{L_{100} - L_0}\right)$ x 100%

Note

- L_Θ = length of mercury at Θ
- L_{100} = length mercury at 100^oC
- L_0 = length of mercury at 0^oC

Example

1. At 0^oC , the length of mercury thread in a thermometer is 2cm. at 100^oC, the length of mercury is 22cm. At a temperature, Θ, the length of mercury thread is 18cm. Determine the temperature, Θ.

Solution

$\Theta = \left(\frac{L_\Theta - L_0}{L_{100} - L_0}\right) \text{ x } 100^oC$

$\Theta = \left(\frac{18cm-2cm}{22cm-2cm}\right) \text{ x } 100^oC$

$\Theta = \left(\frac{16cm}{20cm}\right) \text{ x } 100^oC$

$\Theta = 80^oC$

2. The table below shows information about a mercury – in – glass thermometer

Length of mercury thread / cm	Temperature / °C
1	−10
25	110

At a temperature, Θ, the length of the mercury thread is 13cm. determine the temperature, Θ.

Solution

$\Theta = \left(\frac{L_\Theta - L_{-10}}{L_{110} - L_{-10}}\right) \text{ x } 120^oC$

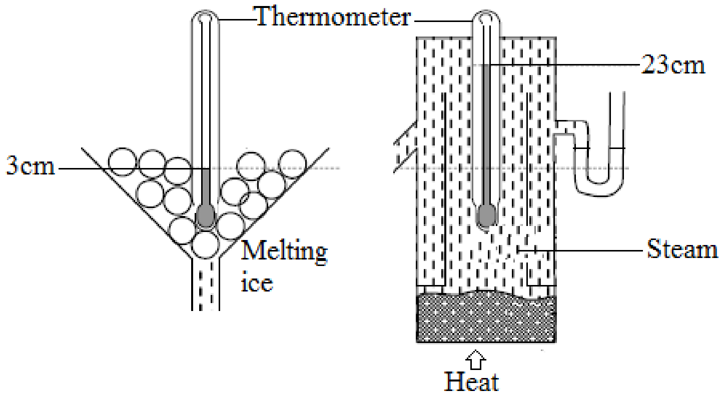
$\Theta = \left(\frac{13cm-1cm}{25cm-1cm}\right) \text{ x } 120^oC$

$\Theta = \left(\frac{12cm}{24cm}\right) \text{ x } 120^oC$

$\Theta = 60^oC$

Exercise

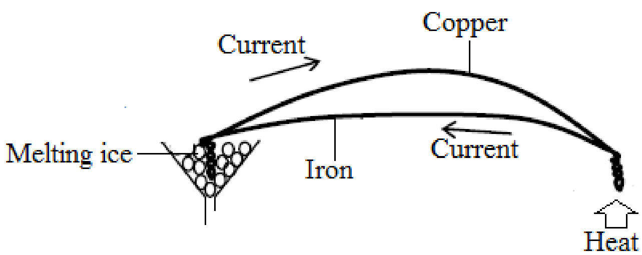
1. The diagrams below show a mercury thermometer whose bulb is immersed in pure melting ice and then in steam from boiling water.



The columns of mercury are 3cm and 23cm in the ice and steam, respectively. What is the temperature when the column of mercury is 8cm long?

[C] Thermocouple thermometer

A thermocouple is made from wires of two different materials e.g. copper and iron. The wires are soldered or just twisted tightly together at the ends. When two junctions are placed in different temperatures, an electric current flows around the circuit. The amount of current depends on the differences in temperatures. If one of the junctions is placed into the known temperature, e.g. melting ice (0°C), and the other junction is placed into the measured object, e.g. fire, it is possible to measure the temperature by reading the current. The thermocouple is very sensitive and it can measure high temperatures because of melting points of metals.



Advantages of the thermocouple thermometer

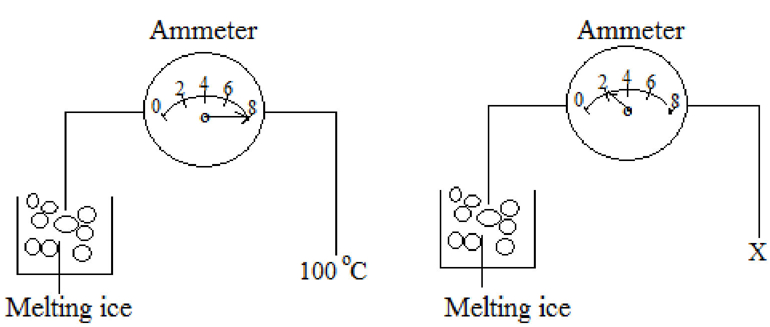
It can measure temperature at a point.

It can measure very high temperatures.

It can measure rapidly changing temperatures.

Examples

1. Refer to the diagram below.



Determine temperature x.

Solution

8A → 100°C

2A → x

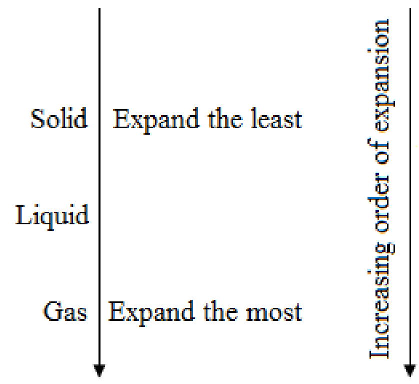
$x = \frac{2\text{A x } 100^{o}\text{C}}{8\text{A}}$

x = 25°C

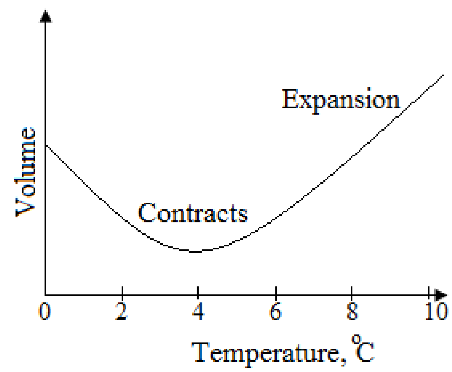
Thermal properties
When a body of something is heated, the body increases in size. It is called ***thermal expansion***.

- When molecules get heat energy, they have more kinetic energy. They move or vibrate more. Then they need larger spaces between them.

Thermal expansion
Definition: Thermal expansion is the increase in volume of a material resulting from the application of heat.
Relative order of expansion in solids, liquids and gases



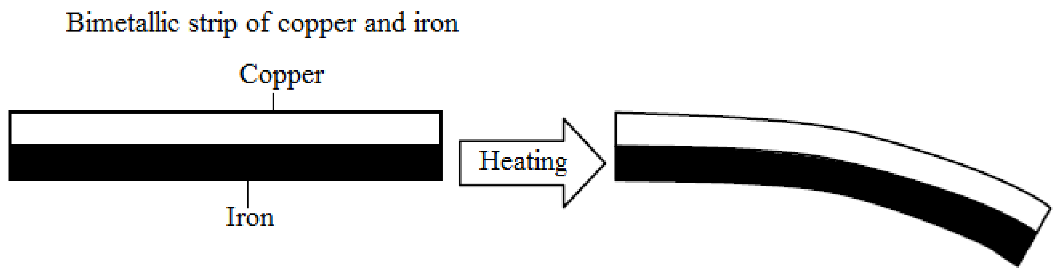
Anomalous expansion of water



When temperature of water is rising from a value lower than 4°C, water contracts
When temperature rises from 4°C to higher value, the water expands.
The anomalous behavior of water explains why its density is highest at 4°C.
Applications of thermal expansion

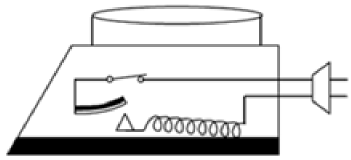
- Thermometer
- Bimetallic strip

Bimetallic strip
This is a compound bar made from two metals riveted together. When it is heated, it bends because the metals expand differently.



Copper expands more than iron when they are headed. It causes the bimetallic strip to bend.
This is used as a thermostat which is a switch to keep the temperature in electrical circuits e.g. a pressing iron, refrigerator and fire alarm.

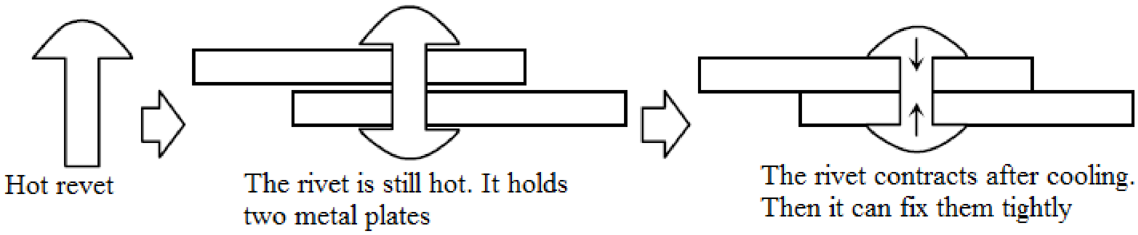
Electric pressing iron



- When a bimetallic strip is bent by heat, the circuit is disconnected.
- This system controls the heat on the iron. It is called a thermostat.

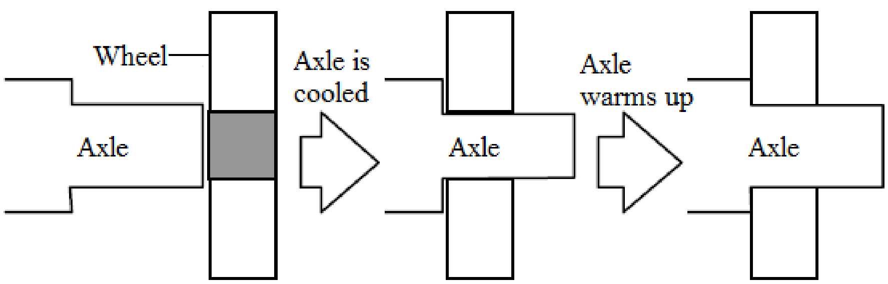
Rivets

- Rivets are a form of nail used to hold two metal plates tightly together. If hot rivet is used, the rivet contracts after cooling. It can fix plates tightly.



Wheel fitting

- A slightly larger axles does not fit into the wheel. However it can fit into the wheel by cooling because it is contracted.
- As it warms up again, it will expand back to its natural size. It causes a tight fit between axle and wheel.

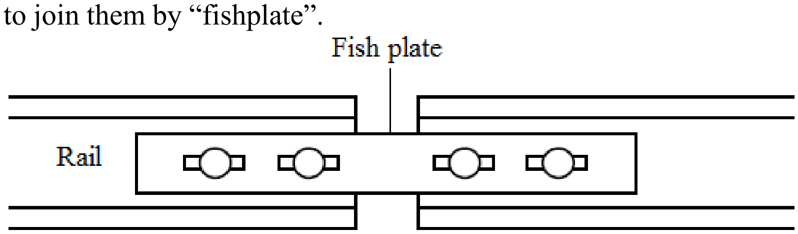


Consequences of thermal expansion

The expansion of materials may cause bad influences. If a solid or a liquid is prevented from expanding, very large forces are exerted. The forces may destroy something. The effects of expansion must be remembered when you design anything.

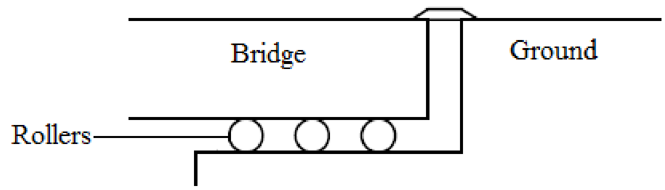
Railway

- The rails of railway lines expand when the temperature rises. The expansion can cause bending of the rails. One way to prevent the rails from bending due to expansion is to leave gaps between the ends of the rails and to join them by “fishplate”.



Bridge

- Metal bridges must be made to allow for expansion on one end of the bridge. One end is usually fixed while the other end rests on rollers to allow movement due to expansion.



Transfer of thermal energy

Heat transfer

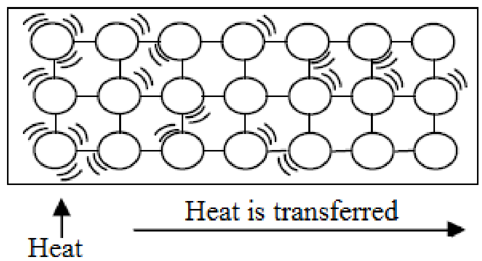
Heat energy is transferred from a higher temperature region to a lower temperature region until the temperatures are balanced. There are three methods by which heat is transmitted from one place to another.

- Conduction
- Convection
- Radiation

Conduction

Definition: **Conduction** is defined as the process by which heat is transmitted through a medium from its hotter part to its colder part.

- At the hotter part, the molecules vibrate actively. They collide with neighbors. Then, the vibration is transferred from the hotter part to the colder part.



Conductivity

Thermal conductivity depends on the materials. For example;

- Air, wool, cotton, wood, water, glass and plastic are bad conductors.
- Metals (e.g. steel, iron, and copper, silver) are generally good conductors.

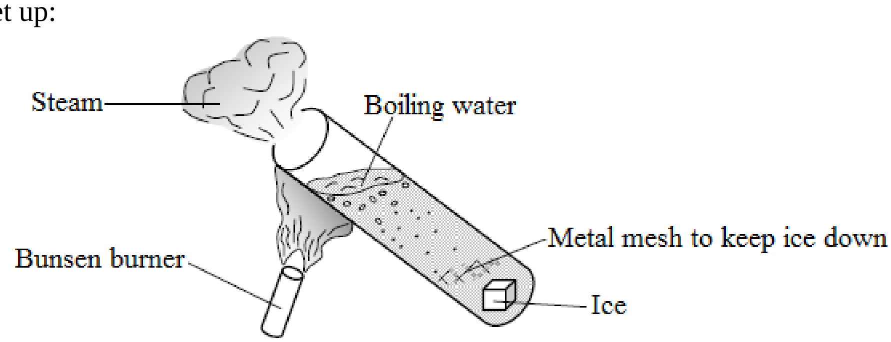
The relative order is as follows;

(Higher conductivity) Metal > Nonmetal solid > Liquid > Gas (lower conductivity)

Experiment

Aim: To show that water is a bad conductor of heat.

Experimental set up:



Cautions for the setting:

- Ice block must be at the bottom of the test tube. (The wire gauze prevents the ice from floating.)
- The flame from the Bunsen burner must be placed on the top of water. (It prevents the convection of water.)

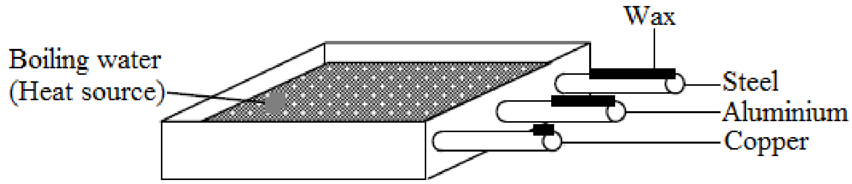
Result

When the water on the top begins to boil, the ice block at the bottom does not melt immediately.

Conclusion

Water is a bad conductor.

Experiment
Aim: To compare the thermal conductivities of metals.
Experimental set up

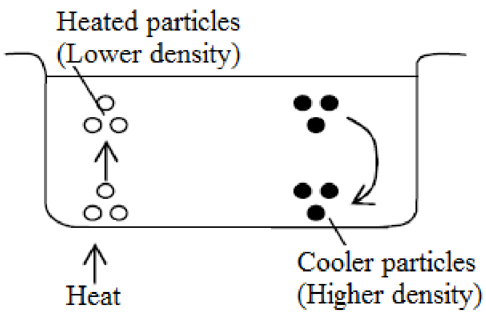


Cautions for the setting

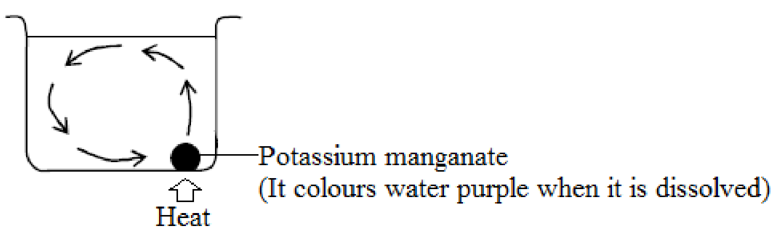
- These bars must be the same size (length and diameter).
- They begin to heat at the same time.

Result
Melting speeds of the waxes on the bars are in the following order.
Copper → Aluminium → Steel
Conclusion
The copper is the best conductor of the three.
Convection
Definition: **Convection** is defined as the process by which heat is transmitted from one place to another by the movement of heated particles of a gas or liquid.

- When we heat a fluid (a gas or a liquid), it expands and its volume increases. Its density is therefore reduced. Hotter fluid surrounded by cooler fluid (higher density) will tend to float. The warmed fluid will move upwards, and it carries heat energy with it.
- The movement of fluid is called convection current.
- Convection only occurs in a fluid.
- Convection current rises vertically from the source of heat where the fluid is hottest.

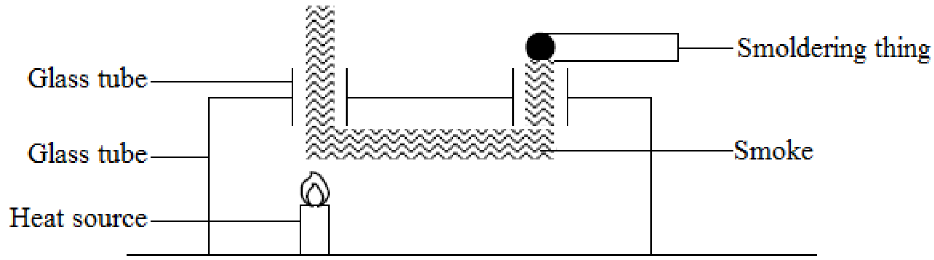


Experiment
Aim: To observe convection current in water
Experimental set up



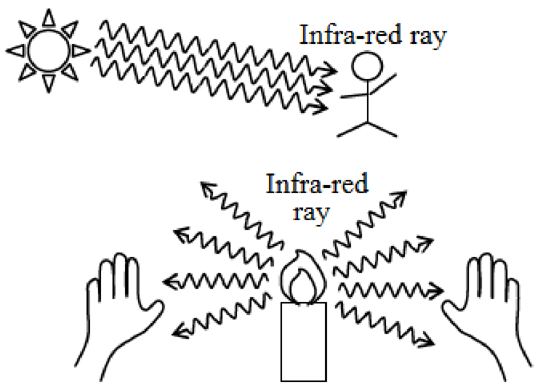
Cautions for the setting
It must be heated at the bottom of the potassium manganate.
Result
The colour changes as shown by arrows.
Conclusion
The coloured path shows the convection current in water.

Experiment
Aim: To observe convection current in air.
Experimental set up

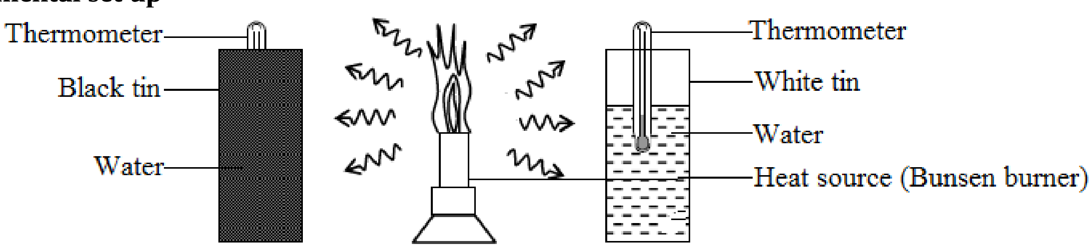


Cautions for the setting
- Smoldering thing must be placed just above the glass tube.
- The heat source must be placed under another glass tube.

Result
The smoke produced by the smoldering thing goes down through the tube, flows through the top of the box and comes out the other tube.
Conclusion
The smoke shows the convection currents in air.
Radiation
Definition: **Radiation** is defined as the flow of heat energy in the form of electromagnetic waves.
This process does not require any medium.
- It can occur in a vacuum space.
- These electromagnetic waves are called infra-red ray.
- Infra-red rays are invisible.
- An object which receives infra-red rays is called an Absorber.
- An object which releases infra-red rays is called an Emitter.



Experiment
Aim: To show which surface absorbs heat better (black and white).
Experimental set up



Cautions for the setting
- These tins must be the same size.
- They must be at the same distance from the heat source.

- They must have the same amount of water.

Result

The thermometer in the black tin shows a higher temperature reading than the one in the white tin.

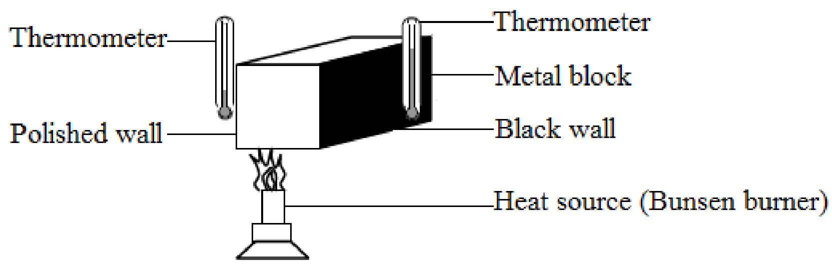
Conclusion

The black (dark) surface is a good absorber of heat.

Experiment

Aim: To show which surface emits heat better (black and white).

Experimental setup



Cautions for the setting

- The thermometers must be at the same distance from each wall (don't touch the walls).
- The thermometers must be far from the heat source.

(It prevents the thermometers from heating by the heat source directly.)

Result

The thermometer near the black wall shows a higher temperature reading than the other.

Conclusion

The black (dark) surface is a good emitter of heat.

- The dark colour emits and absorbs the heat well, otherwise the bright shiny colour can prevent the heat loss by radiation.

Applications of heat transfer

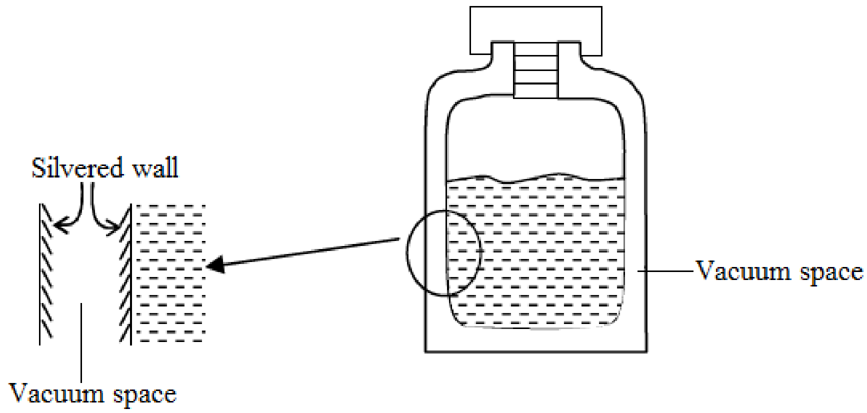
Vacuum flask (Thermos flask)

It can keep liquid hot or cold.

- Vacuum space prevents conduction and convection.

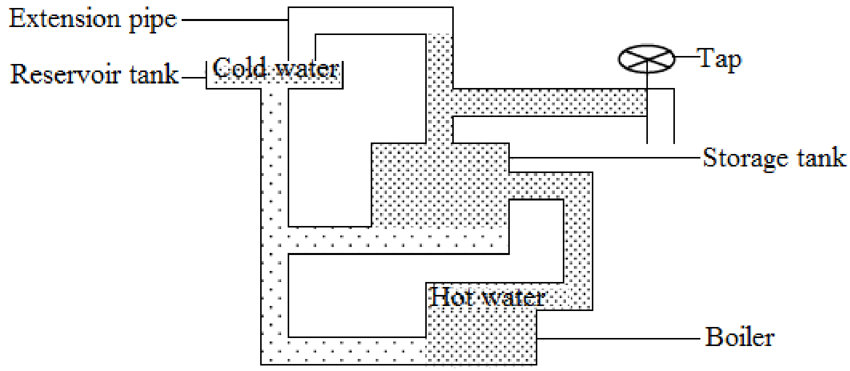
(If there is no particles, conduction and convection don't occur)

- Silvered wall prevents radiation.



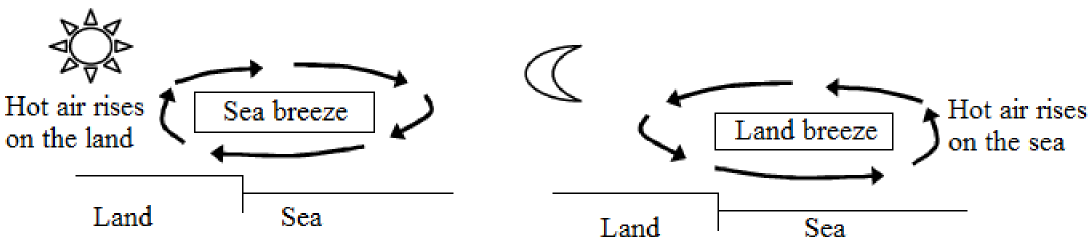
Hot water system

- Water is heated in the boiler. It rises by convection to the hot water tank, while colder water flows from the tank to the boiler. The convection current keeps the water in the tank hot. When you open a tap, hot water flows out.



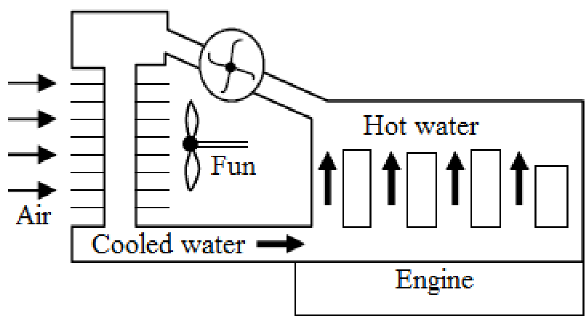
Sea and Land Breeze

- The sun can cause very large convection currents of air. This flow of air is wind. In daytime, the land has higher temperature than the sea. The warm air rises over the land and it is replaced by colder air from the sea. This is called Sea breeze.
  At night, the reverse occurs, because the land cools down faster than the sea. This is called Land breeze.



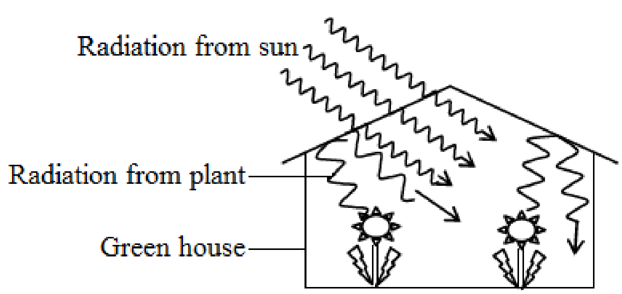
Radiator

- Car engines are cooled by convection current in the water pipe. The radiator is a heat exchanger where water gives up its heat to the air.



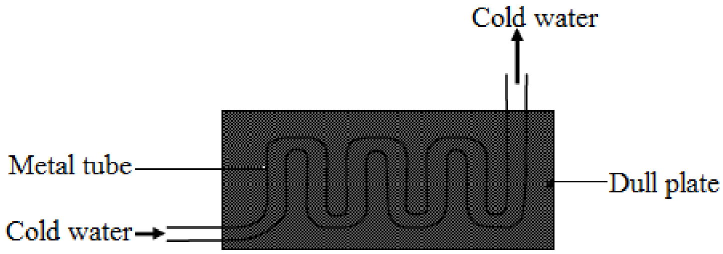
Greenhouse

- A greenhouse is a building made of glass or transparent plastic for growing plants. Radiation from the sun enters through the glass but radiation from the plants can't get out of the glass. It keeps the temperature inside warm.



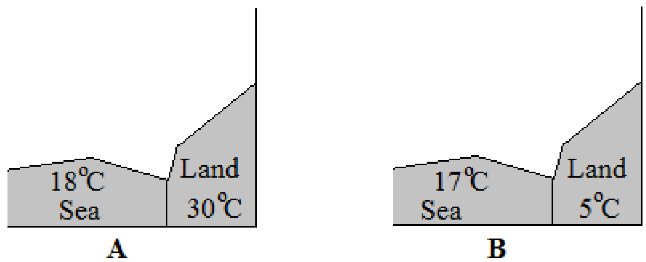
Solar panels

- In sunny countries, warm water can be produced using solar panels. In one type of panel, a metal tube is welded to the metal plate which is painted dull black. The plate absorbs the sun's radiation, and shares this energy with the water by conduction.



Exercise

1. Why are the nails colder than cotton when we touch them?
2. The diagrams A and B show maximum surface temperatures of a region of land and sea during a 24 hour period.



(a) Which of the two diagrams above represents night time coastal temperatures?
(b) Explain your answer in (a) above.
(c) On diagram **A**, show how the coastal breeze will blow.

Internal energy

Definition: Internal energy is the total energy due to the random motion of molecules in a sample of a substance

Note

- When a substance is heated, its internal energy increases.
- An increase in internal energy cause temperature rise.

Heat capacity

Symbol: C

SI unit: Joule per kelvin [J/K]

Definition: Heat capacity is the quantity of thermal energy required to raise the temperature of a body through 1°C or 1K

Formula: $C = \frac{\theta}{\Delta T}$

Specific heat capacity

Symbol: c

SI unit: Joule per kilogram kelvin [J/KgK]

Definition: Specific heat capacity is the quantity of thermal energy required to raise a 1kg substance through 1°C or 1K. Specific heat capacity can also be defined as the heat capacity per unit mass

Formula: $c = \frac{\theta}{m\Delta T}$

Substance	**Specific heat capacity ($JKg^{-1}K^{-1}$)**
Water	4200
Ice	2100
Alcohol	2500
Glycerin	2400
Brine	3900

Paraffin	2200
Glass	670
Aluminium	880
Copper	390
Lead	130
Silver	230
Mercury	140
Iron	460
Air	720

Relationship between heat capacity and specific heat capacity

$c = \frac{C}{m}$

Examples

1. Calculate the quantity of energy required to raise the temperature of 500g of water from 20^{o}C to 30^oC.

Solution

$Q = mc\Delta T$

$Q = 0.5Kg \times 4200JKg^oC^{-1} \times 10^oC$

$Q = 21000J$

2. Determine the heat capacity of a 2Kg aluminium block

Solution

$c = \frac{C}{m}$

$C = mc$

$C = 2Kg \times 880JKg^{-1}K^{-1}$

$C = 1760JK^{-1}$

3. To 5Kg of water at 22^{o}C was added 500g of water at 77^{o}C, calculate the final temperature of water.

Solution

Heat lost = Heat gained

$C \times 0.5Kg \times (77^oC - T_f) = C \times 5Kg \times (T_f - 22^oC)$

$38.5Kg^oC - 0.5TKg = 5TKg - 110Kg^oC$

$38.5Kg^oC + 110Kg^oC = 5TKg + 0.5TKg$

$148.5Kg^oC = 5.5TKg$

$T = \frac{148.5Kg^oC}{5.5Kg}$

$T = 27^oC$

4. Calculate the heat energy required to raise the temperature of 5Kg of water from 20K to 100K.

Solution

$Q = mc\Delta T$

$Q = 5Kg \times 4200JK^{-1}Kg^{-1} \times (100 - 20)K$

$Q = 1.68 \times 10^6$ J

5. In an experiment, 920000J of energy is transferred to 2kg of iron ($c = 460JKg^{-1}K^{-1}$). The initial temperature of iron is 25K. What is the final temperature of iron?

Solution

$Q = mc\Delta T$

$\Delta T = \frac{Q}{cm}$

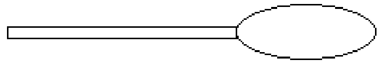
$\Delta T = \frac{920000J}{460J(KKg)^{-1} \times 2Kg}$

$(T - 25K) = 1000K$

$T = 1000K + 25K$

$T = 1025K$

6. The figure below shows a silver spoon



The mass of the spoon is 75.0g.
The spoon is heated using an electric circuit

(a) Determine the quantity of thermal energy needed to raise the temperature of the spoon from 15ºC to 105ºC
(b) It took 40s for the temperature of the spoon to rise from 15ºC to 105ºC. neglecting the heat loss to the surrounding, determine the power of the electric heating system

Solution

(a) $Q = mc\Delta T$
$Q = 0.075Kg \times 230J/(Kg^oC) \times (106 - 15)^oC$
$Q = 1552.5J$
(b) $P = \frac{Q}{t}$
$P = \frac{1552.5J}{40s}$
$P = 38.8125W$

Latent heat

Definition: Latent heat is thermal energy that is transferred to cause phase change of a pure substance without temperature change.

Specific latent heat

Symbol: L
SI unit: Joule per Kilogram [J/Kg]
Definition: Specific latent heat is the quantity of thermal energy that is transferred to cause phase change of 1Kg of a pure substance without temperature change
Formula: $L = \frac{Q}{m}$

Specific latent heat of fusion

Symbol: L_f
SI unit: Joule per Kilogram [J/Kg]
Definition: Specific latent heat of fusion is thermal energy required to change 1Kg of a substance from solid to liquid state without a change in temperature.

Values of specific latent heat of fusion

Substance	J/Kg	Melting point, ºC
Ice	335000	0
Copper	212000	1083
Lead	26200	327
Naphthalene	148000	80

Specific latent heat of vaporization

Symbol: L_V
SI unit: Joule per Kilogram [J/Kg]
Definition: Specific latent heat of vaporization is thermal energy required to change 1Kg of a substance from liquid to gaseous state without a change in temperature. 669-831

Values of specific latent heat of vaporization

Substance	J/Kg	Melting point, ºC
Water	2250000	100
Chloroform	240000	61
Alcohol	850000	78
Ether	350000	34

Example

1. Calculate thermal energy required to convert 5Kg of ice at 0ºC to steam at 100oC [specific heat capacity of ice = $2100JKg^{-1o}C^{-1}$, water = $4200JKg^{-1o}C^{-1}$,]

[$L_V = 2.3 \times 10^6J/Kg$, $L_f = 3.3 \times 10^5J/Kg$]

Solution

$Q_1 = mL_f$

$Q_1 = 5Kg \times 3.3 \times 10^5 J/Kg$

$Q_1 = 1.65 \times 10^6 J$

$Q_2 = mc\Delta T$

$Q_2 = 5Kg \times 4200JKg^{-1o}C^{-} \times 100^oC$

$Q_2 = 2.1 \times 10^6 J$

$Q_3 = mL_V$

$Q_3 = 5Kg \times 2.3 \times 10^6 J/Kg$

$Q_3 = 1.2 \times 10^7 J$

$Q = Q_1 + Q_2 + Q_3$

$Q = 1.65 \times 10^6 J + 2.1 \times 10^6 J + 1.2 \times 10^7 J$

$Q = 1.6 \times 10^7 J$

Exercise

1. Define specific latent heat of fusion
2. Calculate the quantity of heat:

(a) Which has to be supplied to melt 5g of ice at 0^oC

(b) Which has to be removed to turn 10g of water into ice at 0^oC

3. Determine the quantity of heat needed to convert 2Kg of ice at 0^oC to water at 50^oC.

UNIT 4: WAVES MOTION

Definition: A wave is a disturbance in a medium which transmits energy. A wave can also be defined as the form that some types of energy take as they move. .e.g. Water wave, Sound wave, Light wave Electromagnetic wave.

A wave front is a line joining points on a wave which are in phase

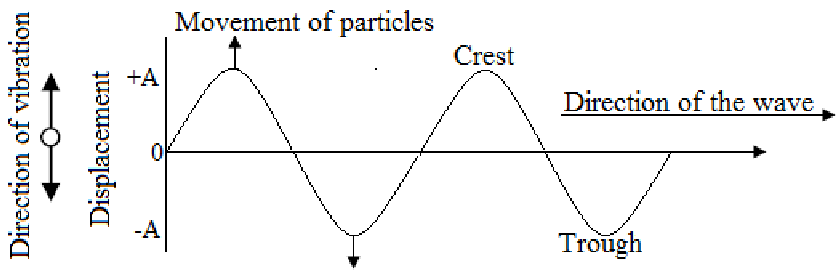
Types of waves

There are two types of waves

- Transverse waves
- Longitudinal waves

Transverse wave

Transverse wave is a wave in which the movement of particles is perpendicular to the direction of travel of the wave



Note

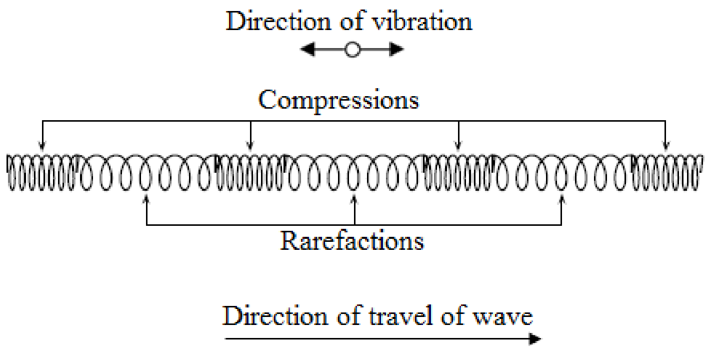
- A crest is a highest point of a wave
- A trough is a lowest point of a wave

Examples of transverse waves

- Water waves
- Light
- A wave in a rope

Longitudinal wave

Longitudinal wave is a wave in which the movement of the particles is parallel to the direction of travel of the wave



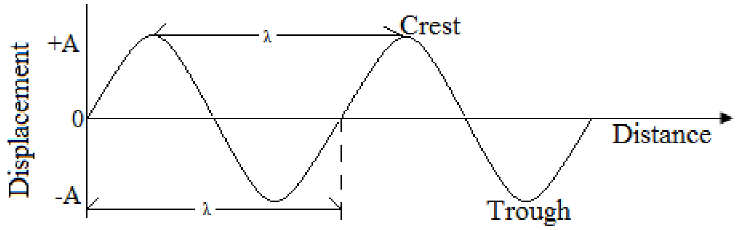
Examples of longitudinal waves

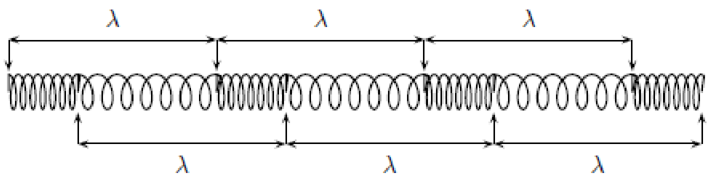
- Sound waves
- Waves on the spring
- Seismic wave

Seismic Waves

Seismic waves are waves from vibrations in the Earth (core, mantle, oceans). Seismic waves also occur on other planets, for example the moon and can be natural (due to earthquakes, volcanic eruptions or meteor strikes) or man-made (due to explosions or anything that hits the earth hard). Seismic P-waves (P for pressure) are longitudinal waves which can travel through solid and liquid.

Graphs of waves
Graph of displacement against distance

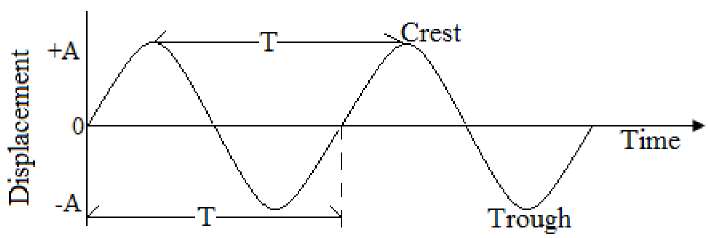




Amplitude
Symbol: A
SI unit: Meter [m]
Definition: Amplitude is the maximum displacement of a wave from its rest position (height of crest or depth of trough)

Wave length
Symbol: λ
SI unit: Meter [m]
Definition: Wave length is distance between two successive similar points on a wave

Graph of displacement against time



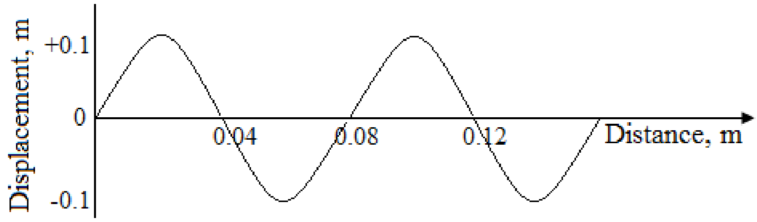
Period
Symbol: T
SI unit: Second [s]
Definition: Period is the time taken for one complete wave to be generated.
Formula: $T = \frac{1}{f}$

Frequency
Symbol: f
SI unit: Hertz [Hz]
Definition: Frequency is the number of waves generated per second
Formula: $f = \frac{1}{T}$
$f = \frac{\text{Number of waves}}{\text{Time}}$

Speed
Symbol: V
SI unit: Meter per second [m/s]
Definition: Speed of a wave is the distance travelled by the wave in one second
Formula: $V = \lambda \times f$

Example

1. Refer to the graph below



(a) What is the amplitude of the wave?
(b) What is the wavelength of the wave?
(c) Given that the speed of the wave is 4m/s, calculate its frequency

Solution

(a) A = 0.1m
(b) λ = 0.08m
(c) $f = \frac{V}{\lambda}$
$f = \frac{4m/s}{0.08m}$
f = 50Hz

2. If 100 waves were produced in 5 seconds, what is the frequency?

$f = \frac{\text{Number of waves}}{\text{Time}}$

$f = \frac{100}{5}$

f = 20Hz

Change of wavelength and speed of water waves

When a wave is travelling from:

(a) Deep water to shallow water:
- Wavelength decreases
- Speed decreases
- Frequency remains the same

(b) Shallow water to deep water
- Wavelength increases
- Speed increases
- Frequency remains the same

UNIT 5: SOUND

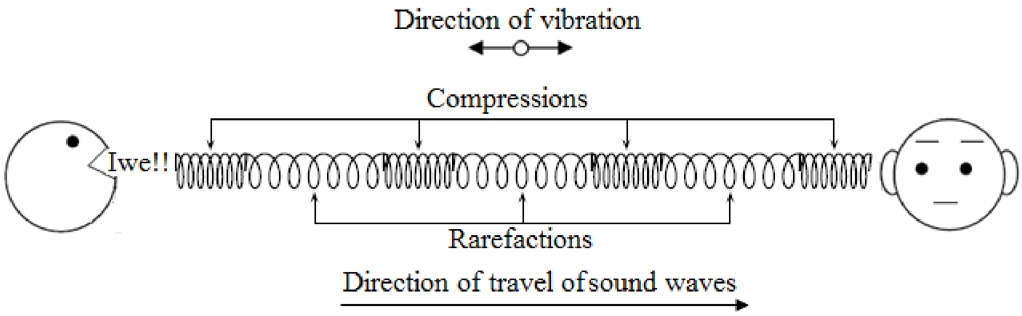
Properties of sound

When we talk to someone, the sound is transmitted in the air. In this case, some molecules of air are pushed, and some molecules of air are farther apart. This motion of molecules transmits the sound.

- Molecules just vibrate to and fro. Molecules do not move across the medium.
- A slightly higher-pressure place is called compression.
- A slightly lower-pressure place is called rarefaction.
- Sound wave is longitudinal wave because the direction of molecular vibration and the direction of traveling sound wave are the same.
- Sound waves need any medium (solids, liquid and gas) when it is transmitted.
- Sound cannot travel through a vacuum.
- Sound travels faster in denser media. (It travels faster in liquids than in gases, and fastest in solids.)

For example

- Air (15ºC) 340m/s
- Water (25ºC) 1500m/s
- Iron 5950m/s



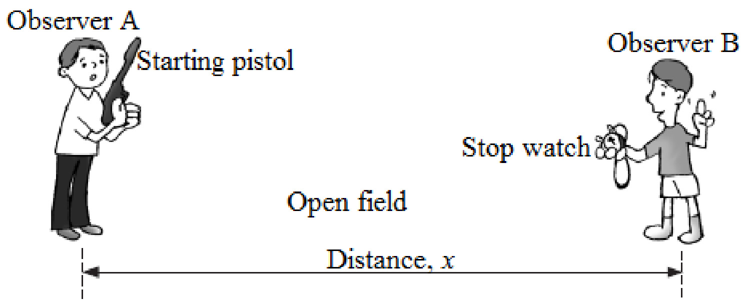
Speed of sound

In the air, sound travels at a speed of about 340m/s (15 ºC).

Experiment

Aim: To determine the speed of sound in air

Direct method

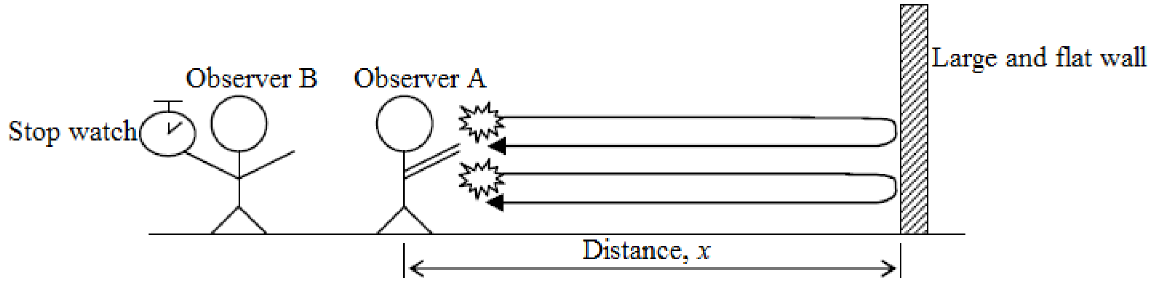


Procedure

- Observer A and B stand at a known distance 'x' apart in an open field. Record the distance 'x' measured by measuring tape. (x must be set as far as possible. e.g. 500m,1000m)
- Observer A fires the starting pistol.
- When observer B sees the flash of the starting pistol, he starts the stopwatch.
- When he hears the sound, he stops the stopwatch. The time taken 't' is recorded.
- The speed of sound v can be calculated by using the formula:

$$v = \frac{x}{t}$$

Echo method



Procedure

- Observer A and B stand at a distance 'x' from a large and flat wall. Measure and record 'x' measured by measuring tape.
- Observer A claps hands and listen to the echo. Repeat the clap on hearing the echo.
- Observer B start the stopwatch and counting from Zero to the 50th clap. The time taken t_1 is recorded.
- The time interval t between each clap can be calculated by; $t = \frac{t_1}{50}$
- The speed of sound, v, can be calculated by; $v = \frac{2x}{t}$

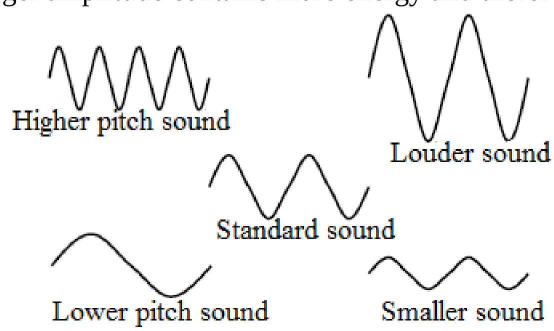
(Total distance covered by the echo is $2x$. Going and coming back)

Pitch of sound

- The pitch of sound shows how high or low is. For example, girl's voice is high pitched but boy's voice is low pitched.
- The pitch of sound depends on the *frequency*. Low pitch is low frequency and high pitch is high frequency.
- A man can listen sound waves with frequencies ranging from 20Hz to 20000Hz (20 kHz). [Audible sound]

Loudness of sound

- The loudness of sound wave depends on the *amplitude* of the wave.
- A sound wave with larger amplitude contains more energy and therefore louder.



Exercise

1. The speed of light in air is 3×10^8m/s. the speed of sound in air is 340m/s. an observer is 500m (5Km) away from a lighting discharge.

(a) Calculate the travel time to the observer of...
   - (I) Light from the lighting flush
   - (II) Sound from the thunder

(b) Describe an experiment you would perform accurately determine the speed of sound in air.

2. A fishing boat on Lake Tanganyika uses ultra sound of frequency 6.0×10^{14} Hz to detect fish directly below. Two echoes of the sound are received one after 0.09 seconds coming from a shoal of fish and the other after 0.12 seconds coming from the lake bed. If the lake bed is 84m below the sound transmitter and receiver,

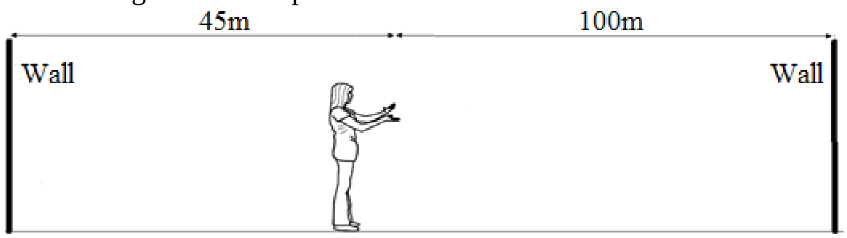
Calculate

(a) The speed of sound waves in water

(b) The wave length of the sound waves in water

(c) The depth of the shoal of fish below the boat.

3. Find the speed of a sound in the air if it has a frequency of 1120Hz and a wavelength of 30cm.
4. A pupil stands 85m in front of a wall. He claps his hands and repeats the claps when he listens the echo. The other pupil who stands with clapping pupil starts a stopwatch at the 10th clap and stops at the 50th clap. The time taken for the all the claps is 20s. Find the speed of sound.
5. Under the temperature of 15°C, the speed of the sound is 340m/s. What if the temperature becomes higher?
6. A student standing between two walls claps once and hears the first echo after 0.60s and the second echo after 1.00s. How long after the clap will she hear the next echo?



UNIT 6: LIGHT

The nature of light

Light consists of streams of tiny wave like packets of energy called photons which travel at a speed of 3 x 10^8m/s. The general outline of the nature of light is listed below.

- **Light transfers energy from one place to another**
  Energy is needed to produce light. Materials gain energy when they absorb light. Mostly this causes an increase in their internal energy. The solar cells change some of the energy in sunlight directly into electrical energy
- **Light is a form of radiation**
  Radiation is a general term applied to almost anything that travels outwards from its source but can't immediately be identified as solid, liquid or gas like the more familiar forms of matter.
- **Light is a form of wave motion**
  The way in which light radiates from its source is similar in many ways to the way in which ripples spread outwards across a pond when a pebble is dropped into the water. In case of light the ripples are electric and magnetic in nature.
- **Light is something detected by the human eye**
  Objects emit many types of radiations, most of which are not detectable by the human eye. Light is the wave given to radiations which the eye can detect.

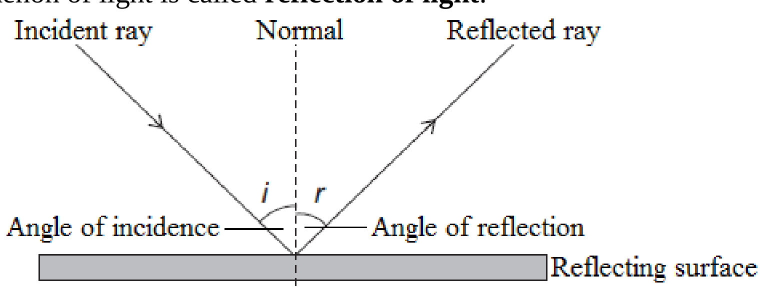
Property of light

- The light from an object travels in a straight line.
- There are two ways to change the direction of light.
- Reflection
- Refraction

Reflection of light

When a light ray from a light source reaches on the surface of an object, the ray bounces off on it. Then the ray enters our eyes. This is the reason why we can see an object. Especially, the ray bounces in a regular way on the polished surface such as a mirror.

This bouncing phenomenon of light is called **reflection of light**.



Medium

A medium is a substance through which light moves.

Examples of media

- Air
- Water
- Glass

Ray

It is a path followed by light.

Beam

It is a group of light rays.

Normal

It is the line drawn at right angles to the reflecting surface at the point of incidence. When the incident ray strikes a medium at right angles, it does not get refracted. It is a perpendicular line drawn to the reflecting or refracting surface of a medium.

Normal ray

It is a ray of light that passes through the normal

Point of incidence
Point of incidence is the point where the ray strikes the reflecting surface.
Incident ray
It is a path followed by light before it strikes the surface of the next medium.
It is a ray that falls on the reflecting surface.
Angle of incidence
Angle of incidence is the angle between the incident ray and the normal.
Angle of reflection
Angle of reflection is the angle between the reflected ray and the normal.
Reflected ray
Reflected ray is a path followed by light after reflection.
Law Laws of Reflection

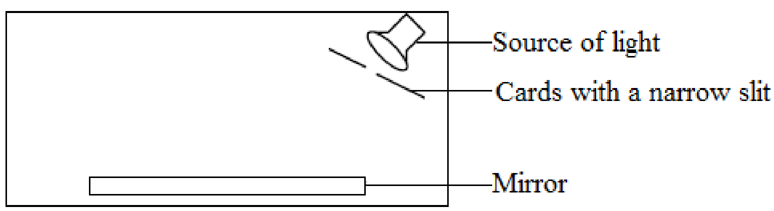
1. *The angle of incidence is equal to the angle of reflection.*
   $i = r$
2. *The incident ray, the reflected ray and the normal all lie in the same plane.*

Experiment
Aim: To verify the raw of reflection, $i = r$

Materials

- Light source (Torch, Sun)
- Plain paper
- flat mirror
- cards (notebooks)

Experimental set up

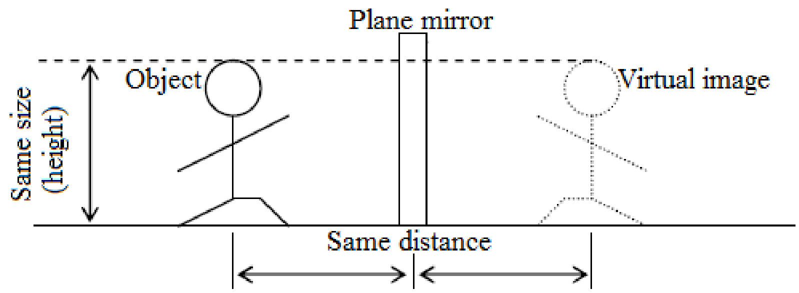


Procedure

- Arrange the apparatus as shown in the diagram.
- Mark the position of the mirror with a straight line.
- Turn on the torch (The light ray comes through the slit to the mirror.)
- Mark the paths of the light before and after reflection by putting a pencil dot at two places each as far apart as possible.
- Join each pair of dots with a straight line and extend the line to the mirror position.
- Draw a normal on meeting the rays.
- Measure and compare the angle of incidence and the angle of reflection.

Conclusion
The angle of incidence and the angle of reflection are equal.
Virtual image (in the plane mirror)
The image which cannot be formed on a screen is called the virtual image. You can see your virtual image in a mirror.
Properties of the image formed by plane mirrors

- The image formed is the same size as the object.
- The image is as far behind the mirror as the object is in front. The virtual image in a mirror appears at the same distance behind the mirror as the object is in front.
- The image is virtual.
- The image is laterally inverted (Right and left are opposite.)
- A line joining any point on the object to the corresponding point on the image cuts the mirror at right angle.

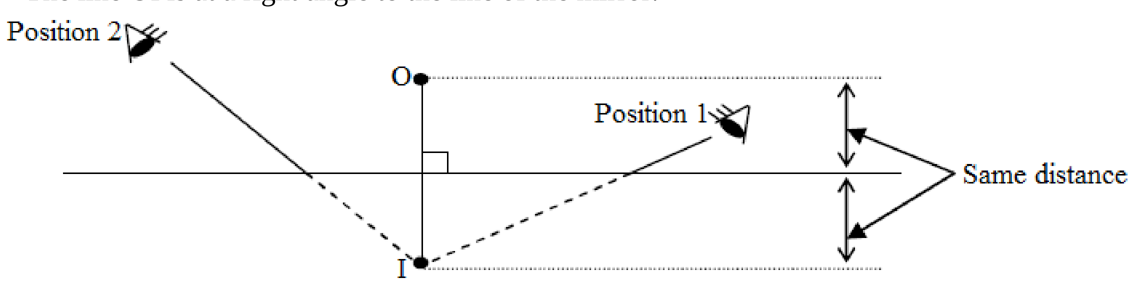


The image which appears on a screen is called the real image. e.g. pine hole camera, movie)

How to locate the virtual image formed by a plane mirror

1. Stand a mirror on a plain paper.
2. Draw the position of the mirror with a straight line.
3. Stand the object (such as pin) O in front of the mirror and locate the image behind the mirror.
4. View the image in the mirror from some convenient position 1.
5. Draw the straight line in front of the mirror from the eye to the image.
6. Repeat the process 4 and 5 from another position 2.
7. Remove the mirror and extend those lines behind the mirror with the dotted line.
8. The intersection of those lines, I, is the position of the image you observed from each position.
9. Join O and I with a straight line and measure the distance from mirror to O and to I.

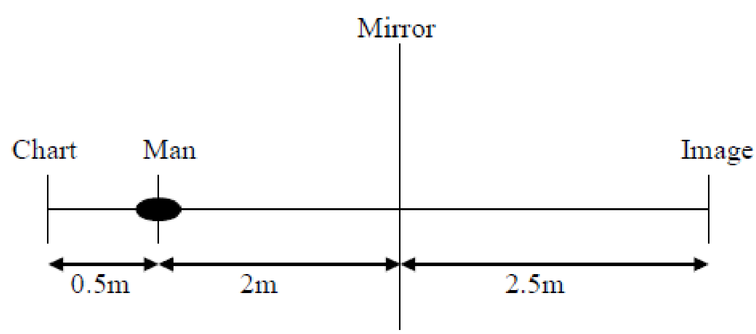
- The distance from the mirror to O is equal to the distance from the mirror to I.
- The line OI is at a right angle to the line of the mirror.



Example

1. A man sits in an optician's chair, looking into a plane mirror which is 2m away from him and views the image of a chart which faces the mirror and is 50cm behind his head. How far away from his eyes does the chart appear to be?

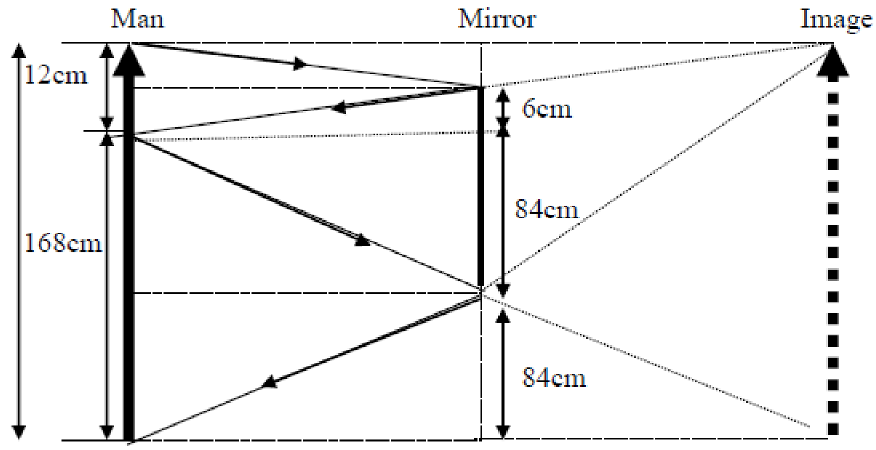
Solution



Distance of image from man is = 2 + 2.5 = 4.5m

2. Draw a ray diagram to show that a vertical mirror need not be 180cm long in order that a man 180cm tall may see a full – length image of himself in it.
   If a man's eyes are 12cm below the top of his head find the shortest length of mirror necessary and the height of its base above the floor level.

Solution



- Shortest length of the mirror = 84 + 6 = 90cm
- Height of its base above floor level = 84cm

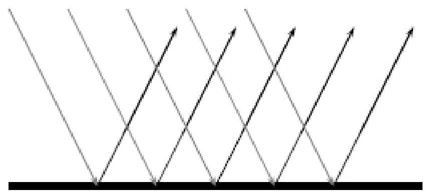
Types of Reflection

Regular Reflection

When a set of parallel light rays strike a plane flat surface such as a mirror or polished sheet of metal, the reflected rays are also parallel. This is called regular or specular reflection.

Parallel incident light rays hit the smooth surface and parallel reflected light rays leave the surface.

The diagram below shows a surface that is flat and even.


Smooth surface

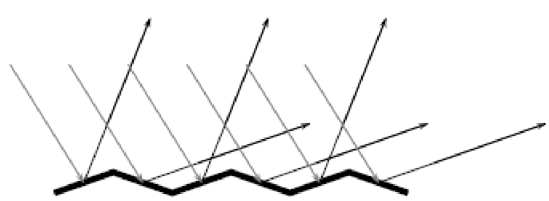
This type of reflection is called specular reflection. Specular reflection occurs when rays are reflected from a smooth, shiny surface. The normal to the surface is the same at every point on the surface. Parallel incident rays become parallel reflected rays. When you look in a mirror, the image you see is formed by specular reflection.

Diffuse Reflection

When a set of parallel light rays strike an irregular surface such as a piece of paper or concrete path, the rays are scattered in different directions. This is called diffuse reflection. We see most of the objects around us because of diffuse reflection.

When multiple rays hit the uneven surface, diffuse reflection occurs. The incident rays are parallel but the reflected rays are not.

The diagram below shows a surface with bumps and curves.


Rough surface

Each point on the surface has a different normal. This means the angle of incidence is different at each point. Then according to the Law of Reflection, each angle of reflection is different.

Irregular reflection occurs when light rays are reflected from bumpy surfaces. You can still see a reflection as long as the surface is not too bumpy. Irregular reflection enables us to see all objects that are not sources of light.

Because of the difference between diffuse and regular refection, images of objects can be seen in plane mirrors or polished surfaces but not in paper or other irregular surfaces.

Exercise

1. The diagram in figure 1.0 below shows a green light represented by the line AB striking the shiny surface.

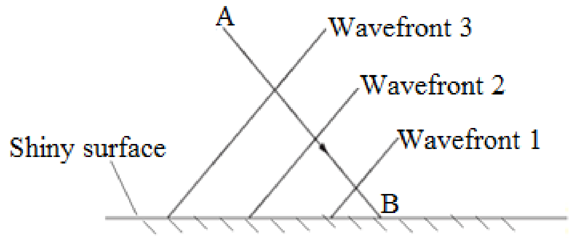


Figure 1.0: The ray of green light striking a shiny surface

(a) On striking the surface at an angle shown above complete the diagram of the green light wave front to how it behaves.
(b) From the diagram you have completed for the diagram of figure 1.0 shown above, what principle is depicted by the green light wave front?
(c) State two properties that the green light will satisfy even when it strikes the shiny surface.
(d) In another experiment two rays of green light were used but on a rough surface as shown in figure 1.1 below.

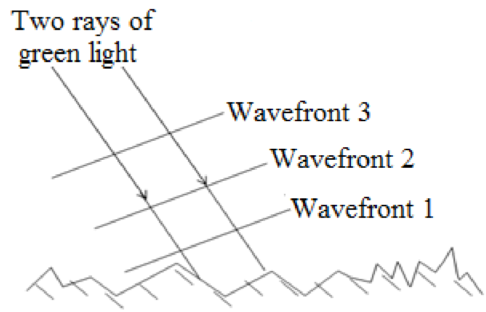


Figure 1.1: The two rays of green light striking in rough surface

Complete the diagram in figure 1.1 to show the behavior of the two rays as they strike the rough surface.

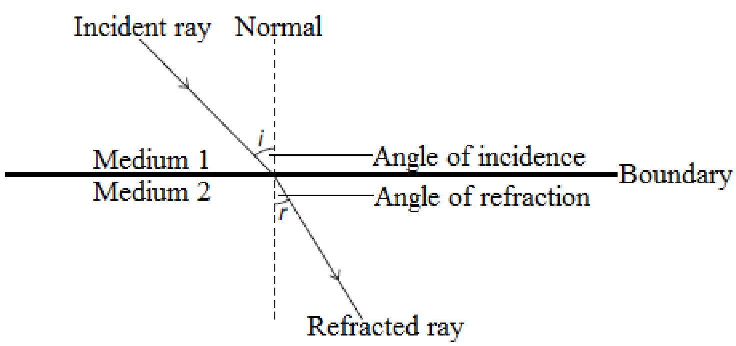
(e) What name is given to the phenomena shown in the diagram in figure 1.1 above called?

Refraction of light

Definition: Refraction is the bending of light rays after passing through different media.

It is the change of direction or bending of light when it passes from one medium to another.

Refraction takes place at the boundary of two media. Refraction is due to the different speeds of light as it travels from one medium to another.



Normal

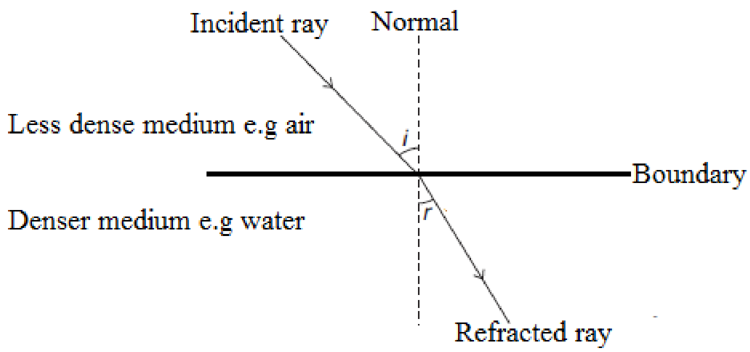
Normal is the line drawn at right angles to the boundary.

Incident ray

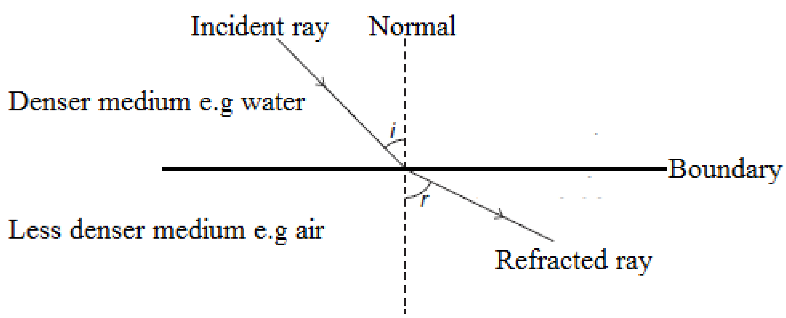
Incident ray is the ray that falls on the boundary between the two media.

Refracted ray
Refracted ray is a path followed by light after refraction. It is a ray that leaves the reflecting surface.
Angle of incidence
Angle of incidence is the angle between the incident ray and the normal.
Angle of refraction
Angle of refraction is the angle between the refracted ray and the normal
Emergent ray
It is a path followed by light as it comes out of glass block

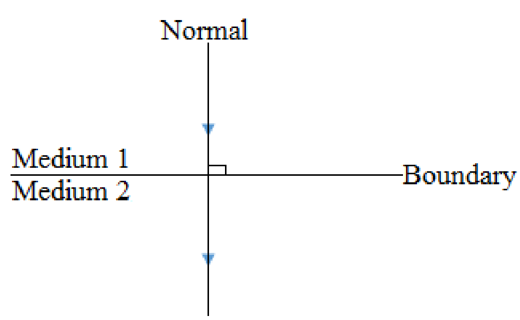
(a) When light travels form less dense to denser medium,
- light is refracted towards the normal
- the speed in the denser medium is slower than that in the less dense medium.



(b) When light travels form denser to less dense medium,
- light is refracted away from the normal
- the speed in the less dense medium is faster than that in the denser medium.



(c) When light enters another medium at right angle to the boundary,
- light is not bent.
- the speed also changes. It depends on media.



Law Laws of Refraction

1. *The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant.*

   Formula: $n = \frac{\text{Sin i}}{\text{Sin r}}$

- The constant number ‘n’ is called the **refractive index.** This law is called Snell’s law

Examples of refractive index

- Water 1.33
- Glass 1.52
- Diamond 2.42

2. *The incident ray, the reflected ray and the normal are in the same plane.*

Experiment 1

Aim: To verify the raw of refraction (Snell's law).

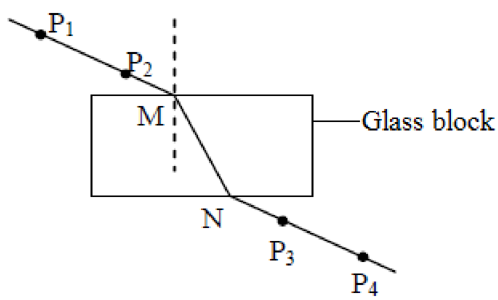
Materials

- 4 pins (pens)
- rectangular glass block
- plain paper

Procedure

1. Put a rectangular glass block on the plain paper.
2. Draw the outline of the block.
3. Put 2 pins on one side of the block. (P_1 and P_2)
4. Connect P_1 and P_2 with a straight line, and extend it to the surface of the block.
5. Look at the pins through the block from the opposite side.
6. Put another pins in line with P_1 and P_2 through the block on the opposite side. (P_3 and P_4)
7. Connect P_3 and P_4 with a straight line, and extend it to the surface of the block.
8. Draw a straight line between two boundaries N and M.
9. Draw the normal at M.
10. Measure the angle of incidence, *i*, and the angle of refraction, *r*.
11. Find sine *i*, sine *r* and calculate the refractive index.
12. Repeat 3 to 11 with different positions.

Experimental set up



Results

(Samples of results are shown below)

i	***r***	***Sin i***	***Sin r***	$\frac{\textbf{Sin } i}{\textbf{Sin } r}$
39°	29°	0.6293	0.4695	1.34
57°	33°	0.8387	0.5446	1.54
17°	12°	0.2924	0.2079	1.41

Conclusion

- The refractive indexes of two fixed materials are almost constant numbers.
- Line P_1P_2 are parallel to the line P_3P_4

Experiment 2

Aim: To show some effects of refraction

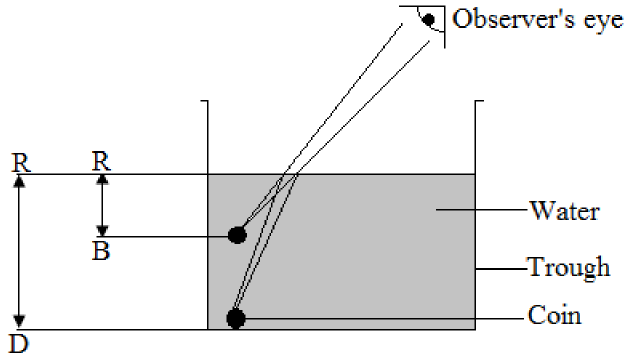
Materials

- Coin
- Trough
- Water

Method

- Put a coin in an empty trough.
- Move back until you just cannot see the coin.

- While standing in that position, let your friend pour water gently into the trough a little bit at a time until the trough is full.
- As your friend is pouring water into the trough, observe what happens.



Observation

The coin in water appears nearer to the surface than it really is. This is because light from the coin in water is refracted before reaching the observer's' eyes.

Explanation

Light from point D is refracted away from the normal at the water surface. Light reaching the eye of the observer appears to come from point B, which is directly above point D. The object D appears to be in position B. Thus RB is its apparent depth. The true or real is RD since the coin has not been raised. The apparent depth is always less than the real depth of the object.

Refractive index $= \frac{\text{Real depth}}{\text{Apparent depth}}$

Note

- *The same reasons can be used to explain why swimming pools or fish in ponds appear shallower than they really are, and why a fish in water appears closer to the surface of the water than it actually is.*

Experiment 3

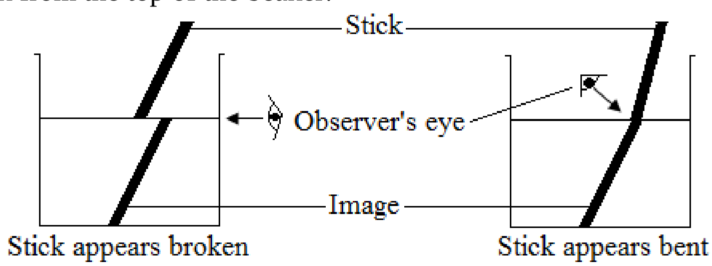
Aim: To show the effect of refraction.

Materials

- Beaker
- Water
- Stick which is about 30cm long

Method

- Fill the beaker with water.
- Dip the stick into water at an angle.
- Stand aside and look at the water surface through the wall of the beaker.
- Look at the stick from the top of the beaker.

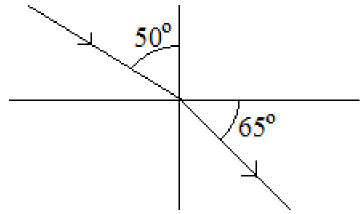


Observations

The stick appears to be broken or bent. This is because the stick in water reflects rays of light to the observers' eye through the air. Since water is denser than air, the rays of light bent away from the normal, hence the stick appears bent or broken at the water surface.

Example

1. Find the refractive index in the diagram below.



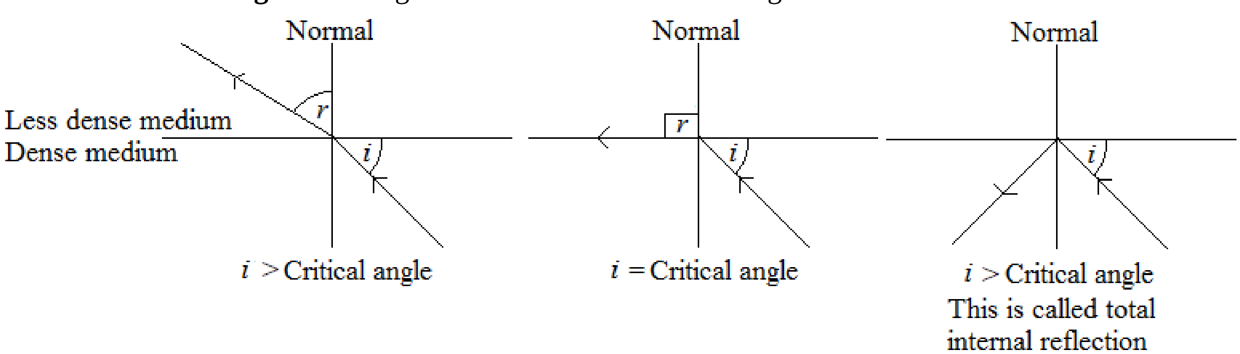
Data	**Solution**
n = ? i = 50 ° r = 90-65 = 25°	$n = \frac{\text{Sin i}}{\text{Sin r}}$ $n = \frac{\text{Sin } 50^o}{\text{Sin } 25^o}$ $n = n = \frac{0.77}{0.42}$ n =1.83

2. A ray of light travels from air into water at an angle of incidence 60 o. Find the angle of refraction if the refractive index of water is 1.33.

Data	**Solution**
r = ? i = 60o n = 1.33	$n = \frac{\text{Sin i}}{\text{Sin r}}$ $sin\ r = \frac{\text{Sin i}}{\text{n}}$ $sin\ r = \frac{\sin 60^o}{1.33}$ $sin\ r = \frac{0.87}{1.33}$ sin r = 0.65 $r = \sin^{-1}(0.65)$ = 41°

Critical angle

Definition**: Critical angle** is the angle of incidence at which the angle of refraction is 90°



Examples of Critical angles

- Water 49°
- Glass 42°
- Diamond 24°

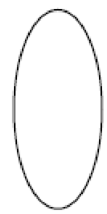
Exercise

1. What is meant by each of the following terms?
   (a) Critical angle
   (b) Refracted index
2. A coin is placed in a bucket full of water and its real depth is 30cm. given that the refracted index of water 1.33, calculate the apparent depth.

Lenses
A lens is a piece of glass or any other transparent material which has curved surfaces.
Types of lenses
There are two main types of lenses

1. **Convex lenses**

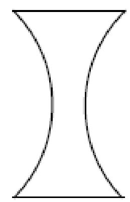
Alternative term: Converging lens
The convex lens is thicker in the middle and thinner at the edges.



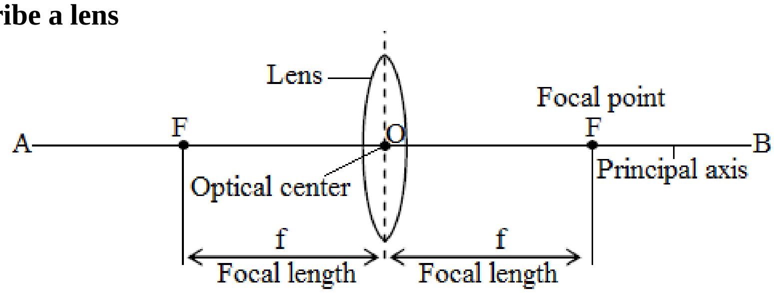
The convex lens converges or draws together light to a point.

2. **Concave lenses**

Alternative term: Diverging lens
The concave lens is thinner in the middle than and thicker at the edges.



The concave lens diverges or spread out light.
Terms used to describe a lens



Focal point
Alternative term: *Principal focus*
Symbol: F
Focal point is the point at which light rays converge or diverge from.
It is the point where rays parallel to the principal axis converge.
A lens has two focal points, one from either side.
Optical centre
Symbol: O
Definition: Optical center is the centre of lens.
Principal axis
Definition: Principal axis is the line passing through center and perpendicular to the plane of the lens.
It is the line drawn through the optical center and is perpendicular to the lens.
The line AB is the principal axis
Focal length
Symbol: f
Definition: Focal length is the distance between the optical center and the focal point.

Images formed by a converging lens

1. **Real image**

   A real image is one that can be formed on the screen.

   For the image to be seen clearly, it must be focused properly. Focusing is the process of moving the screen nearer to or further away from the lens in order to obtain a clear and sharp image.

Characteristics of a real image

- It is always upside down
- It is either magnified, diminished or same size as the object
- It is formed on the other side of the lens.

2. **Virtual image**

   A virtual image is one that cannot be produced on the screen. In order for a virtual image to be seen, it is necessary to look through the lens from the side which is always from the object.

Characteristics of a virtual image

- It is always upright
- It is always magnified.
- It is formed on the same side as the object.

Ray diagrams

A ray diagram is a diagram showing the arrangement of the object, the lens, light rays and an image in relation to each other.

From a ray diagram, we can tell whether the image formed is real or virtual. A ray diagram also shows whether the image is magnified, diminished or the same size as the object and whether it is on the same side of the lens as the object or not.

Magnification

Magnification is the ratio of the size of the image to the size of the object.

Formula: $\text{Magnification} = \dfrac{\text{Height of image}}{\text{Height of object}}$

$\text{Magnification} = \dfrac{\text{Distance of image from lens}}{\text{Distance of object from lens}}$

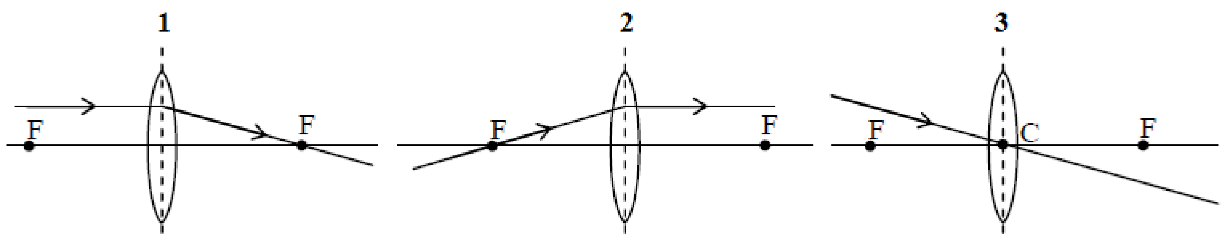
Magnification has no units and represents how big or small the image is compared to the object. The letter M is used for magnification.

When:

- M = 1, then the drawing and the specimen are of the same size
- M > 1, then the drawing is larger than the specimen
- M < 1, then the drawing is smaller the specimen

Rules to draw the ray diagram through lens

1. Rays parallel to principal axis are refracted through F.
2. Rays passing through F are refracted parallel to the principal axis.
3. Rays passing through C continue in a straight.



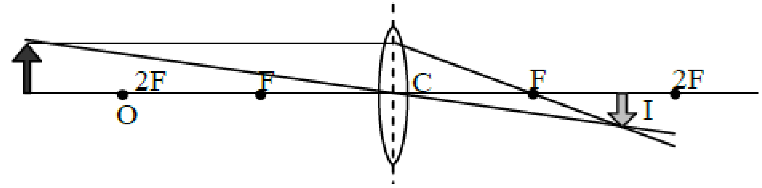
How to locate the image through a lens

1. Use any two of the rules. (Any two of the rules are enough for locating the image.)
2. Find the intersection of the refracted rays. It gives the position of the image.

- If the refracted rays intersect at any point, the image is real.
- If the refracted rays extend behind the object and the extended lines intersect at any point, the image is virtual.
- If the refracted rays do not intersect at any point, the image is infinity.

Various images formed by convex lens

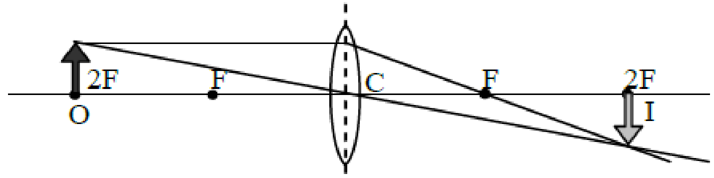
1. Object beyond 2F



Characteristics of the image formed

- The image is between F and 2F.
- The image is real.
- The image is inverted.
- The image is diminished.

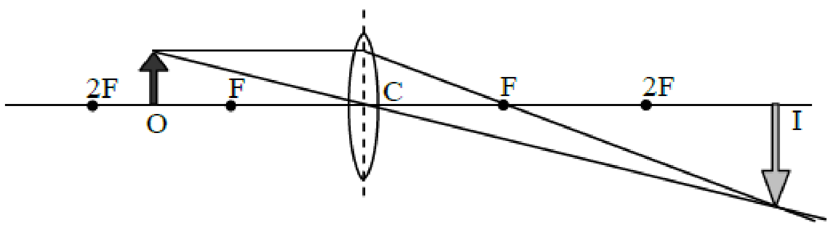
2. **Object at 2F**



Characteristics of the image formed

- The image is at 2F.
- The image is real.
- The image is inverted.
- The image is the same size as the object.

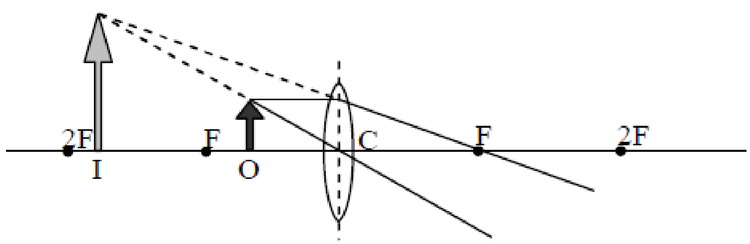
3. Object between 2F and F



Characteristics of the image formed

- The image is beyond 2F.
- The image is real.
- The image is inverted.
- The image is magnified.

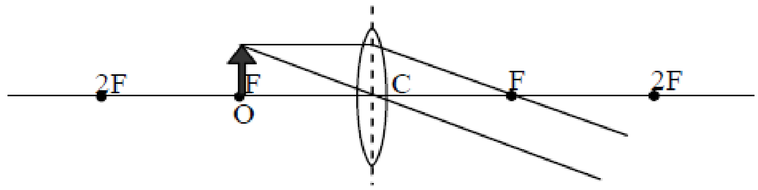
4. Object between F and C



Characteristics of the image formed

- The image is between F and 2F.
- The image is virtual.
- The image is upright.
- The image is magnified.

5. **Object at F**

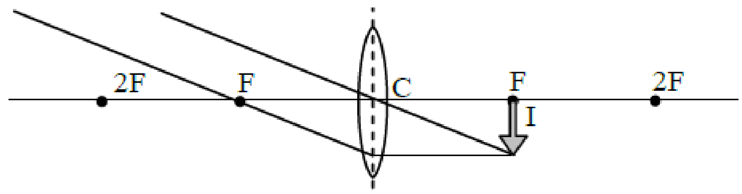


Characteristics of the image formed

- The image is at infinity

6. **Object is at infinity**

Parallel rays from distant object



Characteristics of the image formed

- The image is at F.
- The image is real.
- The image is inverted.
- The image is diminished.

Electromagnetic spectrum

Visible light consists of some colours, red, orange, yellow, green, blue, indigo and violet. When you see the rainbow, you can experience a light band. This band of coloured light is called the spectrum. The colours are in order of their wavelength.

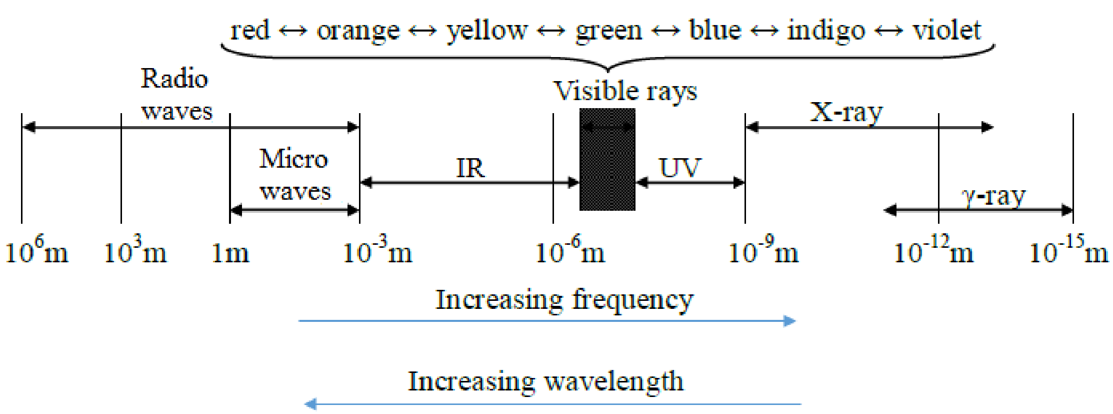
And visible light is one of electromagnetic waves. Electromagnetic waves are named in the order of the ranging of wavelength. They are Radio waves, Microwaves, Infra-red rays (IR), visible rays, Ultra-violet rays (UV), X-rays and γ (gamma)-rays. This band of electromagnetic waves is called the electromagnetic spectrum.

Properties of electromagnetic waves

Electromagnetic waves have some common properties shown below.

- They transfer energy from one place to another.
- They are all transverse waves.
- They can all travel through a vacuum.
- They all travel at 3×10^8m/s (300,000,000 m/s) in a vacuum.
- The equation ($v = f\lambda$) applies to all of them.

But they have different wavelengths, frequencies and applications.



γ – ray

- They are the most energetic and the most penetrating rays. They are dangerous but also useful for the medical treatment such as radiotherapy.

X-ray

- X-rays have a considerable penetrating power through matter and they also affect photographic films. These two properties make them suitable for use in seeing through objects, e.g. X-ray photographs for bones and metallic structure.

Ultraviolet rays

- The sun emits a lot of ultraviolet rays. However, much of them is absorbed by a layer of ozone in the upper atmosphere of the earth. If much of them reaches our bodies, it can cause damages such as heavy sunburn, eye damage and skin cancer.

Infra-red rays

- It transfers heat energy but it is invisible. It can be detected by the infra-red photograph (which is used as Night photography) and the thermopile.

Microwaves

- Microwaves are kinds of radio waves. They are used in radar systems, microwave oven and the communication with satellites.

Radio waves

- Radio waves are used for the broadcasting of radio or TV programmes and cellphones.

Exercise

1. The table below shows the electromagnetic spectrum.

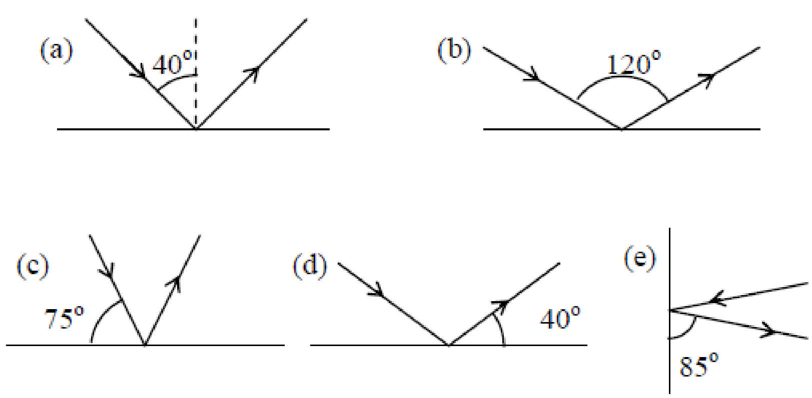
Gamma rays	**P**	Ultra-violet	Visible light	**Q**	microwaves	Radio waves

(a) State the name of
  (I) Component P
  (II) Component Q

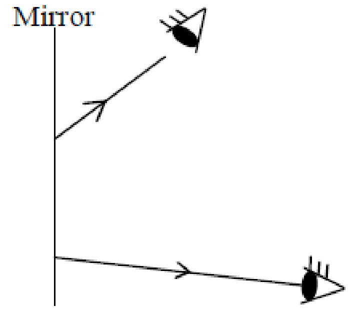
(b) Which kind of radiation is used to:
  (I) Detect an aircraft
  (II) Treat skin cancer
  (III) Study crystal structure

(c) Calculate the frequency of radio waves detected by an object by an aerial of 1.5m long
(Assume that $C = 3 \times 10^8$m/s and the aerial is a quarter wavelength)

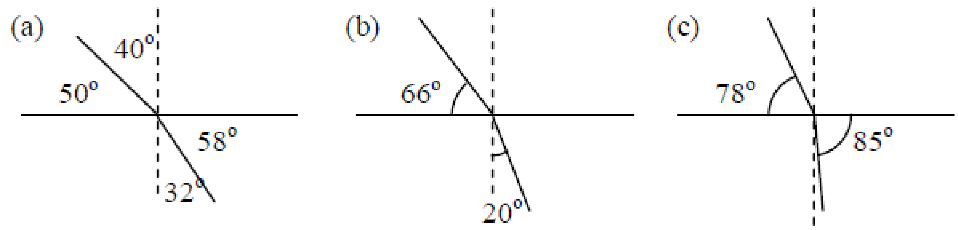
2. Find the angle of reflection in the diagram below.



3. A pupil stands 2m in front of the mirror. Find the distance from the pupil to the image.
4. The diagram below shows reflected rays of an object from a mirror to your eyes at two positions.
   (a) Locate the image of an object in the mirror and label I on it.
   (b) Find the position of the object and label O on it.



5. Find the refractive index in the diagram below.



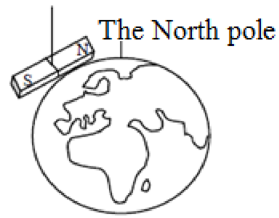
6. Alight ray travels from air into water at an angle of 50° between the boundary and the incident ray. The angle of refraction is 29°.
   (a) Calculate the refractive index
   (b) Calculate the angle of refraction if the angle between the boundary and the incident ray changes into 70°.
7. Describe an experiment you would carry out in the laboratory to find the focal length of a convex lens using the object and image distance method.

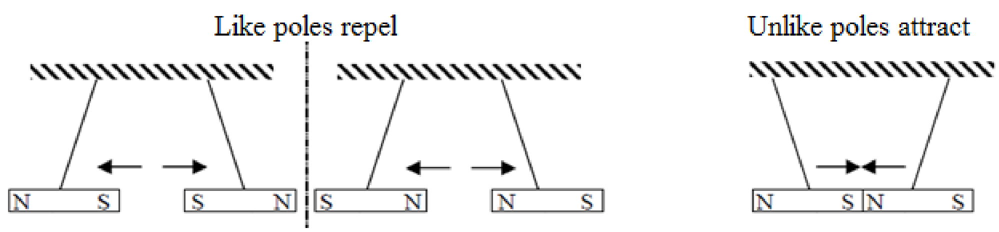
UNIT 7: MAGNETISM

Simple phenomenon of magnetism

Properties of magnetism

- The end of a magnet is called the **pole**.
- If a light magnet hangs on a string, one end points towards the North. This end is called **the North Pole**.
- The other end points towards the South. This end is called **the South Pole**.
- The same poles (North and North Pole, South and South Pole) repel. It is called '**Like poles repel'**.
- The different poles (North and South Pole) attract. '**Unlike poles attract**'.
- Magnets attract some materials, e.g. iron, cobalt, nickel, steel… These materials that can be attracted to a magnet are called **magnetic materials**. Other materials, e.g. copper, plastic, paper, wood, that can't be attracted to a magnet are called **non-magnetic materials**.

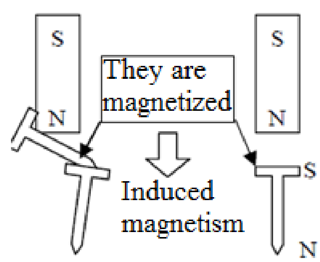




Induced magnetism

Definition: **Induced magnetism** is the temporary magnetization of a magnetic material when it is placed near to or in contact with a magnet.

- If a nail is placed in contact with a permanent magnet, the nail is magnetized. And it also attracts another nail.
- The end of a nail nearer (connected) to the North pole of a permanent magnet becomes the induced South pole, and the other end becomes the induced North pole.

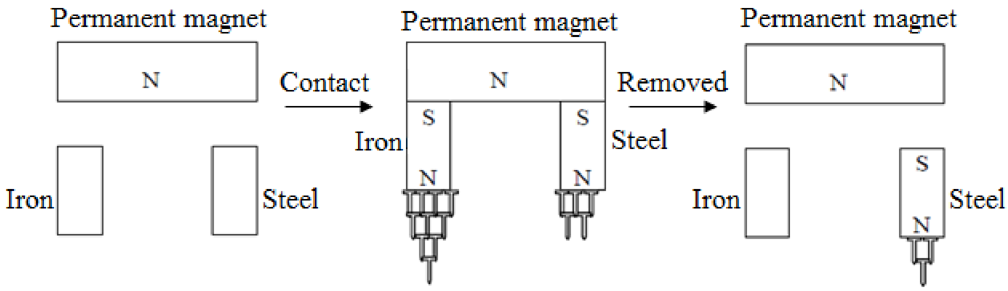


Magnetic materials (Iron and Steel)

Iron and steel are common magnetic materials. But they have different properties as shown below.

Material	Iron	Steel
Magnetize	Easy and strong	Hard and weak
Demagnetize	Easy	Hard
Application	Electromagnet	Permanent magnet
Example	Transformer, Electric bell	D.C. motor Generator
Named	Soft magnetic material	Hard magnetic material

If iron and steel are placed in contact with a permanent magnet, they are induced. Then, they attract some small nails. Iron attracts more nails than steel. And then, if they are removed from the permanent magnet, iron releases the nails soon but steel still attracts some nails.

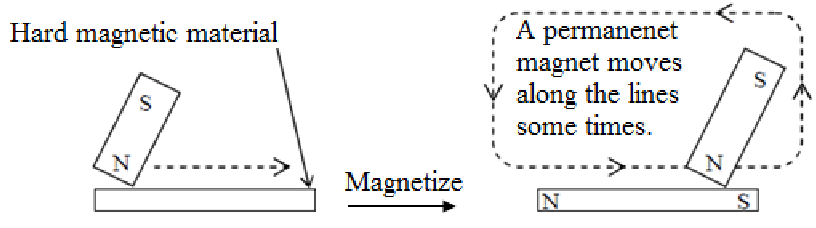


Magnetize

How to make a permanent magnet (How to magnetize a hard magnetic material)

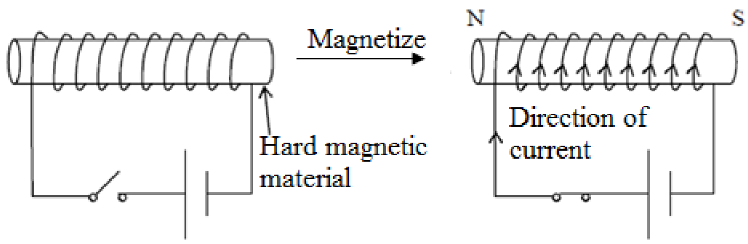
1. **Stroking method**

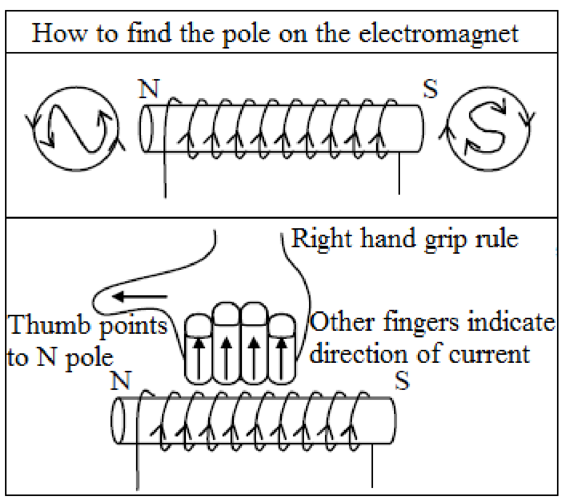
   If permanent magnets are stroked along a hard magnetic material, the hard magnetic material is magnetized and it is changed into a permanent magnet.



2. **Electrical method**

   If a hard magnetic material is placed into a solenoid (Direct current), a hard magnetic material is magnetized and it is changed into a permanent magnet.



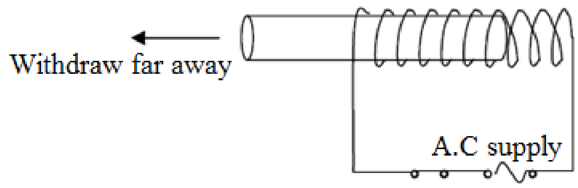


- This is the best method to make powerful magnets.
- The magnet which is made with a solenoid is called the electromagnet.
- In the boxes, it shows how to find (decide) the poles.

Demagnetize

How to demagnetize a hard magnetic material

1. **Heating**
   If a magnetized material (magnet) is heated to a higher temperature, it will lose its magnetism very quickly.
2. **Hammering**
   If a magnetized material (magnet) is hammered many times, the magnetism becomes weaker and weaker.
3. **Use alternating current and the solenoid**
   A magnet is placed inside a solenoid which has an A.C. supply. When the magnet is withdrawn far away from the solenoid, this process is repeated, the magnet is demagnetized.



Electromagnet

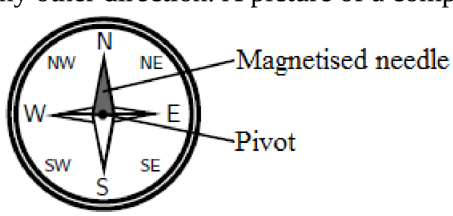
- Iron is commonly used as the core of electromagnets because it is easy to magnetize and also demagnetize. (It is easy to control the magnetism.)
- The strength of magnetism depends on:
  (a) the current
  (b) the number of turns per unit length of the solenoid
  (c) the material of core

The compass and the earth's magnetic field

A compass is an instrument which is used to find the direction of a magnetic field. It can do this because a compass consists of a small metal needle which is magnetized itself and which is free to turn in any direction. Therefore, when in the presence of a magnetic field, the needle is able to line up in the same direction as the field.

Compasses are mainly used in navigation to find direction on the earth. This works because the earth itself has a magnetic field which is similar to that of a bar magnet (see the picture below).

The compass needle aligns with the magnetic field direction and points north (or south). Once you know where north is, you can figure out any other direction. A picture of a compass is shown below:



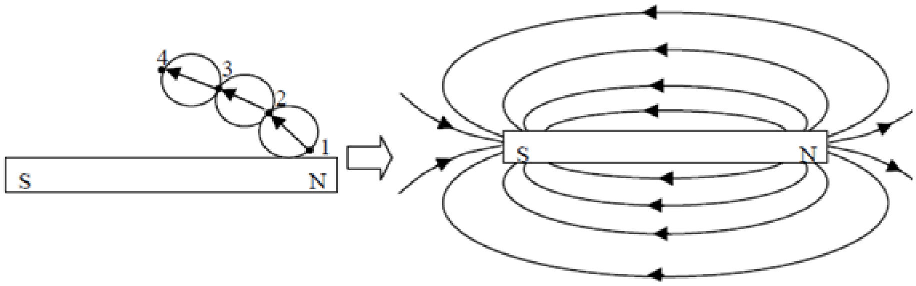
Some animals can detect magnetic fields, which helps them orientate themselves and find direction. Animals which can do this include pigeons, bees, Monarch butterflies, sea turtles and fish.

Magnetic field

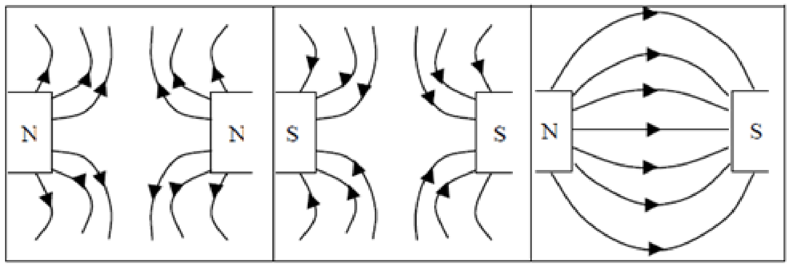
Definition: **Magnetic field** is defined as the region around a magnet where magnetic effect can be detected.

How to draw the magnetic field lines

1. Place a bar magnet on a plane paper.
2. Place the compass near one end of the magnet.
3. Plot two dots (1 and 2) at the ends of the needle.
4. Move the compass to the position where the previous dot 2 is with another pole.
5. Plot one dot 3 at the other end of the needle.
6. Repeat 4 and 5 until the compass reaches to the other pole of the magnet.
7. Connect dots from one end of magnet to another

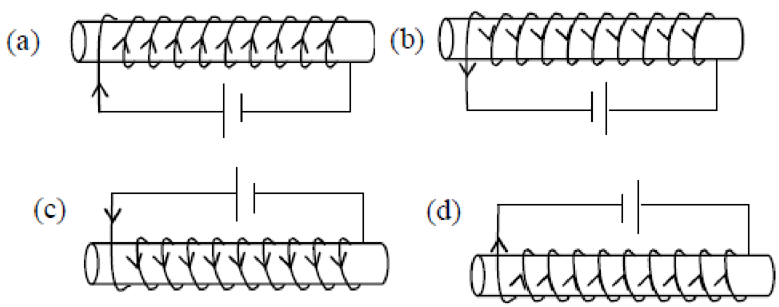


If two poles are close together, magnetic fields are shown below table.



Exercise

1. Draw the Magnetic field of a U-magnet around the poles.
2. Find the North Pole of the electromagnet below. The arrows show the direction of current.



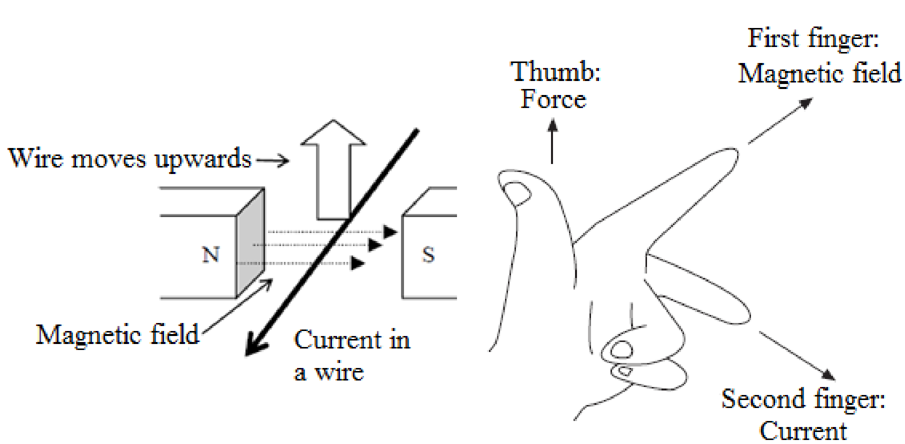
Electromagnetic effect

Fleming's left-hand rule

If a current flows in a wire, in a magnetic field, a force acts on the wire.

- The direction of force can be found by Fleming's left hand rule.

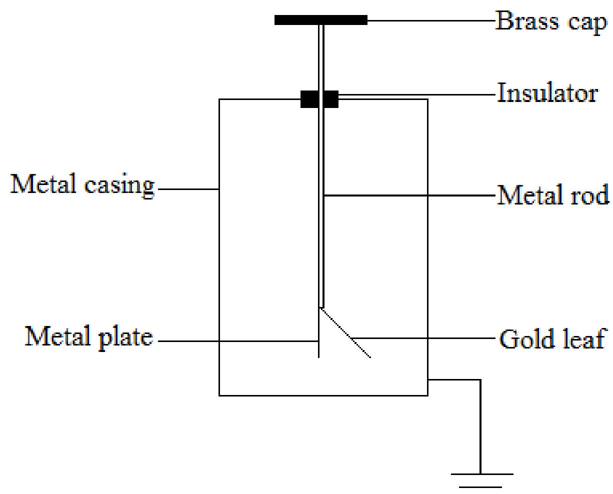
Fleming's left hand rule



UNIT 8: STATIC ELECTRICITY
Static electricity is simply the electricity at rest in form of charge.
Static charge
An object can store electric charges that cannot flow. These charges are called **static charges** (**Static electricity**).
Electric charges
There are two types of electric charges namely
- Positive (+) charge
- Negative (–) charge

The two charges obey the laws of electric charge which states that:
- *Like charges repel*
- *Unlike charges attract*

Leaf electroscope
Gold leaf electroscope
This is a device used to detect the presence of an electric charge. a gold leaf consists of the following:
- A brass rod with a gold leaf at the end
- A brass metal disc or cap on top of the rod
- An insulator surrounding the brass rod
- A metal case with glass window

The bench is sufficiently a good conductor to supply electrons from the ground or remove excess electrons from the metal case


Functions of various components of the leaf electroscope
1. **Brass cap**
   It receives charges. It has free electrons as metal, which can be repelled or attracted to one end
2. **Insulator**
   It prevents unwanted flow of charges between the positive end (brass cap) and the negative end (metal casing)
3. **Metal plate**
   It supports the gold leaf. When charged, it attracts or repels the leaf.
4. **Gold leaf**
   It the sensitive part of the electroscope. It diverges or falls in any presence of charge.
5. **Metal casing**
   It shields the leaf from external factors which can affect the leaf.

Uses of the electroscope
- It used for detecting the presence of charge
- It is used for testing the insulating properties of materials
- It is used for estimating the quantity of charge
- It is used for identifying the type of charge on a body

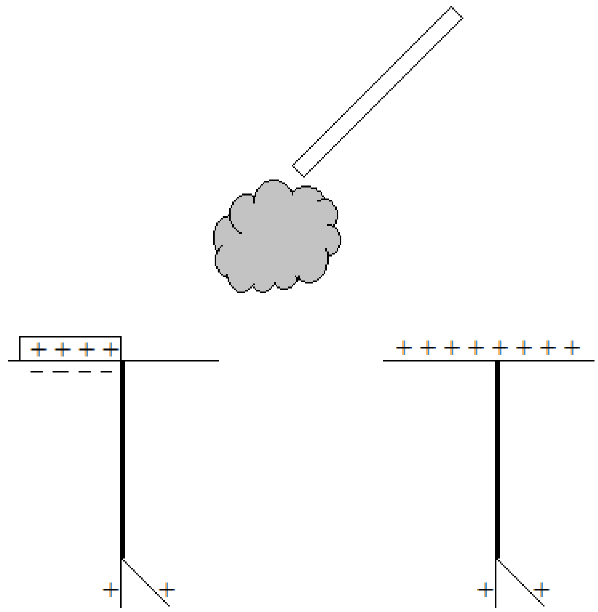
Charging an electroscope

There are methods of charging an electroscope.

- By contact
- By induction

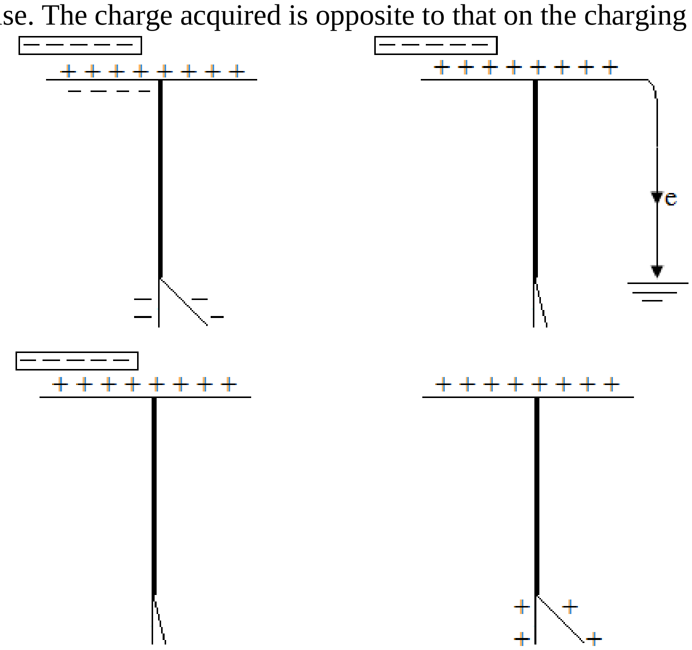
1. Charging by contact

When a glass rod is rubbed with fur, it acquires positive charges. The rod is then rubbed firmly across the metal cap of the electroscope. The negative charges are attracted from the plate and the leaf to the charged rod. The charge acquired by the electroscope is the same as that on the charging rod.



2. Charging by induction

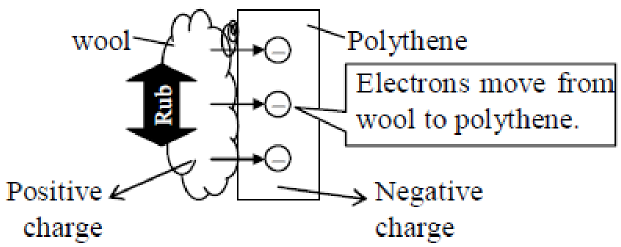
Using the law of electric charges, when a negatively rod is brought near the metal cap (not touching), electrons are repelled to the metal plate and the leaf leaving a net positive charge on the metal cap. Touching the metal cap at this point while still retaining the charged rod at its position ensures that the rod maintains the positive charges at the top of the metal cap due to the attraction between the negatively charged rod and the positive charges. Electrons then flow to the earth through the body and the leaf collapses. The finger is then removed before the rod. A net positive charge is left on the electroscope causing the leaf to rise. The charge acquired is opposite to that on the charging rod.

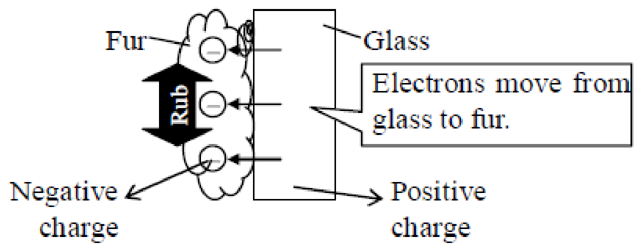


For example, when you wear or take off a sweater in very cold and dry season, you can get small amount of electric shock.
It is caused as a result of the sweater being charged.
When is an object charged?
- When two different substances are rubbed, they are charged.

How are they charged by rubbing?
- The atoms of all substances consist of protons, electrons and neutrons. Usually, the atoms have the same numbers of protons and electrons, therefore are electrically neutral. When one object is rubbed with another object, some electrons escape from one object and move on to the other object. One object decreases in electrons.

It is positively charged. Otherwise, the other object increases in electrons. It is negatively charged

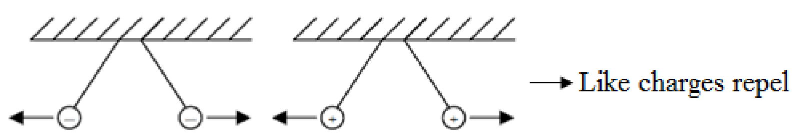




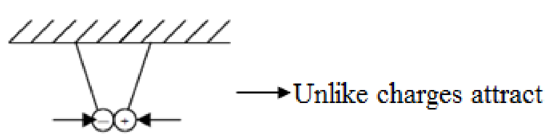
Another example of charge, when you rub a pen by a tissue or your hair, it is also charged. Then, if some small pieces of tissue are placed near the pen, they are attracted.
Why does rubbed pen attract pieces of tissue?
- If two objects have the same charges, they repel each other. It is called ‘like charges repel’ (1). Otherwise, if they have different charges, they attract each other. It is called ‘unlike charges attract’ (2). And any charged object (positive or negative) can attract uncharged objects because charges are induced in an uncharged object. This separate charge in an object is called ‘induced charge’ (3).

If a pen is charged, it induces charges on pieces of tissue. Therefore, the pen attracts pieces of tissue.

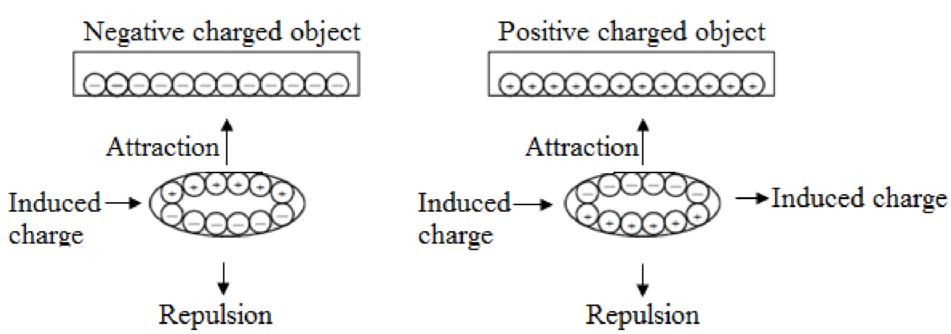
1. Repulsion



2. Attraction



3. Attraction by induced charge

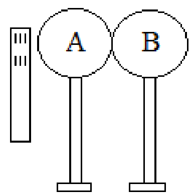


Lightning

In a thunderstorm, the clouds are charged by friction with airflow. Lightning is the discharge of electrons occurring between two charged clouds or between a charged cloud and the earth. Due to the huge amount of charges on the cloud, it can produce heat which can burn forests, damage houses and kill people.

Exercise

1. Two metal spheres A and B each stand on an insulating base and are in contact. A negatively charged rod is brought near to the sphere A as shown below.



   (a) Explain, in terms of electrons, the differences between conductors and insulators.
   (b) What effect does the charged rod have on the electrons in A and B?
   (c) In what way will A and B differ if separated while the rod is near?
   (d) State one use of static electricity.
2. Explain why lightning comes faster than thunder.

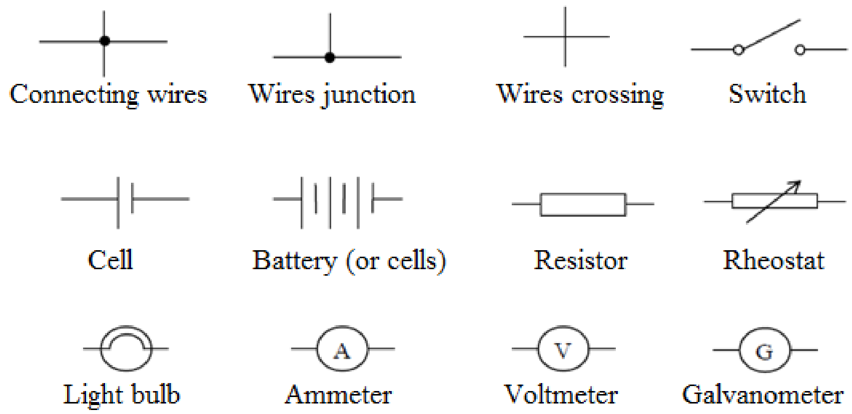
3. Two balls, A and B, are brought near each other with the following static charges. Complete the table.

Charge on ball A	**Charge on ball B**	**Attract of repel**
+	+	(a)
+	—	(b)
—	(c)	Repel
—	(d)	Attract
+	Uncharge	(e)
—	Uncharge	(f)

4. **A** and **B** are two balls which carry electric charges. Initially, A has a charge of +4 units and B has a charge of –2 units.
   (a) If 1 unit of positive charge are added to both of them, what will be the direction of the force on A and B?
   (b) If 4 units of negative charge are added to both of them, what will be the direction of the force on A and B?

UNIT 9: CURRENT ELECTRICITY

Symbols of electrical component



Electric current

Symbol: I

SI unit: Ampere [A]

Definition: **Current** is the rate of the flow of charge.

Formula: $I = \frac{Q}{t}$

I = Current [A]

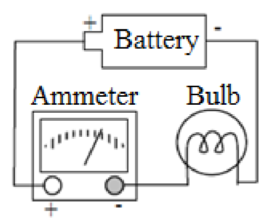
Q = Charge [C]

The unit of charge is coulomb [C]

T = time [s]

When we use electrical appliances, electric current flows in the circuit. The flowing current means that the electrons flow in the circuit.

- Current is measured using an instrument called Ammeter.



- An ammeter must be connected to a component in series.
- The positive terminal (usually red terminal) must be connected to the positive terminal of the battery.
- The negative terminal (usually black terminal) must be connected to the negative terminal of the battery.

Example

1. A motor uses a current of 20A for 10s. How much charge flows through it?

Data	Solution
Q = ? I = 20A t = 10s	Q = It Q = 20A×10s Q = 200C

Electromotive force

Symbol: e.m.f

SI unit: volt [V]

Definition: **Electromotive force (e.m.f.)** of a cell (or battery) is defined as the energy supplied to each coulomb of charge within it.

Formula: $e.m.f = \frac{E}{Q}$

e.m.f = Electromotive force [V]

E = Energy supplied by the cell [J]

Q = Charge flow through the cell [C]

Note

- 1V = 1J/C

Potential difference

Symbol: P.d

SI unit: volt [V]

Definition: **Potential difference (p.d.)** is defined as the energy converted per unit charge passing through a component.

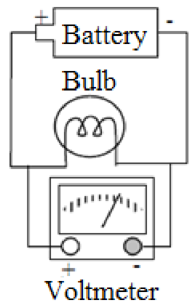
Formula: $V = \frac{E}{Q}$

V = Potential difference [V]

E = Energy converted to other forms in the component [J]

Q = Charge flow through the component [C]

The instrument used for measuring potential difference is called Voltmeter.



- A voltmeter must be connected across a component in a circuit (parallel to a component).
- The positive terminal (usually red terminal) must be connected to the positive terminal of the battery.
- The negative terminal (usually black terminal) must be connected to the negative terminal of the battery.

Note

- Energy carried by charge from a cell or a battery is consumed in electrical components like resistor, lamp, bulb or heater of the circuit. For example, when the charges flow through the bulbs in a circuit, their energy is converted to light and heat energy. This consumed energy is called the potential difference across the component.

Examples

1. 60 C of charge flow through a bulb which transfers 180J of energy into light. What is the potential difference?

Data	Solution
V =? Q = 60C E = 180J	$V = \frac{E}{Q}$ $V = \frac{180J}{60C}$ V = 3V

2. When a current of 2.5A flows for 8s through a bulb, 240J of energy are consumed.
   (a) How much charge flows through the bulb?
   (b) What is the potential difference across the bulb?

	Data	Solution
(a)	Q = ? t = 8s I = 2.5A	Q = It Q = 2.5A×8s Q = 20C
(b)	V = ? Q = 20C E = 240J	$V = \frac{E}{Q}$ $V = \frac{240J}{20C}$ V = 12V

Electrical resistance
Symbol: R
SI unit: Ohm [Ω]
Definition: The **resistance** in a circuit is the opposition to the flow of current.
Formula: $R = \frac{V}{I}$
This relationship is called **Ohm's law**
R = Resistance [Ω]
V = Potential difference [V]
I = Current [A]

Examples

1. A current of 2A flows through a conductor. The conductor has the p.d. of 12V. Find the resistance of the conductor.

Data	Solution
R =? V = 12V I = 2A	$R= \frac{V}{I}$ $R = \frac{12V}{2A}$ 6Ω

2. Find the p.d. across a 1.5Ω resistor when a current of 4A flows through it.

Data	Solution
V =? R = 1.5Ω I = 4A	V= IR V = 4×1.5 V = 6V

3. Find the current flowing through a 5Ω resistor that has 20V across it.

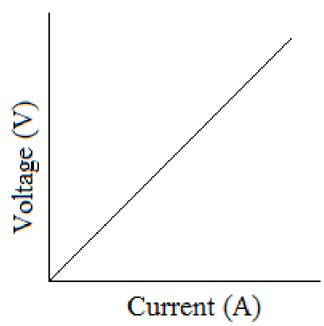
Data	Solution
I =? V= 20V R = 5Ω	$I= \frac{V}{R}$ $I = \frac{20V}{5\Omega}$ = 4A

Ohm's law
Ohm's law states that *the current, I, passing through a conductor is directly proportional to the voltage, V, provided the temperature is constant.*

Note

- *Some components in a circuit such as a bulb or a heater have resistance.*
- *A device which provides some resistance in the circuit is called a* ***resistor****.*
- *A resistor which can vary resistance is called the rheostat.*

The graph of voltage against current is a straight line passing through the origin for an ohmic conductor.



The slope of the graph gives the resistance of the conductor.
$R= \frac{V}{I}$

Conditions under which ohm's law must be obeyed

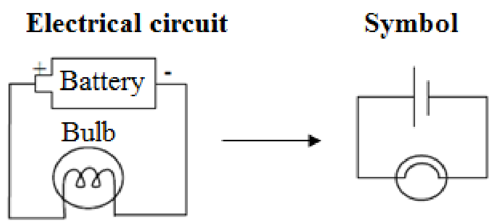
1. The temperature of the conductor should be remain constant. Increase in temperature increases the resistance of the conductor.
2. The conductor must not be subjected to any mechanical strain for example, being stretched or bent. The length of the conductor determines its resistance.
3. The conductor should not be placed at right angles to a straight magnetic field.

Note

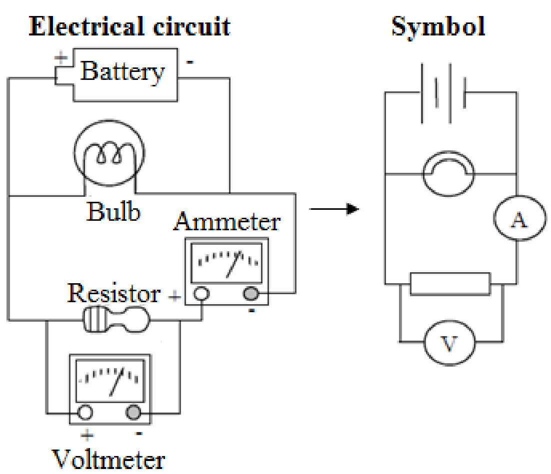
- *A conductor which obeys ohm's law is called ohmic conductor and the one that does not is called a non – ohmic conductor.*

Electrical circuit

Example 1



Example 2

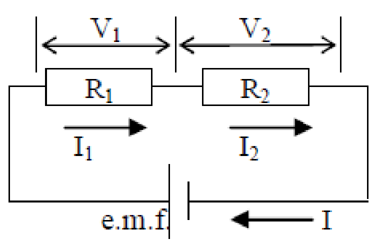


Arrangement of resistors

Resistors can be arranged in various ways:

- In series
- In parallel
- Combining both series and parallel

Resistors in Series



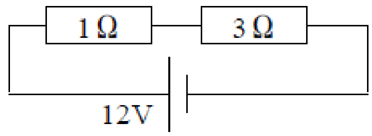
- The current is the same at all points in the series circuit.
  $I = I_1 = I_2$
- The sum of the p.d. (V) across the resistors (the total resistance) is the same as the e.m.f.

$\text{e.m.f} = V_1 + V_2 = V$

- The total resistance, R, of the components connected in series circuit is equal to the sum of the separate resistances.
  $R = R_1 + R_2$

Example

1. Study the circuit diagram below and answer the questions that follow.

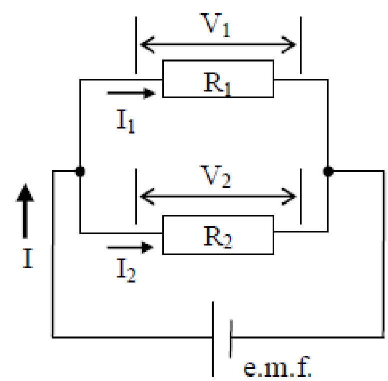


Find

(a) the total resistance.
(b) the current through the cell
(c) the p.d. of 1Ω resistor.
(d) the p.d. of 3Ω resistor.

	Data	**Solution**
(a)	$R = ?$ $R_1 = 1\Omega$ $R_2 = 3\Omega$	$R = R_1 + R_2$ $= 1\Omega + 3\Omega$ $= 4\Omega$
(b)	$I = ?$ $V = V_1 + V_2 = \text{e.m.f.} = 12V$ $R = 4\Omega$	$I = \frac{V}{R}$ $I = \frac{12V}{4\Omega}$ $= 3A$
(c)	$V_1 = ?$ $R_1 = 1\Omega$ $I_1 = I = 3A$	$V_1 = I_1R_1$ $V_1 = 1\Omega \times 3A$ $V_1 = 3V$
(d)	$V_2 = ?$ $R_2 = 3\Omega$ $I_2 = 3A$	$V_2 = I_2R_2$ $V_2 = 3A \times 3\Omega$ $V_2 = 9V$

Resistors in Parallel



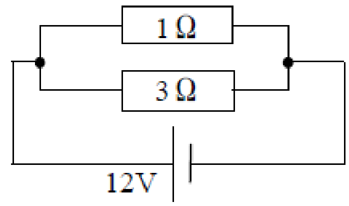
- The current in the main circuit is the sum of the currents in the separate branches.
  $I = I_1 + I_2$
- Each component (resistor) in a parallel arrangement has the same p.d. across it.
  $V = V_1 = V_2$ (= e.m.f.)
- The reciprocal of the total resistance is equal to the sum of the reciprocal of individual resistances.
  $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

Note

- *When there are only two resistors involved, resistance can be found using the formula*: $R = \frac{R_1 \times R_2}{R_1 + R_2}$

Example

1. Study the circuit diagram below and answer the questions that follow.



Find

(a) the total resistance
(b) the p.d. of 1Ωresistor
(c) the p.d. of 3Ω resistor
(d) the current through 1Ω resistor
(e) the current through 3Ω resistor
(f) the current through the cell

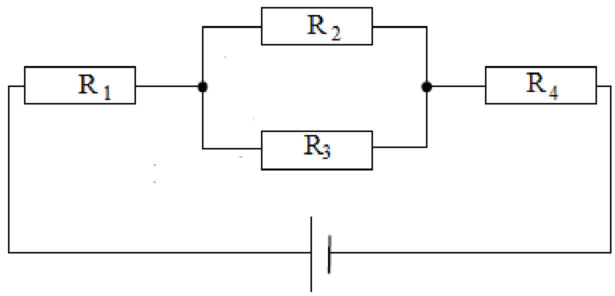
	Data	**Solution**
(a)	R = ? $R_1 = 1\Omega$ $R_2 = 3\Omega$	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ $\frac{1}{R} = \frac{1}{1} + \frac{1}{3}$ $\frac{1}{R} = \frac{4}{3}$ 4R = 3 $R = \frac{3}{4}$ R = 0.75Ω
(b)	e.m.f. =12V	V_1 = e.m.f. = 12V
(c)	e.m.f. =12V	V_2 = e.m.f. = 12V
(d)	$I_1 = ?$ $V_1 = 12V$ $R_1 = 1\Omega$	$I_1 = \frac{V_1}{R_1}$ $I_1 = \frac{12V}{1\Omega}$ $I_1 = 12A$
(e)	$I_2 = ?$ $V_2 = 12V$ $R_2 = 3\Omega$	$I_2 = \frac{V_2}{R_2}$ $I_2 = \frac{12V}{3\Omega}$ $I_2 = 4A$
(f)	I = ? $I_1 = 12A$ $I_2 = 4A$	$I = I_1 + I_2$ I = 12A + 4A I = 16A

Exercise

1. A resistor uses a current of 25A for 6s. Find the charge that flows through the resistor.
2. In a circuit, a charge of 16C passes through a point in 4s. Find the size of the current.
3. A current of 5A flows through a cell which supplies 120J of energy in 16s.
   (a) How much charge flows through the cell?
   (b) What is the e.m.f. of the cell?
4. Find the p.d. of a resistor which consume 690J of energy when a current of 3A flows in 10s.
5. How much current would flow through a resistor of 60Ω if its p.d. is 150V?
6. A light bulb has 4Ωresistance. When 3.5A of current passes through it, what is the p.d. of the bulb?
7. If 1.8A of current flows through a resistor which has the p.d. of 3.6V, what is the resistance?

Combining both series and parallel

When the arrangement of the resistors is a combination of both resistors in series and resistors in parallel, then steps need to be taken to calculate the total resistance. The effective resistance for all resistors in parallel should be determined using the formula for resistors in parallel, then the total resistance of the resistors in parallel added to those in series.



The resistors in parallel, R_p

- $R_p = \frac{R_2 R_3}{R_2 + R_3}$

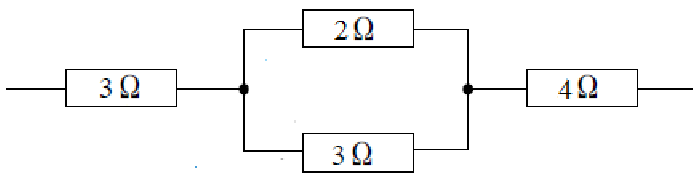
  $R_p = \frac{1}{R_2} + \frac{1}{R_3}$

In series (Total resistance, R_T)

- $R_T = R_1 + R_p + R_4$

Example

1. Calculate the effective resistance in the following arrangement



Solution

Step 1

The resistors in parallel, R_p

$R_p = \frac{R_2 R_3}{R_2 + R_3}$

$R_p = \frac{2\Omega \text{ x } 3\Omega}{2\Omega + 3\Omega}$

$R_p = \frac{6\Omega}{5\Omega}$

$R_p = 1.2\ \Omega$

Step 2

In series (Total resistance, R_T)

$R_T = R_1 + R_p + R_4$

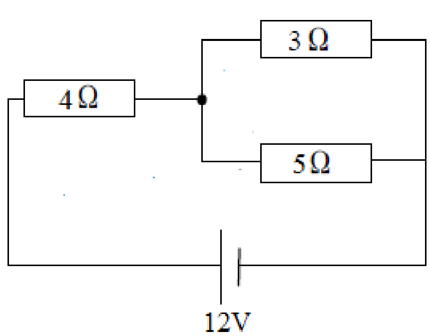
$R_T = 3\ \Omega + 1.2\ \Omega + 4\ \Omega$

$R_T = 8.2\ \Omega$

The total resistance in the circuit is 8.2 Ω

Exercise

1. Study the circuit diagram below and answer the questions that follow



Calculate

(a) The total resistance

(b) The total current flowing in the circuit

(c) The current through 3 Ω resistor

(d) The current through 5 Ω resistor

Electrical power

Symbol: P

SI unit: watt [W]

Definition: Electrical power is the rate of using electrical energy.

Formula: P = VI

P = Electrical power [W]

V = Potential difference [V]

I = Current [A]

Note

- $P = \frac{E}{t}$
- E = VIt
- A bulb of 60W converts 60J of electrical energy into light and heat energy per second.

Example

1. A 12V battery is giving off a current of 2A to a resistor. Find the power dissipated in the resistor.

Data	Solution
P =? V = e.m.f. = 12V I = 2A	P = VI P = 12V×2A P = 24W

2. A p.d. of 12V is applied across the 4Ωresistor. Find the power dissipated in the resistor.

Data	Solution
I = ? P =? V = 12V R = 4Ω	$I = \frac{V}{R}$ $I = \frac{12V}{4\Omega}$ I = 3A P = VI P = 12V×3A P = 36W

Cost of electrical energy

When you use the electricity supplied by ZESCO, you have the electricity meter. In the meter, you can find the unit of

Kilowatt-hours (kWh). By using this unit, the cost of electrical energy is calculated. 1kWh is called 1 unit.

How to calculate the cost of electrical energy

- Calculate the energy consumed in kWh.
  E= Pt
  E = Energy consumed [kWh]
  P = Power of electrical components [kW]
  t = time taken [hr]
- Calculate the cost of electrical energy by cross multiplication.

Example

1. If electrical energy costs K50 per unit, what is the total cost by using 1000kwh?

Solution

1unit → K50

1000 unit → x

$x = \frac{1000 \text{ units x K50}}{1 \text{ unit}}$

x = K50, 000

2. A light bulb of 100W is used for 7hours. What is the energy cost if the energy costs K50 per unit?

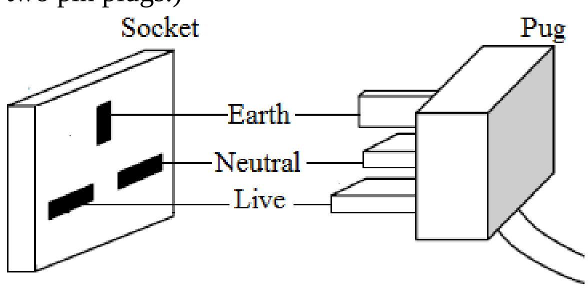
Data	Solution
E = ? P = 100W = 0.1kW t = 7h	E = Pt E = 0.1kWh×7h E = 0.7kWh 1unit → K50 0.7unit → x $x = \frac{0.7 \text{ units x K50}}{1 \text{ unit}}$ x = K35

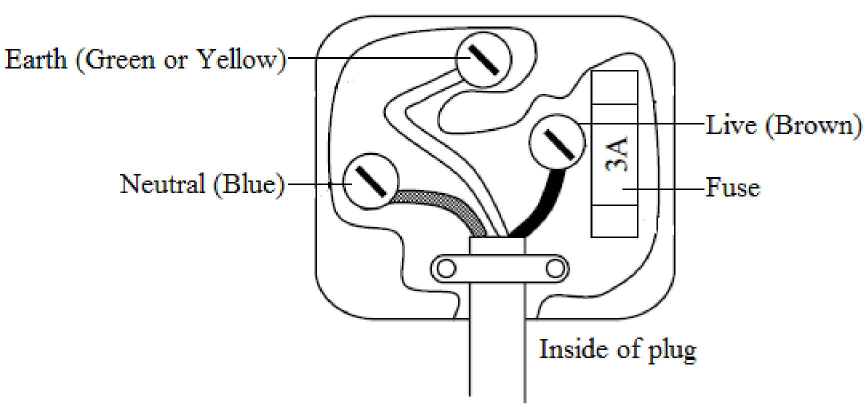
3. 4-security lights of 120W are turned on for 30days. What is the energy cost if it costs K60 per unit?

Data	Solution
E = ? P = 120W×4 = 480W P = 0.48kW t = 30 days = 30 days×24hrs t = 720hrs	E = Pt E = 0.48kW×720h E = 345.60kWh 1unit → K60 345.6units → x $x = \frac{345.6 \text{ units x K60}}{1 \text{ unit}}$ x = K20,736

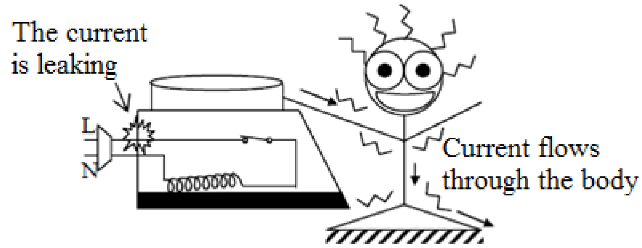
Use of electricity in the house

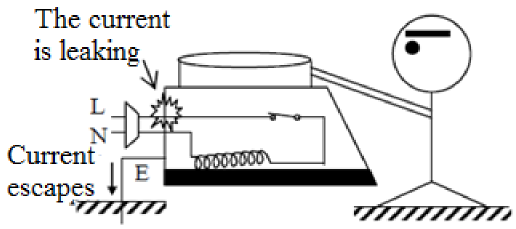
When we use an electrical appliance in our house, a plug of the appliance is connected to a socket. Zambia uses three pin plugs. (Japan uses two pin plugs.)





- Live wire is a brown wire. It supplies the electrical energy to the appliance. The line has a high voltage. If you touch this line, it is dangerous because the current flows through your body. You may die of electric shock.
- Neutral wire is a blue wire. It makes the circuit complete in the electrical appliance.
- Earth wire is a green or yellow wire. This wire is connected to the metal casing of an electrical appliance. If live wire is in contact with the metal casing due to any accident, the user gets the electric shock from the metal casing. If earth wire is connected to the metal casing, the current escapes from the earth wire. It protects the user from the electric shock.





Dangers of electricity

Contacting electricity (especially the live wire) is dangerous and causes some accidents.

- It causes the electric shock to human beings. A large current can be fatal.
- It may cause fires or burns in an electrical appliance, the plug and the socket.

Dangers of electricity can be caused by three cases shown below.

1. **Damaged insulation**
   The electrical wires (cables) are insulated. If those insulators are removed by the deterioration, the live wire can be contacted to somewhere and it can cause electric shock and fire.
2. **Overheating of cables**
   If a large current flows in the wires or components, it can cause overheating. Then it can melt the insulation and start a fire. A short circuit or overloading is easy to cause this accident.
3. **Damp condition**
   In damp condition such as a wet bathroom, the current flows through the human body easily. Because the body's resistance depends on whether the skin is wet or dry.

Safe use of electricity in the house

To use the electricity safely, there are some electrical components. They are shown below.

Earth wire

- It protects the user from the electric shock.

Double insulation

- Some electrical appliances are double insulated. It makes the leakage of current difficult.

Switch

- The function of switch is to turn on or off the electrical appliance. In the case of leakage, the switch can be used as the safety device to cut off the current. The switch should be installed on the live wire so that the electrical appliance is disconnected from high voltage when the switch is open.

Fuse

- If too much current flows through an electrical component, the component can overheat or start a fire.
- The fuse prevents too much current from flowing through it. If too much current flows through a fuse, a wire in the fuse melts and it intercepts too much current from live wire. Therefore it is installed on the live wire.

Electrical appliances

An appliance is a device that uses electricity.

Examples of electrical appliances

- Electric heater
- Electric kettle
- Electric fan
- Pressing iron
- Washing machine

Fuse rating

The fuse rating is the maximum current that the fuse can carry without melting. We should choose a proper fuse rating.

- If we choose a large fuse rating, it allows too much current to flow.
- If we choose a small fuse rating, the electrical appliance doesn't work.
- The fuse rating should be slightly larger than the working current of an appliance under normal operation.
- Available fuse ratings are 3A, 5A, 13A, 15A or 30A.
- Ratings are usually a combination of power and voltage or voltage and current. Sometimes power may be used on its own for the rating.

Example

1. A refrigerator is rated at 240V 480W. Which fuse should be used, 3A or 13A?

Data	**Solution**
I =? V = 240V P = 480W	P = VI $I = \frac{P}{V}$ $I = \frac{480W}{240V}$ I = 2A 3A is a proper fuse rating

Exercise

1. 5A of current flows in a 12V bulb. Find the power of the bulb.
2. Two bulbs with resistance of 4Ω and 6Ω are connected in series. In the circuit, 2A of current flows.
   (a) Calculate the p.d. of each bulbs.
   (b) Calculate the power dissipated by each bulb.
3. A TV of 150W is switched on for 6hours. Calculate the cost assuming it costs K50 per unit.
4. A heater of 5kW and a cooker of 3kW run for 15hours. If a unit costs K60, what is the total cost?
5. An electrical cooker of 1kW uses an electrical supply of 240V. Which fuse should be used, 3A, 5A, 13A or 30A?
6. Two identical bulbs are connected in parallel and series. In which type of circuit are the bulbs brighter? Give a reason in terms of electrical power.
7. When two bulbs, 60W and 100W, are connected in parallel, which one of them is brighter? What if connected in series? Explain the reason in terms of electrical power.

Transformers

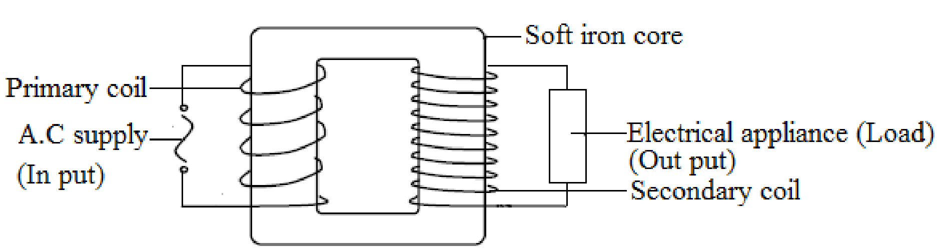
Definition:

Transformer is a device used to vary the voltage of an a.c. supply.

It is a device which is used to change the voltage of an appliance (load) by mutual induction.

Structure of a transformer

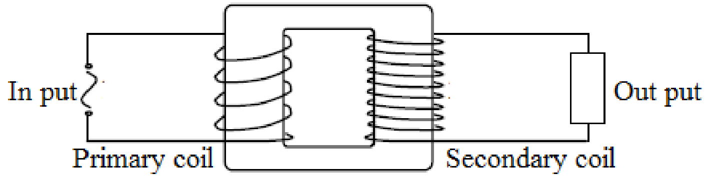
- A transformer consists of two coils (primary coil and secondary coil) wound on a soft iron core.
- The coil that is connected to the alternating current input is called primary coil and the coil that provides the alternating current output is called the secondary coil.



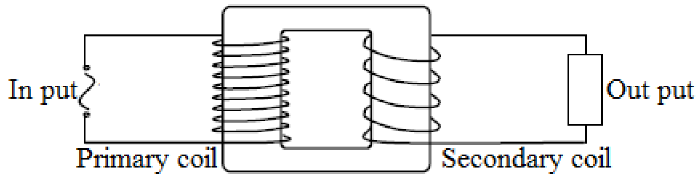
Types of transformers

There are two types of transformers

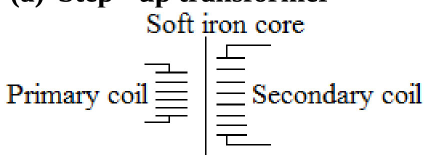
1. **Step - up transformer**
   This is a transformer which increases the voltage of an appliance.
   In the step up transformer:
   - The voltage in the primary coil (input) is lower than the voltage in the secondary coil (output)
   - The number of turns in the primary coil is less than the number of turns in the secondary coil



2. **Step - down transformer**
   This is a transformer which reduces the voltage of an appliance.
   In the step down transformer:
   - The voltage in the primary coil (input) is higher than the voltage in the secondary coil (output)
   - The number of turns in the primary coil is greater than the number of turns in the secondary coil

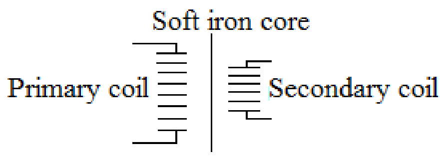


Principle of operation of a basic iron - cored transformer (How a transformer works)
A transformer functions by mutual induction. That is;
An alternating voltage applied to the primary coil causes an alternating current to flow in the coil. The alternating current induces a changing magnetic field.
The changing magnetic field induces an alternating voltage in the secondary coil. This causes flow of alternating current in the secondary coil.
Circuit symbols
(a) Step - up transformer



- $N_p < N_s$
- $V_p < V_s$
- $I_p > I_s$

(b) Step - down transformer



- $N_p > N_s$
- $V_p > V_s$
- $I_p < I_s$

Note
- N_p = Number of turns in primary coil
- N_s = Number of turns in secondary coil
- V_p = Voltage of primary coil
- V_s = Voltage of secondary coil
- I_p = Current in primary coil
- I_s = Current in secondary coil

A transformer will not operate using a direct current input because direct current produces a steady magnetic field which cannot induce a voltage in the secondary coil.
Transformers are used to transmit electricity because they can easily convert the type of voltage needed. For domestic purposes, a step down transformer can be used to drop a very high voltage to a suitable voltage in our homes. A step up transformer can be used to amplify the voltage so that industrial areas can utilize such high voltages

Factors that cause energy loses in a transformer and how this can be minimized

If a transformer has efficiency 100%, it is called ideal transformer.
However, no transformer is ideal. This means that a transformer cannot be 100% perfect. It has energy loses.
The following are factors that can cause energy loses in a transformer and how they can be minimized

1. **The resistance of the coils**
   As the coils have resistance, they give off heat when current flows through. Coil resistance and energy loses can be minimized by making the coils from thick copper because thick copper does not heat up easily.
2. **Magnetization and demagnetization of the core**
   Work has to be done to alter sizes and direction of domains and heat is released in the process. These energy loses are reduced by making the core from soft iron because soft iron is easy to magnetize and easy to demagnetize
3. **Eddy currents in the core**
   Eddy currents are small currents produced within the iron. These occur because the core itself is a conductor in a changing magnetic field. The energy loses are reduced by laminating the iron core.

Advantage of transmitting electrical energy using high voltage

This can reduce energy loses due to long distances since the energy is transmitted from long distances, the wires offer resistance. Some energy will be lost in the cables due to heating effect.

Advantage of transmitting electrical energy using alternating current

Alternating current can be transformed to higher voltage; which is efficient to transmit

Transformer equations

- $\frac{N_p}{N_s} = \frac{V_p}{V_s}$
- $I_pV_p = I_sV_s$

Note

- N_p = Number of turns in primary coil
- N_s = Number of turns in secondary coil
- V_p = Voltage of primary coil
- V_s = Voltage of secondary coil
- I_p = Current in the primary coil
- I_s = Current in secondary coil

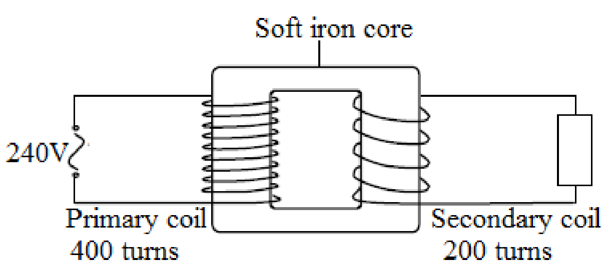
Efficiency of a transformer

- Efficiency = $\frac{V_sI_s}{V_pI_p}$ x 100%

Transformer calculations

Examples

1. The figure below represents a transformer with a primary coil of 400 turns and a secondary coil of 200 turns.



(a) If the primary coil is connected to a 240V a.c mains supply, calculate the secondary voltage
(b) Distinguish between the step-down and step-up transformers

(c) Explain carefully how a transformer works
(d) Why is the core made of iron?

Solution

(a) $\frac{N_p}{N_s} = \frac{V_p}{V_s}$

$V_s = \frac{V_p \text{ x } N_s}{N_p}$

$V_s = \frac{240V \text{ x } 200}{400}$

$V_s = 120V$

(b) A step down transformer reduces the voltage of an appliance while a step up transformer increases the voltage of an appliance.
In the step down transformer, the voltage in the primary coil is higher than the voltage in the secondary coil while in the step up transformer the voltage in the primary coil is lower than the voltage in the secondary coil.
In the step down transformer, the number of turns in the primary coil is greater than the number of turns in the secondary coil while in the step up transformer; the number of turns in the primary coil is less than the number of turns in the secondary coil.

(c) An alternating voltage applied to the primary coil causes an alternating current to flow in the coil. The alternating current induces a changing magnetic field.
The changing magnetic field induces an alternating voltage in the secondary coil. This causes flow of alternating current in the secondary coil.

(d) Because soft iron can magnetize and demagnetize easily.

2. The primary coil of a transformer is connected to a 240V a.c mains and a current of 5A passes through. If the voltage at the secondary coil is 12V, calculate the secondary current.

Data	Solution
I_s =? I_p = 5A V_s = 20V V_p = 240V	$I_S V_S = I_P V_P$ $I_S = \frac{I_p \text{ x } V_p}{V_s}$ $I_S = \frac{5A \text{ x } 240V}{12V}$ I_S = 100A

Exercise

1. The primary coil of a transformer has 800 turns; its secondary coil has 2400 turns. Voltage in the primary coil is 50V;
   (a) Calculate voltage in the secondary coil;
   (b) Given that the current in the secondary coil is 12A, determine the current in the primary coil.
2. A step down transformer is required to transform 240V a.c to 12V a.c for a model railway. If the primary coil has 1000 turns. How many turns should the secondary coil have?
3. A transformer has a primary coil of 8400 turns and a secondary coil of 3500 turns. Find the output voltage if 240V is supplied to the primary coil.
4. A power plant supplies 25kV voltage of a.c. supply. The voltage increases to 230kV through a step-up transformer.
(a) If 15000 turns coil is in the primary, calculate the number of turns in the secondary.
(b) 230kV of voltage is transformed again through a step-down transformer. The turns ratio of the primary coil to the secondary coil in the transformer is 115 to 6. Calculate the voltage of the secondary coil.
5. 240V of voltage is supplied to the primary coil and 5A of current flows through it. Find the current flowing through the secondary coil if the output voltage is 120V and the efficiency is 100%.
6. A refrigerator that is rated at 120V 480W is connected to the transformer. The transformer is connected to the power supply of 240V. Assuming that the efficiency is 100%.
   Calculate;
(a) the current through the refrigerator.
(b) the current from the power supply.
(c) If the turns in the primary are 8500, how many turns are in the secondary?

UNIT 10: ELECTROMAGNETIC INDUCTION

Definition: If any electrical conductor (e.g. copper wire) move in the magnetic field and cut the magnetic flux, an e.m.f. is induced in the conductor. This is called *Electromagnetic induction.*

Faraday's law of electromagnetic induction

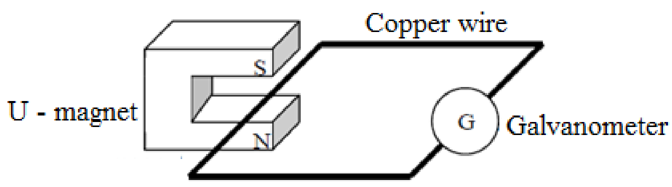
The law states that *the strength of the induced e.m.f. is proportional to the rate of change of the magnetic flux.*

This law is called Faraday's law of electromagnetic induction.

Experiment

Aim: To observe the electromagnetic induction

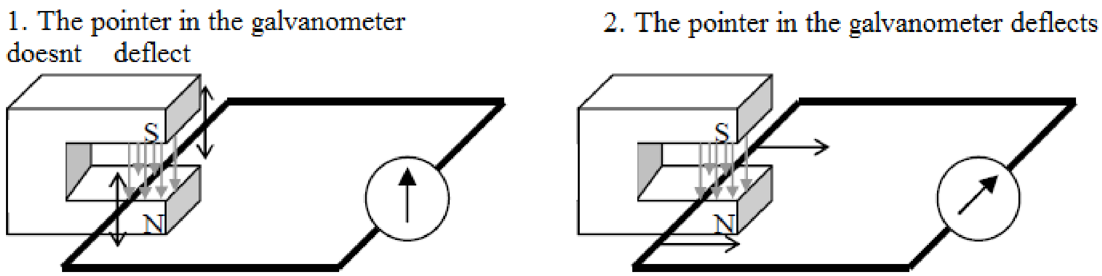
Experimental set up



Operations

1. Move the wire up and down between ends of U-magnet.
2. Withdraw the wire from U-magnet.

Results



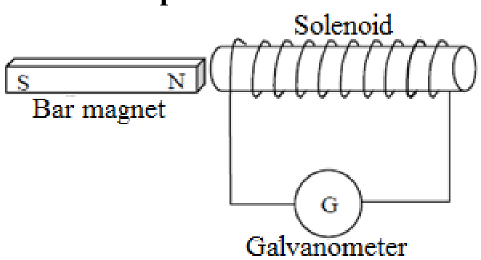
Conclusion

When the copper wire cut the magnetic flux (the wire moves perpendicular to the direction of magnetic field), an e.m.f. is induced in the wire.

Experiment

Aim: To observe the electromagnetic induction by the solenoid.

Experimental set up

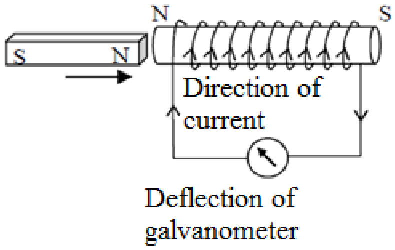


Operations

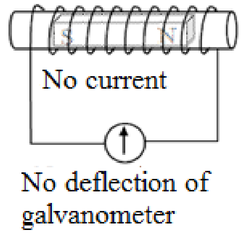
1. North Pole of the magnet move into the solenoid.
2. The magnet is stationary in the solenoid.
3. North Pole of the magnet move away from the solenoid.
4. South Pole of the magnet move into the solenoid.
5. South Pole of the magnet move away from the solenoid.

Results

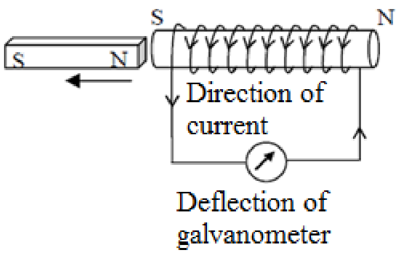
1. North pole of the magnet moves into the selonoid.



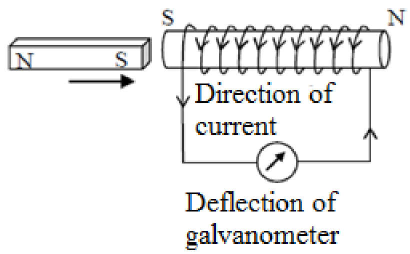
2. The magnet is stationary in the selonoid.



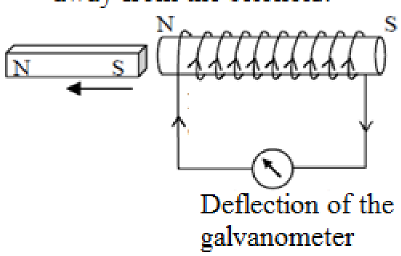
3. North pole of the magnet moves away from the selonoid.



4. South pole of the magnet moves into the selonoid.



5. South pole of the magnet moves away from the selonoid.



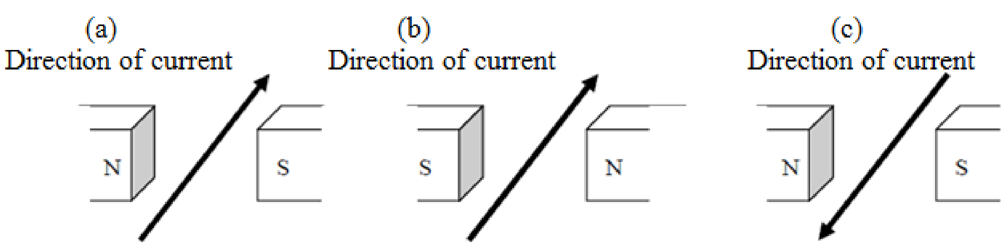
Conclusion

If a magnet moves towards a solenoid, the solenoid makes a magnetic field tending to repel it.
If a magnet moves away from a solenoid, the solenoid makes a magnetic field tending to attract it.
Then current flows according to their magnetic field.
This law is called Lenz's law.
To increase the e.m.f (induced current),

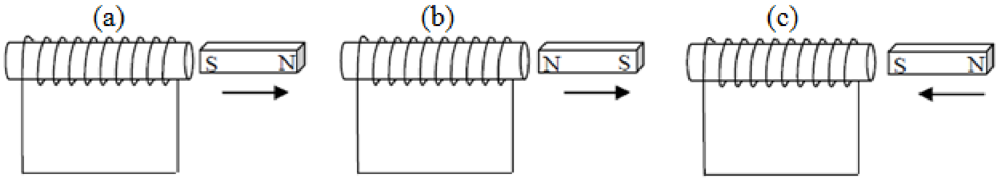
- Move the magnet at higher speed.
- Use a stronger magnet.
- Increase the number of turns in the coil.

Exercise

1. Find the direction of the force action on a wire in each situation.



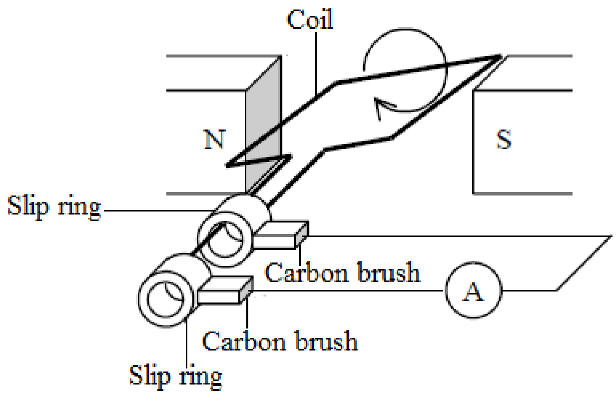
2. Draw the direction of an induced current in each situation.



Simple A.C. generator

- A.C. Generators generate alternate current by electromagnetic induction.
- Alternate current is the current which changes its direction a number of times per second.

Structure of a simple A.C. generator



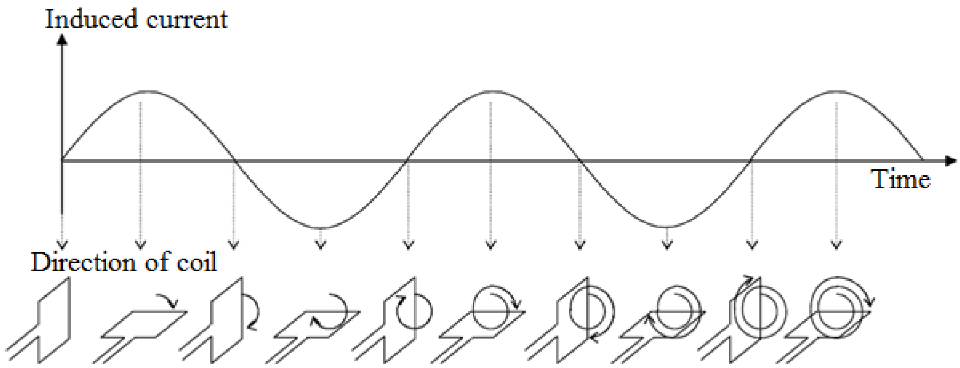
Functions of the components of the A.C. generator

1. Coil
Coil produces electricity by electromagnetic induction.

2. **Permanent magnet**
Permanent magnets produce the magnetic field.

3. Carbon brush
Carbon brushes keep the contact with the rings continuously.
Then coil is electrically connected to the outer circuit.

4. Slip ring
Slip rings are fixed to the coil.
All of them are drivers rotating in a body by the outside power source like an engine or a turbine.
If slip rings are replaced with split ring commutators, the generator becomes a direct current generator.
The diagram below shows the relationship between the induced current and directions of the coil in the A.C. generator.



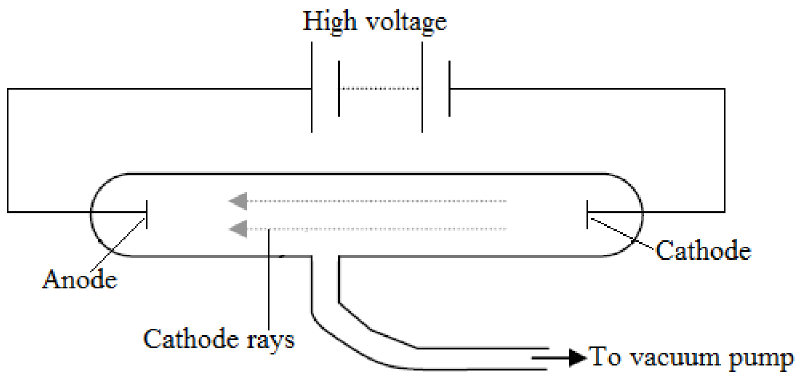
To increase the induced current (e.m.f.),

- the coil should rotate faster.
- a stronger magnet should be used.
- the number of turns in the coil should be increased.
- the coil should be wound around a soft iron core.

The frequency of the induced current is the number of revolutions of the coil per second.

- Zambia uses the power supply of 240V 50Hz. This means the induced current has 50 waves per second. This also means that the coil rotates 50 times in one second.

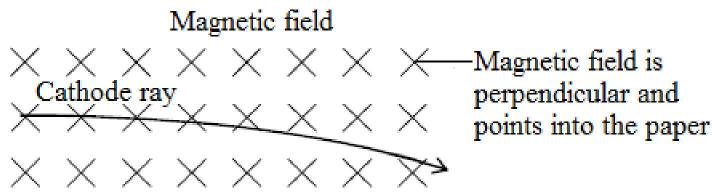
UNIT 11: BASIC ELECTRONICS
INTRODUCTORY ELECTRONICS
Thermionic emission and electrons
Thermionic emission is simply the release of electrons from a heated cathode. It is the emission of electrons from a hot surface.
A hot surface emits electrons. This emission of electrons from a hot surface is called **thermionic emission**.
Cathode rays
Cathode rays are a stream of high velocity electrons emitted from the surface of a heated cathode (negative electrode) inside a vacuum tube.
The production of cathode rays
- The discharge tube (cathode ray tube) allows electric current to pass through a gas at low pressure. The discharge tube consists of two electrodes: the anode (positively charged), and the cathode (negatively charged), which are connected to a high voltage source. The discharge tube is connected to a vacuum pump.

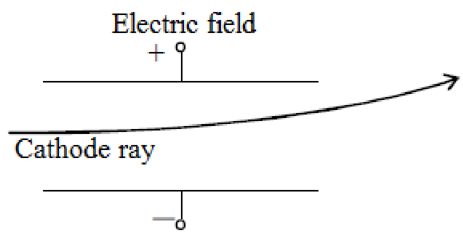


- Initially when the tube is still filled with air at atmospheric pressure, and a high positive potential applied to the electrode no current is detected.
- As the air is pumped out of the tube, the pressure decreases, and the gas becomes ionized, the electron escaping from it, ionizes other gas atoms. Therefore creating a stream of positive ions and negative electrons which move towards the cathode and anode respectively generating a current. At this stage, a bluish glow is noticed which turns pink and as the pressure is further decreased, the space between the electrodes turns dark (Faraday's space). Further reduction of pressure causes this dark space to expand and a greenish glow to appear behind the anode and the sides of the tube to fluoresce (green). The atoms move from the cathode towards the anode giving them the name cathode rays.

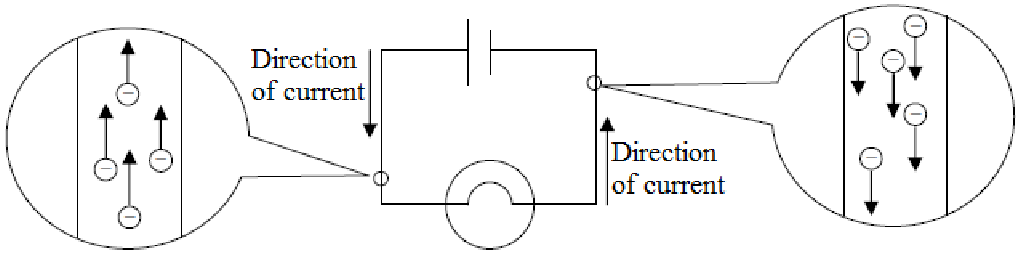
Note
- *If electrodes (cathode and anode) are set in vacuum and connected to high voltage of power supply, **electrons** are emitted from the **cathode** (Negative side of circuit) and flow to the **anode** (Positive side of circuit). The flowing of electrons from cathode to anode is called the **cathode ray.***
- *The position of the anode does not affect the direction of cathode rays.*
- *As air is pumped out of the discharge tube, the electrons at the cathode get attracted to the anode due to the high potential difference.*

Properties of cathode rays
1. They travel in a straight line and are perpendicular to the surface of the cathode in the absence of magnetic or electric fields.
   If an opaque object is placed on the path of the cathode rays, a sharp shadow of the object is cast on the Fluorescent screen.
2. Cathode rays are a beam of negatively charged electrons. Electrons are negatively charged so that they attract to anode (positive side).
3. Cathode rays have energy (kinetic) and can exert mechanical force on objects.
4. Cathode rays are deflected by magnetic and electric fields. (They have a negative charge)



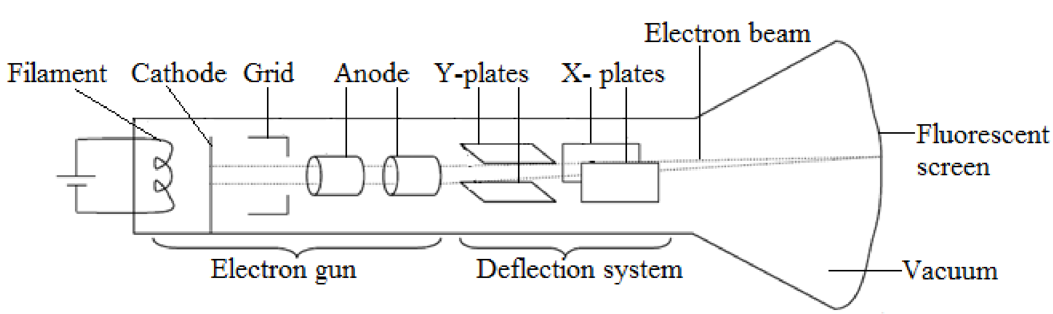


5. Cathode rays produce fluoresces (glow) when they strike certain substances
6. When very high velocity cathode rays are suddenly stopped by a metal surface they produce X- rays.
7. The flow of electrons is opposite to the direction of current in the circuit. The current is called conventional current.



Cathode Ray Oscilloscope (C.R.O.)

- This is an instrument used to investigate the voltage and current waveforms in electronic and electric circuits.
- It is an electrical instrument used to display and analyze wave forms as well as to measure the electrical potentials or voltages that vary over time. It can display alternating current or direct current waveforms. These are shown as visible graphical patterns representing the electrical signals. The cathode ray oscilloscope is made up of a cathode ray tube (CRT) that produces cathode rays, thus its name cathode ray oscilloscope.
  The evacuated cathode ray tube is the most important part of the oscilloscope.



A Cathode Ray Oscilloscope consists of three main parts.

- Electron gun
- Deflection system
- Display system (Fluorescent screen)

1. Electron gun

- An electron gun sends electrons through the vacuum to a fluorescent screen and a light spot appears on the screen.

- The electron gun produces an electron beam, which is a highly concentrated stream of high speed electrons.
- It consists of a cathode, a cylindrical grid and two anodes. The purpose of the cathode is to:
  (I) Emit electrons when heated
  (II) Concentrate emitted electrons into a tight beam (the grid is used for this purpose)
- The negative voltage on the grid can be varied to control the number of electrons reaching the anodes thereby altering the brightness of the spot on the screen. The grid is at a negative potential and it controls the rate of electron flow to the screen by repelling them. The high momentum of electrons carries them through past the anodes.
- The anodes have positive potential relative to the cathode, so they attract the electrons from the cathode. Their functions include:
  (I) To accelerate the electrons providing them with enough energy to cause emission of light as they hit the screen.
  (II) Focusing the electron beam (like a lens) as it leaves the grid, converging the electrons to a sharp point on the screen.

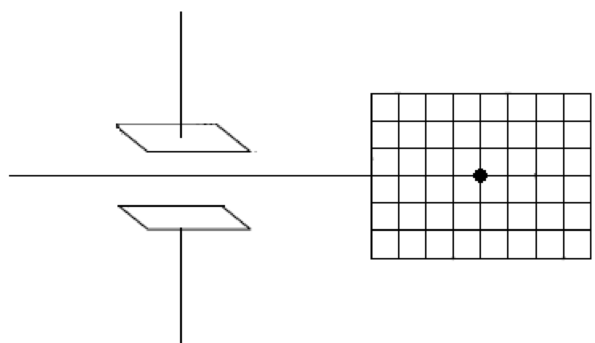
2. Deflecting system

- The deflecting system positions the electron beam on the screen. It consists of two pairs of parallel plates mounted in such a way that they deflect the electron beam (from the electron gun) horizontally and vertically controlling the production of the pattern on the luminous screen. There are two types of deflecting plates namely:
  (I) Y – plates (horizontally aligned)
  (II) X – plates (vertically aligned)

Y – Plates (vertical deflection)

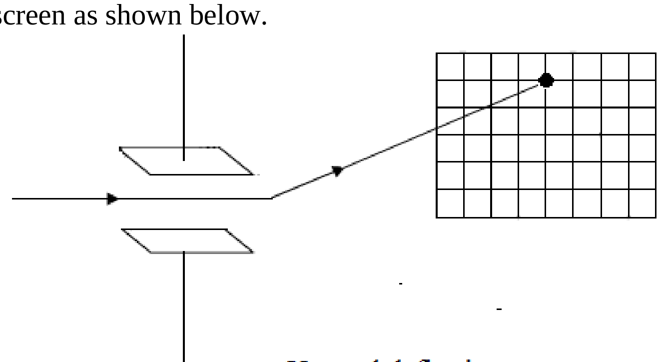
These deflection plates deflect the electron beam vertically across the screen in the following ways:

(I) When the potential difference (d.c) across both plates is zero, there is no deflection and the spot is produced on the screen as shown below.



Zero voltage - no deflection

(II) When the potential difference (d.c) is applied across the Y – plates with the top plate positive, the electrons are deflected upwards. The spot therefore appears on the upper part of the screen as shown below.

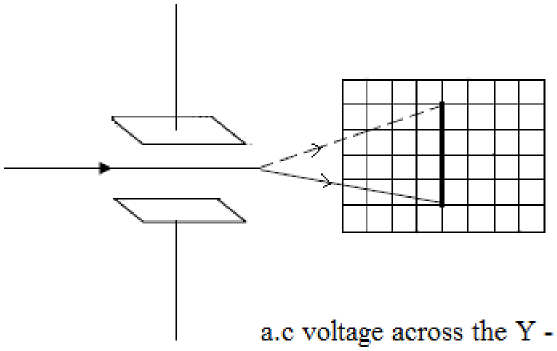


Upward deflection

Note

- *When the lower plate is positive, the deflection is downwards and the spot is on the lower part of the screen.*

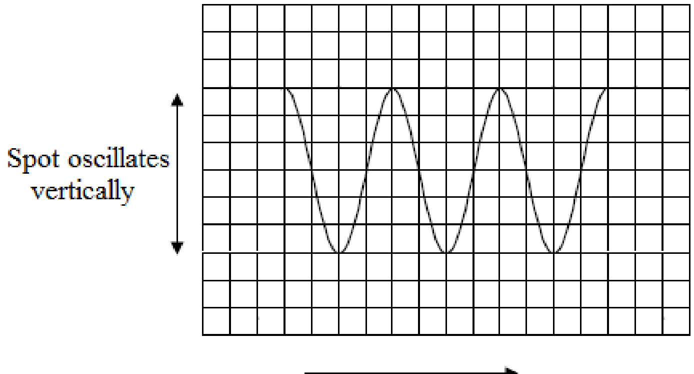
(III) If the a.c voltage is applied across the Y – plates, the spot oscillates up and down rapidly (depending on the frequency) such that what is seen on the screen is a straight line as shown below.



a.c voltage across the Y - plates

X – Plate (horizontal deflection)

- These deflection plates deflect the electron beam horizontally across the screen. If a.c is applied to both plates then the spot would oscillate up and down while at the same time move across the screen resulting in a wave form as shown below.



- When the spot completes one sweep of the screen, it flies back to the starting point and the process is repeated.

3. Display system

- The inside of the screen is coated with phosphor (zinc sulphide), which fluoresces or glows when electrons strike it producing a bright spot that can be seen on the outside surface of the screen. The image formed on the screen represents the voltages that are applied to the deflection plates.
- The inside of the tube near the screen is coated with graphite and it acts as the earth potential conducting electrons that strike the screen to the earth. This helps in preventing a build up of static charge on the tube.

Summary of the functions of the parts of the C.R.O

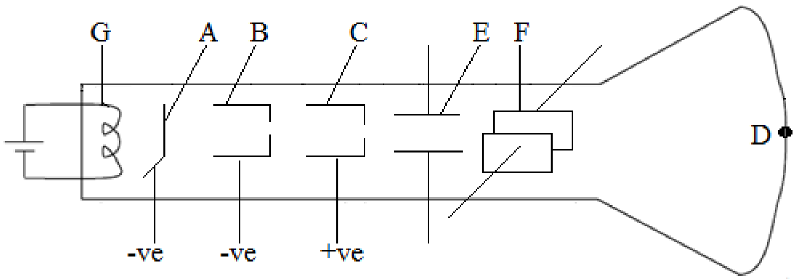
Name	**Function**
Filament	Heating up cathode
Cathode	Emitting electrons by thermionic emission
Grid	Brightness control by controlling amount of electrons passing through it
Anode	Focusing and accelerating of electron beam
Y- plate	Deflecting the electron beam up or down
X- plate	Deflecting the electron beam left or right
Fluorescent screen	Display of waveforms

Precautions

- The oscilloscope has a fragile and expensive vacuum tube. It should be handled gently to prevent it from damage.
- Oscilloscopes use high voltage to create the electron beam. These voltages remain for some time after switching off. Do not attempt to examine the inside of an oscilloscope.

Exercise

1. The diagram below shows the structure of the parts of a Cathode Yay Oscilloscope (C.R.O)



(a) Label parts **E** and **F**

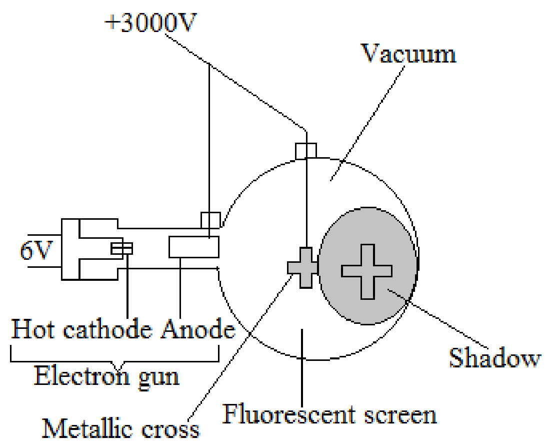
(b) Give the functions of:

(I) C

(II) G

(c) What would be the effect on the electrons of making part **B** very negative?

2. The diagram below shows a glass envelope (vacuum tube) containing a heater, metal cathode, metal anode and aluminium (metallic) cross, all of which are connected to suitable power supplies.



Due to this arrangement, a shadow of the cross appears on the screen.

(I) Explain why the shadow appears on the screen.

(II) What conclusion can draw from your observation

Uses of cathode ray oscilloscopes

1. Measuring potential difference (Voltage)

The cathode ray oscilloscope can be used as a voltameter to measure both d.c and a.c voltages.

2. Measuring d.c voltage

The time base is switched off and the voltage to be measured connected to the Y–input of the oscilloscope. The voltage is then applied to the Y–input displacing the spot on the screen vertically by the amount equivalent to the size of voltage and the Y–GAIN setting. The Y–GAIN control should be used to produce the maximum possible deflection which is then measured (in cm). This length is multiplied by the equilibrium on the Y–GAIN setting (in V/cm) to give potential difference in volts.

3. Measuring a.c voltage

The time base is switched off and the length of the vertical trace taken. This length is the peak- to-peak voltage. To obtain its value, we multiply the length (in cm) by the equilibrium on the Y- GAIN control (in V/cm).

Note

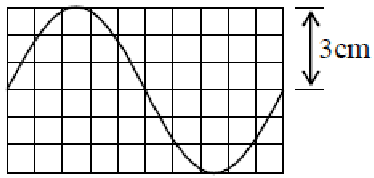
- *The Y-GAIN is the same as sensitivity*
- *Y-gain setting indicates the voltage applied in order to deflect the beam by 1cm on the screen in the vertical direction.*
- *If Y-gain setting is 5V/cm, it means that 1cm of the height on the screen shows 5V of input signal.*

Advantages of C.R.O as a voltmeter

1. It has an extremely high resistance and does not therefore alter the current or voltage in the circuit to which it is connected.
2. Both a.c and d.c voltages can be measured.
3. Responds instantaneously unlike ordinary meters because the electron beam is weightless.
4. Can measure large voltages without being destroyed.

Example

1. The diagram shows the screen of C.R.O. Y-gain setting is 3V/cm.



What is the peak voltage applied to the Y-input of the C.R.O.?

Solution

Y-gain 3V/cm: $3V \rightarrow 1cm$
Peak voltage: $x \rightarrow 3cm$

$3V \rightarrow 1cm$
$x \rightarrow 3cm$
$x = \frac{3V \text{ x } 3cm}{1cm}$
$x = 9$ V

2. The gain control of a C.R.O. is set at 2V/cm. If the horizontal trace is deflected upwards by 5cm, what is the unknown voltage applied to the Y-input of the C.R.O.?

Solution

Y-gain 2V/cm: $2V \rightarrow 1cm$
Unknown voltage: $x \rightarrow 5cm$

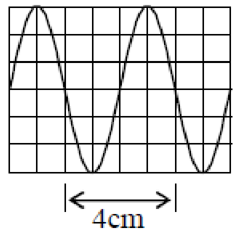
$2V \rightarrow 1cm$
$x \rightarrow 5cm$
$x = \frac{2V \text{ x } 5cm}{1cm}$
$x = 10V$

Measuring short time interval

- C.R.O. can be used to measure short time interval.
- Time base setting indicates the time needed for the light spot to sweep through 1cm on the screen in the horizontal direction.
  If the time base setting is 5ms/cm, it means that 1cm of the horizontal length on the screen shows 5ms (millisecond).

Example

1. The diagram shows the screen of C.R.O. The time base is set to 5ms/cm.



(a) What is the period of the input a.c. signal?
(b) What is the frequency of the input a.c. signal?

	Data	**Solution**
(a)	Time base: 5ms/cm Period : x ms	5ms → 1cm x → 4cm $x = \frac{5\text{ms x }4\text{cm}}{1\text{cm}}$ x = 20ms
(b)	f =? T = 20ms = 0.020s	$f = \frac{1}{T}$ $f = \frac{1}{0.02s}$ f = 50Hz

Exercise

1. The gain control of a C.R.O. is set at 0.2V/cm. If the horizontal trace is deflected upwards by 4cm, what is the unknown voltage applied to the Y-input of the C.R.O.?
2. The time base is set to 2ms/cm in a C.R.O. If one wave has a length of 5cm,
   Find
   (a) the period
   (b) the frequency

THE LOGIC GATES

Logic gates are switching circuits used in digital electronic systems, like computers.

All decisions making processes within the digital computer may be attributed to devices under the common name of **gates**.

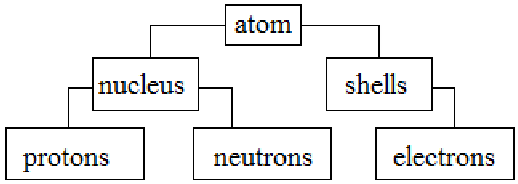
Storage and timing tasks in the digital computer can be carried out by a group of devices called **multivibrators**.

UNIT 12: ATOMIC PHYSICS

Nuclear Atom

Definition: An atom is the smallest particle of an element that can take part in a chemical reaction.

Structure of an atom



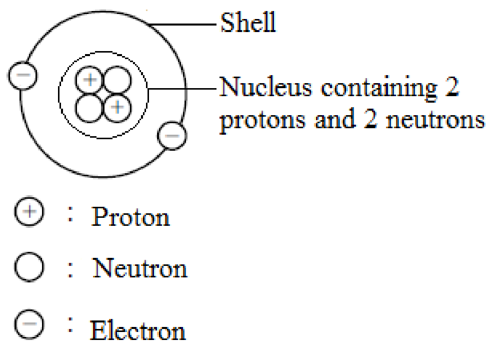
An atom is divided into the following two major parts:

1. The nucleus

The nucleus is in the middle of an atom. This is where protons and neutrons are found. The nucleus is the core of an atom.

2. **The shells or orbitals**

The shells or orbitals are around the nucleus and they contain electrons (also called the electron cloud) The diagram below represents an atom of Helium. The atom of Helium possesses two protons, two neutrons and two electrons.



Neutrality of an atom

- Atoms are electrically neutral or have no overall charge because the number of protons is equal to the number of electrons.

Note

- *In a Neutral particle, Number of electrons = Number of protons = Atomic number*

Fundamental particles of an atom

- There are three fundamentals of an atom. These are electrons, protons and neutrons.

Characteristics of the fundamental particles of an atom

1. Electron

Symbol: e

- It is a negatively charged particle.
- It has a charge of -1
- It is found in shells around the nucleus of an atom. A shell is a concentric ring around the nucleus
- Electrons move round the nucleus in different orbits.
- It has a mass of $\frac{1}{1840}$ atomic mass units (a.m.u)

2. **Proton**

Symbol: p

- It is a positively charged particle
- It has a charge of +1
- It is found in the nucleus of an atom
- It has a mass of 1atomic mass units (a.m.u)

3. **Neutron**

Symbol: n

- It is a neutral particle
- It has no charge
- It is found in the nucleus of an atom
- It has a mass of 1atomic mass units (a.m.u)

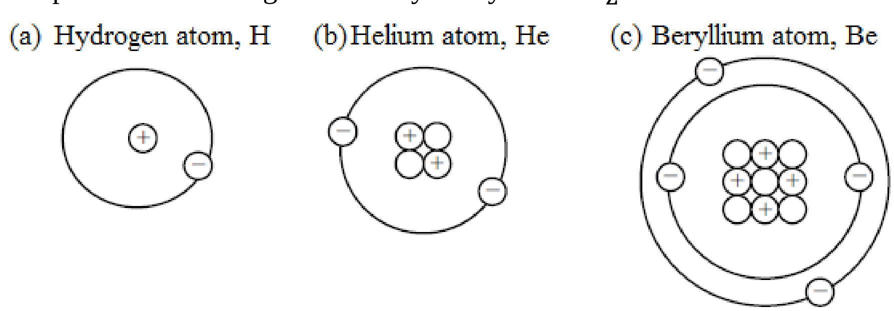
Summary

Particle	Symbol	Relative charge	Position in the atom	Relative mass
Electron	e	−1	Shells around the nucleus	$\frac{1}{1840}$
Proton	p	+1	Nucleus	1
Neutron	n	0	Nucleus	1

Proton number
Alternative term: *Atomic number*
Symbol: Z
Definition: It is the number of protons in the nucleus of an atom.
Neutron number
Symbol: N
Definition: Is the number of neutrons in the nucleus of an atom.
Mass number
Alternative term: *Nucleon number*
Symbol: A
Definition: It the sum of protons and neutrons in the nucleus of an atom
Formula: A = Z + N
A = mass number
Z = proton (atomic) number
N = number of neutrons

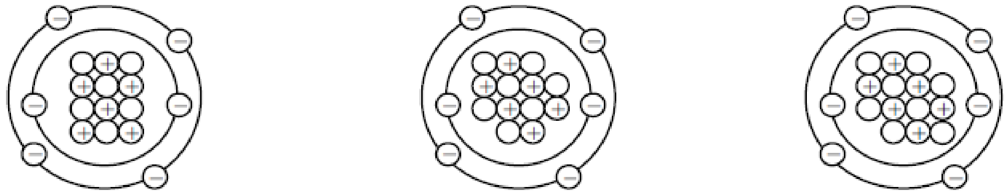
An element of chemical symbol X with a mass number A and an atomic number Z is expressed, ${}^{A}_{Z}X$
Example

1. Express the following elements by the symbol of ${}^{A}_{Z}X$



	Data	Solution
(a)	Mass number : 1 Atomic number : 1	${}^{1}_{1}H$
(b)	Mass number : 4 Atomic number : 2	${}^{4}_{2}He$
(c)	Mass number : 9 Atomic number : 4	${}^{9}_{4}Be$

Nuclide and Isotopes
- Each different form of nucleus is called a **Nuclide**.
- Atoms which have the same atomic number but different mass numbers are called **Isotopes** of an element. e.g. carbon



Atomic number: 6	Atomic Number: 6	Atomic number: 6
Mass number: 12	Mass number: 13	Mass number: 14
${}^{12}_{6}\mathrm{C}$	${}^{13}_{6}\mathrm{C}$	${}^{14}_{6}\mathrm{C}$

Carbon is an element.

Radioactivity

- Some elements which radiate energy of itself without any excitation from outside is called **radioactive elements**, e.g. Uranium, Radium, Thorium and Polonium
- This phenomenon of matter radiating energy of itself is called the **natural radioactivity**.

Radioactive decay

A nucleus, which has too many or too few neutrons gains extra energy, and becomes unstable. It tends to emit radiation such as α-particles, β-particles and γ-rays until a stable atom is reached. This emission of α-particles or β-particles is called **radioactive decay**.

Radioactive elements emit three types of radiation.

- Alpha (α) particles
- Beta (β) particles
- Gamma (γ) rays

Alpha particle

Symbol: α or ${}^{4}_{2}\mathrm{He}$

An alpha particle is a helium nucleus.

Alpha decay

Alpha decay is the emission of an alpha particle.

During alpha decay, the mass number reduces by 4 while the atomic number reduces by 2.

General equation: ${}^{A}_{Z}\mathrm{X} \rightarrow {}^{A-4}_{Z-2}\mathrm{Y} + {}^{4}_{2}\mathrm{He}$

The radioactivity decay emitting α-particle from the nucleus is called α (alpha) decay.

- For example, if ${}^{226}_{88}\mathrm{Ra}$ emits an alpha particle (2 protons and 2 neutrons) from the nucleus, the mass number changes from 226 to 222 and the atomic number changes from 88 to 86. Therefore, Radium changes into Radon (Rn) that has the mass number of 86 from the Periodic Table.

${}^{226}_{88}\mathrm{Ra}$	$\rightarrow$	${}^{222}_{86}\mathrm{Rn}$	$+$	${}^{4}_{2}\mathrm{He}$
Parent nuclide		Daughter nuclide		α-particle

Beta particle

Symbol: β *or* ${}^{0}_{-1}\mathrm{e}$

A beta particle is fast moving electron

Beta decay

Beta decay is the emission of a beta particle.

During beta decay, the mass number remain the same while the atomic number increases by 1.

General equation: ${}^{A}_{Z}\mathrm{X} \rightarrow {}_{Z+1}^{\;\;A}\mathrm{Y} + {}^{0}_{-1}\mathrm{e}$

The radioactivity decay emitting β-particle from the nucleus is called β (beta) decay.

- For example, if ${}^{241}_{94}\mathrm{Pu}$ emits a beta particle (1 electron) from the nucleus, the mass number doesn't change but the atomic number changes from 94 to 95 because 1 neutron changes to a proton. Therefore, Plutonium changes into Am (Americium) that has the mass number of 95 from the Periodic Table.

${}^{241}_{94}\mathrm{Pu}$	$\rightarrow$	${}^{241}_{95}\mathrm{Am}$	$+$	${}^{0}_{-1}\mathrm{e}$
Parent nuclide		Daughter nuclide		β-particle

Gamma rays

Symbol: γ

Gamma rays are electromagnetic waves.

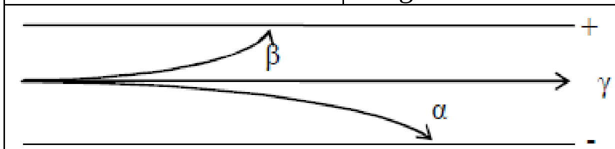
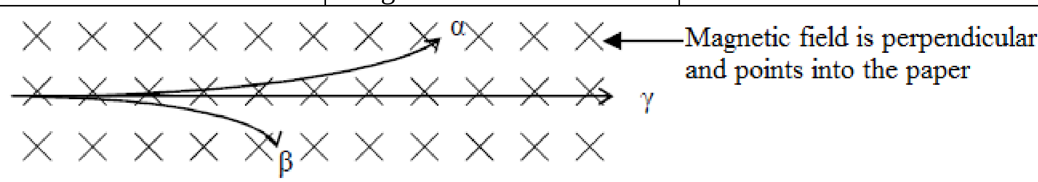
Gamma decay

Gamma decay is the emission of a gamma ray.

During gamma decay, the mass number and atomic number remain the same.

General equation: ${}^{A}_{Z}X \rightarrow {}^{A}_{Z}X + \gamma$

- The atomic number and the mass number do not change, the daughter nuclide is the same element as the parent nuclide.
- When some nucleus emit an α-particle or β-particle, they leave the nucleus in unstable energy condition. Therefore, the nucleus emits an extra energy, a γ-ray.

	α-particle	**β-particle**	**γ-ray**
Nature	Helium nucleus	Electron	Electromagnetic wave
Charge	+2	−1	Zero (Neutral)
Approximate mass	4	$\frac{1}{1840}$	Zero
Ionizing effect	Strong	Weak	very weak
Range in air	Very short	Short	Long
Penetrating power	Very weak	Weak	Strong
Absorbed by	a sheet of paper	5mm of aluminium	5cm of lead
Deflection in electric field	Small	Large	Un deflected
			
Deflection in magnetic field	Small	Large	Un deflected
			

Activity and Half-life

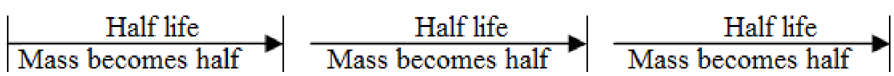
Definition: **Half-life** of a sample of radioactive element is defined as the time taken for half of the unstable nuclei to decay.

- For example, the half-life of radium is 1600 years. If there are 40g of radium initially, half of radium (20g) is decayed in first 1600 years. Next 1600 years, half of 20g radium (10g) is decayed. After next 1600years, half of 10g radium (5g) is decayed.

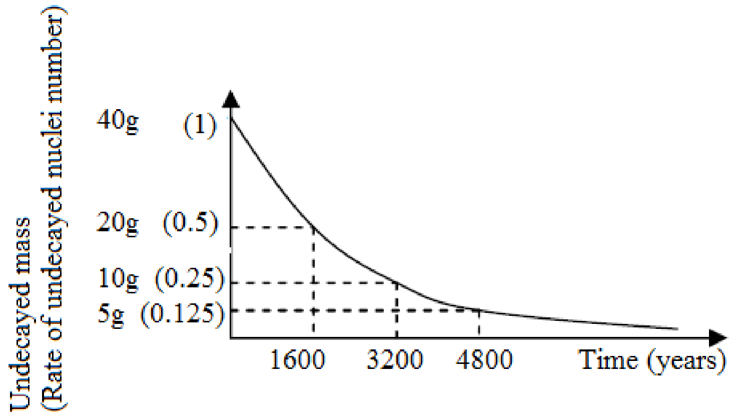
Decay curves

Decay curves are used to represent the decay rate of a radioactive substance. We can find the half-life of the radioactive substance from the decay curve.

Time	0 years later	1600 years later	3200 years later	4800 years later
Undecayed mass of Ra	40g	20g	10g	5g



The graph shows a decay curve for above example.



Example

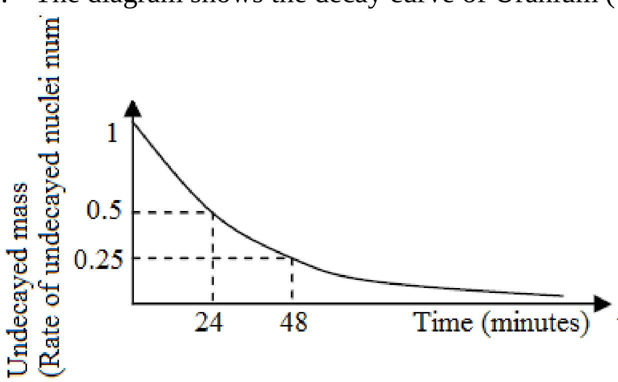
1. There is 1000g of Strontium (Sr) that has the half-life of 30 years.
   (a) How many grams of Strontium are remained after 60 years?
   (b) If 125g of strontium are remained, how long does it take?

Solution

Time	0 years later	30 years later	60 years later	90 years later
Undecayed mass of Sr	1000g	500g	250g	125g

   (a) 125g of strontium are remained 60years later.
   (b) 90years later.

2. The diagram shows the decay curve of Uranium (^{239}U).



What is the half-life?

Solution

Rate of undecayed nucleus number become half in 24 minutes. The half-life is 24 minutes.

Dangers of radiation

Radiation (α-particle, β-particle, γ-ray) can cause the following to human being.

- Damage to living cells
- Genetic changes in living cells.
- Cancer

Safety precautions when handling or storing radioactive sources / materials

- Avoid unnecessary exposure.
- Avoid direct contact with the source.
- Never point the source to any people.
- Radioactive sources must be stored in places out of reach of the public.
- Sources must be stored inside lead box to avoid leakage of radiations.
- Thick lead shields should be installed.
- Radiation symbols must be displayed at all places where radioactive sources are used.
- Radiation workers have to wear special protective clothing and gloves.
- Use film badges to measure the amount of radiation the radiation workers receive during a certain period.

Exercise

1. In an experiment to determine the half-life of a radioactive element, the following results were obtained.

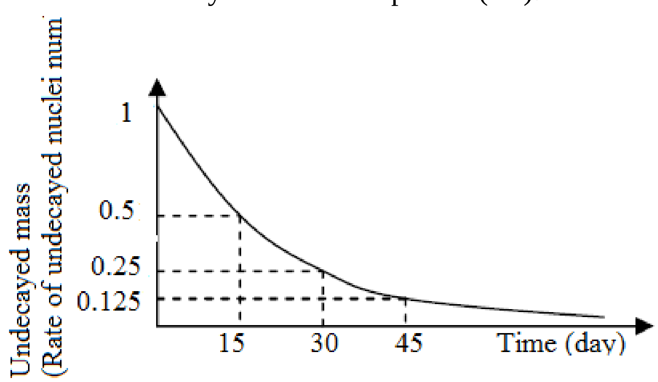
Count rate / counts per minute	1000	250	125
Time / seconds	0	110	160

   (a) State clearly what is meant by the half-life of a radioactive element.
   (b) From the results in the table, calculate
      (I) Two different half-life values of the element and
      (II) The average half-life of the element.

2. The element thorium $^{234}_{90}Th$ is radioactive. It decays by emitting beta particles and has a half-life of 24 days.
      (I) What is a beta particle?
      (II) Calculate the number of protons and neutrons in the nucleus of an atom of thorium.
      (III) Calculate the number of protons and neutrons in the nucleus formed when a thorium atom emits a beta particle.
      (IV) Calculate the time taken for 1g of thorium to decay having $\frac{1}{8}$g unchanged.
      (V) State one use of beta radiation.

3. There are 500g of Cobalt-60 (^{60}Co) that has the half-life of 5years.
   (a) How many grams of Cobalt are remained after 15 years?
   (b) If 125g of cobalt are remained, how long does it take?

4. The diagram below shows the decay curve of Phosphorus (^{30}P).



There are 400g of Phosphorus initially.
(a) What is the half-life?
(b) How many grams of Phosphorus is there 30 days later?
(c) If 25g of Phosphorus is remained, how long does it take?

STUDY AND EXAMINATION SKILLS

How to approach examinations

Planning your time

Read through your examination paper before you start writing. Plan how much time you will spend on each question

How to read questions

Read the instructions twice to make sure you understand what you have to do
Note the mark allocation so that you do not write a paragraph for one mark.

Question types

(a) Multiple – choice questions

- Each multiple – choice question / item will consist of two basic parts: a stem and a list of suggested solutions (alternatives / options). The stem will be in the form of either a question or an incomplete statement, a graph or diagram, and the list of alternatives will contain one correct option (key) and three incorrect alternatives (distracters)
- Each multiple – choice question will carry one mark and each correct answer will score one mark. A mark will **not** be deducted for a wrong answer. Candidates will be required to indicate their choice of answer from the given options by way of crossing or shading the letter indicating the chosen option. Read all the optional answers before you decide. Do not guess

(b) Structured questions

- These are compulsory questions which will comprise questions that will require candidates to supply responses in the form of words, numbers, symbols, phrases, definitions, comparisons, calculations etc. Give one word only if you asked to do so. Do not write full sentences unless you are asked to do so.
- Where the response needs units, candidates are expected to supply the correct units. Failure to do so will result in no mark being awarded even if the numerical is correct.
- Where working is required to be shown, and a candidate supplies a correct numerical response with wrong working or no working shown, no marks will be awarded
- The number of marks available per question will be shown in square brackets [ ] at the end of the question
- Candidates will be required to answer within the question paper in the spaces provided
- Non – programmable calculators are allowed

(c) Semi − structured questions

- These are questions where candidates are required to choose two or three questions and the questions will carry an equal number of marks
- The number of marks available per question will be shown in square brackets [ ] at the end of the question. Look at the mark allocation. The number of marks available per question will depend on the difficulty of the question i.e. the amount of writing required to generate a correct response.
- Candidates will be required to answer within the question paper in the spaces provided on the question paper or on a separate answer booklet provided
- Non – programmable calculators are allowed
- Make sure you understand the instructions and plan your answers.

Note

- *The structured / semi – structured / long compulsory practical questions will consists of sub – components requiring candidates to read instructions carefully, carry out experiments, make observations, draw diagrams, measure, record data, solve problems and make conclusions.*
- *Some questions included in the chemistry practical examination will present contexts and information that may be unfamiliar to candidates. Learners will be required to apply scientific principles and to relate their 'school science' to issues and contexts in the wider world, including the scientific community, in order to generate correct responses to such types of questions.*

GLOSSARY OF TERMS USED IN PHYSICS

The glossary (which is relevant only to physics) will prove helpful to candidates as a guide; it is neither exhaustive nor definitive. The glossary has been deliberately kept brief, not only with respect to the number of terms included, but also to the descriptions of their meanings. Candidates should appreciate that the meaning of a term must depend, in part, on its context.

1. **Define** (the term(s)……) is intended literally, only a formal statement or equivalent paraphrase being required.
2. **What do you understand by / what is meant by** (the term(s)…….) normally implies that a definition should be given, together with some relevant comment on the significance or context of the term(s) concerned, especially where two or more terms are included in the question. The amount of supplementary comment intended should be interpreted in the light of the indicated mark value.
3. **State** implies a concise answer with little or no supporting argument (e.g numerical answer that can readily be obtained by 'inspection'). It may also imply to give, say or write down the information asked for.
4. **List** requires a number of points, generally each of one word, with no elaboration. Where a number of points is specified this should not be exceeded.
5. **Explain** may imply reasoning or reference to theory depending on the context. Give the full details in full sentences and give reasons.
6. **Describe** requires the candidate to state in words (using diagrams where appropriate) the main points of the topic. It is often used with reference either to a particular phenomenon or to particular experiments. In the former instance, the term usually implies that the answer should include reference to (visual) observations associated with the phenomenon. In other contexts, 'describe' should be interpreted more generally (i.e. the candidate has discretion about the nature and organization of the material to be included in the answer). 'Describe' and 'explain' may be coupled, as may 'state and 'explain'.
7. **Discuss** requires the candidate to give a critical account of the points involved. It involves giving different ideas and arguments about the topic.
8. **Outline** implies brevity (e.g restricting the answer to giving essentials).
9. **Predict** implies that the candidate is not expected to produce the required answer by recall but by making a logical connection between other pieces of information. Such information may be wholly given in the question or may depend on answers extracted in an earlier part of the question. 'Predict ' also implies a concise answer with no supporting statement required.
10. **Deduce** is used in a similar way to 'predict' except that some supporting statement is required, e.g reference to law or principle or the necessary reasoning is to be included in the answer.
11. **Suggest** is used in two main contexts, i.e. either to imply that there is no unique answer (e.g in chemistry, two or more substances may satisfy the given conditions describing an 'unknown'), or to imply that candidates are expected to apply their general knowledge of the subject to a 'novel' situation, one that may be formally 'not in the syllabus' – many data response and problem – solving questions are of this type. It may also imply giving ideas, solutions or reasons for something.
12. **Find** is a general term that may variously be interpreted as calculate, measure, determine etc.
13. **Calculate** is used when a numerical answer is required. In general, working should be shown, especially where two or more steps are involved.
14. **Measure** implies that the quantity concerned can be directly obtained from a suitable measuring instrument (e.g length, using a rule, or mass, using a balance).
15. **Determine** often implies that the quantity concerned cannot be measured directly but obtained by calculation, substituting measured or known values of other quantities into a standard formula e.g relative molecular mass. It may also mean to find out.
16. **Estimate** implies a reasoned order of magnitude statement or calculation of the quantity concerned, making such simplifying assumptions as may be necessary about points of principle and about the values of quantities not otherwise included in the question
17. **Sketch**, when applied to graph work, implies that the shape and / or position of the curve need only be qualitatively correct, but candidates should be aware that, depending on the context, some quantitative aspects may be looked for (e.g passing through the origin, having an intercept). In diagrams, 'sketch' implies that simple, freehand drawing is acceptable; nevertheless, care should be taken over proportions and the clear exposition of important details.

FUNDAMENTALS OF PHYSICS

DARLINGTON NAOSA

ABOUT THE AUTHOR

Darlington Naosa started his primary education at Nakatete Primary School in Kafue from 1989 to 1995. He went to Naboye Secondary School for his junior secondary education in Kafue from 1996 to 1997. He continued his senior secondary education at Naboye Secondary School from 1998 to 2000. He went to Nkrumah Teachers College from 2002 to 2003 where he obtained the Secondary Teachers Diploma and graduated as a best student in Science. He went at the University of Zambia from 2012 to 2016 where he obtained the Degree of Bachelor of Education (Chemistry Education).

He has been Head of Department for Natural Sciences at Namushakende Secondary School from 2019. He has been teaching Chemistry at Kambule Technical Secondary School from 2009 to 2019. He has been lecturing Chemistry at Zambian College of Open Learning (ZAMCOL) in Mongu from 2015 and he has also been tutoring Chemistry at the University of Barotseland from 2018. He has been a marker for Integrated Science with the Examinations Council of Zambia (ECZ) from 2009. He previously taught Environmental Science at Kanyonyo Basic School from 2007 to 2009. He also taught Chemistry, Physics and Biology at Naboye Secondary School from 2004 to 2007.